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Peer Review File

Plasma proteomic profiles identify biomarkers predicting Crohn's disease up to 16 years before onset

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Feng et al. conducted an analysis using proteomics data from the UK Biobank, investigating the association between proteomics and incident Crohn's Disease (CD). Their study is notable, with data from 52,896 adults using the Olink Explore panel, among whom 139 developed CD over a median follow-up of 13.6 years. This is an interesting body of work that might provide new insights to the preclinical phase of CD. While the study is interesting, several concerns arise regarding the analyses and interpretation of the results:

-The average age at baseline for this cohort was 56.8 years (SD 8.2). However, it is unclear what the duration of follow-up is for the 139 individuals who developed CD. Summary statistics (minimum, maximum, IQR, median, mean) for the follow-up duration (from baseline to diagnosis) among the 139 cases should be provided. Additionally, the age at diagnosis for the 139 subjects should also be shown.
-Presumably, the age at diagnosis is close to or above 60 years, indicating that the CD cases in this cohort are predominantly elderly onset. This represents a relatively small subgroup of incident CD cases in the general population, which may limit the generalizability of the findings to the broader incident CD population. This limitation should be clearly acknowledged.
-The diagnosis date was determined by the earliest recorded appearance of ICD-10 codes. However, a likely discrepancy exists between the actual diagnosis date and the date of ICD-10 code appearance. Some cases diagnosed near the time of code appearance may have had earlier diagnoses that were not captured by the ICD-10 codes. Were any efforts made to reduce this potential bias? This is distinct from the issue of diagnosis delay. Did the authors set a minimum follow-up duration prior to a CD diagnosis? At a minimum, a sensitivity analysis excluding cases that developed CD within 1, 2, 3, 4, and 5 years before diagnosis should be conducted.
-In the baseline characteristics table, the authors compared those who developed CD over time to those who remained healthy as of October 2022, which should be explicitly stated.
-The authors employed a machine learning algorithm to combine the 10 significant proteins from either model 1 or model 2. It appears that they developed a model to classify a binary outcome and subsequently applied this model to cumulative cases over time, up to 16 years. Have the authors considered using a machine learning algorithm with a time-to-event extension? Additionally, can they provide calibration data for the model in the testing set?
-There is significant class imbalance in the dataset. The authors mention the use of balanced class weighting in the figure legend, but further details on how this was implemented should be provided.
-The authors seem to have used an optimal cut-off based on the Youden index and then reapplied the same cut-off to the same cohort from which it was derived. This raises concerns about potential overfitting. It is unclear whether the cut-off was derived from a training set and then applied to a testing set.

Minor Points:
-The authors used the optimal proteins from the PREDICTS study and applied them to the UK Biobank cohort. However, it should be emphasized that the SOMAscan assay and the Olink Explore panel are two different assays with limited overlapping proteins.
-The authors stated that the time before diagnosis in the GEM cohort was relatively short compared to the UK Biobank cohort. It should be noted that these two cohorts differ in how the CD diagnosis date was defined (physician-confirmed in GEM vs. ICD code appearance in UK Biobank) and represent different populations (at-risk first-degree relatives in GEM vs. an elderly general population in the UK Biobank).

-Please use the TRIPOD+AI reporting guideline for the machine learning model that was used in this manuscript.

Reviewer #2

(Remarks to the Author)
The core objective of this study is to develop a panel of serum protein biomarkers to predict Crohn's disease prior to disease onset. The study analyzes data from the UK Biobank (n > 52K), where baseline plasma proteomics measurements were taken, and patients were followed for up to 16 years. Additional variables, including clinical data, serum assays, polygenic risk scores, and demographic information, were also incorporated to compare the model based on the protein signature alone and in combination with these variables.

The methodology is standard, starting with feature selection based on univariate statistical analyses (Cox proportional hazards model) and proceeding to feature importance ranking with SHAP analysis (resulting in a selection of 10 proteins). These selected proteins were then used as inputs in the LightGBM algorithm (for protein-only, combined indicators, and other combinations). Validation was conducted using a random data split or 5-fold cross-validation (internal validation).

Overall, while the study may have some clinical merit, more robust validation is needed to clarify its potential utility. Here are some specific comments:

1) It is unclear whether feature selection was performed on the training data alone or on the entire dataset. Based on the description, it appears that feature selection may have been conducted on the entire dataset, with data splitting only performed for model training and validation. If true, this would introduce information leakage, compromising the model's generalizability.

2)The initial feature selection approach is univariate (Cox model), followed by SHAP analysis to filter out less contributive features based on correlation analysis. While this is a reasonable approach, it is fairly basic. There have been several advances in machine learning feature selection methods that could be leveraged to enhance model performance.

3) Only one machine learning model, LightGBM, was employed. Although LightGBM is a valid choice, the approach could be broadened by exploring multiple models or ensemble methods, such as meta-learning or superlearners, to integrate predictions from various algorithms and potentially improve performance.

4) Multi-modal integration: The study uses only a simple concatenation of features (early fusion) for multi-modal integration. This approach is valid but basic.

5) Internal validation: The study uses a single run of 5-fold cross-validation. To increase rigour and reduce selection bias, multiple runs of 5-fold cross-validation could be implemented, accounting for potential bias introduced by initial random ordering.

6) Lack of external validation: The absence of external validation significantly impacts the generalisability and clinical relevance of the findings. The authors could consider testing their signature panel in an external dataset, such as the PREDICT study (Torres, 2020). Notably, the PREDICT study reported an AUROC of 0.76, comparable to the findings here. However, upon validation within the UK Biobank samples, the AUROC in this study was observed to decrease, highlighting the importance of independent validation to confirm model robustness and applicability.

Reviewer #3

(Remarks to the Author)
The authors did thorough work with suitable statistical modelling.
The noteworthy results are the identification of the serum proteins most associated with the onset of CD with an identification cohort and a replication cohort. This might give significant insight into the biological processes underlying the onset of CD.
As a predictor for CD in high risk populations this is not clinically useful: there are no protective measures against the development of CD. Especially when the signal appears 16 yrs ahead of the actual development of CD, it's not much of a clinical predictor, but more an opportunity to gain insight in pathomechanisms.
The AUC of these 11 proteins plus known disease predictors is bigger than that from similar studies.
For the most extent the work supports the conclusions. The only issue I have is with their conclusion that their predictive model is reliable 16 years before the onset of CD. With only 159 CD patients in the UK Biobank with a large variation in the time to onset I think that this is a bit of a stretch.
With the caveat mentioned above, I think that the methodology is otherwise sound. I again think one should not interpret along the outer borders (outliers) of a small case group, but otherwise the combined phenotype+PRS+protein model seem to have an exceptionally good AUC.
The methodology is sound: machine learning with capped iterations would be the way to go for this work. The data could be reproduced since UK Biokbank is available on request and authors are transparent in their methods.
My main issue is with the fact that a prediction model such a s proposed here is of extremely limited clinical use, since there are no therapies to prevent CD. This becomes an ever bigger issue if this protein signature arises >10 yrs before disease onset; if it were closer to disease onset one could start monitoring calprotectin on a 6-monthly basis to detect active disease early and start treatment before complications arise.
This protein prediction model is however a gold mine for the biological mechanisms that role into action even before the true

onset of CD. I completely understand that the research group is not specialised in mucosal immunology or other wet lab work, but with the UK Biobank data and the other phenotypes present in this dataset and phenotypes associated with CD risk (depression, obesity etc) it would be worthwhile to check to which extend these proteins also predict depression or whether they are associated to smoking or obesity. To have a protein profile that predicts CD, and that is specifically associated with a "modifiable" phenotype could point into an actual preventative measure for CD with biological underpinnings.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors thoroughly responded to the comments that were raised. However, some concerns remain.

The follow-up duration from baseline to Crohn's disease (CD) diagnosis includes cases with a duration of 0.12 years (approximately 44 days) prior to diagnosis. Given the limitations of relying on the appearance of ICD codes—which may be biased toward a later diagnosis date—it is possible that some of these subjects had already been diagnosed. Therefore, caution is warranted when interpreting the results. The authors performed additional sensitivity analyses excluding individuals with shorter follow-up durations, which showed slightly lower predictive performance. The distribution of follow-up duration should be presented separately for both the training and testing cohorts.

Regarding the external validation cohort, it is unclear whether the Southern China cohort represents a new-onset inception cohort. A description of how this cohort was identified should be included. Additionally, it is important to report the disease duration and medication use for patients with CD. Regardless, it appears that this cohort does not provide external validation that the biomarkers precede CD diagnosis. As such, the validity of this external validation cohort is questionable.

As demonstrated by the calibration curves, there appears to be substantial overestimation, likely due to the use of balanced class weighting to address class imbalance. Building the model without adjusting for class imbalance would more accurately reflect predictive performance in the actual target population. This is important contextual information for readers and should be included as supplemental material and acknowledged in the discussion as a limitation, warranting caution when interpreting the results.

(Remarks on code availability)
Yes, they have provided the code.

Reviewer #2

(Remarks to the Author)
The authors have made a commendable effort in addressing the reviewers' comments. While the response letter was somewhat dense, it is clear that substantial effort went into revising the manuscript and improving its overall quality. In particular, the addition of external validation significantly strengthens the study by supporting its generalisability. I have no further comments.

(Remarks on code availability)
I reviewed the code to assess the approach used for external validation, specifically to verify whether the models were retrained on the validation data—a common pitfall in such evaluations. I can confirm that the external validation is implemented appropriately, with previously trained models being loaded and tested without retraining, which supports the robustness of the reported results.

That said, the codebase is somewhat disorganized and not easily reusable. Making it more accessible—e.g., through packaging for installation via pip—would improve usability. However, I do not consider this a major limitation.

Reviewer #4

(Remarks to the Author)
Authors' response to Reviewer 3.

The comments from reviewer 3 have been addressed sufficiently by the authors, especially with respect to the associated phenotypes, e.g., obesity and smoking. However, these results generate one important question as the nine proteins seem to be part of general immunological processes and homeostasis and not specific to Crohn's disease.

1. Patients with CD are more likely to develop other chronic inflammation-driven disease such as rheumatoid arthritis, diabetes, asthma, psoriasis and many more. These diseases can occur years before the CD diagnosis and are typically seen with increasing prevalence in cohorts of elderly patients, like this one. Do the authors have access to this kind of information on potential significant co-morbidities in the UK Biobank? If so, this should be account and adjusted for. If not, this should be mentioned and discussed accordingly, as other chronic inflammatory disease might impose a significant bias in the association between CD and the nine identified proteins.

(Remarks on code availability)

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have made substantial efforts and responded sufficiently to the comments that were raised. I appreciate the effort they have put into addressing these points. The addition of the EPIC-Norfolk cohort strengthens the paper and enhances the generalizability of predicting the onset of CD. I also thank the authors for showing the calibration curves, presenting the model performance without adjustment for class imbalance, and adding the discussion point regarding the caution needed in interpreting the results.

I have one remaining comment regarding the Southern China cohort. I do not think it is appropriate to state that biomarkers preceding CD were validated in a cross-sectional cohort with no prediagnostic samples. I suggest removing the label of “external validation” for the Southern China cohort, or considering removing this cohort altogether. The EPIC-Norfolk cohort already serves the purpose of external validation.

(Remarks on code availability)
The authors provided the R and Python scripts that were used for developing and validating the model.

Reviewer #4

(Remarks to the Author)
Review comments have been sufficiently addressed, thank you!

(Remarks on code availability)

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

Feng et al. conducted an analysis using proteomics data from the UK Biobank, investigating the association between proteomics and incident Crohn's Disease (CD). Their study is notable, with data from 52,896 adults using the Olink Explore panel, among whom 139 developed CD over a median follow-up of 13.6 years. This is an interesting body of work that might provide new insights to the preclinical phase of CD. While the study is interesting, several concerns arise regarding the analyses and interpretation of the results:

1. The average age at baseline for this cohort was 56.8 years (SD 8.2). However, it is unclear what the duration of follow-up is for the 139 individuals who developed CD. Summary statistics (minimum, maximum, IQR, median, mean) for the follow-up duration (from baseline to diagnosis) among the 139 cases should be provided. Additionally, the age at diagnosis for the 139 subjects should also be shown.

Authors' response:

We thank for the reviewer's constructive suggestions. We have now supplemented the description in the main text by providing the follow-up duration from baseline to CD diagnosis for the 139 individuals who developed CD, including the minimum, maximum, median, IQR, and mean. Additionally, we have included the age at CD diagnosis in baseline characteristics table. These modifications have been made in the corresponding parts of the manuscript, specifically in **2.1 Study Population** and **Supplementary Table 1**.

Noteworthy, our manuscript now includes a Southern China external cohort for model validation, and we performed testing by splitting UK Biobank participants based on recruitment centers, following the Type 2b design from the TRIPOD statement[1]. Compared to random splitting (Type 2a), this approach leverages meaningful geographic divisions, providing a more realistic assessment of model performance and enhancing robustness by accounting for site-specific variations. Therefore, Table 1 has been revised to reflect the baseline characteristics table divided by UKB training and testing sets. We have also attached the relevant modifications below for reference (**2.1 Study Population, Table 1, and Table 2**).

*"**Result 2.1 Study population** …The follow-up duration from baseline to CD diagnosis ranged from 0.12 to 14.88 years (median: 7.84 years; IQR: 4.37-10.74; mean: 7.47 years) (Supplementary Table 1).…"*

Supplementary Table 1: Baseline characteristics of UK Biobank participants by Crohn's disease case status.

	CD n=139	Control n=52,757	P value
Age at recruitment, mean (SD), years	58.76 (7.71)	56.81 (8.21)	0.005
Age at diagnosis, mean (SD), years	66.27 (8.82)	-	
Sex			0.678
Female, n(%)	72 (51.8)	28,445 (53.9)	
Male, n(%)	67 (48.2)	24,312 (46.1)	
Ethnicity			0.134
Other, n(%)	4 (2.9)	3,344 (6.3)	

White, n(%)	135 (97.1)	49,413 (93.7)	
TD index, mean (SD)	-0.88 (3.13)	-1.18 (3.19)	0.264
BMI, mean (SD), kg/m2	27.90 (4.81)	27.47 (4.80)	0.292
Physical activity			0.661
Physically inactive, n(%)	29 (20.9)	10,046 (19.0)	
Physically active, n(%)	110 (79.1)	42,711 (81.0)	
Smoking status			0.044
Never smoker, n(%)	61 (43.9)	28,670 (54.3)	
Previous smoker, n(%)	61 (43.9)	18,470 (35.0)	
Current smoker, n(%)	17 (12.2)	5,617 (10.6)	
Alcohol consumption			0.759
Daily or almost daily, n(%)	28 (20.1)	10,668 (20.2)	
Three or four times a week, n(%)	32 (23.0)	11,869 (22.5)	
Once or twice a week, n(%)	29 (20.9)	13,691 (26.0)	
One to three times a month, n(%)	16 (11.5)	5,760 (10.9)	
Special occasions only, n(%)	19 (13.7)	6,214 (11.8)	
Never, n(%)	15 (10.8)	4,555 (8.6)	
Dietary habit			0.088
Poor diet	97 (69.8)	32,928 (62.4)	
Healthy diet	42 (30.2)	19,829 (37.6)	
Depression (%)	10 (7.2)	3,043 (5.8)	0.591
Anxiety (%)	4 (2.9)	730 (1.4)	0.254
History use of antibiotics (%)	3 (2.2)	732 (1.4)	0.68
History use of NSAIDs (%)	1 (0.7)	844 (1.6)	0.244

Note: Data are presented as mean (SD) or n (%). Differences between participants who developed Crohn's disease over time and those who remained healthy as of October 2022 were assessed using Student's t-test for continuous variables and Pearson's chi-squared test for discrete variables.

Abbreviations: BMI, body mass index; TD index, Townsend deprivation index; CD, Crohn's disease; NSAIDs, nonsteroidal anti-inflammatory drugs.

*"**Results 2.1 Study population** After excluding participants diagnosed with CD at baseline and those with missing proteomic data, a total of 52,896 individuals from the UKB were finally included (**Fig. 1**). Of these, 39,634 participants were assigned to the UKB training cohort, while 13,262 participants from geographically distinct recruitment centers were assigned to the UKB testing cohort for validation. **Table 1**presents the baseline characteristics of the training (n = 39,634) and testing (n = 13,262) cohorts. We observed significant differences in key population characteristics, including age, ethnicity, Townsend deprivation index (TDI), BMI, alcohol consumption, and dietary habits (P<0.05), highlighting the demographic and lifestyle heterogeneity between training and testing cohorts. Over a median follow-up of 13.6 years (interquartile range [IQR]: 12.9-14.4; maximum: 16.6 years), 139 (0.26%) incident CD cases were identified. The follow-up duration from baseline to CD diagnosis ranged from 0.12 to 14.88 years (median: 7.84 years; IQR: 4.37-10.74; mean: 7.47 years) (**Supplementary Table 1**). Details of the validation cohort in China are shown in **Table 2**. In this validation cohort, 37 of 74 participants were diagnosed with CD and 46% were male, and the overall average*

age was 44 years old. Compared to participants without CD, those with CD were more frequently younger (P < 0.001).”

Table 1: Baseline characteristics of UK Biobank participants in the training and testing cohorts.

	Training cohort	Testing cohort	*P* value
	n=39,634	n=13,262	
Age, mean (SD), years	57.06 (8.13)	56.06 (8.41)	<0.001
Sex			0.173
Female, n(%)	21299 (53.7)	7218 (54.4)	
Male, n(%)	18335 (46.3)	6044 (45.6)	
Ethnicity			<0.001
Other, n(%)	2253 (5.7)	1095 (8.3)	
White, n(%)	37381 (94.3)	12167 (91.7)	
TD index, mean (SD)	-1.36 (3.03)	-0.65 (3.57)	<0.001
BMI, mean (SD), kg/m2	27.54 (4.79)	27.28 (4.84)	<0.001
Physical activity			0.289
Physically inactive, n(%)	7591 (19.2)	2484 (18.7)	
Physically active, n(%)	32043 (80.8)	10778 (81.3)	
Smoking status			0.014
Never smoker, n(%)	21565 (54.4)	7166 (54.0)	
Previous smoker, n(%)	13937 (35.2)	4594 (34.6)	
Current smoker, n(%)	4132 (10.4)	1502 (11.3)	
Alcohol consumption			<0.001
Daily or almost daily, n(%)	7798 (19.7)	2898 (21.9)	
Three or four times a week, n(%)	8931 (22.5)	2970 (22.4)	
Once or twice a week, n(%)	10456 (26.4)	3264 (24.6)	
One to three times a month, n(%)	4360 (11.0)	1416 (10.7)	
Special occasions only, n(%)	4680 (11.8)	1553 (11.7)	
Never, n(%)	3409 (8.6)	1161 (8.8)	
Dietary habit			0.022
Poor diet	24856 (62.7)	8169 (61.6)	
Healthy diet	14778 (37.3)	5093 (38.4)	
Depression (%)	2315 (5.8)	738 (5.6)	0.246
Anxiety (%)	548 (1.4)	186 (1.4)	0.899
History use of antibiotics (%)	553 (1.4)	182 (1.4)	0.879
History use of NSAIDs (%)	669 (1.7)	176 (1.3)	0.005

Note: Data are presented as mean (SD) or n (%). The 52,896 participants were randomly divided into training and testing cohorts according to the UK Biobank recruitment centers in a ratio of 75%:25% roughly. The training set included participants from Cardiff, Glasgow, Stoke, Reading, Bury, Newcastle, Leeds, Nottingham, Sheffield, Liverpool, Middlesborough, Hounslow, and Croydon. The testing set included participants from Stockport (pilot), Manchester, Oxford, Edinburgh, Bristol, Barts, Birmingham, Swansea, and Wrexham (Supplemental Table 18). Differences between training and testing cohorts were assessed using Student's t-test for continuous variables and Pearson's chi-squared test for discrete variables.

Abbreviations: BMI, body mass index; TD index, Townsend deprivation index; NSAIDs, nonsteroidal anti-inflammatory drugs.

Table 2. Baseline characteristics of Southern China participants included in the study.

	Overall	**CD**	**Control**	**P value**
	n=74	n=37	n=37	
Age, mean (SD), years	43.56 (18.31)	34.81 (14.88)	53.49 (16.94)	<0.001
Sex				0.093
Female, n(%)	26 (36.1)	10 (27.0)	18 (48.6)	
Male, n(%)	46 (63.9)	27 (73.0)	19 (51.4)	
BMI, mean (SD), kg/m2	22.28 (3.37)	21.68 (3.61)	22.90 (2.93)	0.115
Smoking status				0.588
Never smoker, n(%)	62 (86.1)	33 (89.2)	31 (83.8)	
Previous smoker, n(%)	4 (5.6)	1 (2.7)	3 (8.1)	
Current smoker, n(%)	6 (8.3)	3 (8.1)	3 (8.1)	
Alcohol consumption	8 (11.1)	2 (5.4)	6 (16.2)	0.261

Note: Data are presented as mean (SD) or n (%). Differences between CD and healthy control groups were assessed using Student's t-test for continuous variables and Pearson's chi-squared test for discrete variables.

Abbreviations: BMI, body mass index; CD, Crohn's disease.

Reference

1. Collins, G. S., Reitsma, J. B., Altman, D. G. & Moons, K. G. M. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. BMJ 350, g7594 (2015).

2. Presumably, the age at diagnosis is close to or above 60 years, indicating that the CD cases in this cohort are predominantly elderly onset. This represents a relatively small subgroup of incident CD cases in the general population, which may limit the generalizability of the findings to the broader incident CD population. This limitation should be clearly acknowledged.

Authors' response:

We thank for the reviewer's valuable suggestions. According to your comments, we have acknowledged the limitation in the **3. Discussion.**

*“**3. Discussion** …Third, our model was based on an older CD population, with a mean age at diagnosis of 66 years. This predominantly elderly onset may limit the generalizability of our findings to a broader age range of incident CD cases.…”*

3. The diagnosis date was determined by the earliest recorded appearance of ICD-10 codes. However, a likely discrepancy exists between the actual diagnosis date and the date of ICD-10 code appearance. Some cases diagnosed near the time of code appearance may have had earlier diagnoses that were not captured by the ICD-10 codes. Were any efforts made to reduce this potential bias? This is distinct from the issue of diagnosis delay. Did the authors set a minimum follow-up duration prior to a CD diagnosis? At a minimum, a sensitivity analysis excluding cases that developed CD within 1, 2, 3, 4, and 5 years before diagnosis should be conducted.

Authors’ response:

We thank the reviewer for the valuable suggestions. To address the potential discrepancy between the actual diagnosis date and the first recorded appearance of ICD-10 codes, we conducted a comprehensive sensitivity analysis excluding cases that developed CD within 1, 2, 3, 4, and 5 years before diagnosis. The results remained consistent with our main analysis, demonstrating the robustness of our findings. The results of excluding cases that developed CD within 2 years before diagnosis are shown in our revised manuscript: **Results of replication validation analyses, Supplementary Tables 13–14** and **Supplementary Fig. 8**. Additionally, we provide below the complete results for all exclusion periods (1–5 years) for reference.

*“**2.6 Results of replication validation analyses**…The results remained consistent after excluding individuals who developed CD within the first two years of follow-up (**Supplementary Tables 13–14** and **Supplementary Fig. 8**) …”*

Supplementary Table 13: Results of ROC analyses after excluding individuals who developed Crohn’s disease in the first two years of follow-up.

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.68 (0.67-0.70)*	*0.70 (0.69-0.71)*	*0.66 (0.64-0.67)*	*0.70 (0.69-0.71)*
Demographic	*0.56 (0.55-0.57)*	*0.56 (0.55-0.57)*	*0.55 (0.55-0.55)*	*0.60 (0.60-0.60)*
Serum	*0.59 (0.57-0.61)*	*0.62 (0.60-0.65)*	*0.62 (0.60-0.63)*	*0.67 (0.65-0.68)*
PRS	*0.64 (0.62-0.65)*	*0.61 (0.60-0.62)*	*0.66 (0.65-0.66)*	*0.66 (0.66-0.66)*
Protein+Demographic	*0.70 (0.69-0.71)*	*0.70 (0.69-0.71)*	*0.71 (0.70-0.72)*	*0.72 (0.70-0.73)*
Protein+Serum	*0.69 (0.68-0.71)*	*0.72 (0.71-0.73)*	*0.71 (0.69-0.73)*	*0.74 (0.73-0.75)*
Protein+PRS	*0.67 (0.65-0.69)*	*0.69 (0.67-0.71)*	*0.71 (0.70-0.73)*	*0.71 (0.70-0.73)*
Protein+Demographic+Serum	*0.71 (0.70-0.72)*	*0.71 (0.69-0.73)*	*0.73 (0.71-0.74)*	*0.72 (0.70-0.74)*
Protein+Demographic+PRS	*0.70 (0.69-0.72)*	*0.71 (0.70-0.73)*	*0.73 (0.72-0.74)*	*0.76 (0.75-0.77)*
Protein+Serum+PRS	*0.73 (0.71-0.75)*	*0.71 (0.69-0.74)*	*0.75 (0.74-0.76)*	*0.73 (0.72-0.73)*
Combined	*0.73 (0.72-0.74)*	*0.73 (0.71-0.74)*	*0.70 (0.67-0.73)*	*0.74 (0.71-0.77)*
Training set				

9-protein panel	*0.92 (0.89-0.95)*	*0.92 (0.88-0.96)*	*0.78 (0.75-0.81)*	*0.77 (0.75-0.79)*
Demographic	*0.63 (0.63-0.63)*	*0.63 (0.63-0.63)*	*0.63 (0.63-0.63)*	*0.62 (0.62-0.62)*
Serum	*0.86 (0.80-0.93)*	*0.88 (0.80-0.96)*	*0.81 (0.74-0.89)*	*0.70 (0.69-0.70)*
PRS	*0.62 (0.59-0.64)*	*0.63 (0.61-0.65)*	*0.67 (0.65-0.68)*	*0.60 (0.60-0.60)*
Protein+Demographic	*0.90 (0.87-0.92)*	*0.85 (0.83-0.87)*	*0.85 (0.80-0.89)*	*0.82 (0.76-0.87)*
Protein+Serum	*0.98 (0.96-1.00)*	*0.96 (0.94-0.99)*	*0.95 (0.92-0.98)*	*0.87 (0.82-0.91)*
Protein+PRS	*0.97 (0.94-1.00)*	*0.95 (0.91-0.99)*	*0.95 (0.91-0.99)*	*0.82 (0.78-0.87)*
Protein+Demographic+Serum	*0.98 (0.97-0.99)*	*0.94 (0.91-0.98)*	*0.84 (0.79-0.88)*	*0.81 (0.76-0.86)*
Protein+Demographic+PRS	*0.96 (0.92-0.99)*	*0.94 (0.90-0.97)*	*0.90 (0.86-0.94)*	*0.81 (0.79-0.84)*
Protein+Serum+PRS	*0.98 (0.96-1.00)*	*0.98 (0.96-0.99)*	*0.94 (0.91-0.98)*	*0.79 (0.78-0.79)*
Combined	*0.99 (0.98-1.00)*	*0.97 (0.95-0.98)*	*0.87 (0.83-0.90)*	*0.80 (0.78-0.82)*

After excluding individuals who developed Crohn's disease in the first two years of follow-up, we randomly split the participants into a training and a testing set according to the recruitment centers in a ratio of 3:1 roughly. To further reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. The combined model included protein, demographics, serum, and PRS of Crohn's disease.

Supplementary Table 14: Comparison between AUCs for all-year incident CD after excluding individuals who developed Crohn's disease in the first two year of follow-up.

LGBM											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.77E-04	1.83E-04	7.00E-04	3.12E-02	3.85E-01	3.45E-01	2.57E-02	3.12E-02	1.31E-03	1.01E-03
Demographic	1.77E-04		2.08E-02	1.57E-04	1.77E-04	1.77E-04	1.77E-04	1.77E-04	1.77E-04	1.77E-04	1.77E-04
Serum	1.83E-04	2.08E-02		2.62E-03	1.83E-04	1.83E-04	4.40E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
PRS	7.00E-04	1.57E-04	2.62E-03		2.21E-04	5.29E-04	4.35E-02	2.21E-04	1.63E-04	1.63E-04	1.63E-04
Protein+ Demographic	3.12E-02	1.77E-04	1.83E-04	2.21E-04		4.73E-01	1.04E-01	4.27E-01	9.10E-01	9.11E-03	4.59E-03
Protein+ Serum	3.85E-01	1.77E-04	1.83E-04	5.29E-04	4.73E-01		1.21E-01	3.07E-01	5.21E-01	1.73E-02	4.59E-03
Protein+ PRS	3.45E-01	1.77E-04	4.40E-04	4.35E-02	1.04E-01	1.21E-01		4.52E-02	1.04E-01	3.61E-03	1.31E-03
Protein+ Demographic+ Serum	2.57E-02	1.77E-04	1.83E-04	2.21E-04	4.27E-01	3.07E-01	4.52E-02		4.27E-01	3.76E-02	2.11E-02
Protein+ Demographic+ PRS	3.12E-02	1.77E-04	1.83E-04	1.63E-04	9.10E-01	5.21E-01	1.04E-01	4.27E-01		3.76E-02	2.11E-02

Protein+ Serum+ PRS	1.31E-03	1.77E-04	1.83E-04	1.63E-04	9.11E-03	1.73E-02	3.61E-03	3.76E-02	3.76E-02		9.70E-01
Combined	1.01E-03	1.77E-04	1.83E-04	1.63E-04	4.59E-03	4.59E-03	1.31E-03	2.11E-02	2.11E-02	9.70E-01	

XGBoost											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.73E-04	4.40E-04	1.32E-04	5.21E-01	1.73E-02	7.34E-01	1.04E-01	1.21E-01	2.41E-01	7.28E-03
Demographic	1.73E-04		3.13E-04	1.69E-04	1.73E-04	1.73E-04	1.73E-04	1.73E-04	1.73E-04	1.73E-04	1.73E-04
Serum	4.40E-04	3.13E-04		1.13E-01	1.83E-04	1.83E-04	1.01E-03	3.30E-04	3.30E-04	1.31E-03	1.83E-04
PRS	1.32E-04	1.69E-04	1.13E-01		1.32E-04	1.32E-04	1.32E-04	1.32E-04	1.32E-04	1.32E-04	1.32E-04
Protein+ Demographic	5.21E-01	1.73E-04	1.83E-04	1.32E-04		5.80E-03	6.23E-01	6.40E-02	8.90E-02	2.41E-01	3.61E-03
Protein+ Serum	1.73E-02	1.73E-04	1.83E-04	1.32E-04	5.80E-03		1.62E-01	4.27E-01	8.50E-01	7.91E-01	3.85E-01
Protein+ PRS	7.34E-01	1.73E-04	1.01E-03	1.32E-04	6.23E-01	1.62E-01		4.27E-01	3.45E-01	1.40E-01	4.52E-02
Protein+ Demographic+ Serum	1.04E-01	1.73E-04	3.30E-04	1.32E-04	6.40E-02	4.27E-01	4.27E-01		7.91E-01	6.78E-01	1.86E-01

Protein+ Demographic+ PRS	1.21E-01	1.73E-04	3.30E-04	1.32E-04	8.90E-02	8.50E-01	3.45E-01	7.91E-01		6.78E-01	3.45E-01
Protein+ Serum+ PRS	2.41E-01	1.73E-04	1.31E-03	1.32E-04	2.41E-01	7.91E-01	1.40E-01	6.78E-01	6.78E-01		6.78E-01
Combined	7.28E-03	1.73E-04	1.83E-04	1.32E-04	3.61E-03	3.85E-01	4.52E-02	1.86E-01	3.45E-01	6.78E-01	

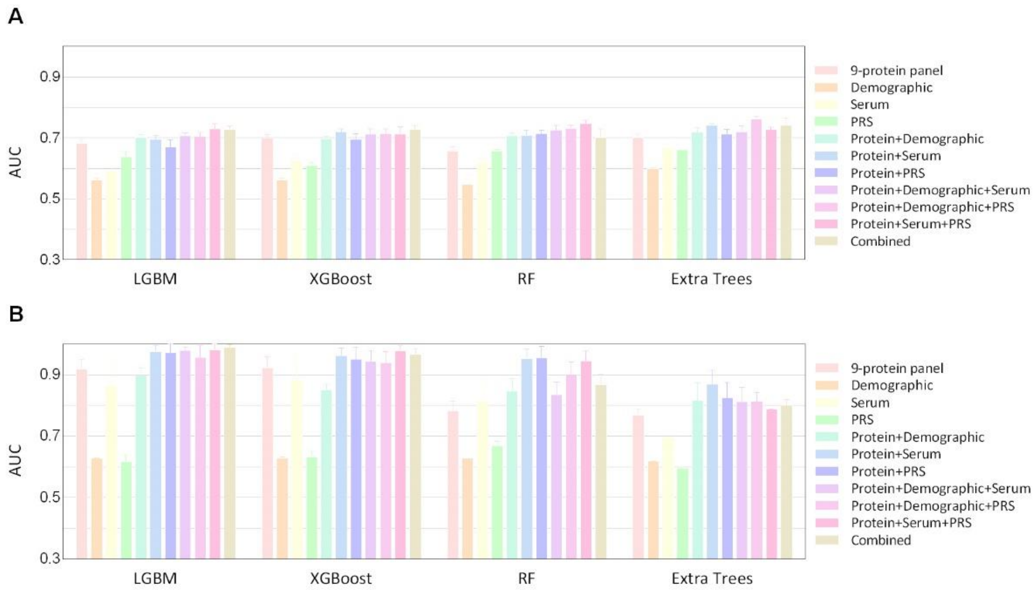
RF											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		4.17E-05	1.32E-04	1.97E-01	4.39E-04	5.88E-04	4.41E-04	3.25E-04	2.42E-04	1.80E-04	3.75E-03
Demographic	4.17E-05		6.39E-05	5.11E-05	6.34E-05	6.39E-05	6.39E-05	6.29E-05	6.34E-05	6.39E-05	6.29E-05
Serum	1.32E-04	6.39E-05		2.09E-04	1.82E-04	1.83E-04	1.83E-04	1.81E-04	1.82E-04	1.83E-04	1.81E-04
PRS	1.97E-01	5.11E-05	2.09E-04		1.53E-04	1.54E-04	5.03E-04	1.52E-04	1.53E-04	1.54E-04	1.36E-01
Protein+ Demographic	4.39E-04	6.34E-05	1.82E-04	1.53E-04		5.71E-01	1.62E-01	2.72E-01	7.24E-03	1.70E-03	1.00E+00
Protein+ Serum	5.88E-04	6.39E-05	1.83E-04	1.54E-04	5.71E-01		9.10E-01	2.73E-01	3.76E-02	2.20E-03	8.50E-01
Protein+ PRS	4.41E-04	6.39E-05	1.83E-04	5.03E-04	1.62E-01	9.10E-01		6.23E-01	3.76E-02	2.20E-03	1.00E+00

Protein+ Demographic+ Serum	3.25E-04	6.29E-05	1.81E-04	1.52E-04	2.72E-01	2.73E-01	6.23E-01		6.23E-01	8.87E-02	2.72E-01
Protein+ Demographic+ PRS	2.42E-04	6.34E-05	1.82E-04	1.53E-04	7.24E-03	3.76E-02	3.76E-02	6.23E-01		5.38E-02	3.84E-01
Protein+ Serum+ PRS	1.80E-04	6.39E-05	1.83E-04	1.54E-04	1.70E-03	2.20E-03	2.20E-03	8.87E-02	5.38E-02		1.72E-02
Combined	3.75E-03	6.29E-05	1.81E-04	1.36E-01	1.00E+00	8.50E-01	1.00E+00	2.72E-01	3.84E-01	1.72E-02	

Extra trees											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.76E-04	5.76E-03	6.34E-05	2.73E-01	3.28E-04	1.62E-01	8.87E-02	1.82E-04	4.05E-03	5.76E-03
Demographic	1.76E-04		1.76E-04	6.11E-05	1.77E-04	1.77E-04	1.76E-04	1.76E-04	1.77E-04	1.44E-04	1.76E-04
Serum	5.76E-03	1.76E-04		1.15E-01	1.82E-04	1.82E-04	1.69E-03	2.81E-03	1.82E-04	1.49E-04	2.19E-03
PRS	6.34E-05	6.11E-05	1.15E-01		6.39E-05	6.39E-05	1.41E-03	6.34E-05	6.39E-05	4.92E-05	1.41E-03
Protein+ Demographic	2.73E-01	1.77E-04	1.82E-04	6.39E-05		1.13E-02	5.71E-01	4.27E-01	1.71E-03	2.90E-02	2.57E-02
Protein+ Serum	3.28E-04	1.77E-04	1.82E-04	6.39E-05	1.13E-02		3.60E-03	1.40E-01	2.20E-03	2.38E-02	6.23E-01

Protein+ PRS	1.62E-01	1.76E-04	1.69E-03	1.41E-03	5.71E-01	3.60E-03		2.73E-01	2.45E-04	5.07E-02	2.56E-02
Protein+ Demographic+ Serum	8.87E-02	1.76E-04	2.81E-03	6.34E-05	4.27E-01	1.40E-01	2.73E-01		7.65E-04	9.08E-01	8.87E-02
Protein+ Demographic+ PRS	1.82E-04	1.77E-04	1.82E-04	6.39E-05	1.71E-03	2.20E-03	2.45E-04	7.65E-04		4.91E-04	2.12E-01
Protein+ Serum+ PRS	4.05E-03	1.44E-04	1.49E-04	4.92E-05	2.90E-02	2.38E-02	5.07E-02	9.08E-01	4.91E-04		3.01E-01
Combined	5.76E-03	1.76E-04	2.19E-03	1.41E-03	2.57E-02	6.23E-01	2.56E-02	8.87E-02	2.12E-01	3.01E-01	

Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn’s disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease.



Supplementary Fig. 8: Predictive accuracy after excluding individuals who developed Crohn's disease in the first two years of follow-up. Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease in the UKB testing set (A) and training set (B), evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Nineteen individuals were excluded for developing Crohn's disease within the first two years. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease.

Table. The results of excluding cases that developed CD within 1, 2, 3, 4, and 5 years before diagnosis after excluding individuals who developed Crohn's disease in the first one year of follow-up

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.70 (0.69-0.71)*	*0.69 (0.68-0.70)*	*0.73 (0.73-0.74)*	*0.72 (0.72-0.72)*
Demographic	*0.56 (0.55-0.56)*	*0.56 (0.55-0.57)*	*0.56 (0.55-0.56)*	*0.59 (0.59-0.60)*
Serum	*0.60 (0.58-0.63)*	*0.60 (0.57-0.63)*	*0.66 (0.65-0.67)*	*0.63 (0.60-0.66)*
PRS	*0.63 (0.62-0.65)*	*0.66 (0.65-0.66)*	*0.64 (0.64-0.65)*	*0.66 (0.66-0.66)*
Protein+Demographic	*0.71 (0.69-0.72)*	*0.71 (0.71-0.72)*	*0.70 (0.69-0.71)*	*0.71 (0.69-0.73)*
Protein+Serum	*0.72 (0.71-0.73)*	*0.72 (0.71-0.73)*	*0.72 (0.70-0.73)*	*0.73 (0.71-0.75)*
Protein+PRS	*0.73 (0.72-0.74)*	*0.72 (0.71-0.73)*	*0.74 (0.73-0.74)*	*0.75 (0.73-0.76)*
Protein+Demographic+Serum	*0.72 (0.72-0.73)*	*0.73 (0.72-0.74)*	*0.74 (0.73-0.74)*	*0.69 (0.66-0.72)*
Protein+Demographic+PRS	*0.73 (0.71-0.74)*	*0.71 (0.70-0.73)*	*0.74 (0.73-0.75)*	*0.75 (0.74-0.76)*
Protein+Serum+PRS	*0.75 (0.74-0.76)*	*0.76 (0.75-0.77)*	*0.76 (0.75-0.76)*	*0.73 (0.71-0.76)*
Combined	*0.76 (0.75-0.77)*	*0.75 (0.73-0.76)*	*0.76 (0.76-0.77)*	*0.76 (0.75-0.77)*

after excluding individuals who developed Crohn's disease in the first two years of follow-up

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.68 (0.67-0.70)	0.70 (0.69-0.71)	0.66 (0.64-0.67)	0.70 (0.69-0.71)
Demographic	0.56 (0.55-0.57)	0.56 (0.55-0.57)	0.55 (0.55-0.55)	0.60 (0.60-0.60)
Serum	0.59 (0.57-0.61)	0.62 (0.60-0.65)	0.62 (0.60-0.63)	0.67 (0.65-0.68)
PRS	0.64 (0.62-0.65)	0.61 (0.60-0.62)	0.66 (0.65-0.66)	0.66 (0.66-0.66)
Protein+Demographic	0.70 (0.69-0.71)	0.70 (0.69-0.71)	0.71 (0.70-0.72)	0.72 (0.70-0.73)
Protein+Serum	0.69 (0.68-0.71)	0.72 (0.71-0.73)	0.71 (0.69-0.73)	0.74 (0.73-0.75)
Protein+PRS	0.67 (0.65-0.69)	0.69 (0.67-0.71)	0.71 (0.70-0.73)	0.71 (0.70-0.73)
Protein+Demographic+Serum	0.71 (0.70-0.72)	0.71 (0.69-0.73)	0.73 (0.71-0.74)	0.72 (0.70-0.74)
Protein+Demographic+PRS	0.70 (0.69-0.72)	0.71 (0.70-0.73)	0.73 (0.72-0.74)	0.76 (0.75-0.77)
Protein+Serum+PRS	0.73 (0.71-0.75)	0.71 (0.69-0.74)	0.75 (0.74-0.76)	0.73 (0.72-0.73)
Combined	0.73 (0.72-0.74)	0.73 (0.71-0.74)	0.70 (0.67-0.73)	0.74 (0.71-0.77)

after excluding individuals who developed Crohn's disease in the first three years of follow-up

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.66 (0.64-0.68)	0.67 (0.65-0.69)	0.69 (0.68-0.70)	0.69 (0.68-0.70)
Demographic	0.60 (0.59-0.60)	0.60 (0.59-0.60)	0.60 (0.59-0.61)	0.63 (0.63-0.63)
Serum	0.62 (0.60-0.64)	0.58 (0.55-0.62)	0.68 (0.67-0.70)	0.68 (0.67-0.68)
PRS	0.59 (0.57-0.61)	0.59 (0.59-0.59)	0.57 (0.55-0.59)	0.64 (0.64-0.64)
Protein+Demographic	0.67 (0.66-0.69)	0.67 (0.65-0.69)	0.69 (0.68-0.71)	0.70 (0.69-0.71)
Protein+Serum	0.69 (0.67-0.71)	0.67 (0.65-0.69)	0.71 (0.69-0.72)	0.70 (0.68-0.71)
Protein+PRS	0.68 (0.67-0.70)	0.69 (0.67-0.70)	0.70 (0.68-0.71)	0.74 (0.73-0.75)
Protein+Demographic+Serum	0.68 (0.66-0.70)	0.70 (0.68-0.72)	0.69 (0.66-0.71)	0.68 (0.67-0.69)
Protein+Demographic+PRS	0.71 (0.71-0.72)	0.71 (0.70-0.72)	0.75 (0.74-0.76)	0.72 (0.71-0.73)
Protein+Serum+PRS	0.73 (0.71-0.74)	0.73 (0.71-0.75)	0.71 (0.69-0.72)	0.73 (0.71-0.74)
Combined	0.72 (0.71-0.72)	0.73 (0.71-0.74)	0.70 (0.67-0.73)	0.71 (0.67-0.74)

after excluding individuals who developed Crohn's disease in the first four years of follow-up

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.69 (0.67-0.70)	0.69 (0.67-0.71)	0.69 (0.67-0.71)	0.69 (0.67-0.71)
Demographic	0.60 (0.59-0.61)	0.59 (0.58-0.60)	0.62 (0.61-0.62)	0.64 (0.64-0.64)
Serum	0.62 (0.59-0.66)	0.66 (0.60-0.71)	0.71 (0.70-0.73)	0.64 (0.60-0.68)
PRS	0.64 (0.63-0.66)	0.62 (0.61-0.64)	0.63 (0.62-0.65)	0.64 (0.64-0.64)
Protein+Demographic	0.68 (0.67-0.70)	0.66 (0.64-0.68)	0.63 (0.63-0.64)	0.69 (0.66-0.71)
Protein+Serum	0.69 (0.68-0.70)	0.70 (0.68-0.72)	0.71 (0.70-0.72)	0.73 (0.71-0.75)
Protein+PRS	0.71 (0.69-0.72)	0.71 (0.69-0.72)	0.68 (0.66-0.70)	0.69 (0.68-0.71)

Protein+Demographic+Serum	*0.69 (0.69-0.70)*	*0.71 (0.70-0.73)*	*0.73 (0.73-0.74)*	*0.72 (0.71-0.74)*
Protein+Demographic+PRS	*0.71 (0.70-0.72)*	*0.71 (0.70-0.73)*	*0.71 (0.70-0.73)*	*0.74 (0.72-0.76)*
Protein+Serum+PRS	*0.70 (0.68-0.73)*	*0.73 (0.70-0.76)*	*0.72 (0.70-0.74)*	*0.72 (0.70-0.74)*
Combined	*0.72 (0.70-0.74)*	*0.73 (0.70-0.75)*	*0.76 (0.75-0.77)*	*0.78 (0.76-0.80)*

after excluding individuals who developed Crohn’s disease in the first five years of follow-up

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.67 (0.65-0.68)*	*0.66 (0.64-0.68)*	*0.64 (0.60-0.67)*	*0.66 (0.65-0.67)*
Demographic	*0.62 (0.61-0.63)*	*0.62 (0.61-0.63)*	*0.65 (0.64-0.65)*	*0.64 (0.64-0.64)*
Serum	*0.61 (0.58-0.64)*	*0.63 (0.59-0.67)*	*0.65 (0.63-0.67)*	*0.68 (0.65-0.70)*
PRS	*0.57 (0.54-0.61)*	*0.60 (0.57-0.64)*	*0.63 (0.61-0.65)*	*0.65 (0.65-0.66)*
Protein+Demographic	*0.66 (0.64-0.67)*	*0.68 (0.67-0.69)*	*0.67 (0.64-0.69)*	*0.64 (0.61-0.66)*
Protein+Serum	*0.68 (0.67-0.70)*	*0.69 (0.66-0.71)*	*0.64 (0.60-0.68)*	*0.67 (0.65-0.69)*
Protein+PRS	*0.69 (0.69-0.69)*	*0.68 (0.66-0.70)*	*0.70 (0.68-0.73)*	*0.64 (0.61-0.66)*
Protein+Demographic+Serum	*0.69 (0.67-0.71)*	*0.67 (0.64-0.71)*	*0.69 (0.67-0.71)*	*0.70 (0.67-0.72)*
Protein+Demographic+PRS	*0.67 (0.63-0.71)*	*0.68 (0.67-0.70)*	*0.70 (0.68-0.71)*	*0.72 (0.69-0.75)*
Protein+Serum+PRS	*0.74 (0.73-0.75)*	*0.73 (0.71-0.74)*	*0.70 (0.67-0.72)*	*0.72 (0.70-0.74)*
Combined	*0.72 (0.72-0.73)*	*0.73 (0.71-0.75)*	*0.71 (0.69-0.73)*	*0.74 (0.73-0.75)*

4. In the baseline characteristics table, the authors compared those who developed CD over time to those who remained healthy as of October 2022, which should be explicitly stated.

Authors’ response:

We thank the reviewer for this valuable suggestion. In response to your comment, we have explicitly noted this detail in the annotation of the baseline characteristics table, as detailed in **Supplementary Table 1** and **Supplementary Table 21**.

Supplementary Table 1: Baseline characteristics of UK Biobank participants by Crohn's disease case status.

	CD	**Control**	**P value**
	n=139	n=52,757	
Age at recruitment, mean (SD), years	58.76 (7.71)	56.81 (8.21)	0.005
Age at diagnosis, mean (SD), years	66.27 (8.82)	-	
Sex			0.678
Female, n(%)	72 (51.8)	28,445 (53.9)	
Male, n(%)	67 (48.2)	24,312 (46.1)	
Ethnicity			0.134
Other, n(%)	4 (2.9)	3,344 (6.3)	
White, n(%)	135 (97.1)	49,413 (93.7)	
TD index, mean (SD)	-0.88 (3.13)	-1.18 (3.19)	0.264

BMI, mean (SD), kg/m2	27.90 (4.81)	27.47 (4.80)	0.292
Physical activity			0.661
Physically inactive, n(%)	29 (20.9)	10,046 (19.0)	
Physically active, n(%)	110 (79.1)	42,711 (81.0)	
Smoking status			0.044
Never smoker, n(%)	61 (43.9)	28,670 (54.3)	
Previous smoker, n(%)	61 (43.9)	18,470 (35.0)	
Current smoker, n(%)	17 (12.2)	5,617 (10.6)	
Alcohol consumption			0.759
Daily or almost daily, n(%)	28 (20.1)	10,668 (20.2)	
Three or four times a week, n(%)	32 (23.0)	11,869 (22.5)	
Once or twice a week, n(%)	29 (20.9)	13,691 (26.0)	
One to three times a month, n(%)	16 (11.5)	5,760 (10.9)	
Special occasions only, n(%)	19 (13.7)	6,214 (11.8)	
Never, n(%)	15 (10.8)	4,555 (8.6)	
Dietary habit			0.088
Poor diet	97 (69.8)	32,928 (62.4)	
Healthy diet	42 (30.2)	19,829 (37.6)	
Depression (%)	10 (7.2)	3,043 (5.8)	0.591
Anxiety (%)	4 (2.9)	730 (1.4)	0.254
History use of antibiotics (%)	3 (2.2)	732 (1.4)	0.68
History use of NSAIDs (%)	1 (0.7)	844 (1.6)	0.244

Note: Data are presented as mean (SD) or n (%). Differences between participants who developed Crohn's disease over time and those who remained healthy as of October 2022 were assessed using Student's t-test for continuous variables and Pearson's chi-squared test for discrete variables.

Abbreviations: BMI, body mass index; TD index, Townsend deprivation index; CD, Crohn's disease; NSAIDs, nonsteroidal anti-inflammatory drugs.

Supplementary Table 21: Baseline clinical demographics for training set.

	CD	Control	P value
	n=102	n=39,532	
Age, mean (SD), years	58.75 (8.21)	57.06 (8.13)	0.035
Sex			1
Female, n(%)	55 (53.9)	21,244 (53.7)	
Male, n(%)	47 (46.1)	18,288 (46.3)	
Ethnicity			0.578
Other, n(%)	4 (3.9)	2,249 (5.7)	
White, n(%)	98 (96.1)	37,283 (94.3)	
TD index, mean (SD)	-1.12 (2.92)	-1.36 (3.03)	0.426
BMI, mean (SD), kg/m2	27.56 (4.36)	27.54 (4.79)	0.965
Physical activity			0.621
Physically inactive, n(%)	22 (21.6)	7,569 (19.1)	

Physically active, n(%)	80 (78.4)	31,963 (80.9)	
Smoking status			0.549
Never smoker, n(%)	50 (49.0)	21,515 (54.4)	
Previous smoker, n(%)	40 (39.2)	13,897 (35.2)	
Current smoker, n(%)	12 (11.8)	4,120 (10.4)	
Alcohol consumption			0.558
Daily or almost daily, n(%)	16 (15.7)	7,782 (19.7)	
Three or four times a week, n(%)	25 (24.5)	8,906 (22.5)	
Once or twice a week, n(%)	23 (22.5)	10,433 (26.4)	
One to three times a month, n(%)	11 (10.8)	4,349 (11.0)	
Special occasions only, n(%)	14 (13.7)	4,666 (11.8)	
Never, n(%)	13 (12.7)	3,396 (8.6)	
Dietary habit			0.257
Poor diet	70 (68.6)	24,786 (62.7)	
Healthy diet	32 (31.4)	14,746 (37.3)	
Depression (%)	8 (7.8)	2,307 (5.8)	0.514
Anxiety (%)	2 (2.0)	546 (1.4)	0.939
History use of antibiotics (%)	3 (2.9)	550 (1.4)	0.363
History use of NSAIDs (%)	0 (0.0)	669 (1.7)	0.347

Note: Data are presented as mean (SD) or n (%). Differences between participants who developed Crohn's disease over time and those who remained healthy as of October 2022 were assessed using Student's t-test for continuous variables and Pearson's chi-squared test for discrete variables. Variables that showed significant differences between groups, including age, were finally included in the demographic model. Abbreviations: BMI, body mass index; TDI, Townsend deprivation index; CD, Crohn's disease; NSAIDs, nonsteroidal anti-inflammatory drugs.

5. The authors employed a machine learning algorithm to combine the 10 significant proteins from either model 1 or model 2. It appears that they developed a model to classify a binary outcome and subsequently applied this model to cumulative cases over time, up to 16 years. Have the authors considered using a machine learning algorithm with a time-to-event extension? Additionally, can they provide calibration data for the model in the testing set?

Authors' response:

We sincerely appreciate the reviewer's thorough review and insightful suggestions, which have been invaluable in refining our manuscript. Below, we provide detailed responses to your two queries:

1. Use of Time-to-Event Models

We are truly grateful for your suggestion regarding the use of machine learning algorithms with time-to-event extensions. We completely agree that survival analysis models, which account for the time-to-event nature of the data, could offer additional insights. To explore this approach, we implemented a Random Survival Forest (RSF) model to evaluate the predictive performance of the protein panel for Crohn's disease (CD) onset. The RSF model achieved an average AUC of 0.70 (0.67–0.73) across 10 repetitions in the UK Biobank testing set, aligning with the results from our primary analysis.

However, incorporating additional survival models into our final analyses posed several challenges. First, our revised manuscript already includes four machine learning algorithms with extensive 10-fold cross-validation, and adding time-to-event models would require a significantly higher computational workload. Second, the automated hyperparameter tuning framework we utilized (FLAML) does not support survival models, making it difficult to ensure consistency and comparability across different modeling approaches. Lastly, our current models have been rigorously assessed across multiple timeframes (10–16 years) through sensitivity analyses, allowing for a comprehensive evaluation of their predictive performance over different follow-up periods.

Given these considerations, we did not include the Random Survival Forest algorithm in our final analyses. We truly appreciate the reviewer's insightful suggestion and hope for your understanding of our approach. If the reviewer has any concerns or would like to discuss this further, we are more than willing to engage in additional communication.

2. Calibration Data

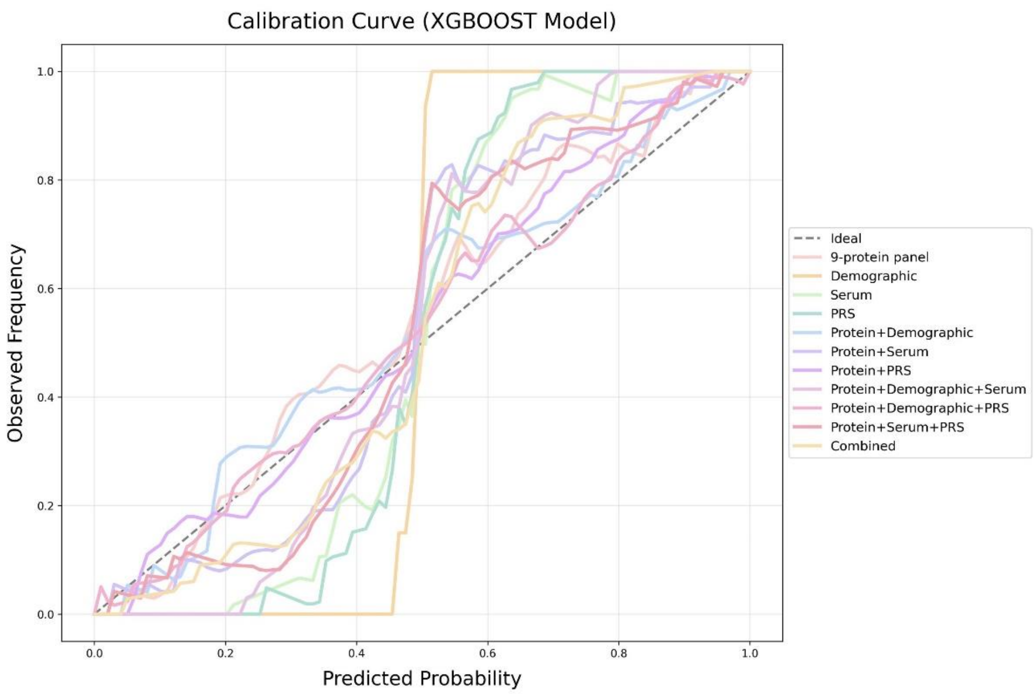
We sincerely appreciate the reviewer's inquiry regarding calibration performance. To ensure a robust evaluation, we have evaluated calibration performance (e.g. calibration curves, Brier score, calibration slope) on both the geographically distinct UKB testing set and an external testing set. These analyses were conducted separately for models trained on unmatched populations and matched populations.

For the matched population, the protein-related models demonstrated strong calibration performance, with calibration slopes close to 1 and Brier scores consistently below 0.25.

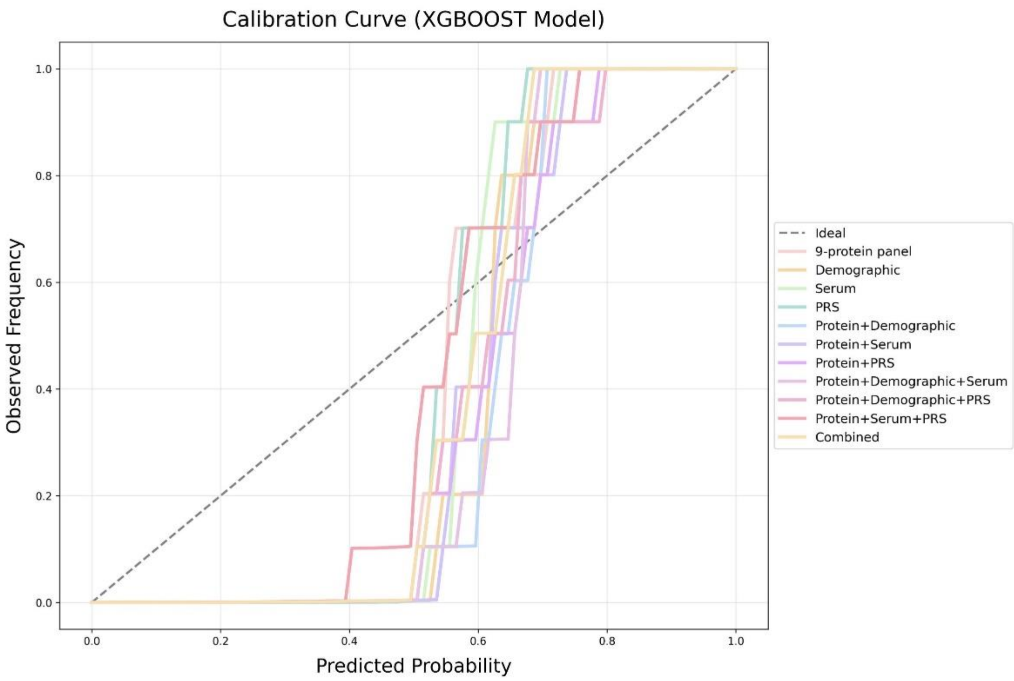
For the unmatched population, the calibration curves indicated that in the lower predicted probability range (0.0–0.6), the models tended to overestimate risk, as evidenced by the calibration curves falling below the diagonal. However, in the higher predicted probability range (>0.6), the models exhibited better calibration. We hypothesize that this pattern may be influenced by the use of class weighting during model training to address class imbalance. Given that Crohn's disease cases are significantly rarer than non-cases, class weighting increases the importance of correctly identifying cases, which may lead to slightly inflated predicted probabilities in lower-risk individuals. However, in the higher probability range, class weighting helps the model focus on case identification, ultimately improving calibration performance. Notably, similar calibration patterns have been observed in prior machine learning studies for disease prediction, including Schalkamp et al., *Nature Medicine* (2023)[1].

Given the long-term progression of Crohn's disease, our model is primarily designed to identify high-risk individuals, thereby providing valuable insights for potential clinical management. While it may slightly overestimate risk in the lower probability ranges, its strong calibration in high-risk ranges aligns well with the practical needs of early risk assessment. Moreover, the protein-related models consistently achieved Brier scores below 0.25, further supporting their good performance.

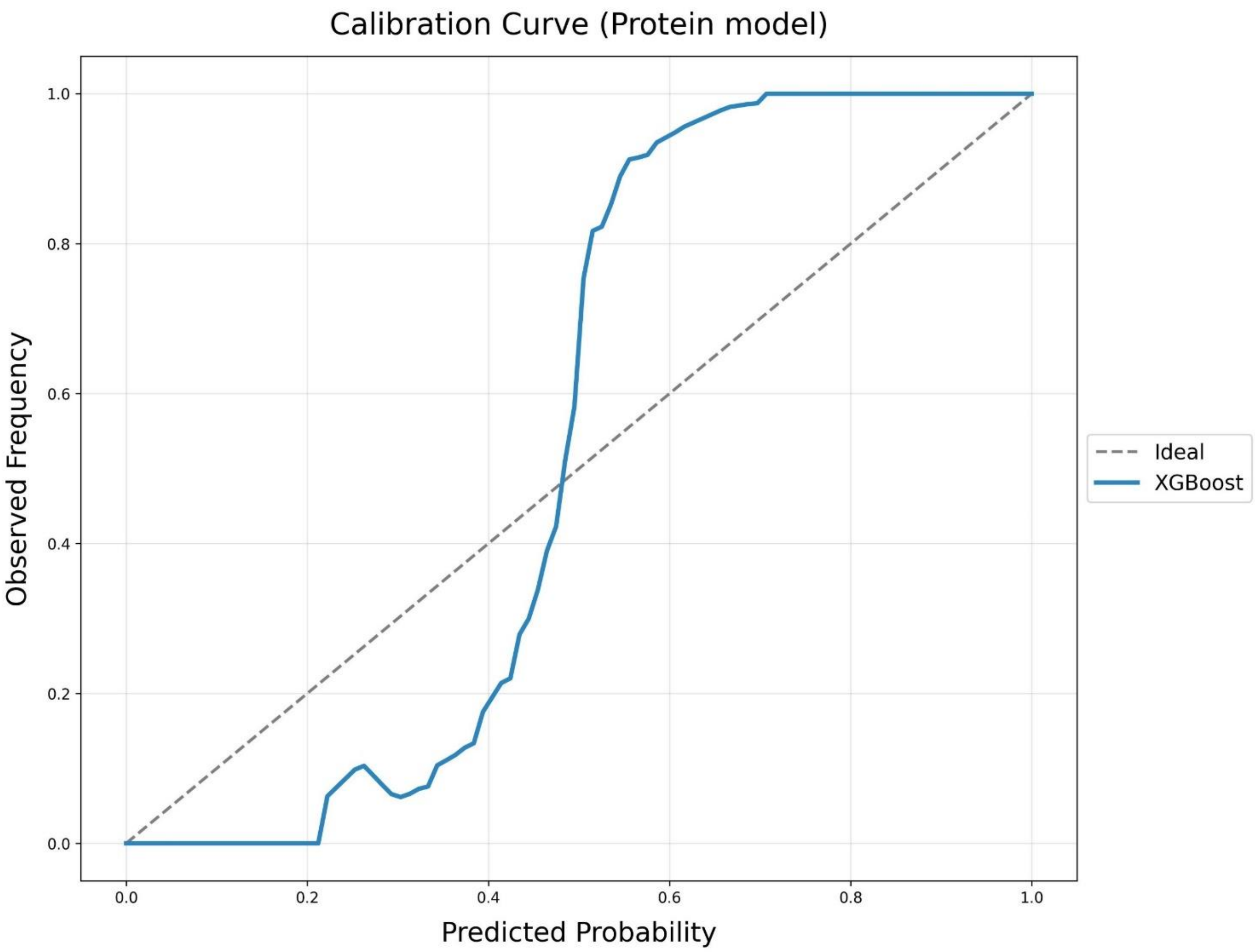
We are deeply grateful for the reviewer's thoughtful feedback, which has greatly enhanced the clarity and robustness of our work. The corresponding calibration data are provided below for your reference. Thank you again for your invaluable input and constructive suggestions. If the reviewer has any further concerns or wishes to explore this issue in more detail, we are open to further discussions.



UKB testing set (matched controls)



UKB testing set (unmatched controls)



External testing set (unmatched controls)

Brier Scores Across Different Testing Sets

	Mean Brier Score	*95% CI*
UKB testing set (Matched controls)		
9-protein panel	*0.2293*	*(0.2136 - 0.2609)*
Demographic	*0.2515*	*(0.2500 - 0.2550)*
Serum	*0.2486*	*(0.2443 - 0.2564)*
PRS	*0.2286*	*(0.2173 - 0.2406)*
Protein+Demographic	*0.239*	*(0.2143 - 0.2879)*
Protein+Serum	*0.2415*	*(0.2270 - 0.2593)*
Protein+PRS	*0.2011*	*(0.1711 - 0.2349)*
Protein+Demographic+Serum	*0.2248*	*(0.2005 - 0.2488)*
Protein+Demographic+PRS	*0.2071*	*(0.1846 - 0.2436)*
Protein+Serum+PRS	*0.2203*	*(0.1928 - 0.2494)*
Combined	*0.2107*	*(0.1826 - 0.2476)*
UKB testing set (Unmatched controls)		
9-protein panel	*0.206*	*(0.1629 - 0.2465)*
Demographic	*0.2285*	*(0.2175 - 0.2412)*

Serum	*0.2151*	*(0.1911 - 0.2393)*
PRS	*0.2402*	*(0.2249 - 0.2484)*
Protein+Demographic	*0.1856*	*(0.1621 - 0.2237)*
Protein+Serum	*0.1775*	*(0.1208 - 0.2180)*
Protein+PRS	*0.154*	*(0.0869 - 0.2255)*
Protein+Demographic+Serum	*0.1616*	*(0.1029 - 0.2332)*
Protein+Demographic+PRS	*0.1682*	*(0.0726 - 0.2391)*
Protein+Serum+PRS	*0.1608*	*(0.0800 - 0.2397)*
Combined	*0.1583*	*(0.1087 - 0.2248)*
External testing set (Unmatched controls)		
9-protein panel	*0.2266*	*(0.1982 - 0.2478)*

Calibration Slopes for Matched Controls in the UKB Testing Set

	Mean Calibration Slope	***95% CI***
9-protein panel	*0.6497*	*(0.5454 - 0.7390)*
Demographic	*0.2868*	*(0.2758 - 0.2988)*
Serum	*0.3283*	*(0.2886 - 0.4051)*
PRS	*0.4525*	*(0.3499 - 0.5787)*
Protein+Demographic	*0.6061*	*(0.3228 - 0.7360)*
Protein+Serum	*0.4794*	*(0.2943 - 0.7272)*
Protein+PRS	*0.8466*	*(0.6289 - 1.0840)*
Protein+Demographic+Serum	*0.523*	*(0.2951 - 0.7272)*
Protein+Demographic+PRS	*0.7936*	*(0.6071 - 0.9272)*
Protein+Serum+PRS	*0.6382*	*(0.2907 - 0.9710)*
Combined	*0.6635*	*(0.3025 - 0.9343)*

Reference

1. Schalkamp, A.-K., Peall, K. J., Harrison, N. A. & Sandor, C. Wearable movement-tracking data identify Parkinson’s disease years before clinical diagnosis. Nat Med 29, 2048–2056 (2023).

6. There is significant class imbalance in the dataset. The authors mention the use of balanced class weighting in the figure legend, but further details on how this was implemented should be provided.

Authors’ response:

We thank for the reviewer’s valuable suggestions. According to your comments, we have supplemented the details on how the balanced class weighting was implemented in the section **4.5 Statistical analysis.**

*“**4.5 Statistical analysis …**Class imbalance was addressed using automated balanced class weighting in scikit-learn's 'balanced' mode, with the computed weights incorporated into AutoML training (via the sample_weight parameter)….”*

7. The authors seem to have used an optimal cut-off based on the Youden index and then reapplied the same cut-off to the same cohort from which it was derived. This raises concerns about potential overfitting. It is unclear whether the cut-off was derived from a training set and then applied to a testing set.

Authors' response:

We appreciate the reviewer's valuable suggestion. In response, we have repeated the analyses using the protein model, deriving the cutoff from the training set and applying it to the testing set to reduce potential overfitting. After applying the training-derived cutoff, participants in the high-risk subgroup also exhibited a significantly higher risk of developing CD than those in the low-risk subgroup in the UKB testing set. The detailed revisions can be found in Section **2.7 Risk Stratification for CD Onset, 4.5 Statistical Analysis, Fig. 5, and Supplementary Table 15.**

*"**2.7 Risk stratification for CD onset** To further assess how the 9-protein model stratifies the risk of CD onset, participants were classified into high- and low-risk subgroups using the optimal probability cutoff (0.484 for the XGBoost model), determined by maximizing the Youden index in the training cohort. Kaplan–Meier survival curves illustrated distinct cumulative risk patterns between the stratified subgroups (**Fig. 5A**). Participants in the high-risk subgroup exhibited a significantly higher risk of developing CD than those in low-risk subgroup, both in the training set (HR 11.6, P = 1.44 × 10-22) and in the geographically distinct UKB testing set (HR 4.23, P = 3.26 × 10-5), after adjusting for age, gender, ethnicity in model 1 (**Supplementary Table 15**). Similarly, Cox regression analyses for individual plasma proteins also demonstrated significant associations with CD onset (**Supplementary Table 16** and **Supplementary Fig. 9**)."*

*"**4.5 Statistical analysis…**Subsequently, Kaplan–Meier survival curves were generated to visualize how the protein model stratifies the risk of CD onset. The model probability cutoff value for the protein model was determined by maximizing the Youden index using the training dataset. The same cutoff was applied to the testing dataset. Cox proportional hazard models were then applied to assess the associations between the protein model and CD risk in both the training and testing sets.….."*

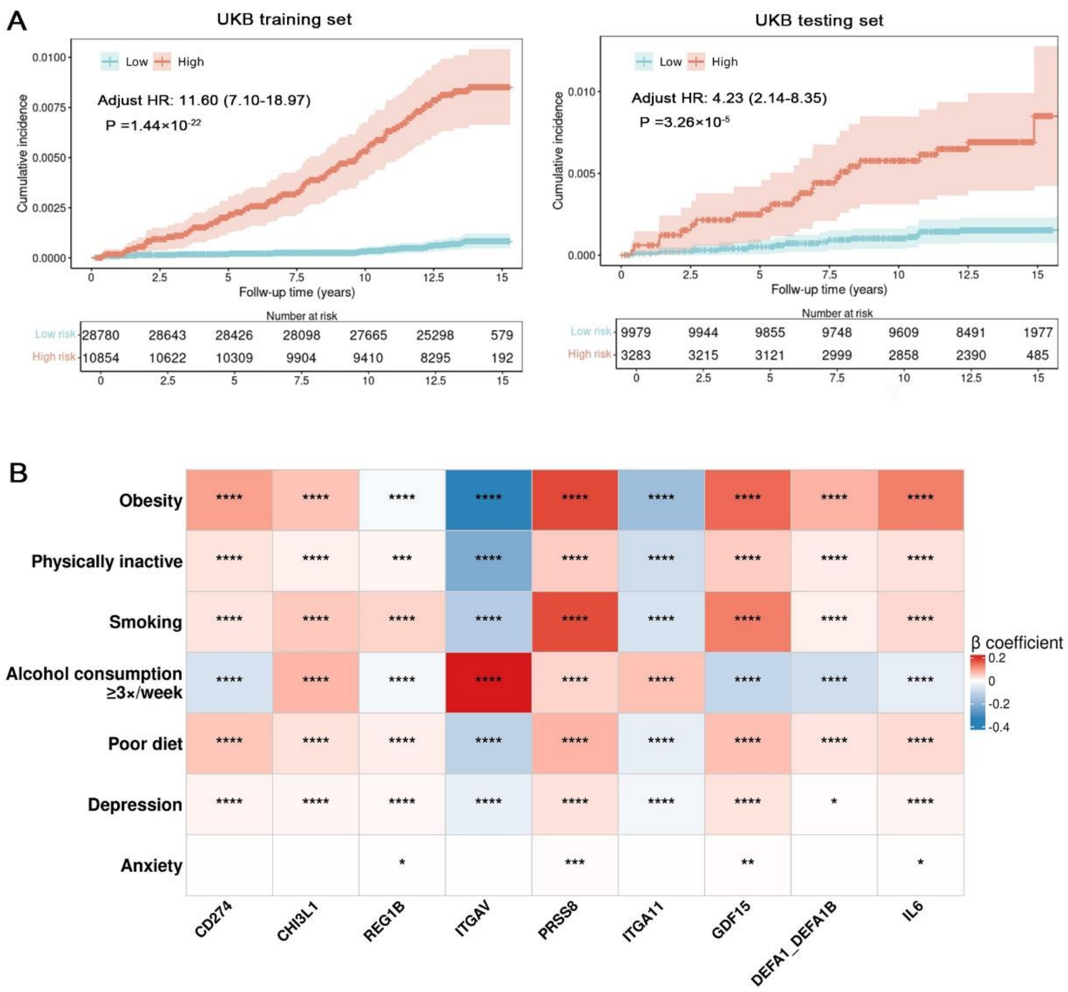


Fig. 5 Risk stratification and clinical associations of the 9-protein model with Crohn' s disease (CD) and CD-related phenotypes. A, *Protein model stratifies the risk of CD onset. Unadjusted Kaplan–Meier survival curves illustrate distinct cumulative risk trajectories for incident CD between the stratified subgroups (high-risk subgroup: red line; low-risk subgroup: blue line). The optimal cutoff (0.484) was determined using the Youden index in the training cohort and applied to both the training and testing sets. Cox proportional hazards models were used to estimate the association between the protein model and disease risk, adjusting for age, sex, and ethnicity. Hazard ratios (HRs) and P values are presented. The shaded area represents the 95% confidence interval (CI) of the survival curves.* ***B,*** *Clinical association between protein levels and CD-related phenotypes. Heatmap shows the associations between nine proteins in the protein model and modifiable risk factors for CD, including obesity, physical inactivity, smoking, alcohol consumption (≥3 times per week), poor diet, depression, and anxiety. Linear regression models were used, adjusting for age, sex, and ethnicity. Protein levels were treated as outcome variables, while CD-related phenotypes served as explanatory variables. The β coefficient represents the effect size, with positive associations in red and negative associations in blue. Statistical significance was determined using the false discovery rate (FDR) correction (*P<0.05, **P<0.01, ***P<0.001, ****P<0.0001).*

Supplementary Table 15: Results of protein model stratifying the risk of CD onset.

UKB testing set

	HR	*LCI*	*UCI*	*P value*
Model 1				
9-protein panel	*4.23*	*2.14*	*8.35*	*3.26E-05*
Model 2				
9-protein panel	*3.64*	*1.78*	*7.45*	*4.19E-04*

UKB training set

	HR	*LCI*	*UCI*	*P value*
Model 1				
9-protein panel	*11.60*	*7.10*	*18.97*	*1.44E-22*
Model 2				
9-protein panel	*13.14*	*7.90*	*21.84*	*2.99E-23*

Participants were classified into high - and low - risk subgroups using the optimal probability cutoff (0.484 for the XGBoost model), determined by maximizing the Youden index in the training cohort, and applied to both the training and testing sets. Cox proportional-hazard protein model and Crohn's disease risk. Model 1 was adjusted for age, sex and ethnicity. Model 2 was additionally adjusted for BMI, Townsend deprivation index, smoking, alcohol intake frequency, physical activity, dietary habits, anxiety, depression, medication history of antibiotics and medication history of NSAIDs.

Minor Points:

-The authors used the optimal proteins from the PREDICTS study and applied them to the UK Biobank cohort. However, it should be emphasized that the SOMAscan assay and the Olink Explore panel are two different assays with limited overlapping proteins.

Authors' response:

We thank the reviewer for highlighting this point. In response, we have added a sentence in the **3. Discussion** section immediately following the comparison between the PREDICTS and our protein models. This addition emphasizes that the SOMAscan assay used in the PREDICTS study and the Olink Explore panel used in our UK Biobank cohort have limited overlapping proteins, which may contribute to the differences in predictive performance.

*"**3. Discussion** …It should be noted, though, that the PREDICTS study utilized the SOMAscan assay, whereas the UK Biobank data were generated using the Olink Explore panel and the limited overlap in the proteins assessed may partly account for the observed differences in model performance...…"*

-The authors stated that the time before diagnosis in the GEM cohort was relatively short compared to the UK Biobank cohort. It should be noted that these two cohorts differ in how the CD diagnosis date was defined (physician-confirmed in GEM vs. ICD code appearance in UK Biobank) and represent different populations (at-risk first-degree relatives in GEM vs. an elderly general population in the UK Biobank).

Authors' response:

We thank the reviewer for this insightful comment. In the **3. Discussion**, we have now briefly clarified that the shorter pre-diagnostic time window in the GEM cohort may partly result from differences in how the diagnosis date is defined and from the differing characteristics of the study populations. Additionally, we have corrected an inconsistency in our reported time window for our cohort, ensuring that it accurately reflects the case population rather than the overall cohort. We apologize for this oversight and appreciate the opportunity to correct it.

*"**Discussion** …Notably, the time window before diagnosis in the GEM cohort was relatively short (IQR 1.0–3.5 years) compared with our cohort (IQR 4.4–10.7 years), though this difference should also be considered in the context of variations in diagnostic definitions (physician-confirmed vs. ICD code-based) and differences in study populations (at-risk first-degree relatives vs. an elderly general population)….."*

-Please use the TRIPOD+AI reporting guideline for the machine learning model that was used in this manuscript.

Authors' response:

We sincerely appreciate the reviewer's suggestion regarding the TRIPOD+AI reporting guideline. We have followed the TRIPOD+AI framework in preparing our manuscript and have ensured that our study adheres to its recommendations. Below, we provide the completed TRIPOD+AI statement for reference.

Section/Topic	Item	Development / evaluation[1]	Checklist item	Reported on page
TITLE				
Title	1	D;E	Identify the study as developing or evaluating the performance of a multivariable prediction model, the target population, and the outcome to be predicted	1
ABSTRACT				
Abstract	2	D;E	See TRIPOD+AI for Abstracts checklist	1-2
INTRODUCTION				
Background	3a	D;E	Explain the healthcare context (including whether diagnostic or prognostic) and rationale for developing or evaluating the prediction model, including references to existing models	3-4
	3b	D;E	Describe the target population and the intended purpose of the prediction model in the context of the care pathway, including its intended users (e.g., healthcare professionals, patients, public)	3-4
	3c	D;E	Describe any known health inequalities between sociodemographic groups	3-4
Objectives	4	D;E	Specify the study objectives, including whether the study describes the development or validation of a prediction model (or both)	4
METHODS				
Data	5a	D;E	Describe the sources of data separately for the development and evaluation datasets (e.g., randomised trial, cohort, routine care or registry data), the rationale for using these data, and representativeness of the data	14-15
	5b	D;E	Specify the dates of the collected participant data, including start and end of participant accrual; and, if applicable, end of follow-up	14-15
Participants	6a	D;E	Specify key elements of the study setting (e.g., primary care, secondary care, general population) including the number and location of centres	14-15
	6b	D;E	Describe the eligibility criteria for study participants	14-15
	6c	D;E	Give details of any treatments received, and how they were handled during model development or evaluation, if relevant	Not Applicable
Data preparation	7	D;E	Describe any data pre-processing and quality checking, including whether this was similar across relevant sociodemographic groups	15-16

Outcome	8a	D;E	Clearly define the outcome that is being predicted and the time horizon, including how and when assessed, the rationale for choosing this outcome, and whether the method of outcome assessment is consistent across sociodemographic groups	16-17
	8b	D;E	If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the outcome assessors	15
	8c	D;E	Report any actions to blind assessment of the outcome to be predicted	17
Predictors	9a	D	Describe the choice of initial predictors (e.g., literature, previous models, all available predictors) and any pre-selection of predictors before model building	16-18
	9b	D;E	Clearly define all predictors, including how and when they were measured (and any actions to blind assessment of predictors for the outcome and other predictors)	16-18
	9c	D;E	If predictor measurement requires subjective interpretation, describe the qualifications and demographic characteristics of the predictor assessors	16-18
Sample size	10	D;E	Explain how the study size was arrived at (separately for development and evaluation), and justify that the study size was sufficient to answer the research question. Include details of any sample size calculation	14-15
Missing data	11	D;E	Describe how missing data were handled. Provide reasons for omitting any data	17
Analytical methods	12a	D	Describe how the data were used (e.g., for development and evaluation of model performance) in the analysis, including whether the data were partitioned, considering any sample size requirements	14, 17-20
	12b	D	Depending on the type of model, describe how predictors were handled in the analyses (functional form, rescaling, transformation, or any standardisation).	15-16
	12c	D	Specify the type of model, rationale[2], all model-building steps, including any hyperparameter tuning, and method for internal validation	17-20
	12d	D;E	Describe if and how any heterogeneity in estimates of model parameter values and model performance was handled and quantified across clusters (e.g., hospitals, countries). See TRIPOD-Cluster for additional considerations[3]	15-16, 17-20
	12e	D;E	Specify all measures and plots used (and their rationale) to evaluate model performance (e.g., discrimination, calibration, clinical utility) and, if relevant, to compare multiple models	17-20
	12f	E	Describe any model updating (e.g., recalibration) arising from the model evaluation, either overall or for particular sociodemographic groups or settings	17-20
	12g	E	For model evaluation, describe how the model predictions were calculated (e.g., formula, code, object, application programming interface)	17-20
Class imbalance	13	D;E	If class imbalance methods were used, state why and how this was done, and any subsequent methods to recalibrate the model or the model predictions	19
Fairness	14	D;E	Describe any approaches that were used to address model fairness and their rationale	17-20
Model output	15	D	Specify the output of the prediction model (e.g., probabilities, classification). Provide details and rationale for any classification and how the thresholds were identified	17-20

Reviewer #2 (Remarks to the Author):

The core objective of this study is to develop a panel of serum protein biomarkers to predict Crohn's disease prior to disease onset. The study analyzes data from the UK Biobank (n > 52K), where baseline plasma proteomics measurements were taken, and patients were followed for up to 16 years. Additional variables, including clinical data, serum assays, polygenic risk scores, and demographic information, were also incorporated to compare the model based on the protein signature alone and in combination with these variables.

The methodology is standard, starting with feature selection based on univariate statistical analyses (Cox proportional hazards model) and proceeding to feature importance ranking with SHAP analysis (resulting in a selection of 10 proteins). These selected proteins were then used as inputs in the LightGBM algorithm (for protein-only, combined indicators, and other combinations). Validation was conducted using a random data split or 5-fold cross-validation (internal validation).

Overall, while the study may have some clinical merit, more robust validation is needed to clarify its potential utility. Here are some specific comments:

1) It is unclear whether feature selection was performed on the training data alone or on the entire dataset. Based on the description, it appears that feature selection may have been conducted on the entire dataset, with data splitting only performed for model training and validation. If true, this would introduce information leakage, compromising the model's generalizability.

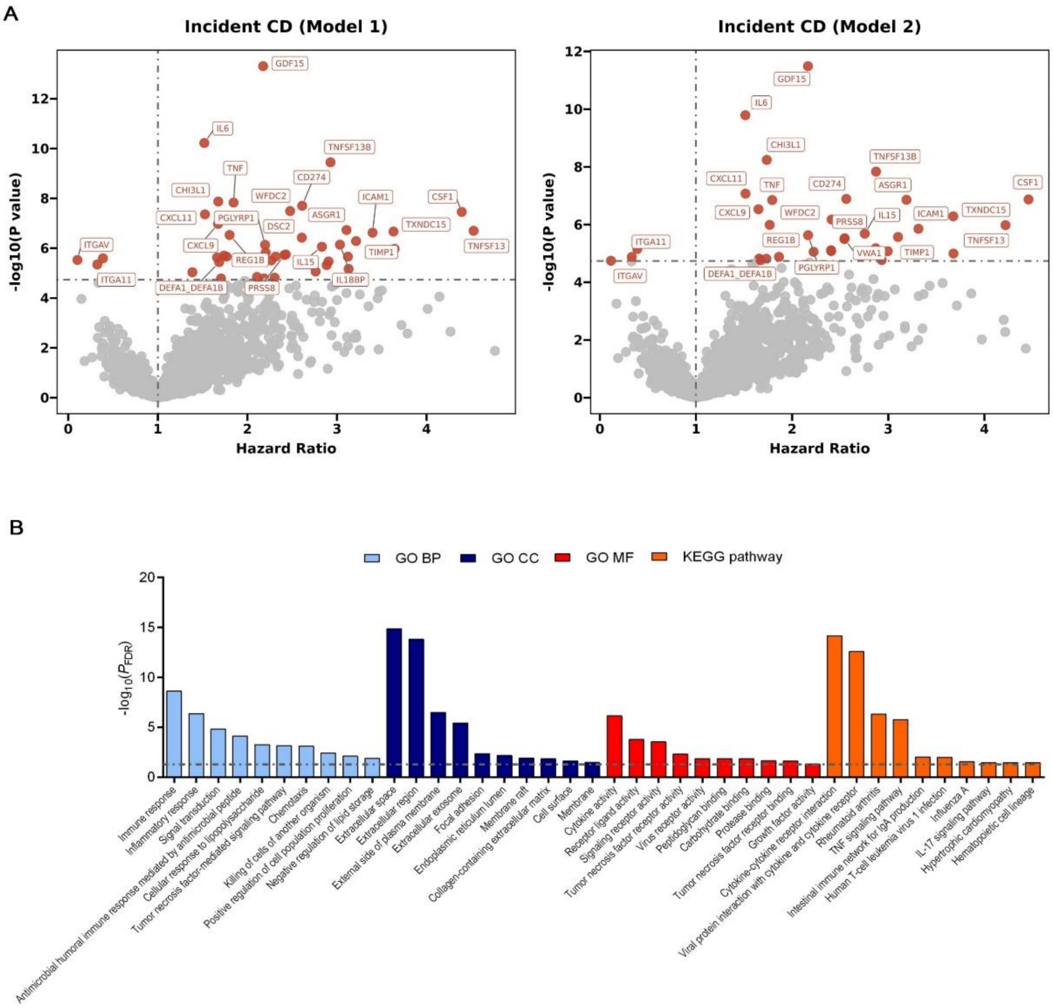
Authors' response:

We sincerely appreciate the time and effort you have invested in reviewing our manuscript. According to your comments, we have repeated the analyses, ensuring that feature selection was performed on the training set alone. This applies to both protein selection and the selection of demographic and serum prediction factors, all of which have now been revised to be based solely on the training set in order to avoid compromising the model's generalizability. Specific details of these revisions can be found in **4.5 Statistical analysis**, **Fig. 2, Supplementary Table 2, Supplementary Tables 20-21, and Supplementary Fig. 1.**

*"**4.5 Statistical analysis…**We conducted Cox proportional hazards model to figure out the association between each protein and incident CD in the training set, using hazard ratios (HRs) with 95% confidence intervals (CIs)….*

The important proteins were then identified using machine learning algorithms. Proteins that remained significant after Bonferroni corrections in both Model 1 and Model 2 were input into a FLAML AutoML framework, wherein LGBM was applied using 5-fold cross-validation with 10 repetitions on the training set. The significant proteins were subsequently ranked according to their mean SHAP values—a measure of feature importance assessing their contribution to model performance. Based on the SHAP ranking, a sequential forward selection approach was employed, where the top 10 proteins were sequentially added, and the protein panel yielding the highest AUC in the training set was ultimately selected, which was then visualized using SHAP plots…

To compare the predictive efficacy of proteins with other clinical indicators and to assess the combined predictive power, we further selected the clinical predictor data and delivered three sets of variables, demographic information (age), serological indicators (11 measures), and PRS of CD. These variables were selected from the training set using t-tests or chi-squared tests for demographic factors and Cox proportional hazard models for serological indicators. More details are shown in ***Supplementary Tables 21-22****…. ”*



"Fig. 2 Associations of plasma proteins with incident Crohn's disease. A, *Volcano plots showing the HR (x axis) and −log10(P value) (y axis) for the global associations of 2,736 proteins with incident Crohn's disease in the training set. All results for both Cox proportional hazard regression models 1 and 2 are shown here. Model 1 was adjusted for age, gender, and ethnicity. Model 2 was additionally adjusted for BMI, Townsend deprivation index, smoking, alcohol intake frequency, physical activity, dietary habits, anxiety, depression, and medication history of antibiotics and NSAIDs. P values were calculated under two-sided tests and no multiple comparisons were applied. Proteins above the horizontal dotted black line were significantly associated with incident Crohn's disease after Bonferroni corrections (P < 0.05) taking into account the number of proteins tested (n = 2,736).* ***B,*** *Enrichment for Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways. Significant proteins after Bonferroni correction* *derived from Cox proportional hazard regressions in model 1 or model 2 were fed into the DAVID website (https://david-d.ncifcrf.gov) for enrichment analysis. P values were calculated under two-sided tests and statistical significance was defined as a false discovery rate corrected P < 0.05 (dotted horizontal line). The number above each bar is the number of observed proteins in each pathway. Detailed results are shown in Supplementary Table 3. Abbreviations: BP, biological process; CC, cellular component; MF, molecular function; TNF, tumour necrosis factor."*

*"**Supplementary Table 2: Associations of plasma proteins with incident Crohn's disease....**" (The detailed table, provided in Supplementary Table 2, presents the full results of the Cox analysis for Model 1 and Model 2, detailing the associations between plasma proteins and incident Crohn's disease. Due to its substantial size—exceeding 5,000 rows—we have opted not to include it directly in this response. We kindly refer you to Supplementary Table 2 for the complete dataset.) "*

"Supplementary Table 20: Associations of serological indicators with incident Crohn's disease.

Model 1

Serological indicators	HR	Lower CI	Upper CI	P value
Leukocyte count, 10^9 cells/L	1.03	0.98	1.08	2.45E-01
Neutrophil count, 10^9 cells/L	1.13	1.00	1.29	**4.68E-02**
Neutrophil percentage, %	1.02	1.00	1.04	9.27E-02
Monocyte count, 10^9 cells/L	0.97	0.35	2.64	9.48E-01
Monocyte percentage, %	0.99	0.91	1.08	8.25E-01
Lymphocyte count, 10^9 cells/L	0.85	0.61	1.18	3.28E-01
Lymphocyte percentage, %	0.97	0.94	1.00	**2.79E-02**
C reactive protein, mg/L	1.03	1.00	1.06	**3.36E-02**
Platelet count, 10^9 cells/L	1.00	1.00	1.01	**1.96E-02**
SII	1.00	1.00	1.00	**5.58E-04**
NLR	1.03	1.01	1.06	**7.06E-03**
PLR	1.00	1.00	1.00	**8.17E-03**
LMR	0.99	0.92	1.07	8.47E-01
Erythrocyte count, 10^12 cells/L	0.79	0.46	1.35	3.90E-01
Reticulocyte count, 10^12 cells/L	3.90	0.66	23.01	1.33E-01
Red blood cell (erythrocyte) distribution width, %	1.21	1.07	1.37	**1.96E-03**
Hematocrit percentage, %	0.94	0.88	1.00	6.17E-02
Hemoglobin concentration, g/dL	0.87	0.72	1.05	1.48E-01
Albumin, g/L	0.86	0.80	0.92	**3.29E-05**
Creatinine, umol/L	0.80	0.67	0.95	**1.19E-02**
Triglycerides, mmol/L	1.17	0.99	1.38	6.74E-02
Cholesterol, mmol/L	0.80	0.67	0.95	**1.19E-02**
LDL Cholesterol, mmol/L	0.78	0.62	0.97	**2.91E-02**
HDL Cholesterol, mmol/L	0.42	0.21	0.81	**9.79E-03**
Apolipoprotein A, g/L	0.39	0.16	0.96	**4.04E-02**
Apolipoprotein B, g/L	0.42	0.18	0.96	**4.06E-02**
Vitamin D, nmol/L	1.00	0.99	1.01	4.09E-01

Model 2

Serological indicators	**HR**	**Lower CI**	**Upper CI**	**P value**
Leukocyte count, 10^9 cells/L	1.02	0.96	1.09	4.89E-01
Neutrophil count, 10^9 cells/L	1.11	0.97	1.26	1.20E-01
Neutrophil percentage, %	1.02	0.99	1.04	1.32E-01
Monocyte count, 10^9 cells/L	0.88	0.31	2.49	8.07E-01
Monocyte percentage, %	1.00	0.92	1.08	9.11E-01
Lymphocyte count, 10^9 cells/L	0.81	0.58	1.14	2.27E-01
Lymphocyte percentage, %	0.97	0.95	1.00	**4.31E-02**
C reactive protein, mg/L	1.03	1.00	1.06	6.25E-02
Platelet count, 10^9 cells/L	1.00	1.00	1.01	**3.02E-02**
SII	1.00	1.00	1.00	**5.97E-04**
NLR	1.03	1.01	1.06	**7.70E-03**
PLR	1.00	1.00	1.00	**5.70E-03**
LMR	0.99	0.92	1.07	8.37E-01
Erythrocyte count, 10^12 cells/L	0.79	0.46	1.35	3.90E-01
Reticulocyte count, 10^12 cells/L	3.67	0.63	21.28	1.47E-01
Red blood cell (erythrocyte) distribution width, %	1.19	1.05	1.35	**6.95E-03**
Hematocrit percentage, %	0.94	0.89	1.01	7.27E-02
Hemoglobin concentration, g/dL	0.88	0.73	1.06	1.87E-01
Albumin, g/L	0.86	0.80	0.92	**4.84E-05**
Creatinine, umol/L	0.81	0.68	0.97	**2.31E-02**
Triglycerides, mmol/L	1.16	0.97	1.38	1.01E-01
Cholesterol, mmol/L	0.81	0.68	0.97	**2.31E-02**
LDL Cholesterol, mmol/L	0.79	0.63	0.99	**4.27E-02**
HDL Cholesterol, mmol/L	0.39	0.19	0.81	**1.19E-02**
Apolipoprotein A, g/L	0.40	0.15	1.05	6.19E-02
Apolipoprotein B, g/L	0.44	0.19	1.01	5.27E-02
Vitamin D, nmol/L	1.00	0.99	1.01	7.17E-01

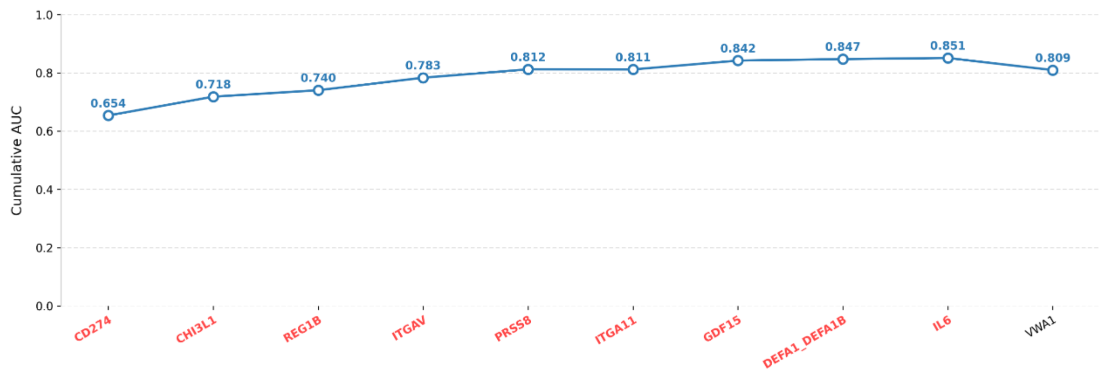
The serologic indicators associated with incident Crohn's disease were determined using Cox proportional hazard regression models in training set. Model 1 was adjusted for age, sex and ethnicity. Model 2 was additionally adjusted for BMI, Townsend deprivation index, smoking, alcohol intake frequency, physical activity, dietary habits, anxiety, depression, medication history of antibiotics and medication history of NSAIDs. P-values were calculated under two-sided tests. The significant serological indicators derived from the Cox proportional hazard regression in both model 1 and model 2 were finally included in the serum model, including lymphocyte percentage, platelet count, SII, NLR, PLR, erythrocyte distribution width, albumin, creatinine, cholesterol, LDL cholesterol, and HDL cholesterol. Abbreviations: SII, systemic immune-inflammation index (neutrophils × platelets/lymphocytes); NLR, neutrophil-to-lymphocyte ratio; PLR, platelet-to-lymphocyte ratio; LMR, lymphocyte-to-monocyte ratio."

"Supplementary Table 21: Baseline clinical demographics for training set.

	CD	**Control**	**P value**
	n=102	n=39,532	

Age, mean (SD), years	*58.75 (8.21)*	*57.06 (8.13)*	*0.035*
Sex			*1*
Female, n(%)	*55 (53.9)*	*21,244 (53.7)*	
Male, n(%)	*47 (46.1)*	*18,288 (46.3)*	
Ethnicity			*0.578*
Other, n(%)	*4 (3.9)*	*2,249 (5.7)*	
White, n(%)	*98 (96.1)*	*37,283 (94.3)*	
TD index, mean (SD)	*-1.12 (2.92)*	*-1.36 (3.03)*	*0.426*
BMI, mean (SD), kg/m2	*27.56 (4.36)*	*27.54 (4.79)*	*0.965*
Physical activity			*0.621*
Physically inactive, n(%)	*22 (21.6)*	*7,569 (19.1)*	
Physically active, n(%)	*80 (78.4)*	*31,963 (80.9)*	
Smoking status			*0.549*
Never smoker, n(%)	*50 (49.0)*	*21,515 (54.4)*	
Previous smoker, n(%)	*40 (39.2)*	*13,897 (35.2)*	
Current smoker, n(%)	*12 (11.8)*	*4,120 (10.4)*	
Alcohol consumption			*0.558*
Daily or almost daily, n(%)	*16 (15.7)*	*7,782 (19.7)*	
Three or four times a week, n(%)	*25 (24.5)*	*8,906 (22.5)*	
Once or twice a week, n(%)	*23 (22.5)*	*10,433 (26.4)*	
One to three times a month, n(%)	*11 (10.8)*	*4,349 (11.0)*	
Special occasions only, n(%)	*14 (13.7)*	*4,666 (11.8)*	
Never, n(%)	*13 (12.7)*	*3,396 (8.6)*	
Dietary habit			*0.257*
Poor diet	*70 (68.6)*	*24,786 (62.7)*	
Healthy diet	*32 (31.4)*	*14,746 (37.3)*	
Depression (%)	*8 (7.8)*	*2,307 (5.8)*	*0.514*
Anxiety (%)	*2 (2.0)*	*546 (1.4)*	*0.939*
History use of antibiotics (%)	*3 (2.9)*	*550 (1.4)*	*0.363*
History use of NSAIDs (%)	*0 (0.0)*	*669 (1.7)*	*0.347*

Note: Data are presented as mean (SD) or n (%). Differences between participants who developed Crohn's disease over time and those who remained healthy as of October 2022 were assessed using Student's t-test for continuous variables and Pearson's chi-squared test for discrete variables. Variables that showed significant differences between groups, including age, were finally included in the demographic model. Abbreviations: BMI, body mass index; TDI, Townsend deprivation index; CD, Crohn's disease; NSAIDs, nonsteroidal anti-inflammatory drugs."



*"**Supplementary Fig. 1: Sequential forward selection from preselected candidate proteins based on SHAP importance ranking in the training set.** The line chart illustrates cumulative AUC values upon the inclusion of proteins one by one in each iteration. The highest AUC was achieved after selecting 9 proteins."*

2)The initial feature selection approach is univariate (Cox model), followed by SHAP analysis to filter out less contributive features based on correlation analysis. While this is a reasonable approach, it is fairly basic. There have been several advances in machine learning feature selection methods that could be leveraged to enhance model performance.

Authors' response:

We sincerely appreciate the reviewer's insightful feedback and valuable suggestions regarding the feature selection methodology. We completely understand the importance of leveraging advanced machine learning feature selection methods to enhance model performance. In response to your comments, we explored and compared multiple advanced feature selection approaches to further optimize and refine our strategy. Specifically, we implemented the following methods:

a) Recursive Feature Elimination (RFE)
b) BorutaSHAP
c) Least Absolute Shrinkage and Selection Operator Cross-Validation (LassoCV)
d) Boruta
e) LightGBM + SHAP
f) Cox Regression + SHAP + Sequential Forward Selection (our refined approach)

For fairness, all feature selection methods were applied to the training data, and subsequent model development/validation followed the same protocol (four machine learning algorithms with 10 repeats of 5-fold cross-validation). Notably, models using features selected by alternative methods (a–e) did not outperform those derived from our optimized Cox + SHAP + Sequential Forward Selection approach in the UK Biobank testing set.

Building on this, we have further refined our feature selection scheme in the revised manuscript. In the refined version, we first two Cox models on the training set to identify 35 intersecting positive proteins. These were then used as input for the LGBM algorithm, which was trained using 5-fold cross-validation with 10 repeats. The significant proteins were ranked based on their mean SHAP values, and a sequential forward selection approach was subsequently employed, wherein the top 10 proteins were sequentially added. Ultimately, the protein panel that yielded the highest AUC in the training set was selected. In the revised manuscript, we expanded the original feature selection scheme by adding repeated cross-validation and sequential forward selection based on SHAP values, resulting in a more robust and better-performing model. The revised method consistently achieved the highest test-set AUC across classifiers, indicating its superior generalizability. These modifications have been made in the corresponding parts of the manuscript, specifically in section **2.2 Proteins associated with incident CD, 4.5 Statistical analysis, Fig. 2, Fig. 3, Supplementary Table 2, and Supplementary Fig. 1.** Moreover, the detailed results of all feature selection methods, including the proteins selected and their corresponding test-set AUCs, are presented in the two tables attached below for your reference.

Thank you again for your constructive suggestions, which have significantly contributed to the improvement and rigor of our study.

Feature Selection Methods and Selected Proteins

Feature Selection Method	*Number of Selected Proteins*	*Selected Proteins*
Recursive Feature Elimination (RFE)	*20*	*ENOX2, ADGRG1, ANKRD54, GUCA2A, CD4, CHI3L1, CHL1, CST7, CXADR, DDR1, CAMLG, VTCN1, TFF3, TMEM25, PPCDC, NFKBIE, NTRK2, MRPL46, MSTN, KRT17*
BorutaSHAP	*1*	*CHI3L1*
Least Absolute Shrinkage and Selection Operator Cross-Validation (LassoCV)	*12*	*GDF15, HRG, CHI3L1, CXCL11, C4BPB, VSIG2, TNF, TNFSF13, TXNDC15, REG1B, NTF3, IL6*
Boruta	*13*	*GDF15, CD274, CHI3L1, CST7, CCL11, CCL21, VSIG2, REG1B, PRSS8, MMP10, LBP, IFNG, IL6*
Light Gradient Boosting Machine (LGBM) + SHAP	*10*	*CCL21, CD274, CHI3L1, REG1B, CEACAM21, COL4A4, PRSS8, IFNG, TIMP3, HTR1A*
Cox Regression + SHAP + Sequential Forward Selection	***9***	***CD274, CHI3L1, REG1B, ITGAV, PRSS8, ITGA11, GDF15, DEFA1_DEFA1B, IL6***

Performance of Protein-Based Models with Different Feature Selection Methods

	AUC (95%CI)			
	LGBM	*XGBoost*	*RF*	*Extra trees*
Testing set				
Recursive Feature Elimination (RFE)	*0.59 (0.58-0.60)*	*0.61 (0.59-0.62)*	*0.64 (0.63-0.66)*	*0.67 (0.66-0.68)*
BorutaSHAP	*0.59 (0.58-0.60)*	*0.59 (0.58-0.60)*	*0.59 (0.59-0.59)*	*0.64 (0.64-0.64)*

Least Absolute Shrinkage and Selection Operator Cross-Validation (LassoCV)	*0.64 (0.62-0.65)*	*0.65 (0.64-0.67)*	*0.65 (0.64-0.66)*	*0.68 (0.68-0.69)*
Boruta	*0.66 (0.65-0.67)*	*0.65 (0.64-0.67)*	*0.67 (0.66-0.68)*	*0.68 (0.67-0.70)*
Light Gradient Boosting Machine (LGBM) + SHAP	*0.62 (0.60-0.63)*	*0.64 (0.61-0.66)*	*0.67 (0.65-0.69)*	*0.64 (0.63-0.65)*
Cox Regression + SHAP + Sequential Forward Selection	***0.73 (0.72-0.73)***	***0.72 (0.71-0.73)***	***0.71 (0.70-0.73)***	***0.72 (0.71-0.73)***
Training set				
Recursive Feature Elimination (RFE)	*0.87 (0.86-0.89)*	*0.92 (0.89-0.94)*	*0.83 (0.79-0.87)*	*0.86 (0.81-0.90)*
BorutaSHAP	*0.68 (0.67-0.69)*	*0.70 (0.69-0.71)*	*0.69 (0.69-0.69)*	*0.65 (0.65-0.65)*
Least Absolute Shrinkage and Selection Operator Cross-Validation (LassoCV)	*0.86 (0.84-0.89)*	*0.83 (0.81-0.86)*	*0.82 (0.78-0.86)*	*0.75 (0.75-0.76)*
Boruta	*0.87 (0.84-0.89)*	*0.84 (0.82-0.86)*	*0.86 (0.84-0.89)*	*0.79 (0.77-0.81)*
Light Gradient Boosting Machine (LGBM) + SHAP	*0.85 (0.84-0.86)*	*0.82 (0.81-0.84)*	*0.79 (0.78-0.81)*	*0.76 (0.75-0.77)*
Cox Regression + SHAP + Sequential Forward Selection	*0.85 (0.82-0.88)*	*0.83 (0.80-0.86)*	*0.84 (0.82-0.87)*	*0.76 (0.74-0.78)*

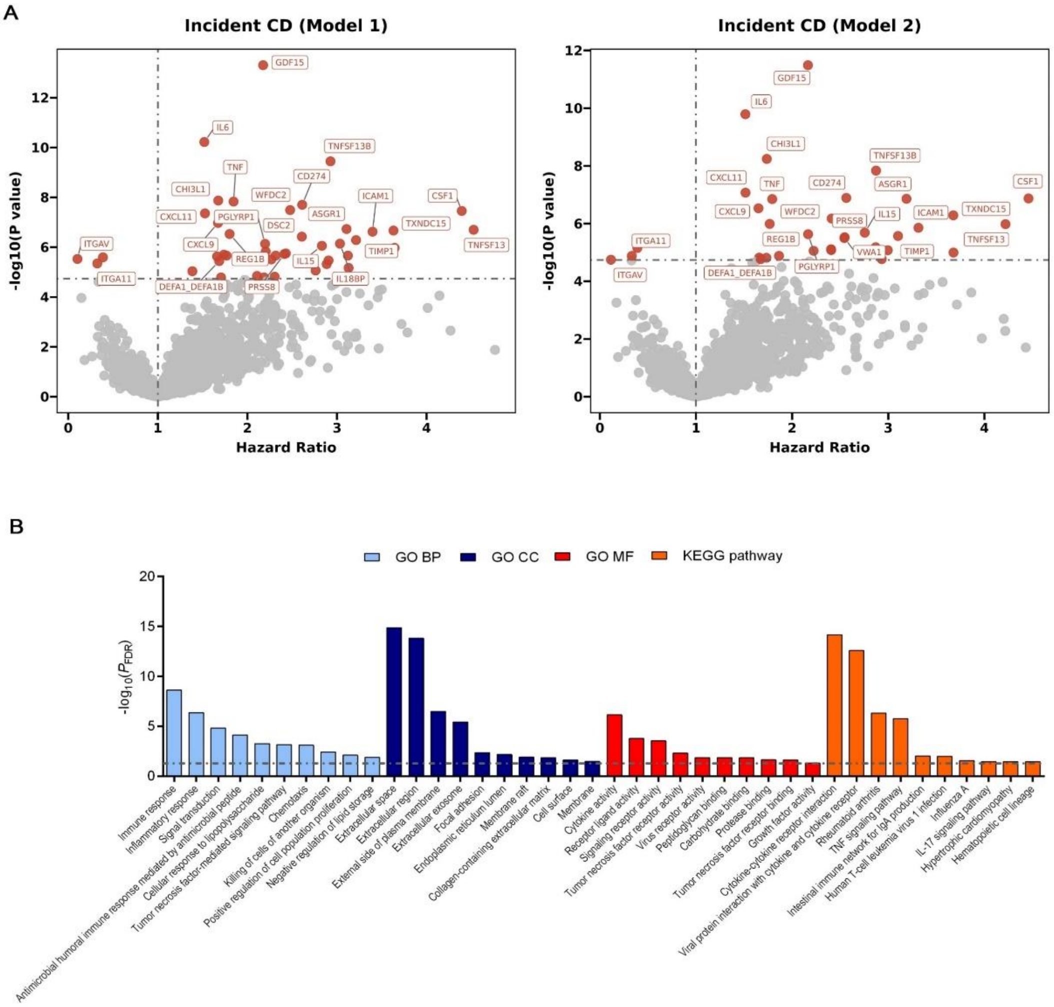
*"**Results 2.2 Proteins associated with incident CD** Among the 2,736 proteomic biomarkers examined, 44 proteins were significantly associated with incident CD after adjusting for age, sex, and ethnicity in model 1 (**Fig. 2A** and **Supplementary Table 2**). Then, we performed a sensitivity analysis using model 2, which additionally adjusted for TDI, BMI, alcohol intake frequency, smoking status, dietary habits, physical activity, comorbidities (depression and anxiety), and history use of antibiotics and NSAIDs, and confirmed that 35 of the significant associations were consistent with those found in model 1 (**Fig. 2A**). Thirty-two proteins (GDF15, IL6, CHI3L1, TNFSF13B, CXCL11, CD274, CSF1, ASGR1, TNF, CXCL9, TXNDC15, WFDC2, REG1B, TNFSF13, ICAM1, IL15, PGLYRP1, TIMP1, PRSS8, VWA1, IL18BP, DSC2, IL10RB, LGALS9, TNFRSF10A, PLAUR, TNFRSF1A, VSIG2, DEFA1_DEFA1B, TNFRSF1B, TNFRSF14, REG3A)*

were positively associated with the risk of CD onset, while three proteins (GSN, ITGA11, ITGAV) were negatively associated. Notably, GDF15 (HR 2.16, $P = 8.86 \times 10^{-9}$) and IL6 (HR 1.51, $P = 4.43 \times 10^{-7}$) had the most significant associations with CD after Bonferroni corrections….

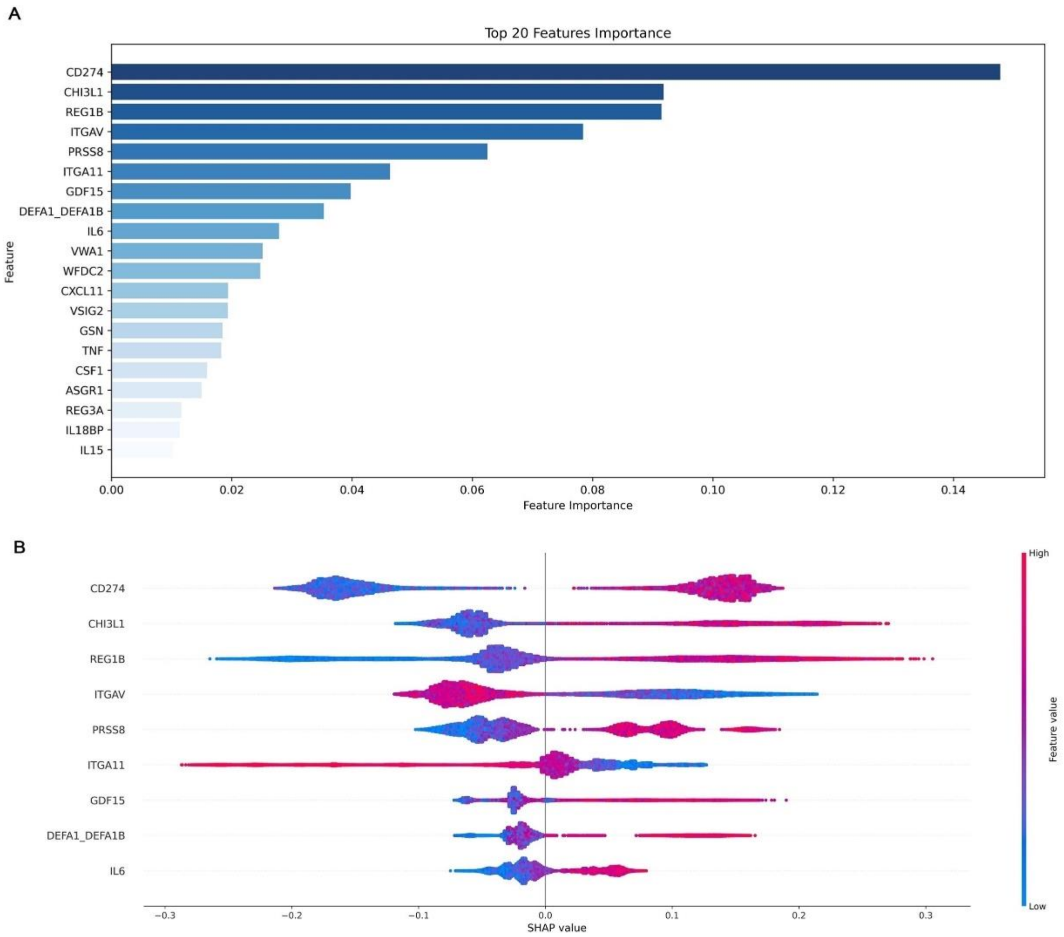
*…**2.4 Protein importance ranking** For the proteins associated with CD in both models 1 and 2, we further ranked them according to their importance in predicting CD. As illustrated in the bar chart (**Fig. 3A**), CD274, CHI3L1, and REG1B ranked top three in protein importance ordering. After using the sequential forward selection scheme, we ultimately selected the top nine proteins (CD274, CHI3L1, REG1B, ITGAV, PRSS8, ITGA11, GDF15, DEFA1_DEFA1B, and IL6) for CD prediction in subsequent analyses (**Supplementary Fig. 1**)…. "*

"Methods 4.5 Statistical analysis *We conducted Cox proportional hazards model to figure out the association between each protein and incident CD in the training set, using hazard ratios (HRs) with 95% confidence intervals (CIs). Model 1 adjusted for age, gender and ethnicity. For model 2, we additionally adjusted for BMI, TDI, alcohol intake frequency, smoking status, dietary habits, physical activity, anxiety, depression, and medication history of antibiotics and NSAIDs.... Bonferroni corrections were employed to evaluate the significance of the associations accounting for the total number of proteins analyzed (n = 2,736)…*

…The important proteins were then identified using machine learning algorithms. Proteins that remained significant after Bonferroni corrections in both Model 1 and Model 2 were input into a FLAML AutoML framework, wherein LGBM was applied using 5-fold cross-validation with 10 repetitions on the training set. The significant proteins were subsequently ranked according to their mean SHAP values—a measure of feature importance assessing their contribution to model performance. Based on the SHAP ranking, a sequential forward selection approach was employed, where the top 10 proteins were sequentially added, and the protein panel yielding the highest AUC in the training set was ultimately selected, which was then visualized using SHAP plots..."

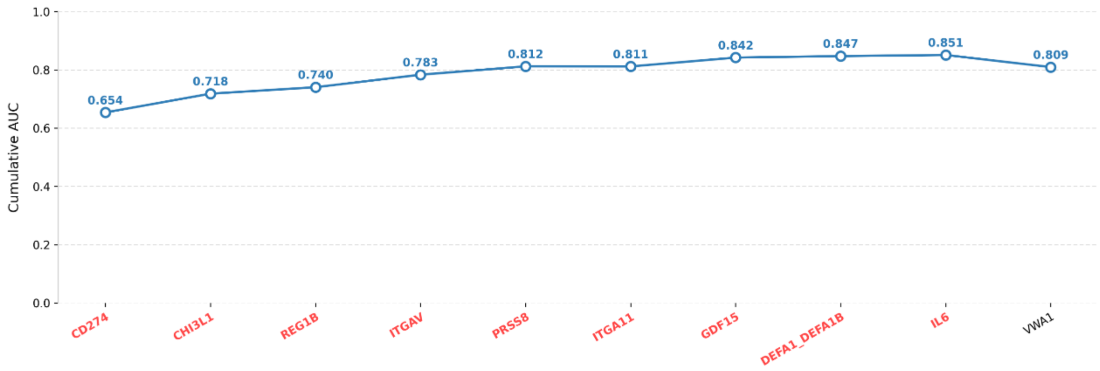


*"**Fig. 2 Associations of plasma proteins with incident Crohn's disease. A,** Volcano plots showing the HR (x axis) and −log10(P value) (y axis) for the global associations of 2,736 proteins with incident Crohn's disease in the training set. All results for both Cox proportional hazard regression models 1 and 2 are shown here. Model 1 was adjusted for age, gender, and ethnicity. Model 2 was additionally adjusted for BMI, Townsend deprivation index, smoking, alcohol intake frequency, physical activity, dietary habits, anxiety, depression, and medication history of antibiotics and NSAIDs. P values were calculated under two-sided tests and no multiple comparisons were applied. Proteins above the horizontal dotted black line were significantly associated with incident Crohn's disease after Bonferroni corrections (P < 0.05) taking into account the number of proteins tested (n = 2,736). **B,** Enrichment for Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways. Significant proteins after Bonferroni correction derived from Cox proportional hazard regressions in model 1 or model 2 were fed into the DAVID website (https://david-d.ncifcrf.gov) for enrichment analysis. P values were calculated under two-sided tests and statistical significance was defined as a false discovery rate corrected P < 0.05 (dotted horizontal line). The number above each bar is the number of observed proteins in each pathway. Detailed results are shown in Supplementary Table 3. Abbreviations: BP, biological process; CC, cellular component; MF, molecular function; TNF, tumour necrosis factor."*



*"**Fig. 3 Protein importance ranking and SHAP visualization of modeling on all-time incident Crohn's disease populations. A**, The top 20 proteins according to the average absolute SHAP value. The bar chart indicates the importance of the sorted proteins based on their contributions to the prediction of future Crohn's disease. **B**, SHAP visualization plot of selected proteins. The width of the range of the horizontal bars can be understood as the extent of the contribution to the prediction of Crohn's disease; the wider their range, the greater the contribution. The color of the horizontal bars denotes the magnitude of plasma proteins, which was coded in a gradient from blue (low) to red (high), shown as the color bar on the right-hand side. The direction on the x axis indicates the likelihood of developing Crohn's disease (right) or being healthy (left)."*

*"**Supplementary Table 2: Associations of plasma proteins with incident Crohn's disease….**" (The detailed table, provided in Supplementary Table 2, presents the full results of the Cox analysis for Model 1 and Model 2, detailing the associations between plasma proteins and incident Crohn's disease. Due to its substantial size—exceeding 5,000 rows—we have opted not to include it directly in this response. We kindly refer you to Supplementary Table 2 for the complete dataset.)"*



*"**Supplementary Fig. 1: Sequential forward selection from preselected candidate proteins based on SHAP importance ranking in the training set.** The line chart illustrates cumulative AUC values upon the inclusion of proteins one by one in each iteration. The highest AUC was achieved after selecting 9 proteins."*

3) Only one machine learning model, LightGBM, was employed. Although LightGBM is a valid choice, the approach could be broadened by exploring multiple models or ensemble methods, such as meta-learning or superlearners, to integrate predictions from various algorithms and potentially improve performance.

Authors' response:

We sincerely appreciate the reviewer's insightful suggestions. To address this valuable feedback, we have repeated all the analyses with four machine learning algorithms: Light Gradient Boosting Machine (LightGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees. Across both the main and sensitivity analyses, these models produced comparable results. Notably, the XGBoost model obtained the optimal AUC in the external testing set (0.79, 95% CI 0.77 – 0.81) and the second-best AUC in the geographically distinct UKB testing set (0.72 for a 75/25 split; 0.76 for an 80/20 split), indicating its robust generalizability across independent cohorts.

To further explore potential improvements, we also implemented a stacking-based ensemble model (meta-learning approach) combining predictions from all four algorithms. However, the stacked model did not outperform the best-performing individual models in our main analysis. Given this, we have retained the results of the multiple machine learning models in the manuscript. The corresponding updates have been made in **Abstract**, **2.4 Protein importance ranking**, **2.5 Predictive accuracy of proteomics-based models, 2.6 Results of replication validation analyses**, **3. Discussion, 4.5 Statistical analysis, Fig. 4, Supplementary Tables 4-14，and Supplementary Fig. 2-8.** We also present the stacking model results below for reference. Once again, we truly appreciate the reviewer's thoughtful suggestions, which have helped strengthen the rigor and clarity of our study.

Results of ROC analyses with additional Stacking model.

	UKB testing set (AUC [95%CI])				
	LGBM	***XGBoost***	***RF***	***Extra trees***	***Stacking***
9-protein panel	*0.73 (0.72-0.73)*	*0.72 (0.71-0.73)*	*0.71 (0.70-0.73)*	*0.72 (0.71-0.73)*	*0.70 (0.68-0.71)*

Demographic	*0.56 (0.55-0.56)*	*0.56 (0.55-0.57)*	*0.58 (0.58-0.58)*	*0.55 (0.55-0.56)*	*0.57 (0.57-0.58)*
Serum	*0.63 (0.62-0.64)*	*0.63 (0.61-0.64)*	*0.66 (0.64-0.67)*	*0.62 (0.59-0.65)*	*0.63 (0.62-0.64)*
PRS	*0.67 (0.66-0.67)*	*0.67 (0.66-0.68)*	*0.63 (0.61-0.65)*	*0.69 (0.69-0.70)*	*0.65 (0.62-0.68)*
Protein+Demographic	*0.72 (0.71-0.73)*	*0.72 (0.72-0.73)*	*0.70 (0.69-0.72)*	*0.73 (0.72-0.74)*	*0.72 (0.70-0.73)*
Protein+Serum	*0.72 (0.70-0.74)*	*0.72 (0.70-0.74)*	*0.73 (0.72-0.74)*	*0.76 (0.75-0.76)*	*0.70 (0.69-0.72)*
Protein+PRS	*0.75 (0.74-0.76)*	*0.74 (0.73-0.75)*	*0.70 (0.68-0.71)*	*0.75 (0.74-0.77)*	*0.73 (0.71-0.74)*
Protein+Demographic+Serum	*0.73 (0.72-0.74)*	*0.73 (0.72-0.74)*	*0.74 (0.72-0.75)*	*0.72 (0.71-0.74)*	*0.70 (0.68-0.72)*
Protein+Demographic+PRS	*0.75 (0.75-0.76)*	*0.73 (0.72-0.75)*	*0.72 (0.70-0.73)*	*0.78 (0.76-0.79)*	*0.73 (0.71-0.75)*
Protein+Serum+PRS	*0.76 (0.75-0.77)*	*0.76 (0.75-0.77)*	*0.75 (0.74-0.77)*	*0.77 (0.76-0.78)*	*0.74 (0.71-0.77)*
Combined	*0.76 (0.75-0.77)*	*0.78 (0.77-0.79)*	*0.75 (0.74-0.76)*	*0.72 (0.68-0.76)*	*0.76 (0.74-0.78)*

	Southern China testing set (AUC [95%CI])				
	LGBM	***XGBoost***	***RF***	***Extra trees***	***Stacking***
9-protein panel	*0.76 (0.73-0.79)*	*0.79 (0.77-0.81)*	*0.77 (0.73-0.80)*	*0.77 (0.76-0.79)*	*0.68 (0.61-0.75)*

	UKB training set (AUC [95%CI])				
	LGBM	***XGBoost***	***RF***	***Extra trees***	***Stacking***
9-protein panel	*0.85 (0.82-0.88)*	*0.83 (0.80-0.86)*	*0.84 (0.82-0.87)*	*0.76 (0.74-0.78)*	*0.89 (0.86-0.92)*
Demographic	*0.66 (0.66-0.67)*	*0.66 (0.66-0.67)*	*0.67 (0.67-0.68)*	*0.66 (0.65-0.67)*	*0.68 (0.68-0.68)*
Serum	*0.77 (0.75-0.78)*	*0.78 (0.75-0.80)*	*0.75 (0.74-0.75)*	*0.72 (0.65-0.78)*	*0.83 (0.79-0.87)*
PRS	*0.61 (0.61-0.61)*	*0.64 (0.62-0.65)*	*0.71 (0.69-0.72)*	*0.64 (0.62-0.66)*	*0.72 (0.70-0.74)*
Protein+Demographic	*0.82 (0.81-0.83)*	*0.82 (0.80-0.83)*	*0.82 (0.78-0.85)*	*0.77 (0.75-0.79)*	*0.86 (0.83-0.89)*
Protein+Serum	*0.90 (0.86-0.93)*	*0.88 (0.85-0.91)*	*0.93 (0.90-0.96)*	*0.79 (0.79-0.80)*	*0.97 (0.94-0.99)*
Protein+PRS	*0.88 (0.86-0.90)*	*0.88 (0.84-0.92)*	*0.85 (0.82-0.88)*	*0.79 (0.78-0.79)*	*0.92 (0.89-0.95)*
Protein+Demographic+Serum	*0.87 (0.85-0.89)*	*0.89 (0.86-0.92)*	*0.88 (0.83-0.93)*	*0.79 (0.78-0.81)*	*0.95 (0.92-0.98)*

Protein+Demographic+PRS	*0.88 (0.85-0.90)*	*0.87 (0.83-0.91)*	*0.80 (0.77-0.82)*	*0.83 (0.81-0.86)*	*0.92 (0.90-0.95)*
Protein+Serum+PRS	*0.91 (0.89-0.93)*	*0.93 (0.89-0.98)*	*0.86 (0.81-0.91)*	*0.80 (0.77-0.82)*	*0.98 (0.95-1.00)*
Combined	*0.92 (0.90-0.95)*	*0.89 (0.85-0.93)*	*0.87 (0.84-0.91)*	*0.80 (0.77-0.83)*	*0.95 (0.93-0.97)*

*“**Abstract** …Machine learning models using four algorithms were constructed based on these proteins, with XGBoost demonstrating the best performance. In the geographically distinct UKB cohort, the protein model (area under the curve [AUC] 0.76) outperformed clinical risk models (highest AUC 0.67; P < 0.05) and was externally validated in the Southern China cohort (AUC = 0.79). Combining these proteins with clinical data enabled desirable predictions up to 16 years pre-diagnosis (AUC 0.78). Our model performed better than existing PREDICTS model (AUC 0.68, P < 0.05) when applied to the UK Biobank population...”*

*“**Results 2.5 Predictive accuracy of proteomics-based models** We evaluated the predictive performance of the 9-protein panel for future CD onset using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (**Fig. 4A-C, Supplementary Tables 4-7,** and **Supplementary Fig. 2-3**). The 9-protein model demonstrated considerable prediction performance across all four algorithms for all-time incident CD. In the UKB testing set (geographically distinct), the area under the curves (AUCs) ranged from 0.71 to 0.73 for a 75/25 training/testing split and 0.71 to 0.77 for an 80/20 split, while in the external Southern China testing set, they ranged from 0.76 to 0.79. Notably, the XGBoost model obtained the optimal AUC in the external testing set (0.79, 95% CI 0.77–0.81) and the second-best AUC in the UKB testing set (0.72 for a 75/25 split, 95% CI 0.71–0.73; 0.76 for an 80/20 split, 95% CI 0.74–0.77), indicating its robust generalizability across independent cohorts. In the UKB testing set, the 9-protein model demonstrated superior predictive performance across four algorithms compared to clinical risk models based on demographics, serum markers, and polygenic risk score (PRS) for CD (highest AUC among clinical risk models for XGBoost: 0.67, Mann‑Whitney U test: P < 0.001). For XGBoost, combining the PRS of CD with the protein panel significantly improved the AUC to 0.74 (95% CI 0.73–0.75), and adding all clinical risk factors further increased the AUC to 0.78 (95% CI 0.77–0.79) for a 75/25 split.*

*We then deployed the six optimal proteins identified in the PREDICTS study into the UKB and compared the performance with our models. The results showed that the predictive performance of the PREDICTS model (AUC range: 0.64-0.69, XGBoost AUC: 0.68) was significantly inferior to our protein model (AUC range: 0.71-0.73, XGBoost AUC: 0.72) in predicting all-time incident CD whether or not it was combined with clinical risk factors (P < 0.05) (**Supplementary Tables 8-9** and **Supplementary Fig. 4**).*

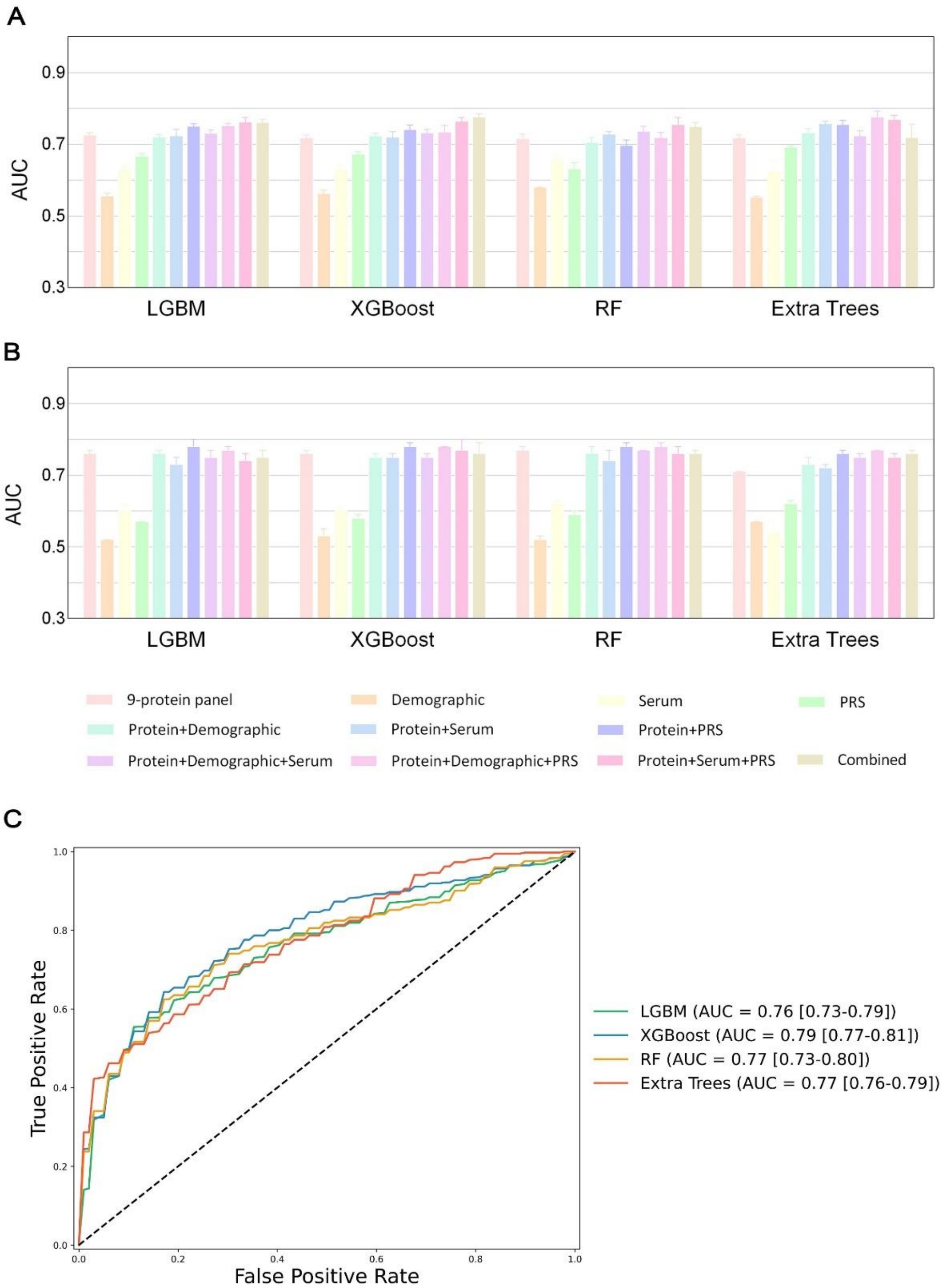
2.6 Results of replication validation analyses** Our findings remained consistent in the sensitivity analyses for XGBoost. At different time points, the protein model still showed higher predictive value (AUC 0.69-0.72) than clinical risk models (AUC 0.53-0.67) (**Supplementary Table 10** and **Supplementary Fig. 5-6**). When randomly subsampling a control group matched 1:1 with CD patients by age, sex, and race, the 9-protein model achieved an AUC of 0.71 (95% CI 0.68–0.73) in predicting all-time incident CD, increasing to 0.76 (95% CI 0.74–0.78) with the addition of all clinical risk factors (**Supplementary Tables 11-12** and **Supplementary Fig. 7**). The results remained consistent after excluding individuals who developed CD within the first two years of follow-up (**Supplementary Tables 13–14** and **Supplementary

***Fig. 8**).*

."

***"3. Discussion** In this study of over 52,000 participants, we applied a large-scale proteomic analysis using four machine learning algorithms to generate a 9-protein model to non-invasively predict future CD onset. The proteomic model (AUC 0.76) significantly outperformed clinical risk models based on demographics, serum biomarkers, and genetics (highest AUC 0.67) in the UKB testing set (geographically distinct). It was further externally validated in a Southern China cohort (AUC 0.79). Combining proteins with clinical risk data enabled better predictions up to 16 years pre-diagnosis (AUC 0.78)..."*

***"Methods 4.5 Statistical analysis…**Based on the optimally selected protein panel, we developed a protein predictive model using four machine-learning algorithms (LGBM, XGBoost, RF, and Extra trees)..."*



"Fig. 4 Predictive accuracy of plasma proteins panel, alone or in combination with clinical variables in region-based UKB and external testing cohorts. *Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease in the geographically distinct UKB testing cohort with 25% (**A**) and 20% (**B**) proportions. Model performance evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum,* *and PRS of Crohn’s disease.* ***C****, Receiver operating curves show the performance of protein model for predicting Crohn’s disease in the external testing set from Southern China, evaluated using the four machine learning algorithms.”*

"Supplementary Table 4: Results of ROC analyses.

	UKB testing set (AUC [95%CI])			
	LGBM	XGBoost	RF	Extra trees
9-protein panel	0.73 (0.72-0.73)	0.72 (0.71-0.73)	0.71 (0.70-0.73)	0.72 (0.71-0.73)
Demographic	0.56 (0.55-0.56)	0.56 (0.55-0.57)	0.58 (0.58-0.58)	0.55 (0.55-0.56)
Serum	0.63 (0.62-0.64)	0.63 (0.61-0.64)	0.66 (0.64-0.67)	0.62 (0.59-0.65)
PRS	0.67 (0.66-0.67)	0.67 (0.66-0.68)	0.63 (0.61-0.65)	0.69 (0.69-0.70)
Protein+Demographic	0.72 (0.71-0.73)	0.72 (0.72-0.73)	0.70 (0.69-0.72)	0.73 (0.72-0.74)
Protein+Serum	0.72 (0.70-0.74)	0.72 (0.70-0.74)	0.73 (0.72-0.74)	0.76 (0.75-0.76)
Protein+PRS	0.75 (0.74-0.76)	0.74 (0.73-0.75)	0.70 (0.68-0.71)	0.75 (0.74-0.77)
Protein+Demographic+Serum	0.73 (0.72-0.74)	0.73 (0.72-0.74)	0.74 (0.72-0.75)	0.72 (0.71-0.74)
Protein+Demographic+PRS	0.75 (0.75-0.76)	0.73 (0.72-0.75)	0.72 (0.70-0.73)	0.78 (0.76-0.79)
Protein+Serum+PRS	0.76 (0.75-0.77)	0.76 (0.75-0.77)	0.75 (0.74-0.77)	0.77 (0.76-0.78)
Combined	0.76 (0.75-0.77)	0.78 (0.77-0.79)	0.75 (0.74-0.76)	0.72 (0.68-0.76)

	Southern China testing set (AUC [95%CI])			
	LGBM	XGBoost	RF	Extra trees
9-protein panel	0.76 (0.73-0.79)	0.79 (0.77-0.81)	0.77 (0.73-0.80)	0.77 (0.76-0.79)

	UKB training set (AUC [95%CI])			
	LGBM	XGBoost	RF	Extra trees
9-protein panel	0.85 (0.82-0.88)	0.83 (0.80-0.86)	0.84 (0.82-0.87)	0.76 (0.74-0.78)
Demographic	0.66 (0.66-0.67)	0.66 (0.66-0.67)	0.67 (0.67-0.68)	0.66 (0.65-0.67)
Serum	0.77 (0.75-0.78)	0.78 (0.75-0.80)	0.75 (0.74-0.75)	0.72 (0.65-0.78)
PRS	0.61 (0.61-0.61)	0.64 (0.62-0.65)	0.71 (0.69-0.72)	0.64 (0.62-0.66)
Protein+Demographic	0.82 (0.81-0.83)	0.82 (0.80-0.83)	0.82 (0.78-0.85)	0.77 (0.75-0.79)
Protein+Serum	0.90 (0.86-0.93)	0.88 (0.85-0.91)	0.93 (0.90-0.96)	0.79 (0.79-0.80)
Protein+PRS	0.88 (0.86-0.90)	0.88 (0.84-0.92)	0.85 (0.82-0.88)	0.79 (0.78-0.79)
Protein+Demographic+Serum	0.87 (0.85-0.89)	0.89 (0.86-0.92)	0.88 (0.83-0.93)	0.79 (0.78-0.81)
Protein+Demographic+PRS	0.88 (0.85-0.90)	0.87 (0.83-0.91)	0.80 (0.77-0.82)	0.83 (0.81-0.86)
Protein+Serum+PRS	0.91 (0.89-0.93)	0.93 (0.89-0.98)	0.86 (0.81-0.91)	0.80 (0.77-0.82)
Combined	0.92 (0.90-0.95)	0.89 (0.85-0.93)	0.87 (0.84-0.91)	0.80 (0.77-0.83)

We randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 3:1 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

“Supplementary Table 5: Comparison between AUCs for all-year incident CD.

	LGBM										
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.83E-04	1.83E-04	1.78E-04	2.12E-01	6.78E-01	1.31E-03	5.21E-01	4.40E-04	2.83E-03	5.83E-04
Demographic	1.83E-04		1.83E-04	1.78E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
Serum	1.83E-04	1.83E-04		2.40E-04	1.83E-04	4.40E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
PRS	1.78E-04	1.78E-04	2.40E-04		1.78E-04	2.78E-03	1.78E-04	1.78E-04	1.78E-04	1.78E-04	1.78E-04
Protein+ Demographic	2.12E-01	1.83E-04	1.83E-04	1.78E-04		6.40E-02	4.40E-04	7.57E-02	2.46E-04	2.20E-03	2.46E-04
Protein+ Serum	6.78E-01	1.83E-04	4.40E-04	2.78E-03	6.40E-02		3.61E-03	9.70E-01	1.01E-03	2.20E-03	1.01E-03
Protein+ PRS	1.31E-03	1.83E-04	1.83E-04	1.78E-04	4.40E-04	3.61E-03		7.28E-03	9.70E-01	1.73E-02	1.62E-01
Protein+ Demographic+ Serum	5.21E-01	1.83E-04	1.83E-04	1.78E-04	7.57E-02	9.70E-01	7.28E-03		1.71E-03	2.83E-03	2.83E-03
Protein+ Demographic+ PRS	4.40E-04	1.83E-04	1.83E-04	1.78E-04	2.46E-04	1.01E-03	9.70E-01	1.71E-03		1.13E-02	2.41E-01
Protein+ Serum+ PRS	2.83E-03	1.83E-04	1.83E-04	1.78E-04	2.20E-03	2.20E-03	1.73E-02	2.83E-03	1.13E-02		5.71E-01

Combined	*5.83E-04*	*1.83E-04*	*1.83E-04*	*1.78E-04*	*2.46E-04*	*1.01E-03*	*1.62E-01*	*2.83E-03*	*2.41E-01*	*5.71E-01*	

XGBoost											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.82E-04*	*1.83E-04*	*4.29E-04*	*3.45E-01*	*7.34E-01*	*2.11E-02*	*2.11E-02*	*8.90E-02*	*1.83E-04*	*1.83E-04*
Demographic	*1.82E-04*		*1.82E-04*	*1.77E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*
Serum	*1.83E-04*	*1.82E-04*		*9.86E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
PRS	*4.29E-04*	*1.77E-04*	*9.86E-04*		*1.78E-04*	*4.29E-04*	*1.78E-04*	*1.78E-04*	*2.40E-04*	*1.78E-04*	*1.78E-04*
Protein+ Demographic	*3.45E-01*	*1.82E-04*	*1.83E-04*	*1.78E-04*		*9.10E-01*	*3.12E-02*	*1.40E-01*	*1.86E-01*	*2.46E-04*	*2.46E-04*
Protein+ Serum	*7.34E-01*	*1.82E-04*	*1.83E-04*	*4.29E-04*	*9.10E-01*		*3.76E-02*	*4.27E-01*	*1.40E-01*	*5.83E-04*	*4.40E-04*
Protein+ PRS	*2.11E-02*	*1.82E-04*	*1.83E-04*	*1.78E-04*	*3.12E-02*	*3.76E-02*		*1.40E-01*	*7.34E-01*	*3.12E-02*	*3.61E-03*
Protein+ Demographic+ Serum	*2.11E-02*	*1.82E-04*	*1.83E-04*	*1.78E-04*	*1.40E-01*	*4.27E-01*	*1.40E-01*		*3.07E-01*	*5.83E-04*	*4.40E-04*
Protein+ Demographic+ PRS	*8.90E-02*	*1.82E-04*	*1.83E-04*	*2.40E-04*	*1.86E-01*	*1.40E-01*	*7.34E-01*	*3.07E-01*		*3.76E-02*	*2.83E-03*

Protein+ Serum+ PRS	1.83E-04	1.82E-04	1.83E-04	1.78E-04	2.46E-04	5.83E-04	3.12E-02	5.83E-04	3.76E-02		2.41E-01
Combined	1.83E-04	1.82E-04	1.83E-04	1.78E-04	2.46E-04	4.40E-04	3.61E-03	4.40E-04	2.83E-03	2.41E-01	

RF											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.62E-04	1.31E-03	1.68E-04	4.26E-01	3.07E-01	8.66E-02	8.90E-02	7.91E-01	2.20E-03	1.71E-03
Demographic	1.62E-04		1.61E-04	1.49E-04	1.55E-04	1.62E-04	1.44E-04	1.62E-04	1.57E-04	1.62E-04	1.62E-04
Serum	1.31E-03	1.61E-04		5.25E-02	9.69E-04	1.82E-04	4.27E-03	1.82E-04	1.76E-04	5.80E-04	1.82E-04
PRS	1.68E-04	1.49E-04	5.25E-02		1.60E-04	1.68E-04	1.49E-04	1.68E-04	1.62E-04	1.68E-04	1.68E-04
Protein+ Demographic	4.26E-01	1.55E-04	9.69E-04	1.60E-04		3.71E-02	5.67E-01	4.46E-03	4.40E-02	4.46E-03	9.73E-04
Protein+ Serum	3.07E-01	1.62E-04	1.82E-04	1.68E-04	3.71E-02		2.04E-03	3.85E-01	5.20E-01	4.59E-03	9.11E-03
Protein+ PRS	8.66E-02	1.44E-04	4.27E-03	1.49E-04	5.67E-01	2.04E-03		8.59E-03	1.99E-02	2.62E-03	1.57E-03
Protein+ Demographic+ Serum	8.90E-02	1.62E-04	1.82E-04	1.68E-04	4.46E-03	3.85E-01	8.59E-03		1.20E-01	3.12E-02	3.45E-01

Protein+ Demographic+ PRS	*7.91E-01*	*1.57E-04*	*1.76E-04*	*1.62E-04*	*4.40E-02*	*5.20E-01*	*1.99E-02*	*1.20E-01*		*5.69E-03*	*7.15E-03*
Protein+ Serum+ PRS	*2.20E-03*	*1.62E-04*	*5.80E-04*	*1.68E-04*	*4.46E-03*	*4.59E-03*	*2.62E-03*	*3.12E-02*	*5.69E-03*		*1.86E-01*
Combined	*1.71E-03*	*1.62E-04*	*1.82E-04*	*1.68E-04*	*9.73E-04*	*9.11E-03*	*1.57E-03*	*3.45E-01*	*7.15E-03*	*1.86E-01*	

Extra trees											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.60E-04*	*1.82E-04*	*1.67E-04*	*1.03E-01*	*3.28E-04*	*1.26E-03*	*7.91E-01*	*4.37E-04*	*3.28E-04*	*5.20E-01*
Demographic	*1.60E-04*		*2.60E-03*	*1.48E-04*	*1.57E-04*	*1.61E-04*	*1.52E-04*	*1.61E-04*	*1.61E-04*	*1.61E-04*	*1.60E-04*
Serum	*1.82E-04*	*2.60E-03*		*1.68E-04*	*1.79E-04*	*1.83E-04*	*1.73E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*9.08E-03*
PRS	*1.67E-04*	*1.48E-04*	*1.68E-04*		*1.64E-04*	*1.68E-04*	*1.58E-04*	*2.67E-03*	*1.68E-04*	*1.68E-04*	*4.70E-01*
Protein+ Demographic	*1.03E-01*	*1.57E-04*	*1.79E-04*	*1.64E-04*		*2.79E-03*	*1.84E-01*	*2.12E-01*	*4.53E-03*	*9.90E-04*	*7.91E-01*
Protein+ Serum	*3.28E-04*	*1.61E-04*	*1.83E-04*	*1.68E-04*	*2.79E-03*		*4.71E-01*	*1.31E-03*	*2.11E-02*	*1.73E-02*	*2.12E-01*
Protein+ PRS	*1.26E-03*	*1.52E-04*	*1.73E-04*	*1.58E-04*	*1.84E-01*	*4.71E-01*		*8.85E-03*	*1.03E-01*	*1.20E-01*	*4.25E-01*

Protein+ Demographic+ Serum	*7.91E-01*	*1.61E-04*	*1.83E-04*	*2.67E-03*	*2.12E-01*	*1.31E-03*	*8.85E-03*		*2.83E-03*	*7.69E-04*	*7.34E-01*
Protein+ Demographic+ PRS	*4.37E-04*	*1.61E-04*	*1.83E-04*	*1.68E-04*	*4.53E-03*	*2.11E-02*	*1.03E-01*	*2.83E-03*		*1.21E-01*	*2.57E-02*
Protein+ Serum+ PRS	*3.28E-04*	*1.61E-04*	*1.83E-04*	*1.68E-04*	*9.90E-04*	*1.73E-02*	*1.20E-01*	*7.69E-04*	*1.21E-01*		*3.76E-02*
Combined	*5.20E-01*	*1.60E-04*	*9.08E-03*	*4.70E-01*	*7.91E-01*	*2.12E-01*	*4.25E-01*	*7.34E-01*	*2.57E-02*	*3.76E-02*	

Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 6: Results of ROC analyses after splitting the participants into a training and a testing set according to the recruitment centers in a ratio of 8:2.

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.76 (0.75-0.77)*	*0.76 (0.74-0.77)*	*0.77 (0.77-0.78)*	*0.71 (0.70-0.71)*
Demographic	*0.52 (0.51-0.52)*	*0.53 (0.52-0.55)*	*0.52 (0.51-0.53)*	*0.57 (0.56-0.57)*
Serum	*0.60 (0.58-0.62)*	*0.60 (0.58-0.61)*	*0.62 (0.61-0.63)*	*0.54 (0.51-0.57)*
PRS	*0.57 (0.56-0.57)*	*0.58 (0.56-0.59)*	*0.59 (0.58-0.60)*	*0.62 (0.61-0.63)*
Protein+Demographic	*0.76 (0.75-0.77)*	*0.75 (0.74-0.76)*	*0.76 (0.75-0.78)*	*0.73 (0.72-0.75)*
Protein+Serum	*0.73 (0.71-0.75)*	*0.75 (0.73-0.76)*	*0.74 (0.72-0.77)*	*0.72 (0.70-0.73)*
Protein+PRS	*0.78 (0.77-0.80)*	*0.78 (0.78-0.79)*	*0.78 (0.76-0.79)*	*0.76 (0.74-0.77)*
Protein+Demographic+Serum	*0.75 (0.73-0.77)*	*0.75 (0.73-0.76)*	*0.77 (0.76-0.77)*	*0.75 (0.74-0.76)*
Protein+Demographic+PRS	*0.77 (0.75-0.78)*	*0.78 (0.77-0.78)*	*0.78 (0.77-0.79)*	*0.77 (0.76-0.77)*
Protein+Serum+PRS	*0.74 (0.73-0.76)*	*0.77 (0.75-0.80)*	*0.76 (0.75-0.78)*	*0.75 (0.74-0.76)*
Combined	*0.75 (0.74-0.77)*	*0.76 (0.74-0.79)*	*0.76 (0.75-0.77)*	*0.76 (0.74-0.77)*
Training set				
9-protein panel	*0.86 (0.83-0.89)*	*0.84 (0.79-0.88)*	*0.81 (0.78-0.83)*	*0.74 (0.73-0.76)*
Demographic	*0.68 (0.68-0.68)*	*0.67 (0.67-0.68)*	*0.67 (0.67-0.67)*	*0.64 (0.64-0.65)*
Serum	*0.81 (0.75-0.88)*	*0.77 (0.73-0.81)*	*0.77 (0.73-0.82)*	*0.70 (0.67-0.73)*
PRS	*0.68 (0.66-0.70)*	*0.69 (0.68-0.70)*	*0.70 (0.69-0.72)*	*0.64 (0.61-0.66)*
Protein+Demographic	*0.88 (0.85-0.91)*	*0.83 (0.79-0.88)*	*0.83 (0.80-0.87)*	*0.75 (0.73-0.78)*
Protein+Serum	*0.91 (0.88-0.94)*	*0.89 (0.84-0.93)*	*0.88 (0.84-0.91)*	*0.78 (0.76-0.81)*
Protein+PRS	*0.94 (0.92-0.97)*	*0.89 (0.85-0.93)*	*0.91 (0.87-0.95)*	*0.76 (0.76-0.76)*
Protein+Demographic+Serum	*0.94 (0.91-0.97)*	*0.87 (0.82-0.92)*	*0.79 (0.78-0.79)*	*0.79 (0.77-0.80)*
Protein+Demographic+PRS	*0.89 (0.86-0.92)*	*0.86 (0.83-0.89)*	*0.84 (0.80-0.87)*	*0.77 (0.76-0.78)*
Protein+Serum+PRS	*0.97 (0.93-1.00)*	*0.95 (0.91-0.98)*	*0.96 (0.94-0.97)*	*0.80 (0.79-0.82)*
Combined	*0.96 (0.93-1.00)*	*0.93 (0.90-0.97)*	*0.91 (0.87-0.95)*	*0.79 (0.79-0.80)*

We randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 8:2 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

“Supplementary Table 7: Comparison between AUCs for all-time incident CD after splitting the participants into a training and a testing set in a ratio of 8:2.

LGBM											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.82E-04*	*1.83E-04*	*1.83E-04*	*4.27E-01*	*2.57E-02*	*9.11E-03*	*6.78E-01*	*4.27E-01*	*5.71E-01*	*9.70E-01*
Demographic	*1.82E-04*		*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*
Serum	*1.83E-04*	*1.82E-04*		*1.13E-02*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
PRS	*1.83E-04*	*1.82E-04*	*1.13E-02*		*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
Protein+ Demographic	*4.27E-01*	*1.82E-04*	*1.83E-04*	*1.83E-04*		*1.40E-02*	*7.28E-03*	*2.41E-01*	*5.71E-01*	*2.12E-01*	*4.73E-01*
Protein+ Serum	*2.57E-02*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*1.40E-02*		*1.31E-03*	*7.57E-02*	*9.11E-03*	*2.12E-01*	*5.39E-02*
Protein+ PRS	*9.11E-03*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*7.28E-03*	*1.31E-03*		*1.40E-02*	*5.39E-02*	*3.61E-03*	*2.11E-02*
Protein+ Demographic+ Serum	*6.78E-01*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*2.41E-01*	*7.57E-02*	*1.40E-02*		*3.85E-01*	*8.50E-01*	*5.71E-01*
Protein+ Demographic+ PRS	*4.27E-01*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*5.71E-01*	*9.11E-03*	*5.39E-02*	*3.85E-01*		*1.40E-01*	*3.85E-01*

Protein+ Serum+ PRS	5.71E-01	1.82E-04	1.83E-04	1.83E-04	2.12E-01	2.12E-01	3.61E-03	8.50E-01	1.40E-01		6.78E-01
Combined	9.70E-01	1.82E-04	1.83E-04	1.83E-04	4.73E-01	5.39E-02	2.11E-02	5.71E-01	3.85E-01	6.78E-01	

XGBoost											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		1.83E-04	1.83E-04	1.83E-04	4.27E-01	4.27E-01	2.57E-02	2.41E-01	7.57E-02	8.90E-02	3.07E-01
Demographic	1.83E-04		4.40E-04	2.20E-03	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
Serum	1.83E-04	4.40E-04		8.90E-02	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
PRS	1.83E-04	2.20E-03	8.90E-02		1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
Protein+ Demographic	4.27E-01	1.83E-04	1.83E-04	1.83E-04		9.10E-01	1.01E-03	8.50E-01	1.40E-02	2.57E-02	1.21E-01
Protein+ Serum	4.27E-01	1.83E-04	1.83E-04	1.83E-04	9.10E-01		5.83E-04	9.10E-01	1.13E-02	3.12E-02	1.04E-01
Protein+ PRS	2.57E-02	1.83E-04	1.83E-04	1.83E-04	1.01E-03	5.83E-04		4.40E-04	1.04E-01	1.00E+00	1.40E-01
Protein+ Demographic+ Serum	2.41E-01	1.83E-04	1.83E-04	1.83E-04	8.50E-01	9.10E-01	4.40E-04		4.59E-03	1.13E-02	4.52E-02

Protein+ Demographic+ PRS	*7.57E-02*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.40E-02*	*1.13E-02*	*1.04E-01*	*4.59E-03*		*6.23E-01*	*7.34E-01*
Protein+ Serum+ PRS	*8.90E-02*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*2.57E-02*	*3.12E-02*	*1.00E+00*	*1.13E-02*	*6.23E-01*		*6.23E-01*
Combined	*3.07E-01*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.21E-01*	*1.04E-01*	*1.40E-01*	*4.52E-02*	*7.34E-01*	*6.23E-01*	

RF											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.61E-04*	*1.82E-04*	*1.77E-04*	*5.20E-01*	*1.04E-01*	*2.41E-01*	*8.49E-01*	*9.10E-01*	*3.45E-01*	*6.39E-02*
Demographic	*1.61E-04*		*1.62E-04*	*1.57E-04*	*1.62E-04*	*1.61E-04*	*1.62E-04*	*1.44E-04*	*1.62E-04*	*1.62E-04*	*1.62E-04*
Serum	*1.82E-04*	*1.62E-04*		*5.70E-03*	*1.83E-04*	*1.82E-04*	*1.83E-04*	*1.63E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
PRS	*1.77E-04*	*1.57E-04*	*5.70E-03*		*1.78E-04*	*1.77E-04*	*1.78E-04*	*1.58E-04*	*1.78E-04*	*1.78E-04*	*1.78E-04*
Protein+ Demographic	*5.20E-01*	*1.62E-04*	*1.83E-04*	*1.78E-04*		*2.41E-01*	*3.07E-01*	*6.75E-01*	*2.73E-01*	*9.10E-01*	*8.50E-01*
Protein+ Serum	*1.04E-01*	*1.61E-04*	*1.82E-04*	*1.77E-04*	*2.41E-01*		*3.76E-02*	*3.41E-01*	*8.89E-02*	*4.27E-01*	*4.27E-01*
Protein+ PRS	*2.41E-01*	*1.62E-04*	*1.83E-04*	*1.78E-04*	*3.07E-01*	*3.76E-02*		*1.83E-01*	*4.27E-01*	*2.41E-01*	*1.04E-01*

Protein+ Demographic+ Serum	8.49E-01	1.44E-04	1.63E-04	1.58E-04	6.75E-01	3.41E-01	1.83E-01		6.75E-01	6.75E-01	1.02E-01
Protein+ Demographic+ PRS	9.10E-01	1.62E-04	1.83E-04	1.78E-04	2.73E-01	8.89E-02	4.27E-01	6.75E-01		1.40E-01	2.57E-02
Protein+ Serum+ PRS	3.45E-01	1.62E-04	1.83E-04	1.78E-04	9.10E-01	4.27E-01	2.41E-01	6.75E-01	1.40E-01		9.10E-01
Combined	6.39E-02	1.62E-04	1.83E-04	1.78E-04	8.50E-01	4.27E-01	1.04E-01	1.02E-01	2.57E-02	9.10E-01	

Extra trees											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.71E-04	1.73E-04	1.70E-04	2.72E-03	6.31E-02	6.90E-04	1.70E-04	1.74E-04	1.74E-04	1.74E-04
Demographic	1.71E-04		8.85E-02	1.76E-04	1.79E-04	1.80E-04	1.66E-04	1.76E-04	1.80E-04	1.80E-04	1.80E-04
Serum	1.73E-04	8.85E-02		5.69E-04	1.81E-04	1.82E-04	1.68E-04	1.78E-04	1.82E-04	1.82E-04	1.82E-04
PRS	1.70E-04	1.76E-04	5.69E-04		1.78E-04	1.79E-04	1.65E-04	1.75E-04	1.79E-04	1.79E-04	1.79E-04
Protein+ Demographic	2.72E-03	1.79E-04	1.81E-04	1.78E-04		2.41E-01	2.04E-02	1.40E-01	1.00E-03	4.51E-02	3.11E-02
Protein+ Serum	6.31E-02	1.80E-04	1.82E-04	1.79E-04	2.41E-01		1.61E-03	2.79E-03	1.83E-04	1.71E-03	7.69E-04

Protein+ PRS	6.90E-04	1.66E-04	1.68E-04	1.65E-04	2.04E-02	1.61E-03		1.19E-01	4.38E-03	8.73E-02	8.49E-01
Protein+ Demographic+ Serum	1.70E-04	1.76E-04	1.78E-04	1.75E-04	1.40E-01	2.79E-03	1.19E-01		7.20E-03	3.07E-01	3.84E-01
Protein+ Demographic+ PRS	1.74E-04	1.80E-04	1.82E-04	1.79E-04	1.00E-03	1.83E-04	4.38E-03	7.20E-03		2.83E-03	2.73E-01
Protein+ Serum+ PRS	1.74E-04	1.80E-04	1.82E-04	1.79E-04	4.51E-02	1.71E-03	8.73E-02	3.07E-01	2.83E-03		5.21E-01
Combined	1.74E-04	1.80E-04	1.82E-04	1.79E-04	3.11E-02	7.69E-04	8.49E-01	3.84E-01	2.73E-01	5.21E-01	

Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

*“**Supplementary Table 8: Results of ROC analyses of the PREDICTS model in UK Biobank.***

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
PREDICTS	0.67 (0.65-0.69)	0.68 (0.67-0.69)	0.69 (0.67-0.70)	0.64 (0.62-0.65)
9-protein panel	0.73 (0.72-0.73)	0.72 (0.71-0.73)	0.71 (0.70-0.73)	0.72 (0.71-0.73)
Protein+Demographic	0.72 (0.71-0.73)	0.72 (0.72-0.73)	0.70 (0.69-0.72)	0.73 (0.72-0.74)
Protein+Serum	0.72 (0.70-0.74)	0.72 (0.70-0.74)	0.73 (0.72-0.74)	0.76 (0.75-0.76)
Protein+PRS	0.75 (0.74-0.76)	0.74 (0.73-0.75)	0.70 (0.68-0.71)	0.75 (0.74-0.77)
Protein+Demographic+Serum	0.73 (0.72-0.74)	0.73 (0.72-0.74)	0.74 (0.72-0.75)	0.72 (0.71-0.74)
Protein+Demographic+PRS	0.75 (0.75-0.76)	0.73 (0.72-0.75)	0.72 (0.70-0.73)	0.78 (0.76-0.79)
Protein+Serum+PRS	0.76 (0.75-0.77)	0.76 (0.75-0.77)	0.75 (0.74-0.77)	0.77 (0.76-0.78)
Combined	0.76 (0.75-0.77)	0.78 (0.77-0.79)	0.75 (0.74-0.76)	0.72 (0.68-0.76)
Training set				
PREDICTS	0.87 (0.81-0.93)	0.78 (0.72-0.83)	0.77 (0.74-0.81)	0.69 (0.66-0.73)
9-protein panel	0.85 (0.82-0.88)	0.83 (0.80-0.86)	0.84 (0.82-0.87)	0.76 (0.74-0.78)
Protein+Demographic	0.82 (0.81-0.83)	0.82 (0.80-0.83)	0.82 (0.78-0.85)	0.77 (0.75-0.79)
Protein+Serum	0.90 (0.86-0.93)	0.88 (0.85-0.91)	0.93 (0.90-0.96)	0.79 (0.79-0.80)
Protein+PRS	0.88 (0.86-0.90)	0.88 (0.84-0.92)	0.85 (0.82-0.88)	0.79 (0.78-0.79)
Protein+Demographic+Serum	0.87 (0.85-0.89)	0.89 (0.86-0.92)	0.88 (0.83-0.93)	0.79 (0.78-0.81)
Protein+Demographic+PRS	0.88 (0.85-0.90)	0.87 (0.83-0.91)	0.80 (0.77-0.82)	0.83 (0.81-0.86)
Protein+Serum+PRS	0.91 (0.89-0.93)	0.93 (0.89-0.98)	0.86 (0.81-0.91)	0.80 (0.77-0.82)
Combined	0.92 (0.90-0.95)	0.89 (0.85-0.93)	0.87 (0.84-0.91)	0.80 (0.77-0.83)

We externally validated the six optimal proteins identified in PREDICTS study (C-reactive protein, C5, PRSS2, APCS, OMD, ACAN) using UK Biobank populations. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn’s disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease.”

"Supplementary Table 9: Comparison between AUCs between PREDICTS model and other models.

	9-protein panel	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
LGBM	1.83E-04	2.46E-04	2.20E-03	1.83E-04	1.83E-04	1.83E-04	2.46E-04	1.83E-04
XGBoost	7.69E-04	2.46E-04	3.61E-03	3.30E-04	4.40E-04	1.71E-03	1.83E-04	1.83E-04
RF	1.72E-02	3.43E-01	5.80E-04	8.49E-01	7.65E-04	2.08E-02	1.31E-03	3.28E-04
Extra trees	1.77E-04	1.74E-04	1.78E-04	1.68E-04	1.78E-04	1.78E-04	1.78E-04	2.77E-03

Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 10: Results of ROC analyses within different time points in the UKB testing set.

10-year incident CD

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.69 (0.68-0.71)	0.69 (0.69-0.70)	0.69 (0.67-0.71)	0.72 (0.71-0.72)
Demographic	0.54 (0.53-0.54)	0.53 (0.52-0.54)	0.56 (0.55-0.57)	0.57 (0.56-0.57)
Serum	0.64 (0.61-0.67)	0.63 (0.62-0.64)	0.61 (0.58-0.63)	0.62 (0.60-0.65)
PRS	0.60 (0.59-0.60)	0.57 (0.56-0.58)	0.59 (0.59-0.60)	0.62 (0.58-0.65)
Protein+Demographic	0.69 (0.68-0.70)	0.70 (0.70-0.71)	0.72 (0.71-0.72)	0.72 (0.71-0.73)
Protein+Serum	0.70 (0.69-0.71)	0.71 (0.69-0.72)	0.71 (0.71-0.72)	0.72 (0.71-0.73)
Protein+PRS	0.72 (0.71-0.72)	0.71 (0.71-0.72)	0.73 (0.72-0.74)	0.75 (0.74-0.77)
Protein+Demographic+Serum	0.71 (0.70-0.71)	0.70 (0.69-0.72)	0.72 (0.71-0.72)	0.72 (0.70-0.73)
Protein+Demographic+PRS	0.71 (0.70-0.72)	0.72 (0.71-0.73)	0.72 (0.70-0.73)	0.75 (0.74-0.77)
Protein+Serum+PRS	0.73 (0.72-0.74)	0.73 (0.72-0.74)	0.73 (0.72-0.74)	0.74 (0.73-0.75)
Combined	0.72 (0.71-0.74)	0.72 (0.71-0.74)	0.71 (0.70-0.73)	0.75 (0.74-0.75)
Training set				
9-protein panel	0.94 (0.93-0.96)	0.95 (0.93-0.96)	0.89 (0.86-0.92)	0.84 (0.82-0.85)
Demographic	0.72 (0.72-0.72)	0.71 (0.70-0.71)	0.71 (0.70-0.72)	0.68 (0.67-0.69)
Serum	0.86 (0.83-0.90)	0.87 (0.82-0.93)	0.84 (0.80-0.88)	0.75 (0.72-0.78)
PRS	0.71 (0.67-0.76)	0.79 (0.74-0.83)	0.71 (0.70-0.73)	0.75 (0.70-0.79)
Protein+Demographic	0.95 (0.93-0.96)	0.92 (0.90-0.95)	0.90 (0.87-0.92)	0.87 (0.84-0.89)
Protein+Serum	0.97 (0.96-0.99)	0.96 (0.94-0.98)	0.95 (0.92-0.97)	0.90 (0.86-0.94)
Protein+PRS	0.95 (0.94-0.96)	0.95 (0.93-0.96)	0.95 (0.93-0.97)	0.87 (0.83-0.90)
Protein+Demographic+Serum	0.96 (0.95-0.98)	0.95 (0.92-0.98)	0.98 (0.98-0.99)	0.87 (0.85-0.88)
Protein+Demographic+PRS	0.93 (0.92-0.95)	0.95 (0.94-0.97)	0.93 (0.91-0.95)	0.84 (0.82-0.86)
Protein+Serum+PRS	0.97 (0.95-0.99)	0.97 (0.95-0.99)	0.93 (0.90-0.96)	0.87 (0.82-0.92)
Combined	0.97 (0.96-0.98)	0.95 (0.93-0.98)	0.91 (0.88-0.95)	0.87 (0.85-0.89)

11-year incident CD

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.73 (0.72-0.73)	0.72 (0.72-0.73)	0.71 (0.70-0.72)	0.72 (0.70-0.73)
Demographic	0.57 (0.56-0.58)	0.57 (0.56-0.58)	0.58 (0.57-0.58)	0.60 (0.59-0.61)
Serum	0.64 (0.62-0.65)	0.64 (0.63-0.65)	0.65 (0.62-0.67)	0.64 (0.63-0.65)
PRS	0.61 (0.61-0.61)	0.66 (0.65-0.67)	0.64 (0.63-0.66)	0.68 (0.68-0.68)
Protein+Demographic	0.72 (0.72-0.73)	0.71 (0.70-0.72)	0.72 (0.71-0.73)	0.70 (0.69-0.72)
Protein+Serum	0.71 (0.71-0.72)	0.72 (0.72-0.73)	0.73 (0.73-0.74)	0.72 (0.71-0.74)
Protein+PRS	0.74 (0.73-0.76)	0.74 (0.73-0.75)	0.74 (0.73-0.76)	0.77 (0.75-0.78)
Protein+Demographic+Serum	0.72 (0.71-0.72)	0.73 (0.72-0.74)	0.70 (0.68-0.72)	0.70 (0.68-0.72)
Protein+Demographic+PRS	0.75 (0.75-0.76)	0.75 (0.74-0.76)	0.75 (0.74-0.76)	0.77 (0.76-0.78)
Protein+Serum+PRS	0.73 (0.73-0.74)	0.75 (0.74-0.76)	0.74 (0.72-0.75)	0.72 (0.70-0.74)

Combined	0.74 (0.73-0.75)	0.75 (0.74-0.76)	0.74 (0.72-0.76)	0.77 (0.76-0.77)
Training set				
9-protein panel	0.88 (0.87-0.89)	0.89 (0.87-0.92)	0.85 (0.84-0.85)	0.83 (0.79-0.87)
Demographic	0.71 (0.70-0.71)	0.70 (0.69-0.70)	0.69 (0.68-0.70)	0.66 (0.64-0.67)
Serum	0.83 (0.79-0.86)	0.86 (0.81-0.91)	0.81 (0.77-0.85)	0.71 (0.70-0.72)
PRS	0.62 (0.62-0.62)	0.66 (0.65-0.68)	0.64 (0.61-0.67)	0.61 (0.60-0.62)
Protein+Demographic	0.88 (0.87-0.89)	0.87 (0.85-0.89)	0.85 (0.83-0.87)	0.82 (0.79-0.84)
Protein+Serum	0.96 (0.95-0.98)	0.94 (0.91-0.96)	0.87 (0.85-0.90)	0.88 (0.84-0.93)
Protein+PRS	0.89 (0.88-0.90)	0.91 (0.87-0.94)	0.87 (0.85-0.89)	0.81 (0.80-0.83)
Protein+Demographic+Serum	0.97 (0.95-0.98)	0.92 (0.91-0.94)	0.92 (0.88-0.97)	0.87 (0.81-0.92)
Protein+Demographic+PRS	0.91 (0.88-0.93)	0.90 (0.88-0.92)	0.93 (0.90-0.95)	0.92 (0.88-0.96)
Protein+Serum+PRS	0.98 (0.96-0.99)	0.95 (0.93-0.97)	0.92 (0.89-0.94)	0.88 (0.82-0.93)
Combined	0.97 (0.96-0.98)	0.95 (0.93-0.98)	0.93 (0.91-0.95)	0.93 (0.88-0.97)

12-year incident CD

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.71 (0.71-0.72)	0.71 (0.71-0.72)	0.73 (0.72-0.73)	0.71 (0.70-0.72)
Demographic	0.57 (0.57-0.57)	0.57 (0.56-0.57)	0.58 (0.58-0.59)	0.58 (0.57-0.59)
Serum	0.63 (0.62-0.65)	0.62 (0.60-0.64)	0.62 (0.60-0.64)	0.63 (0.62-0.64)
PRS	0.64 (0.63-0.65)	0.63 (0.61-0.65)	0.61 (0.60-0.62)	0.69 (0.69-0.69)
Protein+Demographic	0.72 (0.71-0.72)	0.72 (0.71-0.72)	0.72 (0.71-0.73)	0.74 (0.72-0.75)
Protein+Serum	0.72 (0.70-0.73)	0.70 (0.69-0.72)	0.72 (0.71-0.73)	0.74 (0.73-0.75)
Protein+PRS	0.73 (0.71-0.74)	0.73 (0.71-0.75)	0.75 (0.74-0.75)	0.78 (0.77-0.79)
Protein+Demographic+Serum	0.72 (0.71-0.73)	0.71 (0.68-0.73)	0.72 (0.70-0.73)	0.71 (0.68-0.74)
Protein+Demographic+PRS	0.73 (0.72-0.74)	0.74 (0.72-0.76)	0.72 (0.72-0.73)	0.74 (0.71-0.77)
Protein+Serum+PRS	0.74 (0.73-0.76)	0.73 (0.71-0.75)	0.74 (0.72-0.75)	0.74 (0.73-0.76)
Combined	0.75 (0.74-0.76)	0.73 (0.72-0.74)	0.73 (0.73-0.74)	0.79 (0.78-0.80)
Training set				
9-protein panel	0.89 (0.88-0.90)	0.89 (0.87-0.90)	0.85 (0.83-0.86)	0.79 (0.78-0.81)
Demographic	0.70 (0.70-0.70)	0.69 (0.69-0.70)	0.69 (0.69-0.69)	0.67 (0.67-0.68)
Serum	0.86 (0.81-0.91)	0.86 (0.80-0.92)	0.74 (0.73-0.75)	0.69 (0.68-0.71)
PRS	0.69 (0.65-0.72)	0.69 (0.66-0.72)	0.78 (0.68-0.88)	0.60 (0.59-0.60)
Protein+Demographic	0.89 (0.88-0.90)	0.87 (0.86-0.89)	0.90 (0.87-0.93)	0.82 (0.78-0.86)
Protein+Serum	0.92 (0.90-0.94)	0.89 (0.85-0.92)	0.88 (0.85-0.90)	0.87 (0.84-0.90)
Protein+PRS	0.92 (0.90-0.95)	0.90 (0.87-0.93)	0.95 (0.93-0.98)	0.81 (0.80-0.82)
Protein+Demographic+Serum	0.93 (0.90-0.96)	0.89 (0.86-0.92)	0.88 (0.86-0.91)	0.85 (0.83-0.88)
Protein+Demographic+PRS	0.91 (0.90-0.92)	0.89 (0.86-0.91)	0.91 (0.88-0.94)	0.89 (0.84-0.95)
Protein+Serum+PRS	0.94 (0.91-0.97)	0.90 (0.87-0.94)	0.85 (0.83-0.87)	0.84 (0.82-0.86)
Combined	0.93 (0.90-0.96)	0.94 (0.90-0.98)	0.94 (0.91-0.96)	0.82 (0.82-0.83)

13-year incident CD

	AUC (95%CI)

	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.72 (0.71-0.73)	0.71 (0.69-0.72)	0.73 (0.71-0.74)	0.70 (0.70-0.71)
Demographic	0.58 (0.58-0.58)	0.58 (0.57-0.58)	0.59 (0.59-0.59)	0.58 (0.57-0.58)
Serum	0.61 (0.60-0.63)	0.62 (0.61-0.63)	0.62 (0.60-0.64)	0.63 (0.61-0.65)
PRS	0.65 (0.64-0.66)	0.63 (0.61-0.65)	0.62 (0.61-0.64)	0.68 (0.68-0.68)
Protein+Demographic	0.72 (0.70-0.73)	0.73 (0.72-0.73)	0.72 (0.71-0.73)	0.71 (0.70-0.72)
Protein+Serum	0.72 (0.71-0.74)	0.71 (0.70-0.72)	0.70 (0.68-0.72)	0.74 (0.73-0.75)
Protein+PRS	0.73 (0.73-0.74)	0.75 (0.74-0.76)	0.72 (0.71-0.73)	0.75 (0.73-0.77)
Protein+Demographic+Serum	0.71 (0.69-0.73)	0.72 (0.71-0.73)	0.73 (0.72-0.74)	0.73 (0.71-0.74)
Protein+Demographic+PRS	0.74 (0.74-0.75)	0.75 (0.74-0.75)	0.74 (0.72-0.75)	0.77 (0.75-0.78)
Protein+Serum+PRS	0.74 (0.72-0.76)	0.75 (0.74-0.76)	0.75 (0.73-0.77)	0.71 (0.69-0.73)
Combined	0.76 (0.75-0.76)	0.76 (0.75-0.77)	0.76 (0.75-0.76)	0.76 (0.74-0.78)
Training set				
9-protein panel	0.87 (0.85-0.90)	0.83 (0.82-0.84)	0.83 (0.81-0.85)	0.76 (0.76-0.77)
Demographic	0.67 (0.67-0.68)	0.67 (0.67-0.68)	0.68 (0.68-0.68)	0.68 (0.68-0.68)
Serum	0.77 (0.76-0.79)	0.78 (0.75-0.81)	0.76 (0.72-0.80)	0.69 (0.67-0.71)
PRS	0.76 (0.72-0.80)	0.75 (0.70-0.80)	0.78 (0.72-0.84)	0.60 (0.60-0.60)
Protein+Demographic	0.86 (0.85-0.87)	0.83 (0.82-0.85)	0.88 (0.83-0.92)	0.79 (0.76-0.82)
Protein+Serum	0.91 (0.87-0.94)	0.90 (0.86-0.93)	0.82 (0.77-0.87)	0.81 (0.78-0.85)
Protein+PRS	0.94 (0.92-0.97)	0.90 (0.87-0.93)	0.83 (0.78-0.88)	0.78 (0.77-0.79)
Protein+Demographic+Serum	0.95 (0.93-0.97)	0.87 (0.84-0.91)	0.87 (0.84-0.90)	0.84 (0.79-0.90)
Protein+Demographic+PRS	0.92 (0.88-0.95)	0.88 (0.86-0.90)	0.92 (0.88-0.97)	0.81 (0.80-0.82)
Protein+Serum+PRS	0.96 (0.93-0.99)	0.94 (0.90-0.98)	0.89 (0.85-0.92)	0.74 (0.72-0.76)
Combined	0.96 (0.93-0.99)	0.92 (0.90-0.95)	0.90 (0.88-0.93)	0.85 (0.80-0.90)

14-year incident CD

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.72 (0.71-0.73)	0.71 (0.71-0.72)	0.71 (0.70-0.72)	0.71 (0.70-0.71)
Demographic	0.57 (0.56-0.57)	0.57 (0.56-0.58)	0.59 (0.58-0.59)	0.55 (0.55-0.56)
Serum	0.63 (0.62-0.64)	0.62 (0.61-0.64)	0.66 (0.64-0.67)	0.63 (0.60-0.65)
PRS	0.65 (0.65-0.66)	0.66 (0.66-0.67)	0.61 (0.60-0.63)	0.68 (0.68-0.69)
Protein+Demographic	0.71 (0.71-0.72)	0.72 (0.71-0.73)	0.70 (0.69-0.72)	0.73 (0.72-0.74)
Protein+Serum	0.72 (0.70-0.74)	0.72 (0.71-0.74)	0.72 (0.72-0.72)	0.75 (0.75-0.76)
Protein+PRS	0.74 (0.74-0.75)	0.73 (0.72-0.75)	0.70 (0.68-0.72)	0.75 (0.74-0.76)
Protein+Demographic+Serum	0.73 (0.72-0.74)	0.73 (0.72-0.74)	0.73 (0.72-0.75)	0.71 (0.70-0.73)
Protein+Demographic+PRS	0.75 (0.74-0.75)	0.73 (0.71-0.75)	0.71 (0.70-0.73)	0.78 (0.76-0.79)
Protein+Serum+PRS	0.76 (0.75-0.77)	0.75 (0.73-0.77)	0.75 (0.73-0.77)	0.76 (0.75-0.77)
Combined	0.75 (0.75-0.76)	0.77 (0.76-0.77)	0.74 (0.73-0.76)	0.73 (0.69-0.76)
Training set				
9-protein panel	0.81 (0.80-0.82)	0.84 (0.81-0.87)	0.84 (0.82-0.87)	0.75 (0.75-0.75)
Demographic	0.66 (0.66-0.67)	0.66 (0.66-0.67)	0.67 (0.67-0.68)	0.66 (0.65-0.67)

Serum	*0.77 (0.75-0.78)*	*0.78 (0.76-0.81)*	*0.74 (0.73-0.75)*	*0.68 (0.67-0.69)*
PRS	*0.61 (0.61-0.62)*	*0.64 (0.62-0.65)*	*0.71 (0.70-0.73)*	*0.63 (0.61-0.65)*
Protein+Demographic	*0.82 (0.81-0.83)*	*0.82 (0.80-0.83)*	*0.81 (0.78-0.84)*	*0.77 (0.75-0.79)*
Protein+Serum	*0.90 (0.86-0.93)*	*0.89 (0.86-0.92)*	*0.93 (0.90-0.97)*	*0.80 (0.79-0.80)*
Protein+PRS	*0.88 (0.85-0.90)*	*0.89 (0.86-0.93)*	*0.85 (0.82-0.88)*	*0.79 (0.78-0.79)*
Protein+Demographic+Serum	*0.88 (0.85-0.91)*	*0.89 (0.86-0.92)*	*0.88 (0.83-0.93)*	*0.80 (0.78-0.81)*
Protein+Demographic+PRS	*0.87 (0.85-0.90)*	*0.87 (0.83-0.91)*	*0.80 (0.78-0.82)*	*0.82 (0.80-0.85)*
Protein+Serum+PRS	*0.91 (0.88-0.93)*	*0.89 (0.85-0.93)*	*0.86 (0.82-0.91)*	*0.79 (0.77-0.82)*
Combined	*0.92 (0.90-0.95)*	*0.88 (0.85-0.91)*	*0.88 (0.84-0.91)*	*0.80 (0.77-0.82)*

15-year incident CD

	UKB testing set (AUC [95%CI])			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.73 (0.72-0.73)*	*0.72 (0.71-0.73)*	*0.71 (0.70-0.73)*	*0.72 (0.71-0.73)*
Demographic	*0.56 (0.55-0.56)*	*0.56 (0.55-0.57)*	*0.58 (0.58-0.58)*	*0.55 (0.55-0.56)*
Serum	*0.63 (0.62-0.64)*	*0.63 (0.61-0.64)*	*0.66 (0.64-0.67)*	*0.62 (0.59-0.65)*
PRS	*0.67 (0.66-0.67)*	*0.67 (0.66-0.68)*	*0.63 (0.61-0.65)*	*0.69 (0.69-0.70)*
Protein+Demographic	*0.72 (0.71-0.73)*	*0.72 (0.72-0.73)*	*0.70 (0.69-0.72)*	*0.73 (0.72-0.74)*
Protein+Serum	*0.72 (0.70-0.74)*	*0.72 (0.70-0.74)*	*0.73 (0.72-0.74)*	*0.76 (0.75-0.76)*
Protein+PRS	*0.75 (0.74-0.76)*	*0.74 (0.73-0.75)*	*0.70 (0.68-0.71)*	*0.75 (0.74-0.77)*
Protein+Demographic+Serum	*0.73 (0.72-0.74)*	*0.73 (0.72-0.74)*	*0.74 (0.72-0.75)*	*0.72 (0.71-0.74)*
Protein+Demographic+PRS	*0.75 (0.75-0.76)*	*0.73 (0.72-0.75)*	*0.72 (0.70-0.73)*	*0.78 (0.76-0.79)*
Protein+Serum+PRS	*0.76 (0.75-0.77)*	*0.76 (0.75-0.77)*	*0.75 (0.74-0.77)*	*0.77 (0.76-0.78)*
Combined	*0.76 (0.75-0.77)*	*0.78 (0.77-0.79)*	*0.75 (0.74-0.76)*	*0.72 (0.68-0.76)*
Training set				
9-protein panel	*0.85 (0.82-0.88)*	*0.83 (0.80-0.86)*	*0.84 (0.82-0.87)*	*0.76 (0.74-0.78)*
Demographic	*0.66 (0.66-0.67)*	*0.66 (0.66-0.67)*	*0.67 (0.67-0.68)*	*0.66 (0.65-0.67)*
Serum	*0.77 (0.75-0.78)*	*0.78 (0.75-0.80)*	*0.75 (0.74-0.75)*	*0.72 (0.65-0.78)*
PRS	*0.61 (0.61-0.61)*	*0.64 (0.62-0.65)*	*0.71 (0.69-0.72)*	*0.64 (0.62-0.66)*
Protein+Demographic	*0.82 (0.81-0.83)*	*0.82 (0.80-0.83)*	*0.82 (0.78-0.85)*	*0.77 (0.75-0.79)*
Protein+Serum	*0.90 (0.86-0.93)*	*0.88 (0.85-0.91)*	*0.93 (0.90-0.96)*	*0.79 (0.79-0.80)*
Protein+PRS	*0.88 (0.86-0.90)*	*0.88 (0.84-0.92)*	*0.85 (0.82-0.88)*	*0.79 (0.78-0.79)*
Protein+Demographic+Serum	*0.87 (0.85-0.89)*	*0.89 (0.86-0.92)*	*0.88 (0.83-0.93)*	*0.79 (0.78-0.81)*
Protein+Demographic+PRS	*0.88 (0.85-0.90)*	*0.87 (0.83-0.91)*	*0.80 (0.77-0.82)*	*0.83 (0.81-0.86)*
Protein+Serum+PRS	*0.91 (0.89-0.93)*	*0.93 (0.89-0.98)*	*0.86 (0.81-0.91)*	*0.80 (0.77-0.82)*
Combined	*0.92 (0.90-0.95)*	*0.89 (0.85-0.93)*	*0.87 (0.84-0.91)*	*0.80 (0.77-0.83)*

16-year incident CD

	UKB testing set (AUC [95%CI])			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.73 (0.72-0.73)*	*0.72 (0.71-0.73)*	*0.71 (0.70-0.73)*	*0.72 (0.71-0.73)*
Demographic	*0.56 (0.55-0.56)*	*0.56 (0.55-0.57)*	*0.58 (0.58-0.58)*	*0.55 (0.55-0.56)*

Serum	*0.63 (0.62-0.64)*	*0.63 (0.61-0.64)*	*0.66 (0.64-0.67)*	*0.62 (0.59-0.65)*
PRS	*0.67 (0.66-0.67)*	*0.67 (0.66-0.68)*	*0.63 (0.61-0.65)*	*0.69 (0.69-0.70)*
Protein+Demographic	*0.72 (0.71-0.73)*	*0.72 (0.72-0.73)*	*0.70 (0.69-0.72)*	*0.73 (0.72-0.74)*
Protein+Serum	*0.72 (0.70-0.74)*	*0.72 (0.70-0.74)*	*0.73 (0.72-0.74)*	*0.76 (0.75-0.76)*
Protein+PRS	*0.75 (0.74-0.76)*	*0.74 (0.73-0.75)*	*0.70 (0.68-0.71)*	*0.75 (0.74-0.77)*
Protein+Demographic+Serum	*0.73 (0.72-0.74)*	*0.73 (0.72-0.74)*	*0.74 (0.72-0.75)*	*0.72 (0.71-0.74)*
Protein+Demographic+PRS	*0.75 (0.75-0.76)*	*0.73 (0.72-0.75)*	*0.72 (0.70-0.73)*	*0.78 (0.76-0.79)*
Protein+Serum+PRS	*0.76 (0.75-0.77)*	*0.76 (0.75-0.77)*	*0.75 (0.74-0.77)*	*0.77 (0.76-0.78)*
Combined	*0.76 (0.75-0.77)*	*0.78 (0.77-0.79)*	*0.75 (0.74-0.76)*	*0.72 (0.68-0.76)*
Training set				
9-protein panel	*0.85 (0.82-0.88)*	*0.83 (0.80-0.86)*	*0.84 (0.82-0.87)*	*0.76 (0.74-0.78)*
Demographic	*0.66 (0.66-0.67)*	*0.66 (0.66-0.67)*	*0.67 (0.67-0.68)*	*0.66 (0.65-0.67)*
Serum	*0.77 (0.75-0.78)*	*0.78 (0.75-0.80)*	*0.75 (0.74-0.75)*	*0.72 (0.65-0.78)*
PRS	*0.61 (0.61-0.61)*	*0.64 (0.62-0.65)*	*0.71 (0.69-0.72)*	*0.64 (0.62-0.66)*
Protein+Demographic	*0.82 (0.81-0.83)*	*0.82 (0.80-0.83)*	*0.82 (0.78-0.85)*	*0.77 (0.75-0.79)*
Protein+Serum	*0.90 (0.86-0.93)*	*0.88 (0.85-0.91)*	*0.93 (0.90-0.96)*	*0.79 (0.79-0.80)*
Protein+PRS	*0.88 (0.86-0.90)*	*0.88 (0.84-0.92)*	*0.85 (0.82-0.88)*	*0.79 (0.78-0.79)*
Protein+Demographic+Serum	*0.87 (0.85-0.89)*	*0.89 (0.86-0.92)*	*0.88 (0.83-0.93)*	*0.79 (0.78-0.81)*
Protein+Demographic+PRS	*0.88 (0.85-0.90)*	*0.87 (0.83-0.91)*	*0.80 (0.77-0.82)*	*0.83 (0.81-0.86)*
Protein+Serum+PRS	*0.91 (0.89-0.93)*	*0.93 (0.89-0.98)*	*0.86 (0.81-0.91)*	*0.80 (0.77-0.82)*
Combined	*0.92 (0.90-0.95)*	*0.89 (0.85-0.93)*	*0.87 (0.84-0.91)*	*0.80 (0.77-0.83)*

We randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 3:1 roughly. To further reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. Considering the small sample size of the participants converted to Crohn's disease within the 9 years of follow-up (fewer than 90), analyses were conducted only for follow-up periods of 10 years or longer."

“Supplementary Table 11: Results of ROC analyses for all-time incident CD after matching controls.

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.69 (0.67-0.71)	0.71 (0.68-0.73)	0.67 (0.64-0.69)	0.64 (0.62-0.66)
Demographic	0.40 (0.37-0.44)	0.47 (0.44-0.51)	0.32 (0.32-0.32)	0.34 (0.33-0.35)
Serum	0.55 (0.52-0.57)	0.57 (0.56-0.59)	0.63 (0.61-0.65)	0.62 (0.60-0.64)
PRS	0.72 (0.72-0.72)	0.72 (0.71-0.74)	0.74 (0.71-0.76)	0.75 (0.75-0.75)
Protein+Demographic	0.70 (0.68-0.71)	0.68 (0.65-0.71)	0.69 (0.66-0.72)	0.64 (0.62-0.67)
Protein+Serum	0.69 (0.67-0.72)	0.69 (0.66-0.72)	0.72 (0.71-0.73)	0.68 (0.66-0.69)
Protein+PRS	0.78 (0.75-0.80)	0.78 (0.75-0.80)	0.71 (0.66-0.75)	0.72 (0.70-0.74)
Protein+Demographic+Serum	0.69 (0.66-0.72)	0.72 (0.70-0.74)	0.73 (0.70-0.77)	0.68 (0.65-0.70)
Protein+Demographic+PRS	0.78 (0.76-0.81)	0.76 (0.73-0.78)	0.78 (0.77-0.80)	0.74 (0.72-0.76)
Protein+Serum+PRS	0.75 (0.73-0.78)	0.73 (0.69-0.77)	0.75 (0.73-0.77)	0.75 (0.72-0.78)
Combined	0.76 (0.73-0.79)	0.76 (0.74-0.78)	0.71 (0.68-0.73)	0.70 (0.69-0.71)
Training set				
9-protein panel	0.91 (0.88-0.95)	0.92 (0.89-0.95)	0.85 (0.82-0.88)	0.80 (0.79-0.81)
Demographic	0.53 (0.52-0.54)	0.51 (0.50-0.52)	0.55 (0.55-0.55)	0.55 (0.55-0.56)
Serum	0.93 (0.89-0.96)	0.88 (0.84-0.92)	0.88 (0.82-0.94)	0.81 (0.76-0.86)
PRS	0.66 (0.65-0.67)	0.68 (0.66-0.70)	0.75 (0.72-0.77)	0.67 (0.67-0.67)
Protein+Demographic	0.88 (0.85-0.91)	0.86 (0.81-0.91)	0.86 (0.83-0.89)	0.81 (0.79-0.84)
Protein+Serum	0.89 (0.85-0.94)	0.91 (0.86-0.95)	0.88 (0.85-0.91)	0.87 (0.83-0.91)
Protein+PRS	0.92 (0.88-0.95)	0.93 (0.89-0.97)	0.85 (0.81-0.90)	0.83 (0.81-0.84)
Protein+Demographic+Serum	0.87 (0.82-0.93)	0.84 (0.82-0.87)	0.84 (0.83-0.86)	0.82 (0.79-0.84)
Protein+Demographic+PRS	0.87 (0.82-0.91)	0.89 (0.84-0.95)	0.88 (0.86-0.91)	0.87 (0.84-0.89)
Protein+Serum+PRS	0.92 (0.88-0.97)	0.86 (0.80-0.92)	0.89 (0.84-0.94)	0.87 (0.83-0.92)
Combined	0.89 (0.84-0.94)	0.89 (0.85-0.94)	0.96 (0.93-0.99)	0.88 (0.84-0.91)

We match the incident Crohn's disease and control populations by age, sex and ethnicity in a 1:1 ratio, and then randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 3:1 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn’s disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease.”

*“**Supplementary Table 12: Comparison between AUCs for all-time incident CD after matching controls.***

LGBM											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.49E-04	1.83E-04	1.70E-02	4.96E-01	4.27E-01	7.69E-04	6.78E-01	4.40E-04	4.06E-03	2.82E-03
Demographic	1.49E-04		2.03E-04	1.43E-04	1.49E-04	1.49E-04	1.49E-04	1.49E-04	1.49E-04	1.49E-04	1.49E-04
Serum	1.83E-04	2.03E-04		1.76E-04	1.83E-04	2.45E-04	1.83E-04	3.30E-04	1.83E-04	1.83E-04	1.82E-04
PRS	1.70E-02	1.43E-04	1.76E-04		2.54E-02	2.74E-03	2.76E-03	1.20E-01	2.76E-03	2.76E-03	6.32E-02
Protein+ Demographic	4.96E-01	1.49E-04	1.83E-04	2.54E-02		9.10E-01	1.71E-03	9.70E-01	5.83E-04	7.28E-03	5.78E-03
Protein+ Serum	4.27E-01	1.49E-04	2.45E-04	2.74E-03	9.10E-01		1.49E-03	7.91E-01	4.37E-04	3.60E-03	7.24E-03
Protein+ PRS	7.69E-04	1.49E-04	1.83E-04	2.76E-03	1.71E-03	1.49E-03		1.31E-03	9.10E-01	1.40E-01	5.71E-01
Protein+ Demographic+ Serum	6.78E-01	1.49E-04	3.30E-04	1.20E-01	9.70E-01	7.91E-01	1.31E-03		1.01E-03	7.28E-03	9.08E-03
Protein+ Demographic+ PRS	4.40E-04	1.49E-04	1.83E-04	2.76E-03	5.83E-04	4.37E-04	9.10E-01	1.01E-03		8.90E-02	5.20E-01
Protein+ Serum+ PRS	4.06E-03	1.49E-04	1.83E-04	2.76E-03	7.28E-03	3.60E-03	1.40E-01	7.28E-03	8.90E-02		3.45E-01

Combined	*2.82E-03*	*1.49E-04*	*1.82E-04*	*6.32E-02*	*5.78E-03*	*7.24E-03*	*5.71E-01*	*9.08E-03*	*5.20E-01*	*3.45E-01*	

XGBoost											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.11E-04*	*1.83E-04*	*3.45E-01*	*2.41E-01*	*4.27E-01*	*2.20E-03*	*4.06E-01*	*1.40E-02*	*2.41E-01*	*7.28E-03*
Demographic	*1.11E-04*		*1.11E-04*	*1.11E-04*	*1.11E-04*	*1.11E-04*	*1.11E-04*	*1.11E-04*	*1.11E-04*	*1.11E-04*	*1.11E-04*
Serum	*1.83E-04*	*1.11E-04*		*1.83E-04*	*1.01E-03*	*2.46E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
PRS	*3.45E-01*	*1.11E-04*	*1.83E-04*		*2.56E-02*	*6.95E-02*	*2.83E-03*	*1.00E+00*	*6.40E-02*	*3.45E-01*	*4.59E-03*
Protein+ Demographic	*2.41E-01*	*1.11E-04*	*1.01E-03*	*2.56E-02*		*8.50E-01*	*5.83E-04*	*5.39E-02*	*2.83E-03*	*4.93E-02*	*1.83E-04*
Protein+ Serum	*4.27E-01*	*1.11E-04*	*2.46E-04*	*6.95E-02*	*8.50E-01*		*1.01E-03*	*8.90E-02*	*5.80E-03*	*9.62E-02*	*1.31E-03*
Protein+ PRS	*2.20E-03*	*1.11E-04*	*1.83E-04*	*2.83E-03*	*5.83E-04*	*1.01E-03*		*7.28E-03*	*2.73E-01*	*4.93E-02*	*2.73E-01*
Protein+ Demographic+ Serum	*4.06E-01*	*1.11E-04*	*1.83E-04*	*1.00E+00*	*5.39E-02*	*8.90E-02*	*7.28E-03*		*8.90E-02*	*5.71E-01*	*1.13E-02*
Protein+ Demographic+ PRS	*1.40E-02*	*1.11E-04*	*1.83E-04*	*6.40E-02*	*2.83E-03*	*5.80E-03*	*2.73E-01*	*8.90E-02*		*3.45E-01*	*9.70E-01*

Protein+ Serum+ PRS	2.41E-01	1.11E-04	1.83E-04	3.45E-01	4.93E-02	9.62E-02	4.93E-02	5.71E-01	3.45E-01		2.73E-01
Combined	7.28E-03	1.11E-04	1.83E-04	4.59E-03	1.83E-04	1.31E-03	2.73E-01	1.13E-02	9.70E-01	2.73E-01	

RF											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		6.34E-05	1.04E-01	8.98E-03	1.61E-01	6.46E-03	1.85E-01	1.68E-02	1.71E-04	1.69E-03	4.11E-02
Demographic	6.34E-05		6.39E-05	6.20E-05	6.25E-05	6.39E-05	6.11E-05	5.94E-05	5.89E-05	6.29E-05	6.39E-05
Serum	1.04E-01	6.39E-05		1.79E-04	4.54E-03	1.83E-04	5.34E-02	5.55E-04	1.72E-04	1.81E-04	1.71E-03
PRS	8.98E-03	6.20E-05	1.79E-04		2.54E-02	6.23E-01	3.83E-01	1.00E+00	1.66E-02	1.00E+00	2.72E-01
Protein+ Demographic	1.61E-01	6.25E-05	4.54E-03	2.54E-02		1.40E-01	3.06E-01	8.74E-02	1.69E-04	2.78E-03	2.41E-01
Protein+ Serum	6.46E-03	6.39E-05	1.83E-04	6.23E-01	1.40E-01		6.22E-01	7.33E-01	4.16E-04	4.50E-02	9.70E-01
Protein+ PRS	1.85E-01	6.11E-05	5.34E-02	3.83E-01	3.06E-01	6.22E-01		3.82E-01	8.67E-03	2.11E-01	5.70E-01
Protein+ Demographic+ Serum	1.68E-02	5.94E-05	5.55E-04	1.00E+00	8.74E-02	7.33E-01	3.82E-01		5.20E-02	7.33E-01	4.71E-01

	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
Protein+ Demographic+ PRS	1.71E-04	5.89E-05	1.72E-04	1.66E-02	1.69E-04	4.16E-04	8.67E-03	5.20E-02		8.74E-02	5.53E-04
Protein+ Serum+ PRS	1.69E-03	6.29E-05	1.81E-04	1.00E+00	2.78E-03	4.50E-02	2.11E-01	7.33E-01	8.74E-02		2.56E-02
Combined	4.11E-02	6.39E-05	1.71E-03	2.72E-01	2.41E-01	9.70E-01	5.70E-01	4.71E-01	5.53E-04	2.56E-02	

Extra trees											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.25E-04	6.14E-02	5.51E-05	7.31E-01	6.85E-03	9.18E-04	8.60E-02	2.97E-04	5.26E-04	1.57E-03
Demographic	1.25E-04		1.36E-04	4.55E-05	1.37E-04	1.41E-04	1.40E-04	1.37E-04	1.41E-04	1.40E-04	1.40E-04
Serum	6.14E-02	1.36E-04		6.11E-05	3.82E-01	2.77E-03	7.44E-04	2.51E-02	4.27E-04	3.18E-04	2.76E-03
PRS	5.51E-05	4.55E-05	6.11E-05		6.16E-05	6.39E-05	1.71E-02	6.16E-05	1.15E-01	4.43E-01	6.34E-05
Protein+ Demographic	7.31E-01	1.37E-04	3.82E-01	6.16E-05		7.51E-02	3.54E-03	4.43E-02	5.69E-04	2.77E-03	1.67E-03
Protein+ Serum	6.85E-03	1.41E-04	2.77E-03	6.39E-05	7.51E-02		7.20E-03	7.62E-01	6.67E-04	1.72E-02	8.10E-03
Protein+ PRS	9.18E-04	1.40E-04	7.44E-04	1.71E-02	3.54E-03	7.20E-03		6.34E-02	3.45E-01	1.04E-01	5.20E-01

Protein+ Demographic+ Serum	*8.60E-02*	*1.37E-04*	*2.51E-02*	*6.16E-05*	*4.43E-02*	*7.62E-01*	*6.34E-02*		*5.70E-03*	*1.11E-02*	*3.06E-01*
Protein+ Demographic+ PRS	*2.97E-04*	*1.41E-04*	*4.27E-04*	*1.15E-01*	*5.69E-04*	*6.67E-04*	*3.45E-01*	*5.70E-03*		*2.73E-01*	*1.90E-02*
Protein+ Serum+ PRS	*5.26E-04*	*1.40E-04*	*3.18E-04*	*4.43E-01*	*2.77E-03*	*1.72E-02*	*1.04E-01*	*1.11E-02*	*2.73E-01*		*2.56E-02*
Combined	*1.57E-03*	*1.40E-04*	*2.76E-03*	*6.34E-05*	*1.67E-03*	*8.10E-03*	*5.20E-01*	*3.06E-01*	*1.90E-02*	*2.56E-02*	

Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 13: Results of ROC analyses after excluding individuals who developed Crohn's disease in the first two years of follow-up.

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.68 (0.67-0.70)	0.70 (0.69-0.71)	0.66 (0.64-0.67)	0.70 (0.69-0.71)
Demographic	0.56 (0.55-0.57)	0.56 (0.55-0.57)	0.55 (0.55-0.55)	0.60 (0.60-0.60)
Serum	0.59 (0.57-0.61)	0.62 (0.60-0.65)	0.62 (0.60-0.63)	0.67 (0.65-0.68)
PRS	0.64 (0.62-0.65)	0.61 (0.60-0.62)	0.66 (0.65-0.66)	0.66 (0.66-0.66)
Protein+Demographic	0.70 (0.69-0.71)	0.70 (0.69-0.71)	0.71 (0.70-0.72)	0.72 (0.70-0.73)
Protein+Serum	0.69 (0.68-0.71)	0.72 (0.71-0.73)	0.71 (0.69-0.73)	0.74 (0.73-0.75)
Protein+PRS	0.67 (0.65-0.69)	0.69 (0.67-0.71)	0.71 (0.70-0.73)	0.71 (0.70-0.73)
Protein+Demographic+Serum	0.71 (0.70-0.72)	0.71 (0.69-0.73)	0.73 (0.71-0.74)	0.72 (0.70-0.74)
Protein+Demographic+PRS	0.70 (0.69-0.72)	0.71 (0.70-0.73)	0.73 (0.72-0.74)	0.76 (0.75-0.77)
Protein+Serum+PRS	0.73 (0.71-0.75)	0.71 (0.69-0.74)	0.75 (0.74-0.76)	0.73 (0.72-0.73)
Combined	0.73 (0.72-0.74)	0.73 (0.71-0.74)	0.70 (0.67-0.73)	0.74 (0.71-0.77)
Training set				
9-protein panel	0.92 (0.89-0.95)	0.92 (0.88-0.96)	0.78 (0.75-0.81)	0.77 (0.75-0.79)
Demographic	0.63 (0.63-0.63)	0.63 (0.63-0.63)	0.63 (0.63-0.63)	0.62 (0.62-0.62)
Serum	0.86 (0.80-0.93)	0.88 (0.80-0.96)	0.81 (0.74-0.89)	0.70 (0.69-0.70)
PRS	0.62 (0.59-0.64)	0.63 (0.61-0.65)	0.67 (0.65-0.68)	0.60 (0.60-0.60)
Protein+Demographic	0.90 (0.87-0.92)	0.85 (0.83-0.87)	0.85 (0.80-0.89)	0.82 (0.76-0.87)
Protein+Serum	0.98 (0.96-1.00)	0.96 (0.94-0.99)	0.95 (0.92-0.98)	0.87 (0.82-0.91)
Protein+PRS	0.97 (0.94-1.00)	0.95 (0.91-0.99)	0.95 (0.91-0.99)	0.82 (0.78-0.87)
Protein+Demographic+Serum	0.98 (0.97-0.99)	0.94 (0.91-0.98)	0.84 (0.79-0.88)	0.81 (0.76-0.86)
Protein+Demographic+PRS	0.96 (0.92-0.99)	0.94 (0.90-0.97)	0.90 (0.86-0.94)	0.81 (0.79-0.84)
Protein+Serum+PRS	0.98 (0.96-1.00)	0.98 (0.96-0.99)	0.94 (0.91-0.98)	0.79 (0.78-0.79)
Combined	0.99 (0.98-1.00)	0.97 (0.95-0.98)	0.87 (0.83-0.90)	0.80 (0.78-0.82)

After excluding 19 individuals who developed Crohn's disease within the first two years of follow-up, we randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 3:1 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. "

“Supplementary Table 14: Comparison between AUCs for all-year incident CD after excluding individuals who developed Crohn’s disease in the first two year of follow-up.

LGBM											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.77E-04	1.83E-04	7.00E-04	3.12E-02	3.85E-01	3.45E-01	2.57E-02	3.12E-02	1.31E-03	1.01E-03
Demographic	1.77E-04		2.08E-02	1.57E-04	1.77E-04	1.77E-04	1.77E-04	1.77E-04	1.77E-04	1.77E-04	1.77E-04
Serum	1.83E-04	2.08E-02		2.62E-03	1.83E-04	1.83E-04	4.40E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
PRS	7.00E-04	1.57E-04	2.62E-03		2.21E-04	5.29E-04	4.35E-02	2.21E-04	1.63E-04	1.63E-04	1.63E-04
Protein+ Demographic	3.12E-02	1.77E-04	1.83E-04	2.21E-04		4.73E-01	1.04E-01	4.27E-01	9.10E-01	9.11E-03	4.59E-03
Protein+ Serum	3.85E-01	1.77E-04	1.83E-04	5.29E-04	4.73E-01		1.21E-01	3.07E-01	5.21E-01	1.73E-02	4.59E-03
Protein+ PRS	3.45E-01	1.77E-04	4.40E-04	4.35E-02	1.04E-01	1.21E-01		4.52E-02	1.04E-01	3.61E-03	1.31E-03
Protein+ Demographic+ Serum	2.57E-02	1.77E-04	1.83E-04	2.21E-04	4.27E-01	3.07E-01	4.52E-02		4.27E-01	3.76E-02	2.11E-02
Protein+ Demographic+ PRS	3.12E-02	1.77E-04	1.83E-04	1.63E-04	9.10E-01	5.21E-01	1.04E-01	4.27E-01		3.76E-02	2.11E-02
Protein+ Serum+ PRS	1.31E-03	1.77E-04	1.83E-04	1.63E-04	9.11E-03	1.73E-02	3.61E-03	3.76E-02	3.76E-02		9.70E-01

Combined	*1.01E-03*	*1.77E-04*	*1.83E-04*	*1.63E-04*	*4.59E-03*	*4.59E-03*	*1.31E-03*	*2.11E-02*	*2.11E-02*	*9.70E-01*	

XGBoost											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.73E-04*	*4.40E-04*	*1.32E-04*	*5.21E-01*	*1.73E-02*	*7.34E-01*	*1.04E-01*	*1.21E-01*	*2.41E-01*	*7.28E-03*
Demographic	*1.73E-04*		*3.13E-04*	*1.69E-04*	*1.73E-04*	*1.73E-04*	*1.73E-04*	*1.73E-04*	*1.73E-04*	*1.73E-04*	*1.73E-04*
Serum	*4.40E-04*	*3.13E-04*		*1.13E-01*	*1.83E-04*	*1.83E-04*	*1.01E-03*	*3.30E-04*	*3.30E-04*	*1.31E-03*	*1.83E-04*
PRS	*1.32E-04*	*1.69E-04*	*1.13E-01*		*1.32E-04*	*1.32E-04*	*1.32E-04*	*1.32E-04*	*1.32E-04*	*1.32E-04*	*1.32E-04*
Protein+ Demographic	*5.21E-01*	*1.73E-04*	*1.83E-04*	*1.32E-04*		*5.80E-03*	*6.23E-01*	*6.40E-02*	*8.90E-02*	*2.41E-01*	*3.61E-03*
Protein+ Serum	*1.73E-02*	*1.73E-04*	*1.83E-04*	*1.32E-04*	*5.80E-03*		*1.62E-01*	*4.27E-01*	*8.50E-01*	*7.91E-01*	*3.85E-01*
Protein+ PRS	*7.34E-01*	*1.73E-04*	*1.01E-03*	*1.32E-04*	*6.23E-01*	*1.62E-01*		*4.27E-01*	*3.45E-01*	*1.40E-01*	*4.52E-02*
Protein+ Demographic+ Serum	*1.04E-01*	*1.73E-04*	*3.30E-04*	*1.32E-04*	*6.40E-02*	*4.27E-01*	*4.27E-01*		*7.91E-01*	*6.78E-01*	*1.86E-01*
Protein+ Demographic+ PRS	*1.21E-01*	*1.73E-04*	*3.30E-04*	*1.32E-04*	*8.90E-02*	*8.50E-01*	*3.45E-01*	*7.91E-01*		*6.78E-01*	*3.45E-01*

Protein+ Serum+ PRS	*2.41E-01*	*1.73E-04*	*1.31E-03*	*1.32E-04*	*2.41E-01*	*7.91E-01*	*1.40E-01*	*6.78E-01*	*6.78E-01*		*6.78E-01*
Combined	*7.28E-03*	*1.73E-04*	*1.83E-04*	*1.32E-04*	*3.61E-03*	*3.85E-01*	*4.52E-02*	*1.86E-01*	*3.45E-01*	*6.78E-01*	

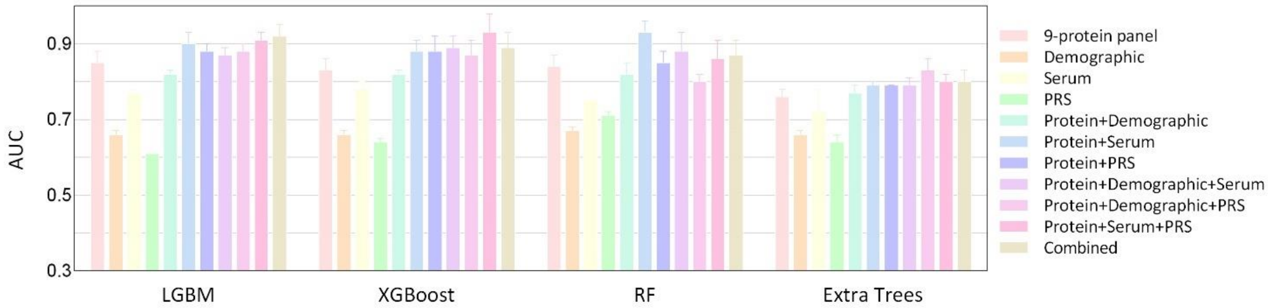
RF											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*4.17E-05*	*1.32E-04*	*1.97E-01*	*4.39E-04*	*5.88E-04*	*4.41E-04*	*3.25E-04*	*2.42E-04*	*1.80E-04*	*3.75E-03*
Demographic	*4.17E-05*		*6.39E-05*	*5.11E-05*	*6.34E-05*	*6.39E-05*	*6.39E-05*	*6.29E-05*	*6.34E-05*	*6.39E-05*	*6.29E-05*
Serum	*1.32E-04*	*6.39E-05*		*2.09E-04*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*1.81E-04*	*1.82E-04*	*1.83E-04*	*1.81E-04*
PRS	*1.97E-01*	*5.11E-05*	*2.09E-04*		*1.53E-04*	*1.54E-04*	*5.03E-04*	*1.52E-04*	*1.53E-04*	*1.54E-04*	*1.36E-01*
Protein+ Demographic	*4.39E-04*	*6.34E-05*	*1.82E-04*	*1.53E-04*		*5.71E-01*	*1.62E-01*	*2.72E-01*	*7.24E-03*	*1.70E-03*	*1.00E+00*
Protein+ Serum	*5.88E-04*	*6.39E-05*	*1.83E-04*	*1.54E-04*	*5.71E-01*		*9.10E-01*	*2.73E-01*	*3.76E-02*	*2.20E-03*	*8.50E-01*
Protein+ PRS	*4.41E-04*	*6.39E-05*	*1.83E-04*	*5.03E-04*	*1.62E-01*	*9.10E-01*		*6.23E-01*	*3.76E-02*	*2.20E-03*	*1.00E+00*
Protein+ Demographic+ Serum	*3.25E-04*	*6.29E-05*	*1.81E-04*	*1.52E-04*	*2.72E-01*	*2.73E-01*	*6.23E-01*		*6.23E-01*	*8.87E-02*	*2.72E-01*

Protein+ Demographic+ PRS	*2.42E-04*	*6.34E-05*	*1.82E-04*	*1.53E-04*	*7.24E-03*	*3.76E-02*	*3.76E-02*	*6.23E-01*		*5.38E-02*	*3.84E-01*
Protein+ Serum+ PRS	*1.80E-04*	*6.39E-05*	*1.83E-04*	*1.54E-04*	*1.70E-03*	*2.20E-03*	*2.20E-03*	*8.87E-02*	*5.38E-02*		*1.72E-02*
Combined	*3.75E-03*	*6.29E-05*	*1.81E-04*	*1.36E-01*	*1.00E+00*	*8.50E-01*	*1.00E+00*	*2.72E-01*	*3.84E-01*	*1.72E-02*	

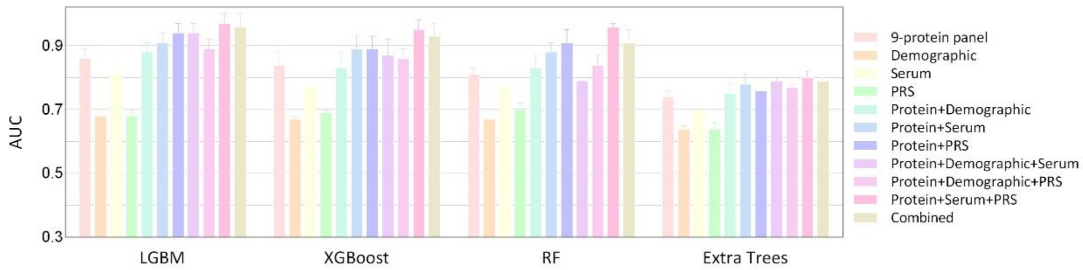
Extra trees											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.76E-04*	*5.76E-03*	*6.34E-05*	*2.73E-01*	*3.28E-04*	*1.62E-01*	*8.87E-02*	*1.82E-04*	*4.05E-03*	*5.76E-03*
Demographic	*1.76E-04*		*1.76E-04*	*6.11E-05*	*1.77E-04*	*1.77E-04*	*1.76E-04*	*1.76E-04*	*1.77E-04*	*1.44E-04*	*1.76E-04*
Serum	*5.76E-03*	*1.76E-04*		*1.15E-01*	*1.82E-04*	*1.82E-04*	*1.69E-03*	*2.81E-03*	*1.82E-04*	*1.49E-04*	*2.19E-03*
PRS	*6.34E-05*	*6.11E-05*	*1.15E-01*		*6.39E-05*	*6.39E-05*	*1.41E-03*	*6.34E-05*	*6.39E-05*	*4.92E-05*	*1.41E-03*
Protein+ Demographic	*2.73E-01*	*1.77E-04*	*1.82E-04*	*6.39E-05*		*1.13E-02*	*5.71E-01*	*4.27E-01*	*1.71E-03*	*2.90E-02*	*2.57E-02*
Protein+ Serum	*3.28E-04*	*1.77E-04*	*1.82E-04*	*6.39E-05*	*1.13E-02*		*3.60E-03*	*1.40E-01*	*2.20E-03*	*2.38E-02*	*6.23E-01*
Protein+ PRS	*1.62E-01*	*1.76E-04*	*1.69E-03*	*1.41E-03*	*5.71E-01*	*3.60E-03*		*2.73E-01*	*2.45E-04*	*5.07E-02*	*2.56E-02*

Protein+ Demographic+ Serum	8.87E-02	1.76E-04	2.81E-03	6.34E-05	4.27E-01	1.40E-01	2.73E-01		7.65E-04	9.08E-01	8.87E-02
Protein+ Demographic+ PRS	1.82E-04	1.77E-04	1.82E-04	6.39E-05	1.71E-03	2.20E-03	2.45E-04	7.65E-04		4.91E-04	2.12E-01
Protein+ Serum+ PRS	4.05E-03	1.44E-04	1.49E-04	4.92E-05	2.90E-02	2.38E-02	5.07E-02	9.08E-01	4.91E-04		3.01E-01
Combined	5.76E-03	1.76E-04	2.19E-03	1.41E-03	2.57E-02	6.23E-01	2.56E-02	8.87E-02	2.12E-01	3.01E-01	

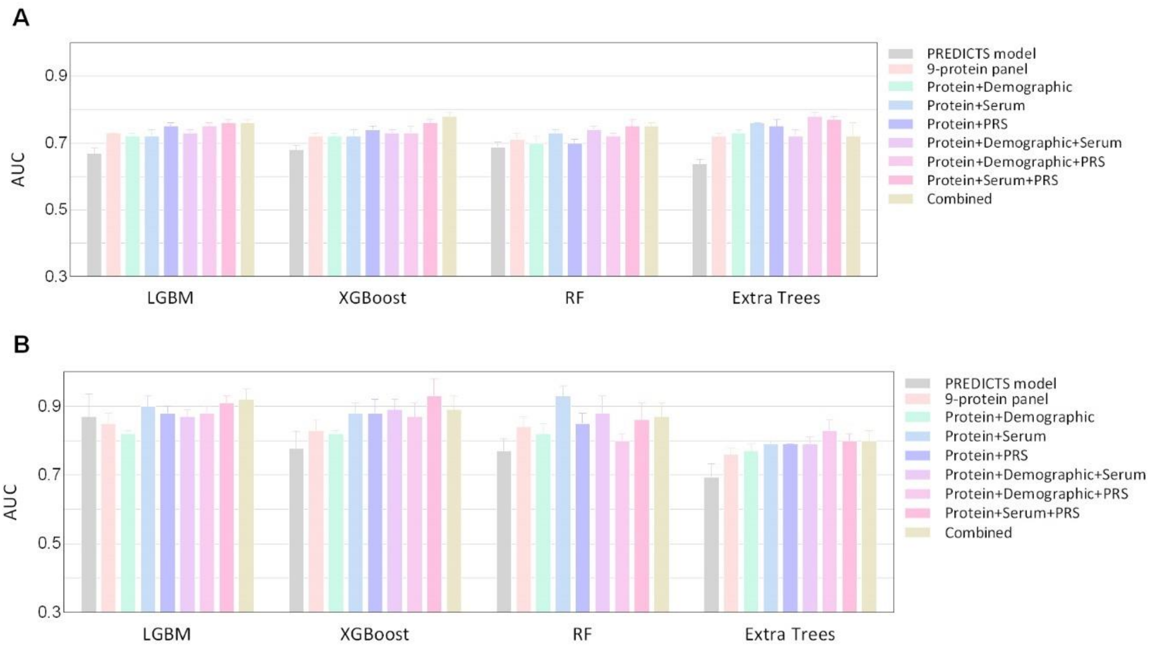
Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."



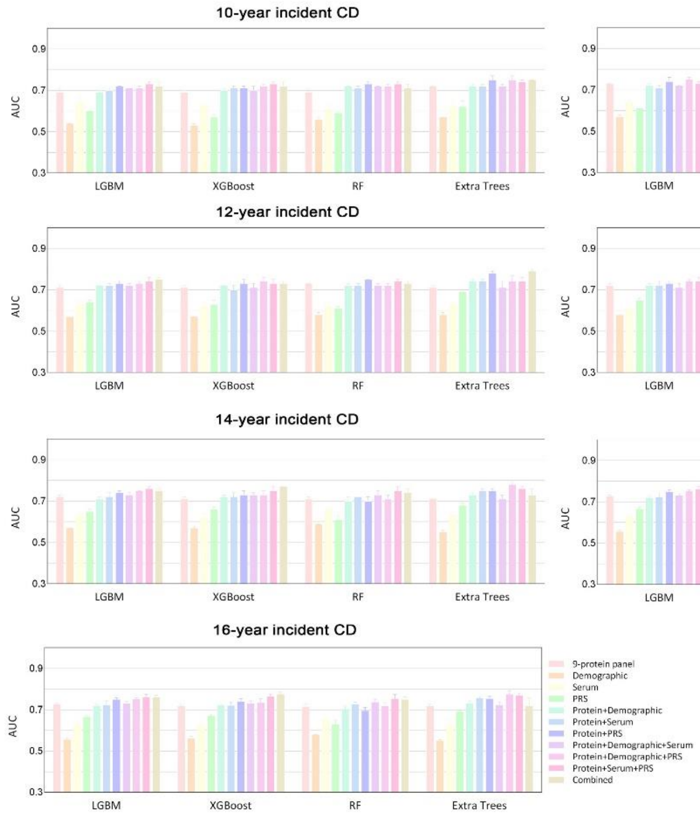
"Supplementary Fig. 2: Predictive accuracy of plasma proteins panel, alone or in combination with clinical variables in the training set with a 75% proportion. Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease in training set with a 75% proportion, evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."



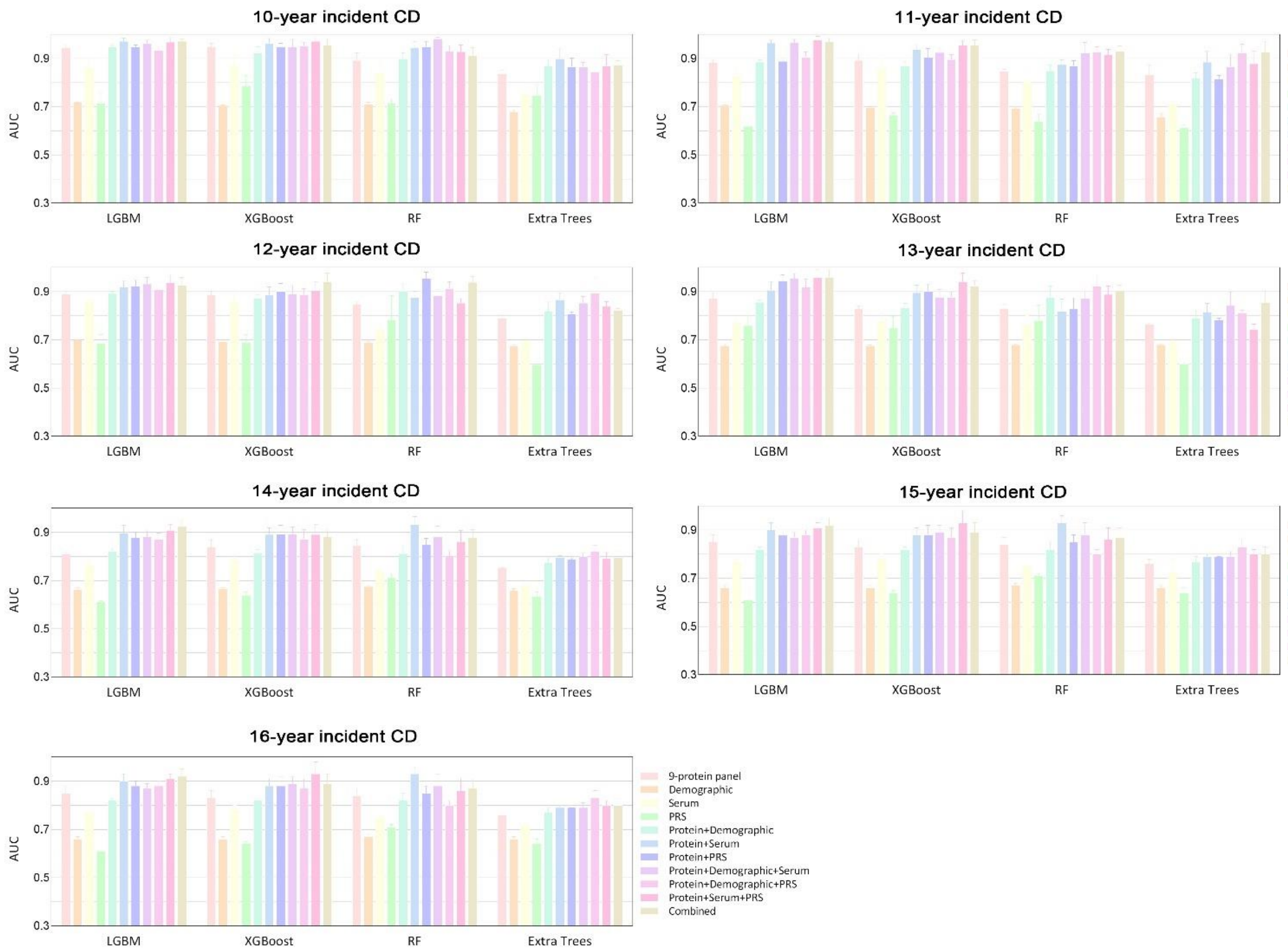
"Supplementary Fig. 3: Predictive accuracy of plasma proteins panel, alone or in combination with clinical variables in the training set with an 80% proportion. Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease in training set with a 80% proportion, evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."



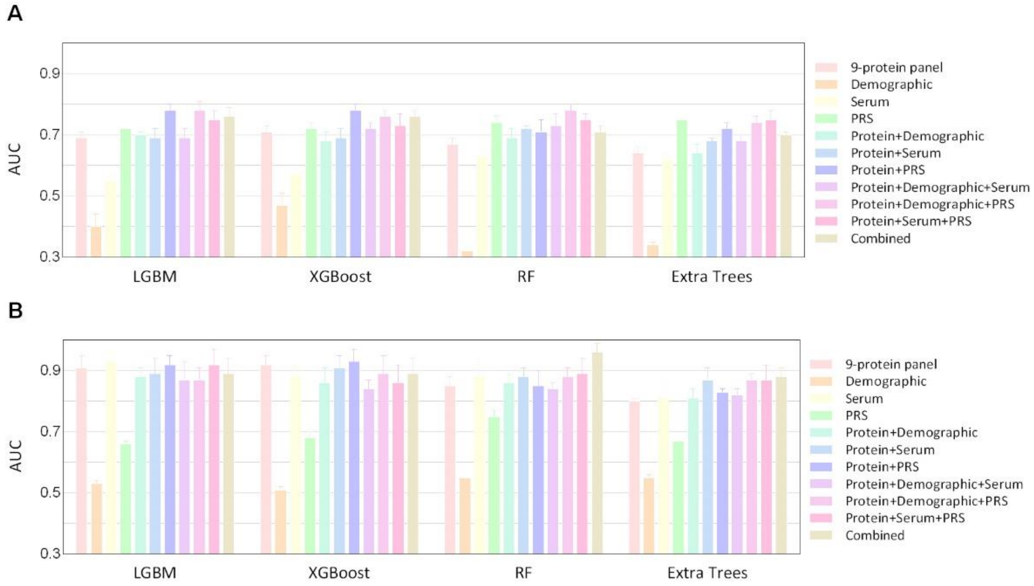
"Supplementary Fig. 4: Predictive accuracy of the PREDICTS model in UK Biobank for predicting all-time incident Crohn's disease. Bar plots show the area under the receiver operating characteristic curve (AUC) of PREDICTS and our protein-based models for predicting all-time incident Crohn's disease in the testing set (A) and training set (B), evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."



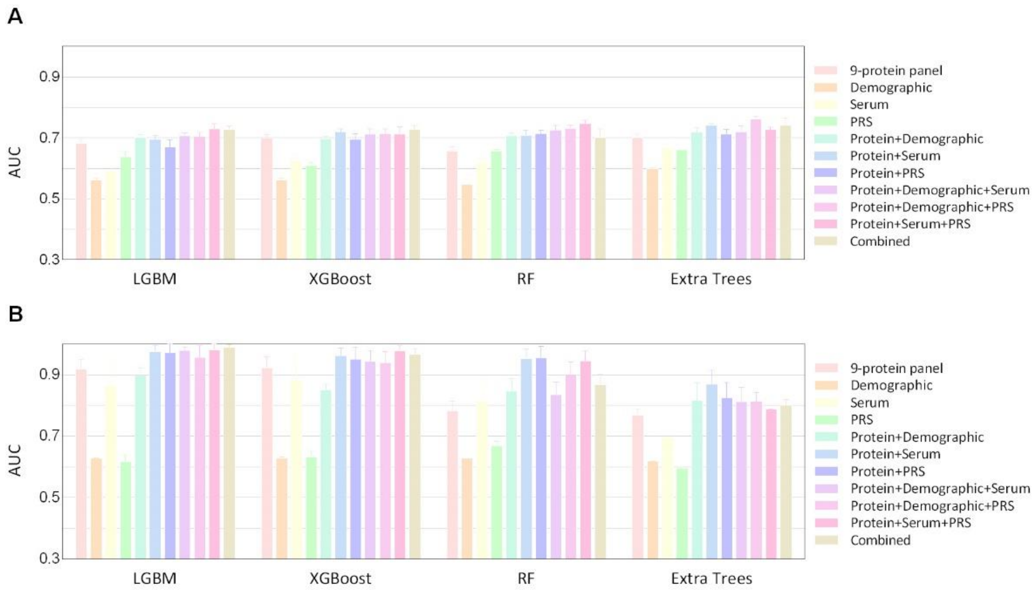
"**Supplementary Fig. 5: Predictive accuracy of plasma proteins panel within different time points in the UKB testing set.** Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models within different time points from 10 to 16 years of incident Crohn's disease in the UKB testing set, evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. Considering the small sample size of the participants converted to Crohn's disease within the 9 years of follow-up (fewer than 90), analyses were conducted only for follow-up periods of 10 years or longer."



"**Supplementary Fig. 6: Predictive accuracy of plasma proteins panel within different time points in the training set.** Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models within different time points from 10 to 16 years of incident Crohn's disease in the training set, evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. Considering the small sample size of the participants converted to Crohn's disease within the 9 years of follow-up (fewer than 90), analyses were conducted only for follow-up periods of 10 years or longer."



"**Supplementary Fig. 7: Predictive accuracy of plasma proteins panel, alone or in combination with matched populations.** Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease with matched populations in the UKB testing set (A) and training set (B), evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."



"Supplementary Fig. 8: Predictive accuracy after excluding individuals who developed Crohn's disease in the first two years of follow-up. Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease in the UKB testing set (A) and training set (B), evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Nineteen individuals were excluded for developing Crohn's disease within the first two years. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

4) Multi-modal integration: The study uses only a simple concatenation of features (early fusion) for multi-modal integration. This approach is valid but basic.

Authors' response:

We thank the reviewer for the insightful comment on our multi-modal integration approach. In addition to using early fusion, we also evaluated a late fusion strategy by employing a stacking method to integrate the outputs of the 9-protein panel, Demographic, Serum, and PRS models across the seven protein combination models (Protein + Demographic, Protein + Serum, Protein + PRS, Protein + Demographic + Serum, Protein + Demographic + PRS, Protein + Serum + PRS, and Combined). Our analyses revealed that while the late fusion strategy provided an alternative perspective, its predictive performance was largely similar to that of the early fusion approach. Therefore, we opted to present the early fusion results in the main manuscript. For completeness, we have included both early and late fusion results from the main analysis below. We sincerely thank the reviewer again for their valuable suggestion.

ROC results for late fusion

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
Protein+Demographic	0.72 (0.70-0.74)	0.71 (0.68-0.74)	0.73 (0.72-0.75)	0.72 (0.71-0.72)
Protein+Serum	0.73 (0.71-0.74)	0.72 (0.70-0.74)	0.74 (0.73-0.74)	0.72 (0.71-0.74)
Protein+PRS	0.75 (0.74-0.76)	0.75 (0.74-0.76)	0.74 (0.72-0.75)	0.77 (0.76-0.78)
Protein+Demographic+Serum	0.73 (0.72-0.75)	0.73 (0.70-0.75)	0.76 (0.75-0.77)	0.72 (0.71-0.74)
Protein+Demographic+PRS	0.75 (0.74-0.76)	0.75 (0.74-0.77)	0.76 (0.75-0.77)	0.78 (0.77-0.78)
Protein+Serum+PRS	0.74 (0.73-0.76)	0.75 (0.73-0.76)	0.76 (0.75-0.77)	0.77 (0.75-0.78)
Combined	0.75 (0.74-0.76)	0.76 (0.74-0.78)	0.78 (0.77-0.79)	0.77 (0.76-0.79)
Training set				
Protein+Demographic	0.82 (0.80-0.83)	0.83 (0.79-0.86)	0.86 (0.84-0.88)	0.78 (0.76-0.80)
Protein+Serum	0.85 (0.83-0.86)	0.86 (0.83-0.89)	0.87 (0.85-0.88)	0.80 (0.75-0.84)
Protein+PRS	0.82 (0.80-0.83)	0.84 (0.81-0.87)	0.88 (0.86-0.89)	0.79 (0.77-0.81)
Protein+Demographic+Serum	0.86 (0.84-0.87)	0.87 (0.84-0.89)	0.88 (0.86-0.90)	0.81 (0.77-0.85)
Protein+Demographic+PRS	0.83 (0.82-0.84)	0.85 (0.82-0.88)	0.89 (0.87-0.90)	0.81 (0.80-0.83)
Protein+Serum+PRS	0.85 (0.84-0.87)	0.87 (0.85-0.90)	0.89 (0.88-0.91)	0.82 (0.78-0.86)
Combined	0.87 (0.85-0.88)	0.88 (0.86-0.91)	0.90 (0.89-0.92)	0.84 (0.80-0.87)

ROC results for early fusion

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
Protein+Demographic	0.72 (0.71-0.73)	0.72 (0.72-0.73)	0.70 (0.69-0.72)	0.73 (0.72-0.74)
Protein+Serum	0.72 (0.70-0.74)	0.72 (0.70-0.74)	0.73 (0.72-0.74)	0.76 (0.75-0.76)
Protein+PRS	0.75 (0.74-0.76)	0.74 (0.73-0.75)	0.70 (0.68-0.71)	0.75 (0.74-0.77)
Protein+Demographic+Serum	0.73 (0.72-0.74)	0.73 (0.72-0.74)	0.74 (0.72-0.75)	0.72 (0.71-0.74)
Protein+Demographic+PRS	0.75 (0.75-0.76)	0.73 (0.72-0.75)	0.72 (0.70-0.73)	0.78 (0.76-0.79)

Protein+Serum+PRS	*0.76 (0.75-0.77)*	*0.76 (0.75-0.77)*	*0.75 (0.74-0.77)*	*0.77 (0.76-0.78)*
Combined	*0.76 (0.75-0.77)*	*0.78 (0.77-0.79)*	*0.75 (0.74-0.76)*	*0.72 (0.68-0.76)*
Training set				
Protein+Demographic	*0.82 (0.81-0.83)*	*0.82 (0.80-0.83)*	*0.82 (0.78-0.85)*	*0.77 (0.75-0.79)*
Protein+Serum	*0.90 (0.86-0.93)*	*0.88 (0.85-0.91)*	*0.93 (0.90-0.96)*	*0.79 (0.79-0.80)*
Protein+PRS	*0.88 (0.86-0.90)*	*0.88 (0.84-0.92)*	*0.85 (0.82-0.88)*	*0.79 (0.78-0.79)*
Protein+Demographic+Serum	*0.87 (0.85-0.89)*	*0.89 (0.86-0.92)*	*0.88 (0.83-0.93)*	*0.79 (0.78-0.81)*
Protein+Demographic+PRS	*0.88 (0.85-0.90)*	*0.87 (0.83-0.91)*	*0.80 (0.77-0.82)*	*0.83 (0.81-0.86)*
Protein+Serum+PRS	*0.91 (0.89-0.93)*	*0.93 (0.89-0.98)*	*0.86 (0.81-0.91)*	*0.80 (0.77-0.82)*
Combined	*0.92 (0.90-0.95)*	*0.89 (0.85-0.93)*	*0.87 (0.84-0.91)*	*0.80 (0.77-0.83)*

5) Internal validation: The study uses a single run of 5-fold cross-validation. To increase rigour and reduce selection bias, multiple runs of 5-fold cross-validation could be implemented, accounting for potential bias introduced by initial random ordering.

Authors' response:

We sincerely appreciate the reviewer's constructive suggestions regarding our internal validation strategy. In response, we repeated all analyses using 10 independent repeats of 5-fold cross-validation across all four machine learning algorithms (LightGBM, XGBoost, Random Forest, and Extra Trees) to enhance the rigor of our analyses and reduce potential selection bias introduced by the initial random ordering. Results were averaged over ten repetitions of fivefold cross validation with different random splits of the data into different folds. These modifications have been incorporated into the revised manuscript, specifically in Section **Abstract**, **2.4 Protein importance ranking**, **2.5 Predictive accuracy of proteomics-based models, 2.6 Results of sensitivity analyses**, **3. Discussion, 4.5 Statistical analysis, Fig. 3-4， Supplementary Tables 4-14， and Supplementary Fig. 2-8.**

*"**Abstract** …Machine learning models using four algorithms were constructed based on these proteins, with XGBoost demonstrating the best performance. In the geographically distinct UKB cohort, the protein model (area under the curve [AUC] 0.76) outperformed clinical risk models (highest AUC 0.67; P < 0.05) and was externally validated in the Southern China cohort (AUC 0.79). Combining these proteins with clinical data enabled desirable predictions up to 16 years pre-diagnosis (AUC 0.78). Our model performed better than existing PREDICTS model (AUC 0.68, P < 0.05) when applied to the UK Biobank population….."*

*"**Results 2.4 Protein importance ranking** For the proteins associated with CD in both models 1 and 2, we further ranked them according to their importance in predicting CD. As illustrated in the bar chart (Fig. 3A), CD274, CHI3L1, and REG1B ranked top three in protein importance ordering. After using the sequential forward selection scheme, we ultimately selected the top nine proteins (CD274, CHI3L1, REG1B, ITGAV, PRSS8, ITGA11, GDF15, DEFA1_DEFA1B, and IL6) for CD prediction in subsequent analyses (Supplementary Fig. 1)…*

2.5 Predictive accuracy of proteomics-based models *We evaluated the predictive performance of the 9-protein panel for future CD onset using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (**Fig. 4A-C, Supplementary Tables 4-7, and Supplementary Fig. 2-3**). The 9-protein model demonstrated* *considerable prediction performance across all four algorithms for all-time incident CD. In the UKB testing set (geographically distinct), the area under the curves (AUCs) ranged from 0.71 to 0.73 for a 75/25 training/testing split and 0.71 to 0.77 for an 80/20 split, while in the external Southern China testing set, they ranged from 0.76 to 0.79. Notably, the XGBoost model obtained the optimal AUC in the external testing set (0.79, 95% CI 0.77 – 0.81) and the second-best AUC in the UKB testing set (0.72 for a 75/25 split, 95% CI 0.71 – 0.73; 0.76 for an 80/20 split, 95% CI 0.74 – 0.77), indicating its robust generalizability across independent cohorts. In the UKB testing set, the 9-protein model demonstrated superior predictive performance across four algorithms compared to clinical risk models based on demographics, serum markers, and polygenic risk score (PRS) for CD (highest AUC among clinical risk models for XGBoost: 0.67, Mann – Whitney U test: P < 0.001). For XGBoost, combining the PRS of CD with the protein panel significantly improved the AUC to 0.74 (95% CI 0.73 – 0.75), and adding all clinical risk factors further increased the AUC to 0.78 (95% CI 0.77 – 0.79) for a 75/25 split.*

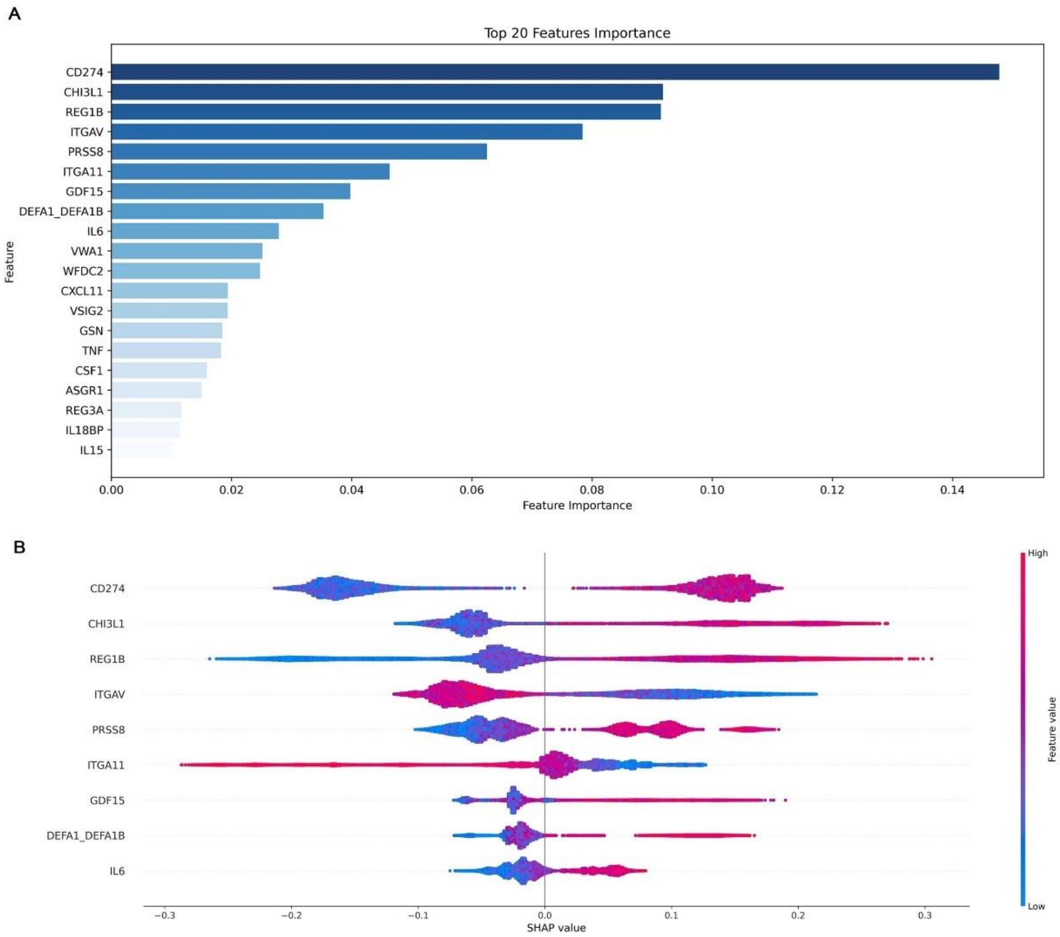
*We then deployed the six optimal proteins identified in the PREDICTS study into the UKB and compared the performance with our models. The results showed that the predictive performance of the PREDICTS model (AUC range: 0.64-0.69, XGBoost AUC: 0.68) was significantly inferior to our protein model (AUC range: 0.71-0.73, XGBoost AUC: 0.72) in predicting all-time incident CD whether or not it was combined with clinical risk factors (P < 0.05) (**Supplementary Tables 8-9** and **Supplementary Fig. 4**).*

***2.6 Results of replication validation analyses** Our findings remained consistent in the sensitivity analyses for XGBoost. At different time points, the protein model still showed higher predictive value (AUC 0.69-0.72) than clinical risk models (AUC 0.53-0.67) (**Supplementary Table 10** and **Supplementary Fig. 5-6**). When randomly subsampling a control group matched 1:1 with CD patients by age, sex, and race, the 9-protein model achieved an AUC of 0.71 (95% CI 0.68 – 0.73) in predicting all-time incident CD, increasing to 0.76 (95% CI 0.74 – 0.78) with the addition of all clinical risk factors (**Supplementary Tables 11-12** and **Supplementary Fig. 7**). The results remained consistent after excluding individuals who developed CD within the first two years of follow-up (**Supplementary Tables 13 – 14** and **Supplementary Fig. 8**).”*

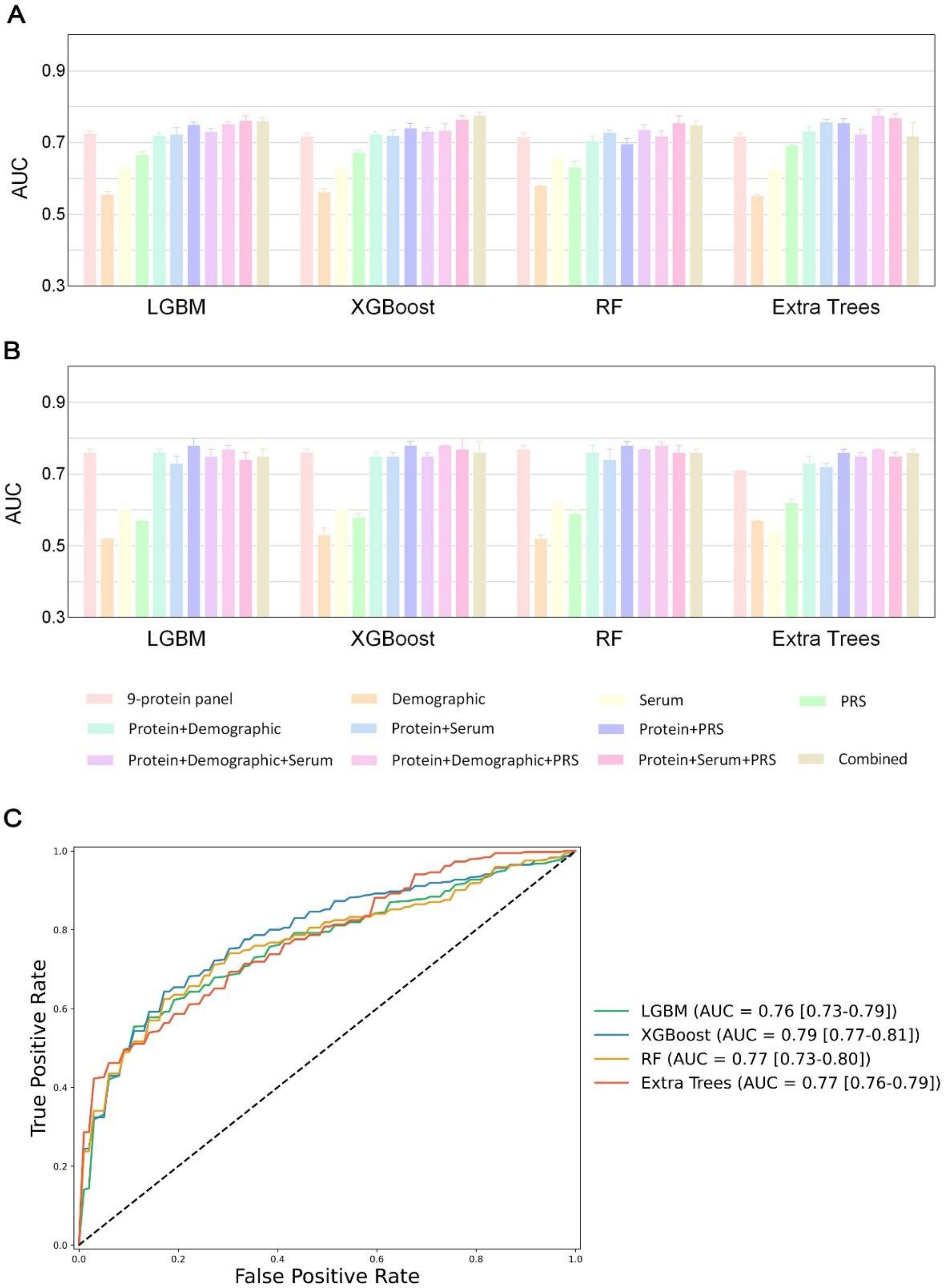
***“3. Discussion** In this study of over 52,000 participants, we applied a large-scale proteomic analysis using four machine learning algorithms to generate a 9-protein model to non-invasively predict future CD onset. The proteomic model (AUC 0.76) significantly outperformed clinical risk models based on demographics, serum biomarkers, and genetics (highest AUC 0.67) in the UKB testing set (geographically distinct). It was further externally validated in a Southern China cohort (AUC 0.79). Combining proteins with clinical risk data enabled better predictions up to 16 years pre-diagnosis (AUC 0.78).....”*

***“Method 4.5 Statistical analysis** The important proteins were then identified using machine learning algorithms. Proteins that remained significant after Bonferroni corrections in both Model 1 and Model 2 were input into a FLAML AutoML framework, wherein LGBM was applied using 5-fold cross-validation with 10 repetitions on the training set…*

…Finally, we established 11 machine-learning models including protein panel, demographics, serological indicators, or PRS of CD, both alone or in combinations. All models were constructed using the same AutoML framework described above. We split the participants into training and testing sets according to the recruitment centers. To further reduce overfitting, internal 5-fold cross-validation (10 repeats) was performed on the training set, and the maximum number of iterations was capped at 100 with early stopping…..”



***"Fig. 3 Protein importance ranking and SHAP visualization of modeling on all-time incident Crohn's disease populations. A**, The top 20 proteins according to the average absolute SHAP value. The bar chart indicates the importance of the sorted proteins based on their contributions to the prediction of future Crohn's disease. **B**, SHAP visualization plot of selected proteins. The width of the range of the horizontal bars can be understood as the extent of the contribution to the prediction of Crohn's disease; the wider their range, the greater the contribution. The color of the horizontal bars denotes the magnitude of plasma proteins, which was coded in a gradient from blue (low) to red (high), shown as the color bar on the right-hand side. The direction on the x axis indicates the likelihood of developing Crohn's disease (right) or being healthy (left)."*



***"Fig. 4 Predictive accuracy of plasma proteins panel, alone or in combination with clinical variables in internal and external testing set. A**, Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease in the UK Biobank (UKB) testing set, evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. **B**, Receiver operating curves show the performance* *of protein model for predicting Crohn's disease in the external testing set from Southern China, evaluated using the four machine learning algorithms."*

*"**Supplementary Table 4: Results of ROC analyses.***

	UKB testing set (AUC [95%CI])			
	LGBM	XGBoost	RF	Extra trees
9-protein panel	0.73 (0.72-0.73)	0.72 (0.71-0.73)	0.71 (0.70-0.73)	0.72 (0.71-0.73)
Demographic	0.56 (0.55-0.56)	0.56 (0.55-0.57)	0.58 (0.58-0.58)	0.55 (0.55-0.56)
Serum	0.63 (0.62-0.64)	0.63 (0.61-0.64)	0.66 (0.64-0.67)	0.62 (0.59-0.65)
PRS	0.67 (0.66-0.67)	0.67 (0.66-0.68)	0.63 (0.61-0.65)	0.69 (0.69-0.70)
Protein+Demographic	0.72 (0.71-0.73)	0.72 (0.72-0.73)	0.70 (0.69-0.72)	0.73 (0.72-0.74)
Protein+Serum	0.72 (0.70-0.74)	0.72 (0.70-0.74)	0.73 (0.72-0.74)	0.76 (0.75-0.76)
Protein+PRS	0.75 (0.74-0.76)	0.74 (0.73-0.75)	0.70 (0.68-0.71)	0.75 (0.74-0.77)
Protein+Demographic+Serum	0.73 (0.72-0.74)	0.73 (0.72-0.74)	0.74 (0.72-0.75)	0.72 (0.71-0.74)
Protein+Demographic+PRS	0.75 (0.75-0.76)	0.73 (0.72-0.75)	0.72 (0.70-0.73)	0.78 (0.76-0.79)
Protein+Serum+PRS	0.76 (0.75-0.77)	0.76 (0.75-0.77)	0.75 (0.74-0.77)	0.77 (0.76-0.78)
Combined	0.76 (0.75-0.77)	0.78 (0.77-0.79)	0.75 (0.74-0.76)	0.72 (0.68-0.76)

	Southern China testing set (AUC [95%CI])			
	LGBM	XGBoost	RF	Extra trees
9-protein panel	0.76 (0.73-0.79)	0.79 (0.77-0.81)	0.77 (0.73-0.80)	0.77 (0.76-0.79)

	UKB training set (AUC [95%CI])			
	LGBM	XGBoost	RF	Extra trees
9-protein panel	0.85 (0.82-0.88)	0.83 (0.80-0.86)	0.84 (0.82-0.87)	0.76 (0.74-0.78)
Demographic	0.66 (0.66-0.67)	0.66 (0.66-0.67)	0.67 (0.67-0.68)	0.66 (0.65-0.67)
Serum	0.77 (0.75-0.78)	0.78 (0.75-0.80)	0.75 (0.74-0.75)	0.72 (0.65-0.78)
PRS	0.61 (0.61-0.61)	0.64 (0.62-0.65)	0.71 (0.69-0.72)	0.64 (0.62-0.66)
Protein+Demographic	0.82 (0.81-0.83)	0.82 (0.80-0.83)	0.82 (0.78-0.85)	0.77 (0.75-0.79)
Protein+Serum	0.90 (0.86-0.93)	0.88 (0.85-0.91)	0.93 (0.90-0.96)	0.79 (0.79-0.80)
Protein+PRS	0.88 (0.86-0.90)	0.88 (0.84-0.92)	0.85 (0.82-0.88)	0.79 (0.78-0.79)
Protein+Demographic+Serum	0.87 (0.85-0.89)	0.89 (0.86-0.92)	0.88 (0.83-0.93)	0.79 (0.78-0.81)
Protein+Demographic+PRS	0.88 (0.85-0.90)	0.87 (0.83-0.91)	0.80 (0.77-0.82)	0.83 (0.81-0.86)
Protein+Serum+PRS	0.91 (0.89-0.93)	0.93 (0.89-0.98)	0.86 (0.81-0.91)	0.80 (0.77-0.82)
Combined	0.92 (0.90-0.95)	0.89 (0.85-0.93)	0.87 (0.84-0.91)	0.80 (0.77-0.83)

We randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 3:1 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 5: Comparison between AUCs for all-year incident CD.

LGBM											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.83E-04*	*1.83E-04*	*1.78E-04*	*2.12E-01*	*6.78E-01*	*1.31E-03*	*5.21E-01*	*4.40E-04*	*2.83E-03*	*5.83E-04*
Demographic	*1.83E-04*		*1.83E-04*	*1.78E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
Serum	*1.83E-04*	*1.83E-04*		*2.40E-04*	*1.83E-04*	*4.40E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
PRS	*1.78E-04*	*1.78E-04*	*2.40E-04*		*1.78E-04*	*2.78E-03*	*1.78E-04*	*1.78E-04*	*1.78E-04*	*1.78E-04*	*1.78E-04*
Protein+ Demographic	*2.12E-01*	*1.83E-04*	*1.83E-04*	*1.78E-04*		*6.40E-02*	*4.40E-04*	*7.57E-02*	*2.46E-04*	*2.20E-03*	*2.46E-04*
Protein+ Serum	*6.78E-01*	*1.83E-04*	*4.40E-04*	*2.78E-03*	*6.40E-02*		*3.61E-03*	*9.70E-01*	*1.01E-03*	*2.20E-03*	*1.01E-03*
Protein+ PRS	*1.31E-03*	*1.83E-04*	*1.83E-04*	*1.78E-04*	*4.40E-04*	*3.61E-03*		*7.28E-03*	*9.70E-01*	*1.73E-02*	*1.62E-01*
Protein+ Demographic+ Serum	*5.21E-01*	*1.83E-04*	*1.83E-04*	*1.78E-04*	*7.57E-02*	*9.70E-01*	*7.28E-03*		*1.71E-03*	*2.83E-03*	*2.83E-03*
Protein+ Demographic+ PRS	*4.40E-04*	*1.83E-04*	*1.83E-04*	*1.78E-04*	*2.46E-04*	*1.01E-03*	*9.70E-01*	*1.71E-03*		*1.13E-02*	*2.41E-01*
Protein+ Serum+ PRS	*2.83E-03*	*1.83E-04*	*1.83E-04*	*1.78E-04*	*2.20E-03*	*2.20E-03*	*1.73E-02*	*2.83E-03*	*1.13E-02*		*5.71E-01*

Combined	*5.83E-04*	*1.83E-04*	*1.83E-04*	*1.78E-04*	*2.46E-04*	*1.01E-03*	*1.62E-01*	*2.83E-03*	*2.41E-01*	*5.71E-01*	

XGBoost											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.82E-04*	*1.83E-04*	*4.29E-04*	*3.45E-01*	*7.34E-01*	*2.11E-02*	*2.11E-02*	*8.90E-02*	*1.83E-04*	*1.83E-04*
Demographic	*1.82E-04*		*1.82E-04*	*1.77E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*
Serum	*1.83E-04*	*1.82E-04*		*9.86E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
PRS	*4.29E-04*	*1.77E-04*	*9.86E-04*		*1.78E-04*	*4.29E-04*	*1.78E-04*	*1.78E-04*	*2.40E-04*	*1.78E-04*	*1.78E-04*
Protein+ Demographic	*3.45E-01*	*1.82E-04*	*1.83E-04*	*1.78E-04*		*9.10E-01*	*3.12E-02*	*1.40E-01*	*1.86E-01*	*2.46E-04*	*2.46E-04*
Protein+ Serum	*7.34E-01*	*1.82E-04*	*1.83E-04*	*4.29E-04*	*9.10E-01*		*3.76E-02*	*4.27E-01*	*1.40E-01*	*5.83E-04*	*4.40E-04*
Protein+ PRS	*2.11E-02*	*1.82E-04*	*1.83E-04*	*1.78E-04*	*3.12E-02*	*3.76E-02*		*1.40E-01*	*7.34E-01*	*3.12E-02*	*3.61E-03*
Protein+ Demographic+ Serum	*2.11E-02*	*1.82E-04*	*1.83E-04*	*1.78E-04*	*1.40E-01*	*4.27E-01*	*1.40E-01*		*3.07E-01*	*5.83E-04*	*4.40E-04*
Protein+ Demographic+ PRS	*8.90E-02*	*1.82E-04*	*1.83E-04*	*2.40E-04*	*1.86E-01*	*1.40E-01*	*7.34E-01*	*3.07E-01*		*3.76E-02*	*2.83E-03*

Protein+ Serum+ PRS	1.83E-04	1.82E-04	1.83E-04	1.78E-04	2.46E-04	5.83E-04	3.12E-02	5.83E-04	3.76E-02		2.41E-01
Combined	1.83E-04	1.82E-04	1.83E-04	1.78E-04	2.46E-04	4.40E-04	3.61E-03	4.40E-04	2.83E-03	2.41E-01	

RF											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.62E-04	1.31E-03	1.68E-04	4.26E-01	3.07E-01	8.66E-02	8.90E-02	7.91E-01	2.20E-03	1.71E-03
Demographic	1.62E-04		1.61E-04	1.49E-04	1.55E-04	1.62E-04	1.44E-04	1.62E-04	1.57E-04	1.62E-04	1.62E-04
Serum	1.31E-03	1.61E-04		5.25E-02	9.69E-04	1.82E-04	4.27E-03	1.82E-04	1.76E-04	5.80E-04	1.82E-04
PRS	1.68E-04	1.49E-04	5.25E-02		1.60E-04	1.68E-04	1.49E-04	1.68E-04	1.62E-04	1.68E-04	1.68E-04
Protein+ Demographic	4.26E-01	1.55E-04	9.69E-04	1.60E-04		3.71E-02	5.67E-01	4.46E-03	4.40E-02	4.46E-03	9.73E-04
Protein+ Serum	3.07E-01	1.62E-04	1.82E-04	1.68E-04	3.71E-02		2.04E-03	3.85E-01	5.20E-01	4.59E-03	9.11E-03
Protein+ PRS	8.66E-02	1.44E-04	4.27E-03	1.49E-04	5.67E-01	2.04E-03		8.59E-03	1.99E-02	2.62E-03	1.57E-03
Protein+ Demographic+ Serum	8.90E-02	1.62E-04	1.82E-04	1.68E-04	4.46E-03	3.85E-01	8.59E-03		1.20E-01	3.12E-02	3.45E-01

Protein+ Demographic+ PRS	*7.91E-01*	*1.57E-04*	*1.76E-04*	*1.62E-04*	*4.40E-02*	*5.20E-01*	*1.99E-02*	*1.20E-01*		*5.69E-03*	*7.15E-03*
Protein+ Serum+ PRS	*2.20E-03*	*1.62E-04*	*5.80E-04*	*1.68E-04*	*4.46E-03*	*4.59E-03*	*2.62E-03*	*3.12E-02*	*5.69E-03*		*1.86E-01*
Combined	*1.71E-03*	*1.62E-04*	*1.82E-04*	*1.68E-04*	*9.73E-04*	*9.11E-03*	*1.57E-03*	*3.45E-01*	*7.15E-03*	*1.86E-01*	

Extra trees											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.60E-04*	*1.82E-04*	*1.67E-04*	*1.03E-01*	*3.28E-04*	*1.26E-03*	*7.91E-01*	*4.37E-04*	*3.28E-04*	*5.20E-01*
Demographic	*1.60E-04*		*2.60E-03*	*1.48E-04*	*1.57E-04*	*1.61E-04*	*1.52E-04*	*1.61E-04*	*1.61E-04*	*1.61E-04*	*1.60E-04*
Serum	*1.82E-04*	*2.60E-03*		*1.68E-04*	*1.79E-04*	*1.83E-04*	*1.73E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*9.08E-03*
PRS	*1.67E-04*	*1.48E-04*	*1.68E-04*		*1.64E-04*	*1.68E-04*	*1.58E-04*	*2.67E-03*	*1.68E-04*	*1.68E-04*	*4.70E-01*
Protein+ Demographic	*1.03E-01*	*1.57E-04*	*1.79E-04*	*1.64E-04*		*2.79E-03*	*1.84E-01*	*2.12E-01*	*4.53E-03*	*9.90E-04*	*7.91E-01*
Protein+ Serum	*3.28E-04*	*1.61E-04*	*1.83E-04*	*1.68E-04*	*2.79E-03*		*4.71E-01*	*1.31E-03*	*2.11E-02*	*1.73E-02*	*2.12E-01*
Protein+ PRS	*1.26E-03*	*1.52E-04*	*1.73E-04*	*1.58E-04*	*1.84E-01*	*4.71E-01*		*8.85E-03*	*1.03E-01*	*1.20E-01*	*4.25E-01*

Protein+ Demographic+ Serum	*7.91E-01*	*1.61E-04*	*1.83E-04*	*2.67E-03*	*2.12E-01*	*1.31E-03*	*8.85E-03*		*2.83E-03*	*7.69E-04*	*7.34E-01*
Protein+ Demographic+ PRS	*4.37E-04*	*1.61E-04*	*1.83E-04*	*1.68E-04*	*4.53E-03*	*2.11E-02*	*1.03E-01*	*2.83E-03*		*1.21E-01*	*2.57E-02*
Protein+ Serum+ PRS	*3.28E-04*	*1.61E-04*	*1.83E-04*	*1.68E-04*	*9.90E-04*	*1.73E-02*	*1.20E-01*	*7.69E-04*	*1.21E-01*		*3.76E-02*
Combined	*5.20E-01*	*1.60E-04*	*9.08E-03*	*4.70E-01*	*7.91E-01*	*2.12E-01*	*4.25E-01*	*7.34E-01*	*2.57E-02*	*3.76E-02*	

Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 6: Results of ROC analyses after splitting the participants into a training and a testing set according to the recruitment centers in a ratio of 8:2.

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.76 (0.75-0.77)*	*0.76 (0.74-0.77)*	*0.77 (0.77-0.78)*	*0.71 (0.70-0.71)*
Demographic	*0.52 (0.51-0.52)*	*0.53 (0.52-0.55)*	*0.52 (0.51-0.53)*	*0.57 (0.56-0.57)*
Serum	*0.60 (0.58-0.62)*	*0.60 (0.58-0.61)*	*0.62 (0.61-0.63)*	*0.54 (0.51-0.57)*
PRS	*0.57 (0.56-0.57)*	*0.58 (0.56-0.59)*	*0.59 (0.58-0.60)*	*0.62 (0.61-0.63)*
Protein+Demographic	*0.76 (0.75-0.77)*	*0.75 (0.74-0.76)*	*0.76 (0.75-0.78)*	*0.73 (0.72-0.75)*
Protein+Serum	*0.73 (0.71-0.75)*	*0.75 (0.73-0.76)*	*0.74 (0.72-0.77)*	*0.72 (0.70-0.73)*
Protein+PRS	*0.78 (0.77-0.80)*	*0.78 (0.78-0.79)*	*0.78 (0.76-0.79)*	*0.76 (0.74-0.77)*
Protein+Demographic+Serum	*0.75 (0.73-0.77)*	*0.75 (0.73-0.76)*	*0.77 (0.76-0.77)*	*0.75 (0.74-0.76)*
Protein+Demographic+PRS	*0.77 (0.75-0.78)*	*0.78 (0.77-0.78)*	*0.78 (0.77-0.79)*	*0.77 (0.76-0.77)*
Protein+Serum+PRS	*0.74 (0.73-0.76)*	*0.77 (0.75-0.80)*	*0.76 (0.75-0.78)*	*0.75 (0.74-0.76)*
Combined	*0.75 (0.74-0.77)*	*0.76 (0.74-0.79)*	*0.76 (0.75-0.77)*	*0.76 (0.74-0.77)*
Training set				
9-protein panel	*0.86 (0.83-0.89)*	*0.84 (0.79-0.88)*	*0.81 (0.78-0.83)*	*0.74 (0.73-0.76)*
Demographic	*0.68 (0.68-0.68)*	*0.67 (0.67-0.68)*	*0.67 (0.67-0.67)*	*0.64 (0.64-0.65)*
Serum	*0.81 (0.75-0.88)*	*0.77 (0.73-0.81)*	*0.77 (0.73-0.82)*	*0.70 (0.67-0.73)*
PRS	*0.68 (0.66-0.70)*	*0.69 (0.68-0.70)*	*0.70 (0.69-0.72)*	*0.64 (0.61-0.66)*
Protein+Demographic	*0.88 (0.85-0.91)*	*0.83 (0.79-0.88)*	*0.83 (0.80-0.87)*	*0.75 (0.73-0.78)*
Protein+Serum	*0.91 (0.88-0.94)*	*0.89 (0.84-0.93)*	*0.88 (0.84-0.91)*	*0.78 (0.76-0.81)*
Protein+PRS	*0.94 (0.92-0.97)*	*0.89 (0.85-0.93)*	*0.91 (0.87-0.95)*	*0.76 (0.76-0.76)*
Protein+Demographic+Serum	*0.94 (0.91-0.97)*	*0.87 (0.82-0.92)*	*0.79 (0.78-0.79)*	*0.79 (0.77-0.80)*
Protein+Demographic+PRS	*0.89 (0.86-0.92)*	*0.86 (0.83-0.89)*	*0.84 (0.80-0.87)*	*0.77 (0.76-0.78)*
Protein+Serum+PRS	*0.97 (0.93-1.00)*	*0.95 (0.91-0.98)*	*0.96 (0.94-0.97)*	*0.80 (0.79-0.82)*
Combined	*0.96 (0.93-1.00)*	*0.93 (0.90-0.97)*	*0.91 (0.87-0.95)*	*0.79 (0.79-0.80)*

We randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 8:2 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 7: Comparison between AUCs for all-time incident CD after splitting the participants into a training and a testing set in a ratio of 8:2.

LGBM											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.82E-04*	*1.83E-04*	*1.83E-04*	*4.27E-01*	*2.57E-02*	*9.11E-03*	*6.78E-01*	*4.27E-01*	*5.71E-01*	*9.70E-01*
Demographic	*1.82E-04*		*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*	*1.82E-04*
Serum	*1.83E-04*	*1.82E-04*		*1.13E-02*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
PRS	*1.83E-04*	*1.82E-04*	*1.13E-02*		*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
Protein+ Demographic	*4.27E-01*	*1.82E-04*	*1.83E-04*	*1.83E-04*		*1.40E-02*	*7.28E-03*	*2.41E-01*	*5.71E-01*	*2.12E-01*	*4.73E-01*
Protein+ Serum	*2.57E-02*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*1.40E-02*		*1.31E-03*	*7.57E-02*	*9.11E-03*	*2.12E-01*	*5.39E-02*
Protein+ PRS	*9.11E-03*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*7.28E-03*	*1.31E-03*		*1.40E-02*	*5.39E-02*	*3.61E-03*	*2.11E-02*
Protein+ Demographic+ Serum	*6.78E-01*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*2.41E-01*	*7.57E-02*	*1.40E-02*		*3.85E-01*	*8.50E-01*	*5.71E-01*
Protein+ Demographic+ PRS	*4.27E-01*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*5.71E-01*	*9.11E-03*	*5.39E-02*	*3.85E-01*		*1.40E-01*	*3.85E-01*
Protein+ Serum+ PRS	*5.71E-01*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*2.12E-01*	*2.12E-01*	*3.61E-03*	*8.50E-01*	*1.40E-01*		*6.78E-01*

Combined	9.70E-01	1.82E-04	1.83E-04	1.83E-04	4.73E-01	5.39E-02	2.11E-02	5.71E-01	3.85E-01	6.78E-01	

XGBoost											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.83E-04	1.83E-04	1.83E-04	4.27E-01	4.27E-01	2.57E-02	2.41E-01	7.57E-02	8.90E-02	3.07E-01
Demographic	1.83E-04		4.40E-04	2.20E-03	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
Serum	1.83E-04	4.40E-04		8.90E-02	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
PRS	1.83E-04	2.20E-03	8.90E-02		1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
Protein+ Demographic	4.27E-01	1.83E-04	1.83E-04	1.83E-04		9.10E-01	1.01E-03	8.50E-01	1.40E-02	2.57E-02	1.21E-01
Protein+ Serum	4.27E-01	1.83E-04	1.83E-04	1.83E-04	9.10E-01		5.83E-04	9.10E-01	1.13E-02	3.12E-02	1.04E-01
Protein+ PRS	2.57E-02	1.83E-04	1.83E-04	1.83E-04	1.01E-03	5.83E-04		4.40E-04	1.04E-01	1.00E+00	1.40E-01
Protein+ Demographic+ Serum	2.41E-01	1.83E-04	1.83E-04	1.83E-04	8.50E-01	9.10E-01	4.40E-04		4.59E-03	1.13E-02	4.52E-02
Protein+ Demographic+ PRS	7.57E-02	1.83E-04	1.83E-04	1.83E-04	1.40E-02	1.13E-02	1.04E-01	4.59E-03		6.23E-01	7.34E-01

Protein+ Serum+ PRS	*8.90E-02*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*2.57E-02*	*3.12E-02*	*1.00E+00*	*1.13E-02*	*6.23E-01*		*6.23E-01*
Combined	*3.07E-01*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.21E-01*	*1.04E-01*	*1.40E-01*	*4.52E-02*	*7.34E-01*	*6.23E-01*	

RF											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.61E-04*	*1.82E-04*	*1.77E-04*	*5.20E-01*	*1.04E-01*	*2.41E-01*	*8.49E-01*	*9.10E-01*	*3.45E-01*	*6.39E-02*
Demographic	*1.61E-04*		*1.62E-04*	*1.57E-04*	*1.62E-04*	*1.61E-04*	*1.62E-04*	*1.44E-04*	*1.62E-04*	*1.62E-04*	*1.62E-04*
Serum	*1.82E-04*	*1.62E-04*		*5.70E-03*	*1.83E-04*	*1.82E-04*	*1.83E-04*	*1.63E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
PRS	*1.77E-04*	*1.57E-04*	*5.70E-03*		*1.78E-04*	*1.77E-04*	*1.78E-04*	*1.58E-04*	*1.78E-04*	*1.78E-04*	*1.78E-04*
Protein+ Demographic	*5.20E-01*	*1.62E-04*	*1.83E-04*	*1.78E-04*		*2.41E-01*	*3.07E-01*	*6.75E-01*	*2.73E-01*	*9.10E-01*	*8.50E-01*
Protein+ Serum	*1.04E-01*	*1.61E-04*	*1.82E-04*	*1.77E-04*	*2.41E-01*		*3.76E-02*	*3.41E-01*	*8.89E-02*	*4.27E-01*	*4.27E-01*
Protein+ PRS	*2.41E-01*	*1.62E-04*	*1.83E-04*	*1.78E-04*	*3.07E-01*	*3.76E-02*		*1.83E-01*	*4.27E-01*	*2.41E-01*	*1.04E-01*
Protein+ Demographic+ Serum	*8.49E-01*	*1.44E-04*	*1.63E-04*	*1.58E-04*	*6.75E-01*	*3.41E-01*	*1.83E-01*		*6.75E-01*	*6.75E-01*	*1.02E-01*

Protein+ Demographic+ PRS	9.10E-01	1.62E-04	1.83E-04	1.78E-04	2.73E-01	8.89E-02	4.27E-01	6.75E-01		1.40E-01	2.57E-02
Protein+ Serum+ PRS	3.45E-01	1.62E-04	1.83E-04	1.78E-04	9.10E-01	4.27E-01	2.41E-01	6.75E-01	1.40E-01		9.10E-01
Combined	6.39E-02	1.62E-04	1.83E-04	1.78E-04	8.50E-01	4.27E-01	1.04E-01	1.02E-01	2.57E-02	9.10E-01	

Extra trees											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.71E-04	1.73E-04	1.70E-04	2.72E-03	6.31E-02	6.90E-04	1.70E-04	1.74E-04	1.74E-04	1.74E-04
Demographic	1.71E-04		8.85E-02	1.76E-04	1.79E-04	1.80E-04	1.66E-04	1.76E-04	1.80E-04	1.80E-04	1.80E-04
Serum	1.73E-04	8.85E-02		5.69E-04	1.81E-04	1.82E-04	1.68E-04	1.78E-04	1.82E-04	1.82E-04	1.82E-04
PRS	1.70E-04	1.76E-04	5.69E-04		1.78E-04	1.79E-04	1.65E-04	1.75E-04	1.79E-04	1.79E-04	1.79E-04
Protein+ Demographic	2.72E-03	1.79E-04	1.81E-04	1.78E-04		2.41E-01	2.04E-02	1.40E-01	1.00E-03	4.51E-02	3.11E-02
Protein+ Serum	6.31E-02	1.80E-04	1.82E-04	1.79E-04	2.41E-01		1.61E-03	2.79E-03	1.83E-04	1.71E-03	7.69E-04
Protein+ PRS	6.90E-04	1.66E-04	1.68E-04	1.65E-04	2.04E-02	1.61E-03		1.19E-01	4.38E-03	8.73E-02	8.49E-01

Protein+ Demographic+ Serum	*1.70E-04*	*1.76E-04*	*1.78E-04*	*1.75E-04*	*1.40E-01*	*2.79E-03*	*1.19E-01*		*7.20E-03*	*3.07E-01*	*3.84E-01*
Protein+ Demographic+ PRS	*1.74E-04*	*1.80E-04*	*1.82E-04*	*1.79E-04*	*1.00E-03*	*1.83E-04*	*4.38E-03*	*7.20E-03*		*2.83E-03*	*2.73E-01*
Protein+ Serum+ PRS	*1.74E-04*	*1.80E-04*	*1.82E-04*	*1.79E-04*	*4.51E-02*	*1.71E-03*	*8.73E-02*	*3.07E-01*	*2.83E-03*		*5.21E-01*
Combined	*1.74E-04*	*1.80E-04*	*1.82E-04*	*1.79E-04*	*3.11E-02*	*7.69E-04*	*8.49E-01*	*3.84E-01*	*2.73E-01*	*5.21E-01*	

Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 8: Results of ROC analyses of the PREDICTS model in UK Biobank.

	AUC (95%CI)			
	LGBM	**XGBoost**	**RF**	**Extra trees**
Testing set				
PREDICTS	*0.67 (0.65-0.69)*	*0.68 (0.67-0.69)*	*0.69 (0.67-0.70)*	*0.64 (0.62-0.65)*
9-protein panel	*0.73 (0.72-0.73)*	*0.72 (0.71-0.73)*	*0.71 (0.70-0.73)*	*0.72 (0.71-0.73)*
Protein+Demographic	*0.72 (0.71-0.73)*	*0.72 (0.72-0.73)*	*0.70 (0.69-0.72)*	*0.73 (0.72-0.74)*
Protein+Serum	*0.72 (0.70-0.74)*	*0.72 (0.70-0.74)*	*0.73 (0.72-0.74)*	*0.76 (0.75-0.76)*
Protein+PRS	*0.75 (0.74-0.76)*	*0.74 (0.73-0.75)*	*0.70 (0.68-0.71)*	*0.75 (0.74-0.77)*
Protein+Demographic+Serum	*0.73 (0.72-0.74)*	*0.73 (0.72-0.74)*	*0.74 (0.72-0.75)*	*0.72 (0.71-0.74)*
Protein+Demographic+PRS	*0.75 (0.75-0.76)*	*0.73 (0.72-0.75)*	*0.72 (0.70-0.73)*	*0.78 (0.76-0.79)*
Protein+Serum+PRS	*0.76 (0.75-0.77)*	*0.76 (0.75-0.77)*	*0.75 (0.74-0.77)*	*0.77 (0.76-0.78)*
Combined	*0.76 (0.75-0.77)*	*0.78 (0.77-0.79)*	*0.75 (0.74-0.76)*	*0.72 (0.68-0.76)*
Training set				
PREDICTS	*0.87 (0.81-0.93)*	*0.78 (0.72-0.83)*	*0.77 (0.74-0.81)*	*0.69 (0.66-0.73)*
9-protein panel	*0.85 (0.82-0.88)*	*0.83 (0.80-0.86)*	*0.84 (0.82-0.87)*	*0.76 (0.74-0.78)*
Protein+Demographic	*0.82 (0.81-0.83)*	*0.82 (0.80-0.83)*	*0.82 (0.78-0.85)*	*0.77 (0.75-0.79)*
Protein+Serum	*0.90 (0.86-0.93)*	*0.88 (0.85-0.91)*	*0.93 (0.90-0.96)*	*0.79 (0.79-0.80)*
Protein+PRS	*0.88 (0.86-0.90)*	*0.88 (0.84-0.92)*	*0.85 (0.82-0.88)*	*0.79 (0.78-0.79)*
Protein+Demographic+Serum	*0.87 (0.85-0.89)*	*0.89 (0.86-0.92)*	*0.88 (0.83-0.93)*	*0.79 (0.78-0.81)*
Protein+Demographic+PRS	*0.88 (0.85-0.90)*	*0.87 (0.83-0.91)*	*0.80 (0.77-0.82)*	*0.83 (0.81-0.86)*
Protein+Serum+PRS	*0.91 (0.89-0.93)*	*0.93 (0.89-0.98)*	*0.86 (0.81-0.91)*	*0.80 (0.77-0.82)*
Combined	*0.92 (0.90-0.95)*	*0.89 (0.85-0.93)*	*0.87 (0.84-0.91)*	*0.80 (0.77-0.83)*

We externally validated the six optimal proteins identified in PREDICTS study (C-reactive protein, C5, PRSS2, APCS, OMD, ACAN) using UK Biobank populations. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

“Supplementary Table 9: Comparison between AUCs between PREDICTS model and other models.

	9-protein panel	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
LGBM	1.83E-04	2.46E-04	2.20E-03	1.83E-04	1.83E-04	1.83E-04	2.46E-04	1.83E-04
XGBoost	7.69E-04	2.46E-04	3.61E-03	3.30E-04	4.40E-04	1.71E-03	1.83E-04	1.83E-04
RF	1.72E-02	3.43E-01	5.80E-04	8.49E-01	7.65E-04	2.08E-02	1.31E-03	3.28E-04
Extra trees	1.77E-04	1.74E-04	1.78E-04	1.68E-04	1.78E-04	1.78E-04	1.78E-04	2.77E-03

Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn’s disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease.”

*“**Supplementary Table 10: Results of ROC analyses within different time points in the UKB testing set.***

10-year incident CD

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.69 (0.68-0.71)*	*0.69 (0.69-0.70)*	*0.69 (0.67-0.71)*	*0.72 (0.71-0.72)*
Demographic	*0.54 (0.53-0.54)*	*0.53 (0.52-0.54)*	*0.56 (0.55-0.57)*	*0.57 (0.56-0.57)*
Serum	*0.64 (0.61-0.67)*	*0.63 (0.62-0.64)*	*0.61 (0.58-0.63)*	*0.62 (0.60-0.65)*
PRS	*0.60 (0.59-0.60)*	*0.57 (0.56-0.58)*	*0.59 (0.59-0.60)*	*0.62 (0.58-0.65)*
Protein+Demographic	*0.69 (0.68-0.70)*	*0.70 (0.70-0.71)*	*0.72 (0.71-0.72)*	*0.72 (0.71-0.73)*
Protein+Serum	*0.70 (0.69-0.71)*	*0.71 (0.69-0.72)*	*0.71 (0.71-0.72)*	*0.72 (0.71-0.73)*
Protein+PRS	*0.72 (0.71-0.72)*	*0.71 (0.71-0.72)*	*0.73 (0.72-0.74)*	*0.75 (0.74-0.77)*
Protein+Demographic+Serum	*0.71 (0.70-0.71)*	*0.70 (0.69-0.72)*	*0.72 (0.71-0.72)*	*0.72 (0.70-0.73)*
Protein+Demographic+PRS	*0.71 (0.70-0.72)*	*0.72 (0.71-0.73)*	*0.72 (0.70-0.73)*	*0.75 (0.74-0.77)*
Protein+Serum+PRS	*0.73 (0.72-0.74)*	*0.73 (0.72-0.74)*	*0.73 (0.72-0.74)*	*0.74 (0.73-0.75)*
Combined	*0.72 (0.71-0.74)*	*0.72 (0.71-0.74)*	*0.71 (0.70-0.73)*	*0.75 (0.74-0.75)*
Training set				
9-protein panel	*0.94 (0.93-0.96)*	*0.95 (0.93-0.96)*	*0.89 (0.86-0.92)*	*0.84 (0.82-0.85)*
Demographic	*0.72 (0.72-0.72)*	*0.71 (0.70-0.71)*	*0.71 (0.70-0.72)*	*0.68 (0.67-0.69)*
Serum	*0.86 (0.83-0.90)*	*0.87 (0.82-0.93)*	*0.84 (0.80-0.88)*	*0.75 (0.72-0.78)*
PRS	*0.71 (0.67-0.76)*	*0.79 (0.74-0.83)*	*0.71 (0.70-0.73)*	*0.75 (0.70-0.79)*
Protein+Demographic	*0.95 (0.93-0.96)*	*0.92 (0.90-0.95)*	*0.90 (0.87-0.92)*	*0.87 (0.84-0.89)*
Protein+Serum	*0.97 (0.96-0.99)*	*0.96 (0.94-0.98)*	*0.95 (0.92-0.97)*	*0.90 (0.86-0.94)*
Protein+PRS	*0.95 (0.94-0.96)*	*0.95 (0.93-0.96)*	*0.95 (0.93-0.97)*	*0.87 (0.83-0.90)*
Protein+Demographic+Serum	*0.96 (0.95-0.98)*	*0.95 (0.92-0.98)*	*0.98 (0.98-0.99)*	*0.87 (0.85-0.88)*
Protein+Demographic+PRS	*0.93 (0.92-0.95)*	*0.95 (0.94-0.97)*	*0.93 (0.91-0.95)*	*0.84 (0.82-0.86)*
Protein+Serum+PRS	*0.97 (0.95-0.99)*	*0.97 (0.95-0.99)*	*0.93 (0.90-0.96)*	*0.87 (0.82-0.92)*
Combined	*0.97 (0.96-0.98)*	*0.95 (0.93-0.98)*	*0.91 (0.88-0.95)*	*0.87 (0.85-0.89)*

11-year incident CD

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.73 (0.72-0.73)*	*0.72 (0.72-0.73)*	*0.71 (0.70-0.72)*	*0.72 (0.70-0.73)*
Demographic	*0.57 (0.56-0.58)*	*0.57 (0.56-0.58)*	*0.58 (0.57-0.58)*	*0.60 (0.59-0.61)*
Serum	*0.64 (0.62-0.65)*	*0.64 (0.63-0.65)*	*0.65 (0.62-0.67)*	*0.64 (0.63-0.65)*
PRS	*0.61 (0.61-0.61)*	*0.66 (0.65-0.67)*	*0.64 (0.63-0.66)*	*0.68 (0.68-0.68)*
Protein+Demographic	*0.72 (0.72-0.73)*	*0.71 (0.70-0.72)*	*0.72 (0.71-0.73)*	*0.70 (0.69-0.72)*
Protein+Serum	*0.71 (0.71-0.72)*	*0.72 (0.72-0.73)*	*0.73 (0.73-0.74)*	*0.72 (0.71-0.74)*
Protein+PRS	*0.74 (0.73-0.76)*	*0.74 (0.73-0.75)*	*0.74 (0.73-0.76)*	*0.77 (0.75-0.78)*
Protein+Demographic+Serum	*0.72 (0.71-0.72)*	*0.73 (0.72-0.74)*	*0.70 (0.68-0.72)*	*0.70 (0.68-0.72)*
Protein+Demographic+PRS	*0.75 (0.75-0.76)*	*0.75 (0.74-0.76)*	*0.75 (0.74-0.76)*	*0.77 (0.76-0.78)*
Protein+Serum+PRS	*0.73 (0.73-0.74)*	*0.75 (0.74-0.76)*	*0.74 (0.72-0.75)*	*0.72 (0.70-0.74)*

Combined	*0.74 (0.73-0.75)*	*0.75 (0.74-0.76)*	*0.74 (0.72-0.76)*	*0.77 (0.76-0.77)*
Training set				
9-protein panel	*0.88 (0.87-0.89)*	*0.89 (0.87-0.92)*	*0.85 (0.84-0.85)*	*0.83 (0.79-0.87)*
Demographic	*0.71 (0.70-0.71)*	*0.70 (0.69-0.70)*	*0.69 (0.68-0.70)*	*0.66 (0.64-0.67)*
Serum	*0.83 (0.79-0.86)*	*0.86 (0.81-0.91)*	*0.81 (0.77-0.85)*	*0.71 (0.70-0.72)*
PRS	*0.62 (0.62-0.62)*	*0.66 (0.65-0.68)*	*0.64 (0.61-0.67)*	*0.61 (0.60-0.62)*
Protein+Demographic	*0.88 (0.87-0.89)*	*0.87 (0.85-0.89)*	*0.85 (0.83-0.87)*	*0.82 (0.79-0.84)*
Protein+Serum	*0.96 (0.95-0.98)*	*0.94 (0.91-0.96)*	*0.87 (0.85-0.90)*	*0.88 (0.84-0.93)*
Protein+PRS	*0.89 (0.88-0.90)*	*0.91 (0.87-0.94)*	*0.87 (0.85-0.89)*	*0.81 (0.80-0.83)*
Protein+Demographic+Serum	*0.97 (0.95-0.98)*	*0.92 (0.91-0.94)*	*0.92 (0.88-0.97)*	*0.87 (0.81-0.92)*
Protein+Demographic+PRS	*0.91 (0.88-0.93)*	*0.90 (0.88-0.92)*	*0.93 (0.90-0.95)*	*0.92 (0.88-0.96)*
Protein+Serum+PRS	*0.98 (0.96-0.99)*	*0.95 (0.93-0.97)*	*0.92 (0.89-0.94)*	*0.88 (0.82-0.93)*
Combined	*0.97 (0.96-0.98)*	*0.95 (0.93-0.98)*	*0.93 (0.91-0.95)*	*0.93 (0.88-0.97)*

12-year incident CD

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.71 (0.71-0.72)*	*0.71 (0.71-0.72)*	*0.73 (0.72-0.73)*	*0.71 (0.70-0.72)*
Demographic	*0.57 (0.57-0.57)*	*0.57 (0.56-0.57)*	*0.58 (0.58-0.59)*	*0.58 (0.57-0.59)*
Serum	*0.63 (0.62-0.65)*	*0.62 (0.60-0.64)*	*0.62 (0.60-0.64)*	*0.63 (0.62-0.64)*
PRS	*0.64 (0.63-0.65)*	*0.63 (0.61-0.65)*	*0.61 (0.60-0.62)*	*0.69 (0.69-0.69)*
Protein+Demographic	*0.72 (0.71-0.72)*	*0.72 (0.71-0.72)*	*0.72 (0.71-0.73)*	*0.74 (0.72-0.75)*
Protein+Serum	*0.72 (0.70-0.73)*	*0.70 (0.69-0.72)*	*0.72 (0.71-0.73)*	*0.74 (0.73-0.75)*
Protein+PRS	*0.73 (0.71-0.74)*	*0.73 (0.71-0.75)*	*0.75 (0.74-0.75)*	*0.78 (0.77-0.79)*
Protein+Demographic+Serum	*0.72 (0.71-0.73)*	*0.71 (0.68-0.73)*	*0.72 (0.70-0.73)*	*0.71 (0.68-0.74)*
Protein+Demographic+PRS	*0.73 (0.72-0.74)*	*0.74 (0.72-0.76)*	*0.72 (0.72-0.73)*	*0.74 (0.71-0.77)*
Protein+Serum+PRS	*0.74 (0.73-0.76)*	*0.73 (0.71-0.75)*	*0.74 (0.72-0.75)*	*0.74 (0.73-0.76)*
Combined	*0.75 (0.74-0.76)*	*0.73 (0.72-0.74)*	*0.73 (0.73-0.74)*	*0.79 (0.78-0.80)*
Training set				
9-protein panel	*0.89 (0.88-0.90)*	*0.89 (0.87-0.90)*	*0.85 (0.83-0.86)*	*0.79 (0.78-0.81)*
Demographic	*0.70 (0.70-0.70)*	*0.69 (0.69-0.70)*	*0.69 (0.69-0.69)*	*0.67 (0.67-0.68)*
Serum	*0.86 (0.81-0.91)*	*0.86 (0.80-0.92)*	*0.74 (0.73-0.75)*	*0.69 (0.68-0.71)*
PRS	*0.69 (0.65-0.72)*	*0.69 (0.66-0.72)*	*0.78 (0.68-0.88)*	*0.60 (0.59-0.60)*
Protein+Demographic	*0.89 (0.88-0.90)*	*0.87 (0.86-0.89)*	*0.90 (0.87-0.93)*	*0.82 (0.78-0.86)*
Protein+Serum	*0.92 (0.90-0.94)*	*0.89 (0.85-0.92)*	*0.88 (0.85-0.90)*	*0.87 (0.84-0.90)*
Protein+PRS	*0.92 (0.90-0.95)*	*0.90 (0.87-0.93)*	*0.95 (0.93-0.98)*	*0.81 (0.80-0.82)*
Protein+Demographic+Serum	*0.93 (0.90-0.96)*	*0.89 (0.86-0.92)*	*0.88 (0.86-0.91)*	*0.85 (0.83-0.88)*
Protein+Demographic+PRS	*0.91 (0.90-0.92)*	*0.89 (0.86-0.91)*	*0.91 (0.88-0.94)*	*0.89 (0.84-0.95)*
Protein+Serum+PRS	*0.94 (0.91-0.97)*	*0.90 (0.87-0.94)*	*0.85 (0.83-0.87)*	*0.84 (0.82-0.86)*
Combined	*0.93 (0.90-0.96)*	*0.94 (0.90-0.98)*	*0.94 (0.91-0.96)*	*0.82 (0.82-0.83)*

13-year incident CD

	AUC (95%CI)

	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	*0.72 (0.71-0.73)*	*0.71 (0.69-0.72)*	*0.73 (0.71-0.74)*	*0.70 (0.70-0.71)*
Demographic	*0.58 (0.58-0.58)*	*0.58 (0.57-0.58)*	*0.59 (0.59-0.59)*	*0.58 (0.57-0.58)*
Serum	*0.61 (0.60-0.63)*	*0.62 (0.61-0.63)*	*0.62 (0.60-0.64)*	*0.63 (0.61-0.65)*
PRS	*0.65 (0.64-0.66)*	*0.63 (0.61-0.65)*	*0.62 (0.61-0.64)*	*0.68 (0.68-0.68)*
Protein+Demographic	*0.72 (0.70-0.73)*	*0.73 (0.72-0.73)*	*0.72 (0.71-0.73)*	*0.71 (0.70-0.72)*
Protein+Serum	*0.72 (0.71-0.74)*	*0.71 (0.70-0.72)*	*0.70 (0.68-0.72)*	*0.74 (0.73-0.75)*
Protein+PRS	*0.73 (0.73-0.74)*	*0.75 (0.74-0.76)*	*0.72 (0.71-0.73)*	*0.75 (0.73-0.77)*
Protein+Demographic+Serum	*0.71 (0.69-0.73)*	*0.72 (0.71-0.73)*	*0.73 (0.72-0.74)*	*0.73 (0.71-0.74)*
Protein+Demographic+PRS	*0.74 (0.74-0.75)*	*0.75 (0.74-0.75)*	*0.74 (0.72-0.75)*	*0.77 (0.75-0.78)*
Protein+Serum+PRS	*0.74 (0.72-0.76)*	*0.75 (0.74-0.76)*	*0.75 (0.73-0.77)*	*0.71 (0.69-0.73)*
Combined	*0.76 (0.75-0.76)*	*0.76 (0.75-0.77)*	*0.76 (0.75-0.76)*	*0.76 (0.74-0.78)*
Training set				
9-protein panel	*0.87 (0.85-0.90)*	*0.83 (0.82-0.84)*	*0.83 (0.81-0.85)*	*0.76 (0.76-0.77)*
Demographic	*0.67 (0.67-0.68)*	*0.67 (0.67-0.68)*	*0.68 (0.68-0.68)*	*0.68 (0.68-0.68)*
Serum	*0.77 (0.76-0.79)*	*0.78 (0.75-0.81)*	*0.76 (0.72-0.80)*	*0.69 (0.67-0.71)*
PRS	*0.76 (0.72-0.80)*	*0.75 (0.70-0.80)*	*0.78 (0.72-0.84)*	*0.60 (0.60-0.60)*
Protein+Demographic	*0.86 (0.85-0.87)*	*0.83 (0.82-0.85)*	*0.88 (0.83-0.92)*	*0.79 (0.76-0.82)*
Protein+Serum	*0.91 (0.87-0.94)*	*0.90 (0.86-0.93)*	*0.82 (0.77-0.87)*	*0.81 (0.78-0.85)*
Protein+PRS	*0.94 (0.92-0.97)*	*0.90 (0.87-0.93)*	*0.83 (0.78-0.88)*	*0.78 (0.77-0.79)*
Protein+Demographic+Serum	*0.95 (0.93-0.97)*	*0.87 (0.84-0.91)*	*0.87 (0.84-0.90)*	*0.84 (0.79-0.90)*
Protein+Demographic+PRS	*0.92 (0.88-0.95)*	*0.88 (0.86-0.90)*	*0.92 (0.88-0.97)*	*0.81 (0.80-0.82)*
Protein+Serum+PRS	*0.96 (0.93-0.99)*	*0.94 (0.90-0.98)*	*0.89 (0.85-0.92)*	*0.74 (0.72-0.76)*
Combined	*0.96 (0.93-0.99)*	*0.92 (0.90-0.95)*	*0.90 (0.88-0.93)*	*0.85 (0.80-0.90)*

14-year incident CD

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	*0.72 (0.71-0.73)*	*0.71 (0.71-0.72)*	*0.71 (0.70-0.72)*	*0.71 (0.70-0.71)*
Demographic	*0.57 (0.56-0.57)*	*0.57 (0.56-0.58)*	*0.59 (0.58-0.59)*	*0.55 (0.55-0.56)*
Serum	*0.63 (0.62-0.64)*	*0.62 (0.61-0.64)*	*0.66 (0.64-0.67)*	*0.63 (0.60-0.65)*
PRS	*0.65 (0.65-0.66)*	*0.66 (0.66-0.67)*	*0.61 (0.60-0.63)*	*0.68 (0.68-0.69)*
Protein+Demographic	*0.71 (0.71-0.72)*	*0.72 (0.71-0.73)*	*0.70 (0.69-0.72)*	*0.73 (0.72-0.74)*
Protein+Serum	*0.72 (0.70-0.74)*	*0.72 (0.71-0.74)*	*0.72 (0.72-0.72)*	*0.75 (0.75-0.76)*
Protein+PRS	*0.74 (0.74-0.75)*	*0.73 (0.72-0.75)*	*0.70 (0.68-0.72)*	*0.75 (0.74-0.76)*
Protein+Demographic+Serum	*0.73 (0.72-0.74)*	*0.73 (0.72-0.74)*	*0.73 (0.72-0.75)*	*0.71 (0.70-0.73)*
Protein+Demographic+PRS	*0.75 (0.74-0.75)*	*0.73 (0.71-0.75)*	*0.71 (0.70-0.73)*	*0.78 (0.76-0.79)*
Protein+Serum+PRS	*0.76 (0.75-0.77)*	*0.75 (0.73-0.77)*	*0.75 (0.73-0.77)*	*0.76 (0.75-0.77)*
Combined	*0.75 (0.75-0.76)*	*0.77 (0.76-0.77)*	*0.74 (0.73-0.76)*	*0.73 (0.69-0.76)*
Training set				
9-protein panel	*0.81 (0.80-0.82)*	*0.84 (0.81-0.87)*	*0.84 (0.82-0.87)*	*0.75 (0.75-0.75)*
Demographic	*0.66 (0.66-0.67)*	*0.66 (0.66-0.67)*	*0.67 (0.67-0.68)*	*0.66 (0.65-0.67)*

Serum	*0.77 (0.75-0.78)*	*0.78 (0.76-0.81)*	*0.74 (0.73-0.75)*	*0.68 (0.67-0.69)*
PRS	*0.61 (0.61-0.62)*	*0.64 (0.62-0.65)*	*0.71 (0.70-0.73)*	*0.63 (0.61-0.65)*
Protein+Demographic	*0.82 (0.81-0.83)*	*0.82 (0.80-0.83)*	*0.81 (0.78-0.84)*	*0.77 (0.75-0.79)*
Protein+Serum	*0.90 (0.86-0.93)*	*0.89 (0.86-0.92)*	*0.93 (0.90-0.97)*	*0.80 (0.79-0.80)*
Protein+PRS	*0.88 (0.85-0.90)*	*0.89 (0.86-0.93)*	*0.85 (0.82-0.88)*	*0.79 (0.78-0.79)*
Protein+Demographic+Serum	*0.88 (0.85-0.91)*	*0.89 (0.86-0.92)*	*0.88 (0.83-0.93)*	*0.80 (0.78-0.81)*
Protein+Demographic+PRS	*0.87 (0.85-0.90)*	*0.87 (0.83-0.91)*	*0.80 (0.78-0.82)*	*0.82 (0.80-0.85)*
Protein+Serum+PRS	*0.91 (0.88-0.93)*	*0.89 (0.85-0.93)*	*0.86 (0.82-0.91)*	*0.79 (0.77-0.82)*
Combined	*0.92 (0.90-0.95)*	*0.88 (0.85-0.91)*	*0.88 (0.84-0.91)*	*0.80 (0.77-0.82)*

15-year incident CD

	UKB testing set (AUC [95%CI])			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.73 (0.72-0.73)*	*0.72 (0.71-0.73)*	*0.71 (0.70-0.73)*	*0.72 (0.71-0.73)*
Demographic	*0.56 (0.55-0.56)*	*0.56 (0.55-0.57)*	*0.58 (0.58-0.58)*	*0.55 (0.55-0.56)*
Serum	*0.63 (0.62-0.64)*	*0.63 (0.61-0.64)*	*0.66 (0.64-0.67)*	*0.62 (0.59-0.65)*
PRS	*0.67 (0.66-0.67)*	*0.67 (0.66-0.68)*	*0.63 (0.61-0.65)*	*0.69 (0.69-0.70)*
Protein+Demographic	*0.72 (0.71-0.73)*	*0.72 (0.72-0.73)*	*0.70 (0.69-0.72)*	*0.73 (0.72-0.74)*
Protein+Serum	*0.72 (0.70-0.74)*	*0.72 (0.70-0.74)*	*0.73 (0.72-0.74)*	*0.76 (0.75-0.76)*
Protein+PRS	*0.75 (0.74-0.76)*	*0.74 (0.73-0.75)*	*0.70 (0.68-0.71)*	*0.75 (0.74-0.77)*
Protein+Demographic+Serum	*0.73 (0.72-0.74)*	*0.73 (0.72-0.74)*	*0.74 (0.72-0.75)*	*0.72 (0.71-0.74)*
Protein+Demographic+PRS	*0.75 (0.75-0.76)*	*0.73 (0.72-0.75)*	*0.72 (0.70-0.73)*	*0.78 (0.76-0.79)*
Protein+Serum+PRS	*0.76 (0.75-0.77)*	*0.76 (0.75-0.77)*	*0.75 (0.74-0.77)*	*0.77 (0.76-0.78)*
Combined	*0.76 (0.75-0.77)*	*0.78 (0.77-0.79)*	*0.75 (0.74-0.76)*	*0.72 (0.68-0.76)*
Training set				
9-protein panel	*0.85 (0.82-0.88)*	*0.83 (0.80-0.86)*	*0.84 (0.82-0.87)*	*0.76 (0.74-0.78)*
Demographic	*0.66 (0.66-0.67)*	*0.66 (0.66-0.67)*	*0.67 (0.67-0.68)*	*0.66 (0.65-0.67)*
Serum	*0.77 (0.75-0.78)*	*0.78 (0.75-0.80)*	*0.75 (0.74-0.75)*	*0.72 (0.65-0.78)*
PRS	*0.61 (0.61-0.61)*	*0.64 (0.62-0.65)*	*0.71 (0.69-0.72)*	*0.64 (0.62-0.66)*
Protein+Demographic	*0.82 (0.81-0.83)*	*0.82 (0.80-0.83)*	*0.82 (0.78-0.85)*	*0.77 (0.75-0.79)*
Protein+Serum	*0.90 (0.86-0.93)*	*0.88 (0.85-0.91)*	*0.93 (0.90-0.96)*	*0.79 (0.79-0.80)*
Protein+PRS	*0.88 (0.86-0.90)*	*0.88 (0.84-0.92)*	*0.85 (0.82-0.88)*	*0.79 (0.78-0.79)*
Protein+Demographic+Serum	*0.87 (0.85-0.89)*	*0.89 (0.86-0.92)*	*0.88 (0.83-0.93)*	*0.79 (0.78-0.81)*
Protein+Demographic+PRS	*0.88 (0.85-0.90)*	*0.87 (0.83-0.91)*	*0.80 (0.77-0.82)*	*0.83 (0.81-0.86)*
Protein+Serum+PRS	*0.91 (0.89-0.93)*	*0.93 (0.89-0.98)*	*0.86 (0.81-0.91)*	*0.80 (0.77-0.82)*
Combined	*0.92 (0.90-0.95)*	*0.89 (0.85-0.93)*	*0.87 (0.84-0.91)*	*0.80 (0.77-0.83)*

16-year incident CD

	UKB testing set (AUC [95%CI])			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.73 (0.72-0.73)*	*0.72 (0.71-0.73)*	*0.71 (0.70-0.73)*	*0.72 (0.71-0.73)*
Demographic	*0.56 (0.55-0.56)*	*0.56 (0.55-0.57)*	*0.58 (0.58-0.58)*	*0.55 (0.55-0.56)*

Serum	*0.63 (0.62-0.64)*	*0.63 (0.61-0.64)*	*0.66 (0.64-0.67)*	*0.62 (0.59-0.65)*
PRS	*0.67 (0.66-0.67)*	*0.67 (0.66-0.68)*	*0.63 (0.61-0.65)*	*0.69 (0.69-0.70)*
Protein+Demographic	*0.72 (0.71-0.73)*	*0.72 (0.72-0.73)*	*0.70 (0.69-0.72)*	*0.73 (0.72-0.74)*
Protein+Serum	*0.72 (0.70-0.74)*	*0.72 (0.70-0.74)*	*0.73 (0.72-0.74)*	*0.76 (0.75-0.76)*
Protein+PRS	*0.75 (0.74-0.76)*	*0.74 (0.73-0.75)*	*0.70 (0.68-0.71)*	*0.75 (0.74-0.77)*
Protein+Demographic+Serum	*0.73 (0.72-0.74)*	*0.73 (0.72-0.74)*	*0.74 (0.72-0.75)*	*0.72 (0.71-0.74)*
Protein+Demographic+PRS	*0.75 (0.75-0.76)*	*0.73 (0.72-0.75)*	*0.72 (0.70-0.73)*	*0.78 (0.76-0.79)*
Protein+Serum+PRS	*0.76 (0.75-0.77)*	*0.76 (0.75-0.77)*	*0.75 (0.74-0.77)*	*0.77 (0.76-0.78)*
Combined	*0.76 (0.75-0.77)*	*0.78 (0.77-0.79)*	*0.75 (0.74-0.76)*	*0.72 (0.68-0.76)*
Training set				
9-protein panel	*0.85 (0.82-0.88)*	*0.83 (0.80-0.86)*	*0.84 (0.82-0.87)*	*0.76 (0.74-0.78)*
Demographic	*0.66 (0.66-0.67)*	*0.66 (0.66-0.67)*	*0.67 (0.67-0.68)*	*0.66 (0.65-0.67)*
Serum	*0.77 (0.75-0.78)*	*0.78 (0.75-0.80)*	*0.75 (0.74-0.75)*	*0.72 (0.65-0.78)*
PRS	*0.61 (0.61-0.61)*	*0.64 (0.62-0.65)*	*0.71 (0.69-0.72)*	*0.64 (0.62-0.66)*
Protein+Demographic	*0.82 (0.81-0.83)*	*0.82 (0.80-0.83)*	*0.82 (0.78-0.85)*	*0.77 (0.75-0.79)*
Protein+Serum	*0.90 (0.86-0.93)*	*0.88 (0.85-0.91)*	*0.93 (0.90-0.96)*	*0.79 (0.79-0.80)*
Protein+PRS	*0.88 (0.86-0.90)*	*0.88 (0.84-0.92)*	*0.85 (0.82-0.88)*	*0.79 (0.78-0.79)*
Protein+Demographic+Serum	*0.87 (0.85-0.89)*	*0.89 (0.86-0.92)*	*0.88 (0.83-0.93)*	*0.79 (0.78-0.81)*
Protein+Demographic+PRS	*0.88 (0.85-0.90)*	*0.87 (0.83-0.91)*	*0.80 (0.77-0.82)*	*0.83 (0.81-0.86)*
Protein+Serum+PRS	*0.91 (0.89-0.93)*	*0.93 (0.89-0.98)*	*0.86 (0.81-0.91)*	*0.80 (0.77-0.82)*
Combined	*0.92 (0.90-0.95)*	*0.89 (0.85-0.93)*	*0.87 (0.84-0.91)*	*0.80 (0.77-0.83)*

We randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 3:1 roughly. To further reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. Considering the small sample size of the participants converted to Crohn's disease within the 9 years of follow-up (fewer than 90), analyses were conducted only for follow-up periods of 10 years or longer."

"Supplementary Table 11: Results of ROC analyses for all-time incident CD after matching controls.

	AUC (95%CI)			
	LGBM	XGBoost	RF	Extra trees
Testing set				
9-protein panel	0.69 (0.67-0.71)	0.71 (0.68-0.73)	0.67 (0.64-0.69)	0.64 (0.62-0.66)
Demographic	0.40 (0.37-0.44)	0.47 (0.44-0.51)	0.32 (0.32-0.32)	0.34 (0.33-0.35)
Serum	0.55 (0.52-0.57)	0.57 (0.56-0.59)	0.63 (0.61-0.65)	0.62 (0.60-0.64)
PRS	0.72 (0.72-0.72)	0.72 (0.71-0.74)	0.74 (0.71-0.76)	0.75 (0.75-0.75)
Protein+Demographic	0.70 (0.68-0.71)	0.68 (0.65-0.71)	0.69 (0.66-0.72)	0.64 (0.62-0.67)
Protein+Serum	0.69 (0.67-0.72)	0.69 (0.66-0.72)	0.72 (0.71-0.73)	0.68 (0.66-0.69)
Protein+PRS	0.78 (0.75-0.80)	0.78 (0.75-0.80)	0.71 (0.66-0.75)	0.72 (0.70-0.74)
Protein+Demographic+Serum	0.69 (0.66-0.72)	0.72 (0.70-0.74)	0.73 (0.70-0.77)	0.68 (0.65-0.70)
Protein+Demographic+PRS	0.78 (0.76-0.81)	0.76 (0.73-0.78)	0.78 (0.77-0.80)	0.74 (0.72-0.76)
Protein+Serum+PRS	0.75 (0.73-0.78)	0.73 (0.69-0.77)	0.75 (0.73-0.77)	0.75 (0.72-0.78)
Combined	0.76 (0.73-0.79)	0.76 (0.74-0.78)	0.71 (0.68-0.73)	0.70 (0.69-0.71)
Training set				
9-protein panel	0.91 (0.88-0.95)	0.92 (0.89-0.95)	0.85 (0.82-0.88)	0.80 (0.79-0.81)
Demographic	0.53 (0.52-0.54)	0.51 (0.50-0.52)	0.55 (0.55-0.55)	0.55 (0.55-0.56)
Serum	0.93 (0.89-0.96)	0.88 (0.84-0.92)	0.88 (0.82-0.94)	0.81 (0.76-0.86)
PRS	0.66 (0.65-0.67)	0.68 (0.66-0.70)	0.75 (0.72-0.77)	0.67 (0.67-0.67)
Protein+Demographic	0.88 (0.85-0.91)	0.86 (0.81-0.91)	0.86 (0.83-0.89)	0.81 (0.79-0.84)
Protein+Serum	0.89 (0.85-0.94)	0.91 (0.86-0.95)	0.88 (0.85-0.91)	0.87 (0.83-0.91)
Protein+PRS	0.92 (0.88-0.95)	0.93 (0.89-0.97)	0.85 (0.81-0.90)	0.83 (0.81-0.84)
Protein+Demographic+Serum	0.87 (0.82-0.93)	0.84 (0.82-0.87)	0.84 (0.83-0.86)	0.82 (0.79-0.84)
Protein+Demographic+PRS	0.87 (0.82-0.91)	0.89 (0.84-0.95)	0.88 (0.86-0.91)	0.87 (0.84-0.89)
Protein+Serum+PRS	0.92 (0.88-0.97)	0.86 (0.80-0.92)	0.89 (0.84-0.94)	0.87 (0.83-0.92)
Combined	0.89 (0.84-0.94)	0.89 (0.85-0.94)	0.96 (0.93-0.99)	0.88 (0.84-0.91)

We match the incident Crohn's disease and control populations by age, sex and ethnicity in a 1:1 ratio, and then randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 3:1 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 12: Comparison between AUCs for all-time incident CD after matching controls.

LGBM											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.49E-04*	*1.83E-04*	*1.70E-02*	*4.96E-01*	*4.27E-01*	*7.69E-04*	*6.78E-01*	*4.40E-04*	*4.06E-03*	*2.82E-03*
Demographic	*1.49E-04*		*2.03E-04*	*1.43E-04*	*1.49E-04*	*1.49E-04*	*1.49E-04*	*1.49E-04*	*1.49E-04*	*1.49E-04*	*1.49E-04*
Serum	*1.83E-04*	*2.03E-04*		*1.76E-04*	*1.83E-04*	*2.45E-04*	*1.83E-04*	*3.30E-04*	*1.83E-04*	*1.83E-04*	*1.82E-04*
PRS	*1.70E-02*	*1.43E-04*	*1.76E-04*		*2.54E-02*	*2.74E-03*	*2.76E-03*	*1.20E-01*	*2.76E-03*	*2.76E-03*	*6.32E-02*
Protein+ Demographic	*4.96E-01*	*1.49E-04*	*1.83E-04*	*2.54E-02*		*9.10E-01*	*1.71E-03*	*9.70E-01*	*5.83E-04*	*7.28E-03*	*5.78E-03*
Protein+ Serum	*4.27E-01*	*1.49E-04*	*2.45E-04*	*2.74E-03*	*9.10E-01*		*1.49E-03*	*7.91E-01*	*4.37E-04*	*3.60E-03*	*7.24E-03*
Protein+ PRS	*7.69E-04*	*1.49E-04*	*1.83E-04*	*2.76E-03*	*1.71E-03*	*1.49E-03*		*1.31E-03*	*9.10E-01*	*1.40E-01*	*5.71E-01*
Protein+ Demographic+ Serum	*6.78E-01*	*1.49E-04*	*3.30E-04*	*1.20E-01*	*9.70E-01*	*7.91E-01*	*1.31E-03*		*1.01E-03*	*7.28E-03*	*9.08E-03*
Protein+ Demographic+ PRS	*4.40E-04*	*1.49E-04*	*1.83E-04*	*2.76E-03*	*5.83E-04*	*4.37E-04*	*9.10E-01*	*1.01E-03*		*8.90E-02*	*5.20E-01*
Protein+ Serum+ PRS	*4.06E-03*	*1.49E-04*	*1.83E-04*	*2.76E-03*	*7.28E-03*	*3.60E-03*	*1.40E-01*	*7.28E-03*	*8.90E-02*		*3.45E-01*

Combined	2.82E-03	1.49E-04	1.82E-04	6.32E-02	5.78E-03	7.24E-03	5.71E-01	9.08E-03	5.20E-01	3.45E-01	

XGBoost											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.11E-04	1.83E-04	3.45E-01	2.41E-01	4.27E-01	2.20E-03	4.06E-01	1.40E-02	2.41E-01	7.28E-03
Demographic	1.11E-04		1.11E-04	1.11E-04	1.11E-04	1.11E-04	1.11E-04	1.11E-04	1.11E-04	1.11E-04	1.11E-04
Serum	1.83E-04	1.11E-04		1.83E-04	1.01E-03	2.46E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04	1.83E-04
PRS	3.45E-01	1.11E-04	1.83E-04		2.56E-02	6.95E-02	2.83E-03	1.00E+00	6.40E-02	3.45E-01	4.59E-03
Protein+ Demographic	2.41E-01	1.11E-04	1.01E-03	2.56E-02		8.50E-01	5.83E-04	5.39E-02	2.83E-03	4.93E-02	1.83E-04
Protein+ Serum	4.27E-01	1.11E-04	2.46E-04	6.95E-02	8.50E-01		1.01E-03	8.90E-02	5.80E-03	9.62E-02	1.31E-03
Protein+ PRS	2.20E-03	1.11E-04	1.83E-04	2.83E-03	5.83E-04	1.01E-03		7.28E-03	2.73E-01	4.93E-02	2.73E-01
Protein+ Demographic+ Serum	4.06E-01	1.11E-04	1.83E-04	1.00E+00	5.39E-02	8.90E-02	7.28E-03		8.90E-02	5.71E-01	1.13E-02
Protein+ Demographic+ PRS	1.40E-02	1.11E-04	1.83E-04	6.40E-02	2.83E-03	5.80E-03	2.73E-01	8.90E-02		3.45E-01	9.70E-01

Protein+ Serum+ PRS	2.41E-01	1.11E-04	1.83E-04	3.45E-01	4.93E-02	9.62E-02	4.93E-02	5.71E-01	3.45E-01		2.73E-01
Combined	7.28E-03	1.11E-04	1.83E-04	4.59E-03	1.83E-04	1.31E-03	2.73E-01	1.13E-02	9.70E-01	2.73E-01	

RF											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		6.34E-05	1.04E-01	8.98E-03	1.61E-01	6.46E-03	1.85E-01	1.68E-02	1.71E-04	1.69E-03	4.11E-02
Demographic	6.34E-05		6.39E-05	6.20E-05	6.25E-05	6.39E-05	6.11E-05	5.94E-05	5.89E-05	6.29E-05	6.39E-05
Serum	1.04E-01	6.39E-05		1.79E-04	4.54E-03	1.83E-04	5.34E-02	5.55E-04	1.72E-04	1.81E-04	1.71E-03
PRS	8.98E-03	6.20E-05	1.79E-04		2.54E-02	6.23E-01	3.83E-01	1.00E+00	1.66E-02	1.00E+00	2.72E-01
Protein+ Demographic	1.61E-01	6.25E-05	4.54E-03	2.54E-02		1.40E-01	3.06E-01	8.74E-02	1.69E-04	2.78E-03	2.41E-01
Protein+ Serum	6.46E-03	6.39E-05	1.83E-04	6.23E-01	1.40E-01		6.22E-01	7.33E-01	4.16E-04	4.50E-02	9.70E-01
Protein+ PRS	1.85E-01	6.11E-05	5.34E-02	3.83E-01	3.06E-01	6.22E-01		3.82E-01	8.67E-03	2.11E-01	5.70E-01
Protein+ Demographic+ Serum	1.68E-02	5.94E-05	5.55E-04	1.00E+00	8.74E-02	7.33E-01	3.82E-01		5.20E-02	7.33E-01	4.71E-01

Protein+ Demographic+ PRS	1.71E-04	5.89E-05	1.72E-04	1.66E-02	1.69E-04	4.16E-04	8.67E-03	5.20E-02		8.74E-02	5.53E-04
Protein+ Serum+ PRS	1.69E-03	6.29E-05	1.81E-04	1.00E+00	2.78E-03	4.50E-02	2.11E-01	7.33E-01	8.74E-02		2.56E-02
Combined	4.11E-02	6.39E-05	1.71E-03	2.72E-01	2.41E-01	9.70E-01	5.70E-01	4.71E-01	5.53E-04	2.56E-02	

Extra trees											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.25E-04	6.14E-02	5.51E-05	7.31E-01	6.85E-03	9.18E-04	8.60E-02	2.97E-04	5.26E-04	1.57E-03
Demographic	1.25E-04		1.36E-04	4.55E-05	1.37E-04	1.41E-04	1.40E-04	1.37E-04	1.41E-04	1.40E-04	1.40E-04
Serum	6.14E-02	1.36E-04		6.11E-05	3.82E-01	2.77E-03	7.44E-04	2.51E-02	4.27E-04	3.18E-04	2.76E-03
PRS	5.51E-05	4.55E-05	6.11E-05		6.16E-05	6.39E-05	1.71E-02	6.16E-05	1.15E-01	4.43E-01	6.34E-05
Protein+ Demographic	7.31E-01	1.37E-04	3.82E-01	6.16E-05		7.51E-02	3.54E-03	4.43E-02	5.69E-04	2.77E-03	1.67E-03
Protein+ Serum	6.85E-03	1.41E-04	2.77E-03	6.39E-05	7.51E-02		7.20E-03	7.62E-01	6.67E-04	1.72E-02	8.10E-03
Protein+ PRS	9.18E-04	1.40E-04	7.44E-04	1.71E-02	3.54E-03	7.20E-03		6.34E-02	3.45E-01	1.04E-01	5.20E-01

Protein+ Demographic+ Serum	*8.60E-02*	*1.37E-04*	*2.51E-02*	*6.16E-05*	*4.43E-02*	*7.62E-01*	*6.34E-02*		*5.70E-03*	*1.11E-02*	*3.06E-01*
Protein+ Demographic+ PRS	*2.97E-04*	*1.41E-04*	*4.27E-04*	*1.15E-01*	*5.69E-04*	*6.67E-04*	*3.45E-01*	*5.70E-03*		*2.73E-01*	*1.90E-02*
Protein+ Serum+ PRS	*5.26E-04*	*1.40E-04*	*3.18E-04*	*4.43E-01*	*2.77E-03*	*1.72E-02*	*1.04E-01*	*1.11E-02*	*2.73E-01*		*2.56E-02*
Combined	*1.57E-03*	*1.40E-04*	*2.76E-03*	*6.34E-05*	*1.67E-03*	*8.10E-03*	*5.20E-01*	*3.06E-01*	*1.90E-02*	*2.56E-02*	

Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 13: Results of ROC analyses after excluding individuals who developed Crohn's disease in the first two years of follow-up.

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.68 (0.67-0.70)*	*0.70 (0.69-0.71)*	*0.66 (0.64-0.67)*	*0.70 (0.69-0.71)*
Demographic	*0.56 (0.55-0.57)*	*0.56 (0.55-0.57)*	*0.55 (0.55-0.55)*	*0.60 (0.60-0.60)*
Serum	*0.59 (0.57-0.61)*	*0.62 (0.60-0.65)*	*0.62 (0.60-0.63)*	*0.67 (0.65-0.68)*
PRS	*0.64 (0.62-0.65)*	*0.61 (0.60-0.62)*	*0.66 (0.65-0.66)*	*0.66 (0.66-0.66)*
Protein+Demographic	*0.70 (0.69-0.71)*	*0.70 (0.69-0.71)*	*0.71 (0.70-0.72)*	*0.72 (0.70-0.73)*
Protein+Serum	*0.69 (0.68-0.71)*	*0.72 (0.71-0.73)*	*0.71 (0.69-0.73)*	*0.74 (0.73-0.75)*
Protein+PRS	*0.67 (0.65-0.69)*	*0.69 (0.67-0.71)*	*0.71 (0.70-0.73)*	*0.71 (0.70-0.73)*
Protein+Demographic+Serum	*0.71 (0.70-0.72)*	*0.71 (0.69-0.73)*	*0.73 (0.71-0.74)*	*0.72 (0.70-0.74)*
Protein+Demographic+PRS	*0.70 (0.69-0.72)*	*0.71 (0.70-0.73)*	*0.73 (0.72-0.74)*	*0.76 (0.75-0.77)*
Protein+Serum+PRS	*0.73 (0.71-0.75)*	*0.71 (0.69-0.74)*	*0.75 (0.74-0.76)*	*0.73 (0.72-0.73)*
Combined	*0.73 (0.72-0.74)*	*0.73 (0.71-0.74)*	*0.70 (0.67-0.73)*	*0.74 (0.71-0.77)*
Training set				
9-protein panel	*0.92 (0.89-0.95)*	*0.92 (0.88-0.96)*	*0.78 (0.75-0.81)*	*0.77 (0.75-0.79)*
Demographic	*0.63 (0.63-0.63)*	*0.63 (0.63-0.63)*	*0.63 (0.63-0.63)*	*0.62 (0.62-0.62)*
Serum	*0.86 (0.80-0.93)*	*0.88 (0.80-0.96)*	*0.81 (0.74-0.89)*	*0.70 (0.69-0.70)*
PRS	*0.62 (0.59-0.64)*	*0.63 (0.61-0.65)*	*0.67 (0.65-0.68)*	*0.60 (0.60-0.60)*
Protein+Demographic	*0.90 (0.87-0.92)*	*0.85 (0.83-0.87)*	*0.85 (0.80-0.89)*	*0.82 (0.76-0.87)*
Protein+Serum	*0.98 (0.96-1.00)*	*0.96 (0.94-0.99)*	*0.95 (0.92-0.98)*	*0.87 (0.82-0.91)*
Protein+PRS	*0.97 (0.94-1.00)*	*0.95 (0.91-0.99)*	*0.95 (0.91-0.99)*	*0.82 (0.78-0.87)*
Protein+Demographic+Serum	*0.98 (0.97-0.99)*	*0.94 (0.91-0.98)*	*0.84 (0.79-0.88)*	*0.81 (0.76-0.86)*
Protein+Demographic+PRS	*0.96 (0.92-0.99)*	*0.94 (0.90-0.97)*	*0.90 (0.86-0.94)*	*0.81 (0.79-0.84)*
Protein+Serum+PRS	*0.98 (0.96-1.00)*	*0.98 (0.96-0.99)*	*0.94 (0.91-0.98)*	*0.79 (0.78-0.79)*
Combined	*0.99 (0.98-1.00)*	*0.97 (0.95-0.98)*	*0.87 (0.83-0.90)*	*0.80 (0.78-0.82)*

After excluding 19 individuals who developed Crohn's disease within the first two years of follow-up, we randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 3:1 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. "

“Supplementary Table 14: Comparison between AUCs for all-year incident CD after excluding individuals who developed Crohn’s disease in the first two year of follow-up.

LGBM											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.77E-04*	*1.83E-04*	*7.00E-04*	*3.12E-02*	*3.85E-01*	*3.45E-01*	*2.57E-02*	*3.12E-02*	*1.31E-03*	*1.01E-03*
Demographic	*1.77E-04*		*2.08E-02*	*1.57E-04*	*1.77E-04*	*1.77E-04*	*1.77E-04*	*1.77E-04*	*1.77E-04*	*1.77E-04*	*1.77E-04*
Serum	*1.83E-04*	*2.08E-02*		*2.62E-03*	*1.83E-04*	*1.83E-04*	*4.40E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*	*1.83E-04*
PRS	*7.00E-04*	*1.57E-04*	*2.62E-03*		*2.21E-04*	*5.29E-04*	*4.35E-02*	*2.21E-04*	*1.63E-04*	*1.63E-04*	*1.63E-04*
Protein+ Demographic	*3.12E-02*	*1.77E-04*	*1.83E-04*	*2.21E-04*		*4.73E-01*	*1.04E-01*	*4.27E-01*	*9.10E-01*	*9.11E-03*	*4.59E-03*
Protein+ Serum	*3.85E-01*	*1.77E-04*	*1.83E-04*	*5.29E-04*	*4.73E-01*		*1.21E-01*	*3.07E-01*	*5.21E-01*	*1.73E-02*	*4.59E-03*
Protein+ PRS	*3.45E-01*	*1.77E-04*	*4.40E-04*	*4.35E-02*	*1.04E-01*	*1.21E-01*		*4.52E-02*	*1.04E-01*	*3.61E-03*	*1.31E-03*
Protein+ Demographic+ Serum	*2.57E-02*	*1.77E-04*	*1.83E-04*	*2.21E-04*	*4.27E-01*	*3.07E-01*	*4.52E-02*		*4.27E-01*	*3.76E-02*	*2.11E-02*
Protein+ Demographic+ PRS	*3.12E-02*	*1.77E-04*	*1.83E-04*	*1.63E-04*	*9.10E-01*	*5.21E-01*	*1.04E-01*	*4.27E-01*		*3.76E-02*	*2.11E-02*
Protein+ Serum+ PRS	*1.31E-03*	*1.77E-04*	*1.83E-04*	*1.63E-04*	*9.11E-03*	*1.73E-02*	*3.61E-03*	*3.76E-02*	*3.76E-02*		*9.70E-01*

Combined	1.01E-03	1.77E-04	1.83E-04	1.63E-04	4.59E-03	4.59E-03	1.31E-03	2.11E-02	2.11E-02	9.70E-01	

XGBoost											
	9-protein panel	Demographic	Serum	PRS	Protein+ Demographic	Protein+ Serum	Protein+ PRS	Protein+ Demographic+ Serum	Protein+ Demographic+ PRS	Protein+ Serum+ PRS	Combined
9-protein panel		1.73E-04	4.40E-04	1.32E-04	5.21E-01	1.73E-02	7.34E-01	1.04E-01	1.21E-01	2.41E-01	7.28E-03
Demographic	1.73E-04		3.13E-04	1.69E-04	1.73E-04	1.73E-04	1.73E-04	1.73E-04	1.73E-04	1.73E-04	1.73E-04
Serum	4.40E-04	3.13E-04		1.13E-01	1.83E-04	1.83E-04	1.01E-03	3.30E-04	3.30E-04	1.31E-03	1.83E-04
PRS	1.32E-04	1.69E-04	1.13E-01		1.32E-04	1.32E-04	1.32E-04	1.32E-04	1.32E-04	1.32E-04	1.32E-04
Protein+ Demographic	5.21E-01	1.73E-04	1.83E-04	1.32E-04		5.80E-03	6.23E-01	6.40E-02	8.90E-02	2.41E-01	3.61E-03
Protein+ Serum	1.73E-02	1.73E-04	1.83E-04	1.32E-04	5.80E-03		1.62E-01	4.27E-01	8.50E-01	7.91E-01	3.85E-01
Protein+ PRS	7.34E-01	1.73E-04	1.01E-03	1.32E-04	6.23E-01	1.62E-01		4.27E-01	3.45E-01	1.40E-01	4.52E-02
Protein+ Demographic+ Serum	1.04E-01	1.73E-04	3.30E-04	1.32E-04	6.40E-02	4.27E-01	4.27E-01		7.91E-01	6.78E-01	1.86E-01
Protein+ Demographic+ PRS	1.21E-01	1.73E-04	3.30E-04	1.32E-04	8.90E-02	8.50E-01	3.45E-01	7.91E-01		6.78E-01	3.45E-01

Protein+ Serum+ PRS	*2.41E-01*	*1.73E-04*	*1.31E-03*	*1.32E-04*	*2.41E-01*	*7.91E-01*	*1.40E-01*	*6.78E-01*	*6.78E-01*		*6.78E-01*
Combined	*7.28E-03*	*1.73E-04*	*1.83E-04*	*1.32E-04*	*3.61E-03*	*3.85E-01*	*4.52E-02*	*1.86E-01*	*3.45E-01*	*6.78E-01*	

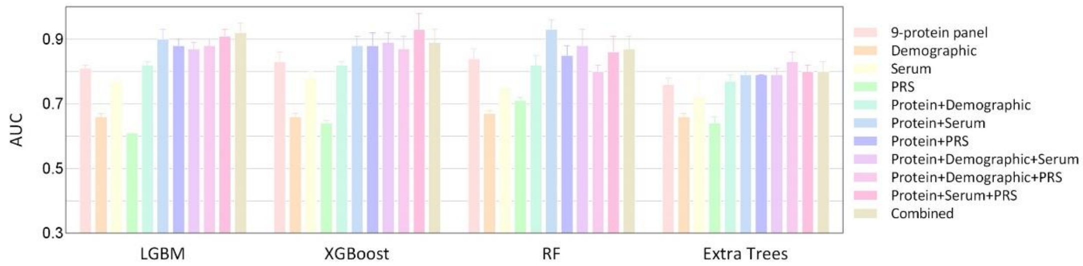
RF											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*4.17E-05*	*1.32E-04*	*1.97E-01*	*4.39E-04*	*5.88E-04*	*4.41E-04*	*3.25E-04*	*2.42E-04*	*1.80E-04*	*3.75E-03*
Demographic	*4.17E-05*		*6.39E-05*	*5.11E-05*	*6.34E-05*	*6.39E-05*	*6.39E-05*	*6.29E-05*	*6.34E-05*	*6.39E-05*	*6.29E-05*
Serum	*1.32E-04*	*6.39E-05*		*2.09E-04*	*1.82E-04*	*1.83E-04*	*1.83E-04*	*1.81E-04*	*1.82E-04*	*1.83E-04*	*1.81E-04*
PRS	*1.97E-01*	*5.11E-05*	*2.09E-04*		*1.53E-04*	*1.54E-04*	*5.03E-04*	*1.52E-04*	*1.53E-04*	*1.54E-04*	*1.36E-01*
Protein+ Demographic	*4.39E-04*	*6.34E-05*	*1.82E-04*	*1.53E-04*		*5.71E-01*	*1.62E-01*	*2.72E-01*	*7.24E-03*	*1.70E-03*	*1.00E+00*
Protein+ Serum	*5.88E-04*	*6.39E-05*	*1.83E-04*	*1.54E-04*	*5.71E-01*		*9.10E-01*	*2.73E-01*	*3.76E-02*	*2.20E-03*	*8.50E-01*
Protein+ PRS	*4.41E-04*	*6.39E-05*	*1.83E-04*	*5.03E-04*	*1.62E-01*	*9.10E-01*		*6.23E-01*	*3.76E-02*	*2.20E-03*	*1.00E+00*
Protein+ Demographic+ Serum	*3.25E-04*	*6.29E-05*	*1.81E-04*	*1.52E-04*	*2.72E-01*	*2.73E-01*	*6.23E-01*		*6.23E-01*	*8.87E-02*	*2.72E-01*

Protein+ Demographic+ PRS	*2.42E-04*	*6.34E-05*	*1.82E-04*	*1.53E-04*	*7.24E-03*	*3.76E-02*	*3.76E-02*	*6.23E-01*		*5.38E-02*	*3.84E-01*
Protein+ Serum+ PRS	*1.80E-04*	*6.39E-05*	*1.83E-04*	*1.54E-04*	*1.70E-03*	*2.20E-03*	*2.20E-03*	*8.87E-02*	*5.38E-02*		*1.72E-02*
Combined	*3.75E-03*	*6.29E-05*	*1.81E-04*	*1.36E-01*	*1.00E+00*	*8.50E-01*	*1.00E+00*	*2.72E-01*	*3.84E-01*	*1.72E-02*	

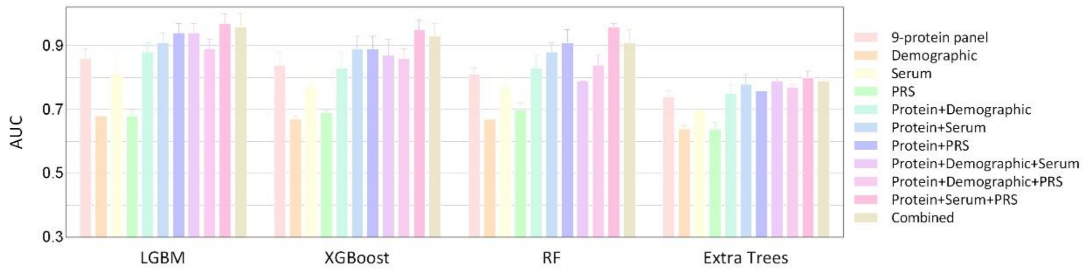
Extra trees											
	9-protein panel	*Demographic*	*Serum*	*PRS*	*Protein+ Demographic*	*Protein+ Serum*	*Protein+ PRS*	*Protein+ Demographic+ Serum*	*Protein+ Demographic+ PRS*	*Protein+ Serum+ PRS*	*Combined*
9-protein panel		*1.76E-04*	*5.76E-03*	*6.34E-05*	*2.73E-01*	*3.28E-04*	*1.62E-01*	*8.87E-02*	*1.82E-04*	*4.05E-03*	*5.76E-03*
Demographic	*1.76E-04*		*1.76E-04*	*6.11E-05*	*1.77E-04*	*1.77E-04*	*1.76E-04*	*1.76E-04*	*1.77E-04*	*1.44E-04*	*1.76E-04*
Serum	*5.76E-03*	*1.76E-04*		*1.15E-01*	*1.82E-04*	*1.82E-04*	*1.69E-03*	*2.81E-03*	*1.82E-04*	*1.49E-04*	*2.19E-03*
PRS	*6.34E-05*	*6.11E-05*	*1.15E-01*		*6.39E-05*	*6.39E-05*	*1.41E-03*	*6.34E-05*	*6.39E-05*	*4.92E-05*	*1.41E-03*
Protein+ Demographic	*2.73E-01*	*1.77E-04*	*1.82E-04*	*6.39E-05*		*1.13E-02*	*5.71E-01*	*4.27E-01*	*1.71E-03*	*2.90E-02*	*2.57E-02*
Protein+ Serum	*3.28E-04*	*1.77E-04*	*1.82E-04*	*6.39E-05*	*1.13E-02*		*3.60E-03*	*1.40E-01*	*2.20E-03*	*2.38E-02*	*6.23E-01*
Protein+ PRS	*1.62E-01*	*1.76E-04*	*1.69E-03*	*1.41E-03*	*5.71E-01*	*3.60E-03*		*2.73E-01*	*2.45E-04*	*5.07E-02*	*2.56E-02*

Protein+ Demographic+ Serum	*8.87E-02*	*1.76E-04*	*2.81E-03*	*6.34E-05*	*4.27E-01*	*1.40E-01*	*2.73E-01*		*7.65E-04*	*9.08E-01*	*8.87E-02*
Protein+ Demographic+ PRS	*1.82E-04*	*1.77E-04*	*1.82E-04*	*6.39E-05*	*1.71E-03*	*2.20E-03*	*2.45E-04*	*7.65E-04*		*4.91E-04*	*2.12E-01*
Protein+ Serum+ PRS	*4.05E-03*	*1.44E-04*	*1.49E-04*	*4.92E-05*	*2.90E-02*	*2.38E-02*	*5.07E-02*	*9.08E-01*	*4.91E-04*		*3.01E-01*
Combined	*5.76E-03*	*1.76E-04*	*2.19E-03*	*1.41E-03*	*2.57E-02*	*6.23E-01*	*2.56E-02*	*8.87E-02*	*2.12E-01*	*3.01E-01*	

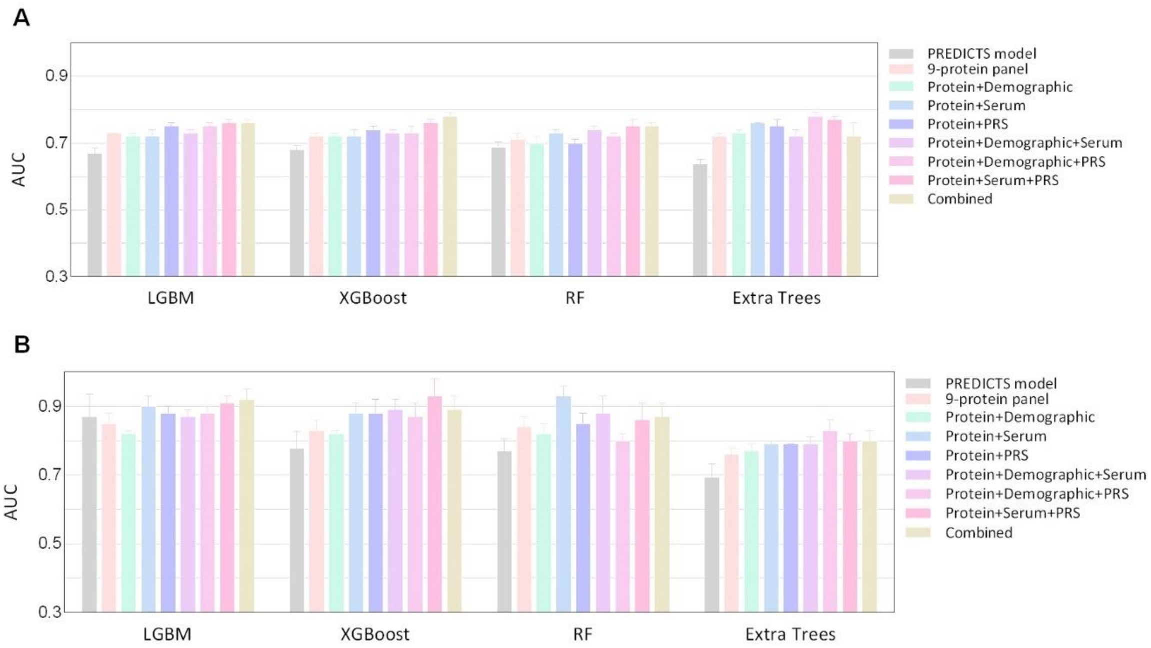
Mann–Whitney U tests were conducted to compare the AUCs between each pair of models in the testing set and the resulting p-values are presented in the table. P-values were calculated under two-sided tests. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."



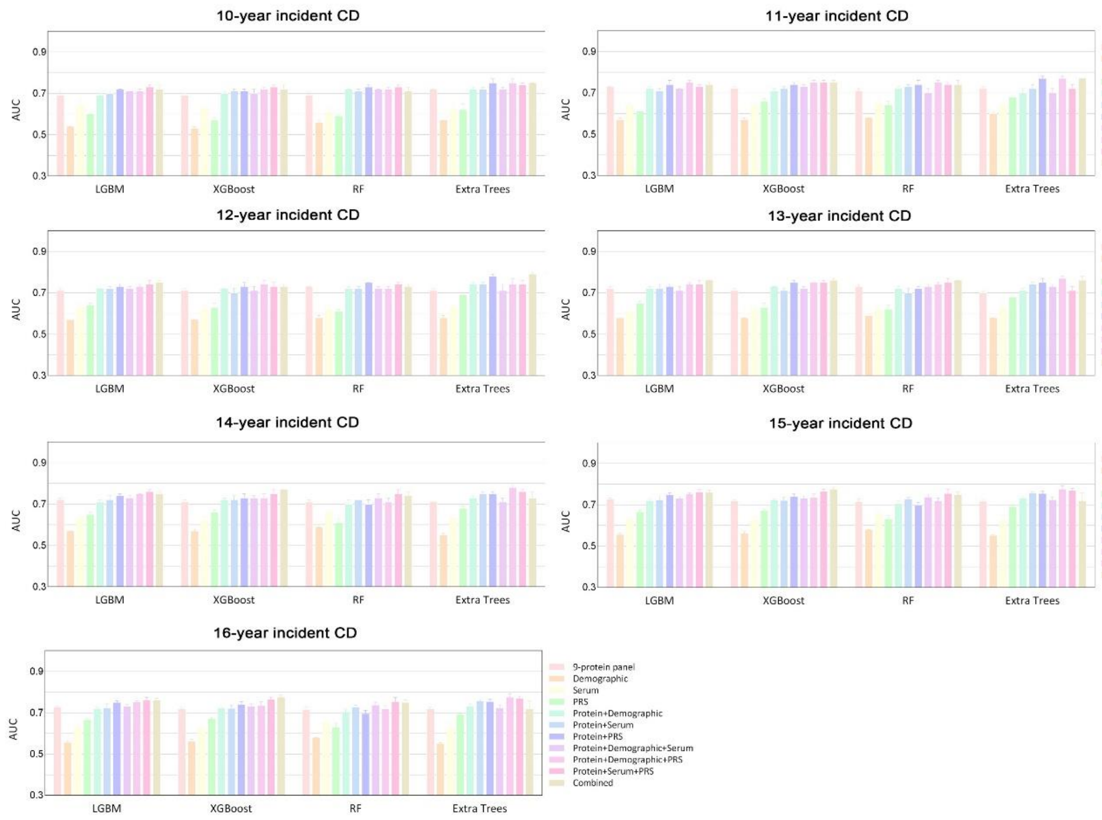
"Supplementary Fig. 2: Predictive accuracy of plasma proteins panel, alone or in combination with clinical variables in the training set with a 75% proportion. Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease in training set with a 75% proportion, evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."



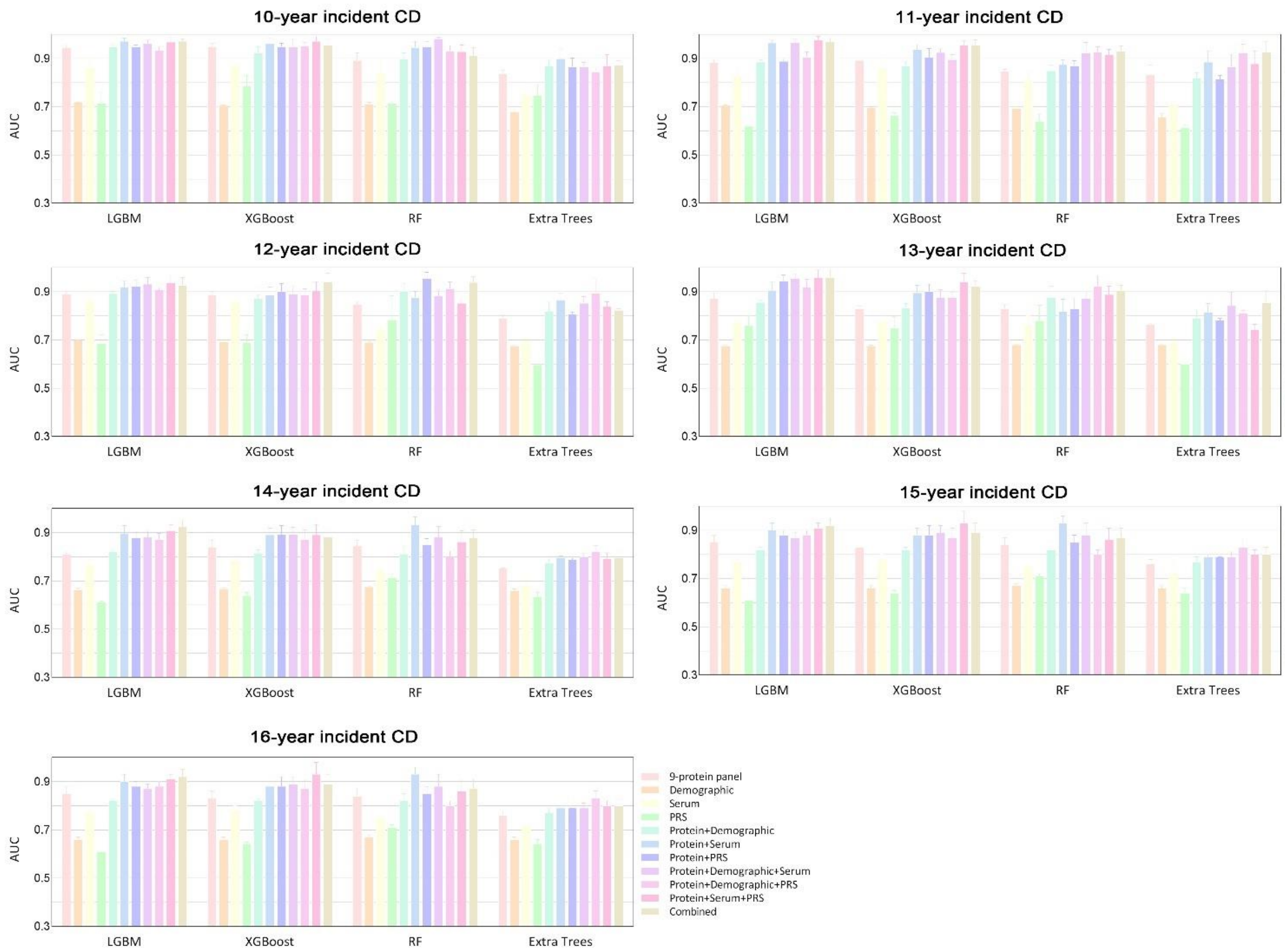
"Supplementary Fig. 3: Predictive accuracy of plasma proteins panel, alone or in combination with clinical variables in the training set with an 80% proportion. Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease in training set with a 80% proportion, evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."



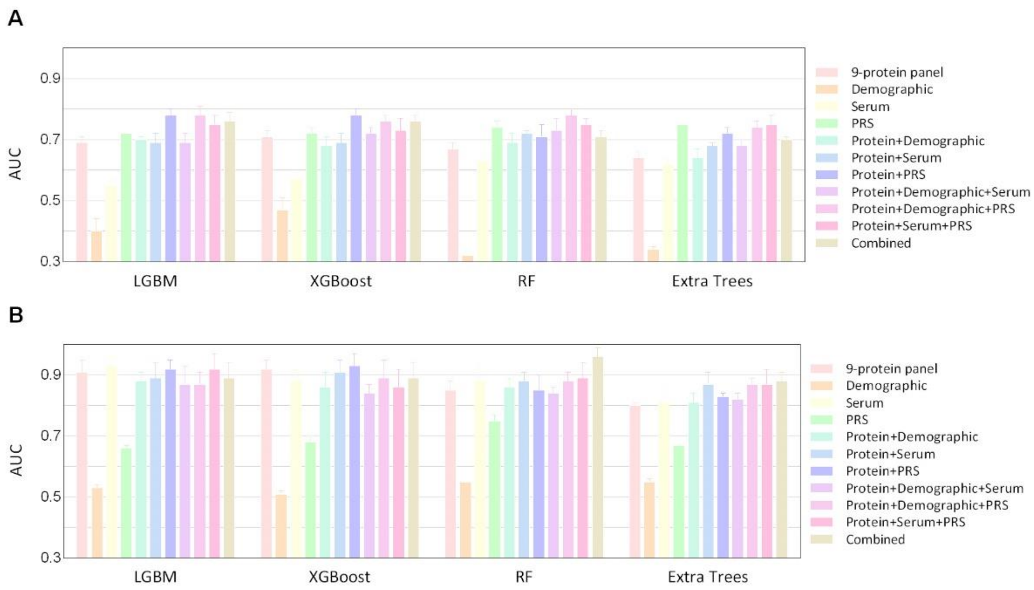
"Supplementary Fig. 4: Predictive accuracy of the PREDICTS model in UK Biobank for predicting all-time incident Crohn's disease. Bar plots show the area under the receiver operating characteristic curve (AUC) of PREDICTS and our protein-based models for predicting all-time incident Crohn's disease in the testing set (A) and training set (B), evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."



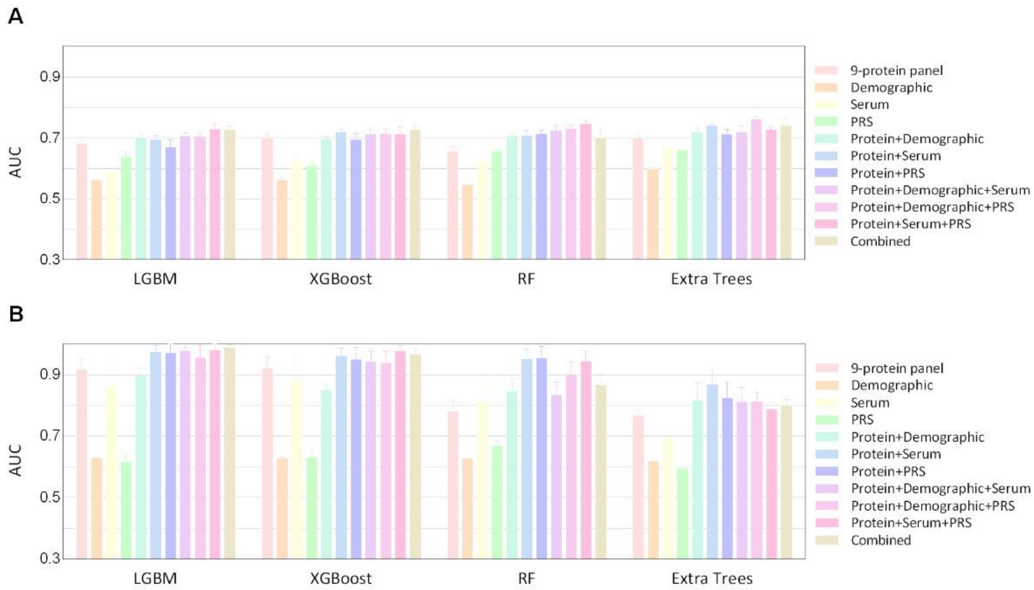
"**Supplementary Fig. 5: Predictive accuracy of plasma proteins panel within different time points in the UKB testing set.** Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models within different time points from 10 to 16 years of incident Crohn's disease in the UKB testing set, evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. Considering the small sample size of the participants converted to Crohn's disease within the 9 years of follow-up (fewer than 90), analyses were conducted only for follow-up periods of 10 years or longer."



"**Supplementary Fig. 6: Predictive accuracy of plasma proteins panel within different time points in the training set.** Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models within different time points from 10 to 16 years of incident Crohn's disease in the training set, evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease. Considering the small sample size of the participants converted to Crohn's disease within the 9 years of follow-up (fewer than 90), analyses were conducted only for follow-up periods of 10 years or longer."



"Supplementary Fig. 7: Predictive accuracy of plasma proteins panel, alone or in combination with matched populations. Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease with matched populations in the UKB testing set (A) and training set (B), evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."



"Supplementary Fig. 8: Predictive accuracy after excluding individuals who developed Crohn's disease in the first two years of follow-up. Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease in the UKB testing set (A) and training set (B), evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees (ET). Nineteen individuals were excluded for developing Crohn's disease within the first two years. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

6) Lack of external validation: The absence of external validation significantly impacts the generalisability and clinical relevance of the findings. The authors could consider testing their signature panel in an external dataset, such as the PREDICT study (Torres, 2020). Notably, the PREDICT study reported an AUROC of 0.76, comparable to the findings here. However, upon validation within the UK Biobank samples, the AUROC in this study was observed to decrease, highlighting the importance of independent validation to confirm model robustness and applicability.

Authors' response:

We sincerely appreciate the reviewer's critical and constructive feedback regarding the external validation of our findings. We fully acknowledge that independent validation is essential for ensuring the generalizability and clinical relevance of predictive models. In response to this concern, we have undertaken extensive efforts to validate our model in external datasets and strengthen our validation framework. Below, we detail the steps taken:

In alignment with your suggestion, we proactively sought to validate our model in external cohorts, including:

- **PREDICTS Study**: We contacted the corresponding author, *Prof. Jean-Frédéric Colombel*, to explore the possibility of accessing proteomic data from the PREDICTS study. However, we were informed that access to proteomic data and biological samples is restricted due to Department of Defense Repository governance regulations.

- **CCC-GEM Cohort**: We contacted the Crohn’s and Colitis Canada Genetics, Environment, and Microbial (CCC-GEM) Project team but did not receive a response despite multiple attempts.
- **Public Biobanks (e.g. Lifelines, All of Us, CKB)**: Most large-scale biobanks do not currently have high-throughput proteomic data available for Crohn’s disease. While the China Kadoorie Biobank (CKB) includes ~4,000 plasma proteomic samples, fewer than five incident Crohn’s disease cases were identified in the available dataset, making meaningful validation infeasible.

Given the challenges in accessing large-scale CD-related high-throughput proteomics data, we adopted two alternative strategies to further assess the generalizability and robustness of our model:

- **Geographically Distinct Validation within the UK Biobank:** We performed testing by splitting UK Biobank participants based on recruitment centers, following the Type 2b design from the TRIPOD statement[1]. Compared to random splitting (Type 2a), this approach leverages meaningful geographic divisions, providing a more realistic assessment of model performance and enhancing robustness by accounting for site-specific variations. When we compared the training and testing sets stratified by geographic regions, we observed significant differences in key population characteristics, including age, ethnicity, Townsend Deprivation Index, BMI, alcohol consumption, and dietary habits, highlighting the demographic and lifestyle heterogeneity across recruitment centers; nevertheless, our model maintained stable performance despite these variations, further supporting its robustness and generalizability.
- **External Validation in a Chinese Cohort:** We actively established an independent clinical cohort at ×× in southern China. Blood samples were collected from Crohn’s disease patients and matched healthy controls in a cross-sectional study design. We then validated our protein signature using ELISA.

These validation strategies provide strong supporting evidence for the generalizability and robustness of our proteomic model across geographically and ethnically distinct populations. Specifically, in the geographically distinct UKB validation cohort, our best proteomic model (AUC = 0.76) outperformed models based solely on demographics, serum markers, or genetic data (highest AUC = 0.67). Notably, external validation in the Southern China cohort demonstrated an excellent performance (AUC = 0.79), further supporting its potential clinical applicability. We believe that the revised approach provides a more rigorous and scientifically sound evaluation of our model’s generalizability. The main modifications can be found in the following sections: **Abstract, 2.1 Study population, 2.5 Predictive accuracy of proteomics-based models, 2.6 Results of replication validation analyses, 3. Discussion, 4.1 Studying population, 4.2 Plasma proteomics, 4.4 Definition of CD, 4.5 Statistical analysis, Fig. 1, Fig. 3-4, Table 1, Table 2, Supplementary Table 3,6,18-19, and 22.**

Once again, we sincerely thank the reviewer for proposing a series of valuable suggestions to enhance the performance, robustness, and generalizability of our model, which have significantly contributed to improving the quality of our manuscript.

*“**Abstract** …In this study, we performed feature selection and model development in a training cohort of 39,634 participants without baseline CD from the UK Biobank (UKB), with internal validation through five-fold cross-validation (10 repeats). Independent validation was conducted in a geographically distinct UKB cohort (n = 13,262) and an external Southern China cohort (n =74)…Machine learning models using four algorithms were constructed based on these proteins, with XGBoost demonstrating the best performance. In the geographically distinct UKB cohort, the protein model (area under the curve [AUC] 0.76) outperformed clinical risk models (highest AUC 0.67; P < 0.05) and was externally validated in the Southern China cohort*

(AUC 0.79). Combining these proteins with clinical data enabled desirable predictions up to 16 years pre-diagnosis (AUC 0.78)…"

"Results 2.1 Study population *After excluding participants diagnosed with CD at baseline and those with missing proteomic data, a total of 52,896 individuals from the UKB were finally included (Fig. 1). Of these, 39,634 participants were assigned to the UKB training cohort, while 13,262 participants from geographically distinct recruitment centers were assigned to the UKB testing cohort for validation. Table 1 presents the baseline characteristics of the training (n = 39,634) and testing (n = 13,262) cohorts. We observed significant differences in key population characteristics, including age, ethnicity, Townsend deprivation index (TDI), BMI, alcohol consumption, and dietary habits (P<0.05), highlighting the demographic and lifestyle heterogeneity between training and testing cohorts. Over a median follow-up of 13.6 years (interquartile range [IQR]: 12.9-14.4; maximum: 16.6 years), 139 (0.26%) incident CD cases were identified. The follow-up duration from baseline to CD diagnosis ranged from 0.12 to 14.88 years (median: 7.84 years; IQR: 4.37-10.74; mean: 7.47 years) (Supplementary Table 1). Details of the validation cohort in China are shown in Table 2. In this validation cohort, 37 of 74 participants were diagnosed with CD and 46% were male, and the overall average age was 44 years old. Compared to participants without CD, those with CD were more frequently younger (P < 0.001).*
2.5 Predictive accuracy of proteomics-based models …*In the UKB testing set (geographically distinct), the area under the curves (AUCs) ranged from 0.71 to 0.73 for a 75/25 training/testing split and 0.71 to 0.77 for an 80/20 split, while in the external Southern China testing set, they ranged from 0.76 to 0.79. Notably, the XGBoost model obtained the optimal AUC in the external testing set (0.79, 95% CI 0.77–0.81) and the second-best AUC in the UKB testing set (0.72 for a 75/25 split, 95% CI 0.71–0.73; 0.76 for an 80/20 split, 95% CI 0.74–0.77), indicating its robust generalizability across independent cohorts. In the UKB testing set, the 9-protein model demonstrated superior predictive performance across four algorithms compared to clinical risk models based on demographics, serum markers, and polygenic risk score (PRS) for CD (highest AUC among clinical risk models for XGBoost: 0.67, Mann–Whitney U test: P < 0.001). For XGBoost, combining the PRS of CD with the protein panel significantly improved the AUC to 0.74 (95% CI 0.73–0.75), and adding all clinical risk factors further increased the AUC to 0.78 (95% CI 0.77–0.79) for a 75/25 split...*
2.6 Results of replication validation analyses *Our findings remained consistent in the sensitivity analyses for XGBoost. At different time points, the protein model still showed higher predictive value (AUC 0.69-0.72) than clinical risk models (AUC 0.53-0.67) (**Supplementary Table 10** and **Supplementary Fig. 5-6**). When randomly subsampling a control group matched 1:1 with CD patients by age, sex, and race, the 9-protein model achieved an AUC of 0.71 (95% CI 0.68–0.73) in predicting all-time incident CD, increasing to 0.76 (95% CI 0.74–0.78) with the addition of all clinical risk factors (**Supplementary Tables 11-12** and **Supplementary Fig. 7**). The results remained consistent after excluding individuals who developed CD within the first two years of follow-up (**Supplementary Tables 13–14** and **Supplementary Fig. 8**)..."*

"3. Discussion …*The proteomic model (AUC 0.76) significantly outperformed clinical risk models based on demographics, serum biomarkers, and genetics (highest AUC 0.67) in the UKB testing set (geographically distinct). It was further externally validated in a Southern China cohort (AUC 0.79). Combining proteins with clinical risk data enabled better predictions up to 16 years pre-diagnosis (AUC 0.78)…*

…A simple model based on the nine most important proteins achieved great long-term predictive performance (0.72 for a 75/25 training/testing split; 0.76 for an 80/20 split) in the UKB testing set (geographically distinct), outperforming models based on demographic, serologic, and PRS data (highest AUC 0.67), and was externally validated in a Southern China cohort (AUC 0.79). When the protein panel was combined with clinical risk factors, the model's predictive performance was further enhanced, with AUCs of 0.78 in predicting all-time incident CD...

…Remarkably, the robustness of our protein model was validated in a geographically distinct UKB cohort and an external clinical cohort from Southern China.…Forth, most UKB participants are of white European descent. Although our protein model was externally validated in the Southern China cohort and demonstrated consistent performance across extensive cross-validation and replication analyses, further validation in large-scale, prospective cohorts with long-term follow-up is warranted…"

"Method 4.1 Studying population…*The 52,896 participants were randomly divided into training and testing cohorts according to the UKB recruitment centers in a ratio of 75%:25% roughly. The UKB training cohort included participants from Cardiff, Glasgow, Stoke, Reading, Bury, Newcastle, Leeds, Nottingham, Sheffield, Liverpool, Middlesborough, Hounslow, and Croydon. The UKB testing cohort included participants from Stockport (pilot), Manchester, Oxford, Edinburgh, Bristol, Barts, Birmingham, Swansea, and Wrexham* ***(Supplementary Table 18).***

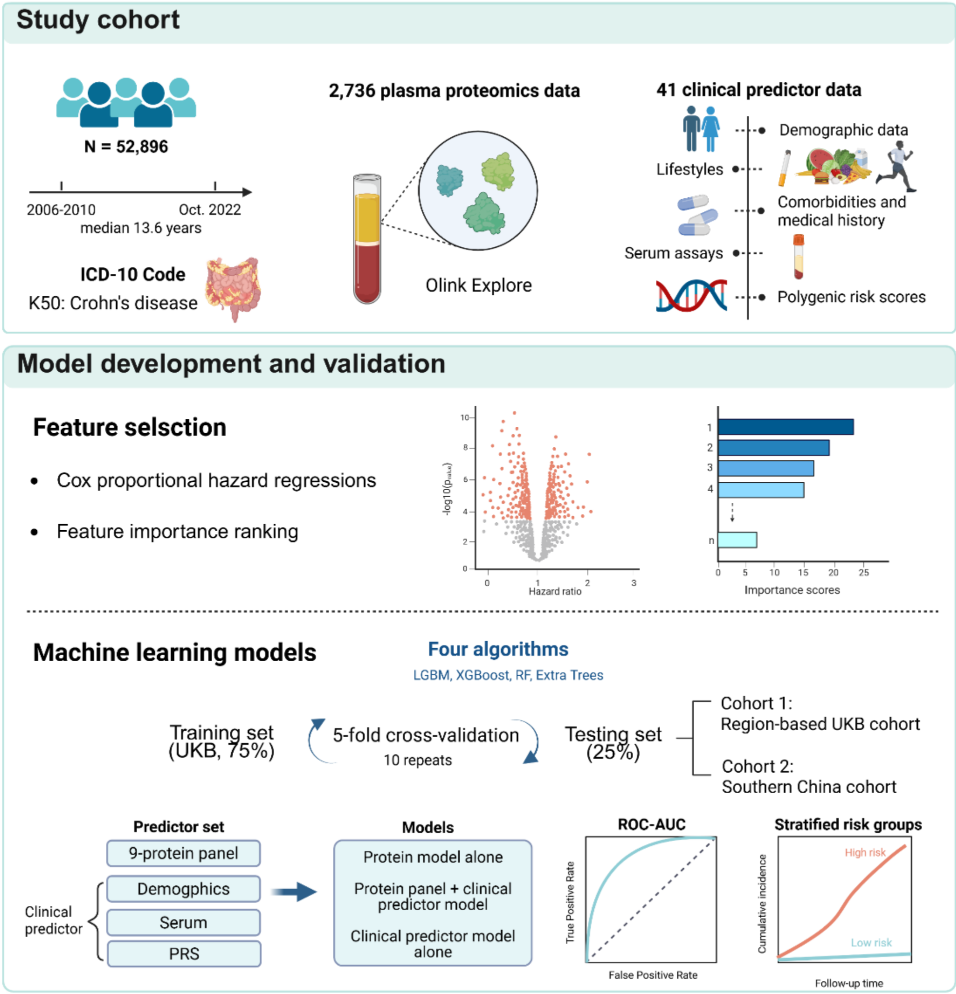
The external validation cohort included control (n = 37) and CD participants (n = 37) from ×× in southern China between 2024 and 2025. Individuals under 16 years of age and those who did not provide consent were excluded. The participants were recruited by two clinicians with over 10 years of experience. Ultimately, 74 participants were enrolled in the study. Population characteristics are described in ***Table 2.***

4.2 Plasma proteomics …*To evaluate the reproducibility of the protein model developed in the training set, Enzyme-linked immunosorbent assay (ELISA) analyses were conducted in the independent Southern China cohort. Blood samples from this cohort were collected at baseline. All blood samples were collected using EDTA tubes, and centrifuged at 1000g for 15 min, and the plasma supernatants were aliquoted and stored at −80°C. After the outcome of the patients in the cohort was determined through follow-up, plasma samples of the case and control groups were used for ELISA analyses within 3 months after collection. ELISA assays were performed per the manufacturer's protocol. The resulting protein data were log2-transformed and standardized. A list of the ELISA kits used is provided in* ***Supplementary Table 19****…*

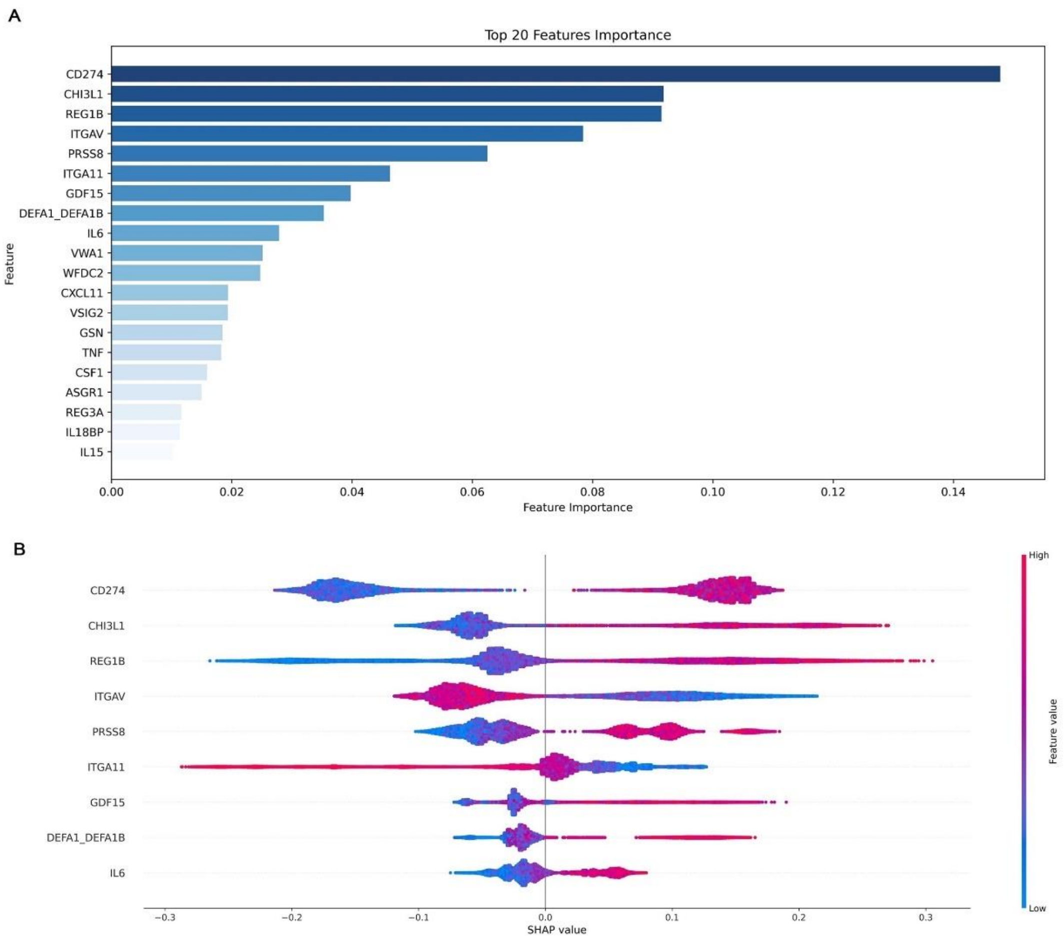
4.4 Definition of CD…*For the Southern China cohort, all patients with CD were clinically diagnosed by physicians who were blinded to the model predictions.*

4.5 Statistical analysis…*We split the participants into a training and a testing set according to the recruitment centers…. Receiver operating characteristic (ROC) area under the curve (AUC) analyses were calculated to assess the predictive accuracy of the model in predicting future CD using both the UKB testing set (geographically distinct) and the Southern China external validation cohort.*

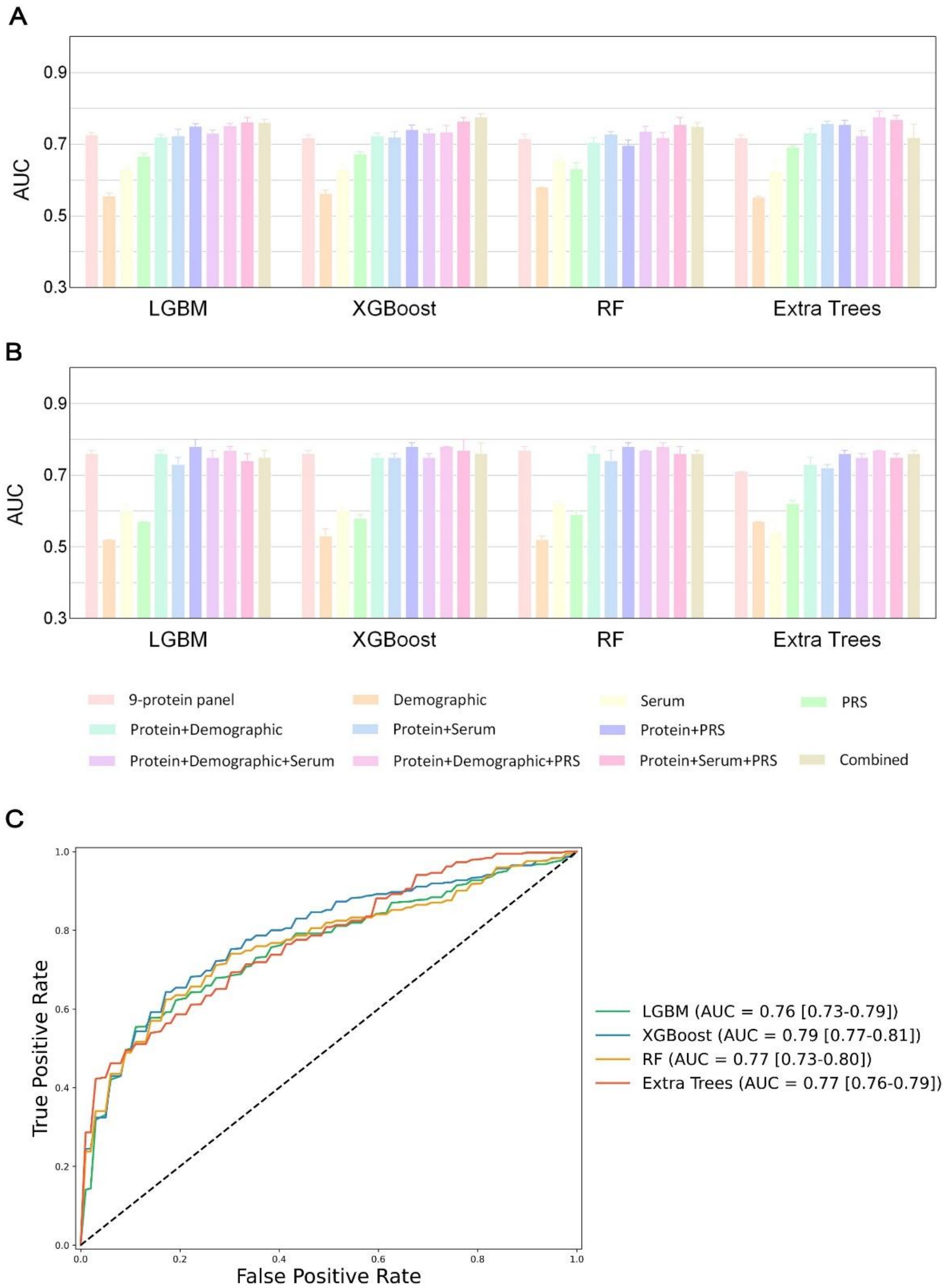
…(1) randomly splitting the studied population into 80% training (42,264 samples) and 20% (10,632 samples) testing set according to the recruitment centers ***(Supplementary Table 22)*** *…"*



*"**Fig. 1 Study overview.** First, we extracted data from 52,896 UK Biobank participants with a median follow-up time of 13.6 years, including Crohn's disease endpoint defined by ICD10 codes, 2,736 plasma proteomics, and 41 clinical predictors spanning demographic, lifestyles, comorbidity and medication history, serum assays, and polygenic risk scores for Crohn's disease. Next, Cox proportional hazard models and machine learning-based feature importance ranking were performed for further feature selection. Model development proceeded in three phases: 10×5-fold cross-validation on the training set, independent validation in a geographically distinct UK Biobank cohort, and external validation in the Southern China cohort. Finally, we investigated the predictive performance of the protein model and its risk stratification for Crohn's disease onset."*



*"**Fig. 3 Protein importance ranking and SHAP visualization of modeling on all-time incident Crohn's disease populations. A**, The top 20 proteins according to the average absolute SHAP value. The bar chart indicates the importance of the sorted proteins based on their contributions to the prediction of future Crohn's disease. **B**, SHAP visualization plot of selected proteins. The width of the range of the horizontal bars can be understood as the extent of the contribution to the prediction of Crohn's disease; the wider their range, the greater the contribution. The color of the horizontal bars denotes the magnitude of plasma proteins, which was coded in a gradient from blue (low) to red (high), shown as the color bar on the right-hand side. The direction on the x axis indicates the likelihood of developing Crohn's disease (right) or being healthy (left). "*



*“**Fig. 4 Predictive accuracy of plasma proteins panel, alone or in combination with clinical variables in region-based UKB and external testing cohorts.** Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn’s disease in the geographically distinct UKB testing cohort with 25% (**A**) and 20% (**B**) proportions. Model performance evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn’s disease. The combined model included protein, demographics, serum,*

and PRS of Crohn's disease. ***C****, Receiver operating curves show the performance of protein model for predicting Crohn's disease in the external testing set from Southern China, evaluated using the four machine learning algorithms."*

"Table 1: Baseline characteristics of UK Biobank participants in the training and testing cohorts.

	Training cohort	***Testing cohort***	***P value***
	n=39,634	*n=13,262*	
Age, mean (SD), years	*57.06 (8.13)*	*56.06 (8.41)*	*<0.001*
Sex			*0.173*
Female, n(%)	*21299 (53.7)*	*7218 (54.4)*	
Male, n(%)	*18335 (46.3)*	*6044 (45.6)*	
Ethnicity			*<0.001*
Other, n(%)	*2253 (5.7)*	*1095 (8.3)*	
White, n(%)	*37381 (94.3)*	*12167 (91.7)*	
TD index, mean (SD)	*-1.36 (3.03)*	*-0.65 (3.57)*	*<0.001*
BMI, mean (SD), kg/m2	*27.54 (4.79)*	*27.28 (4.84)*	*<0.001*
Physical activity			*0.289*
Physically inactive, n(%)	*7591 (19.2)*	*2484 (18.7)*	
Physically active, n(%)	*32043 (80.8)*	*10778 (81.3)*	
Smoking status			*0.014*
Never smoker, n(%)	*21565 (54.4)*	*7166 (54.0)*	
Previous smoker, n(%)	*13937 (35.2)*	*4594 (34.6)*	
Current smoker, n(%)	*4132 (10.4)*	*1502 (11.3)*	
Alcohol consumption			*<0.001*
Daily or almost daily, n(%)	*7798 (19.7)*	*2898 (21.9)*	
Three or four times a week, n(%)	*8931 (22.5)*	*2970 (22.4)*	
Once or twice a week, n(%)	*10456 (26.4)*	*3264 (24.6)*	
One to three times a month, n(%)	*4360 (11.0)*	*1416 (10.7)*	
Special occasions only, n(%)	*4680 (11.8)*	*1553 (11.7)*	
Never, n(%)	*3409 (8.6)*	*1161 (8.8)*	
Dietary habit			*0.022*
Poor diet	*24856 (62.7)*	*8169 (61.6)*	
Healthy diet	*14778 (37.3)*	*5093 (38.4)*	
Depression (%)	*2315 (5.8)*	*738 (5.6)*	*0.246*
Anxiety (%)	*548 (1.4)*	*186 (1.4)*	*0.899*
History use of antibiotics (%)	*553 (1.4)*	*182 (1.4)*	*0.879*
History use of NSAIDs (%)	*669 (1.7)*	*176 (1.3)*	*0.005*

Note: Data are presented as mean (SD) or n (%). The 52,896 participants were randomly divided into training and testing cohorts according to the UK Biobank recruitment centers in a ratio of 75%:25% roughly. The training set included participants from Cardiff, Glasgow, Stoke, Reading, Bury, Newcastle, Leeds, Nottingham, Sheffield, Liverpool, Middlesborough, Hounslow, and Croydon. The testing set included participants from Stockport (pilot), Manchester, Oxford, Edinburgh, Bristol, Barts, Birmingham, Swansea, Wrexham (Supplemental Table 18). Differences between training and testing cohorts were assessed using Student's t-test for continuous variables and Pearson's chi-squared test for discrete variables.
Abbreviations: BMI, body mass index; TD index, Townsend deprivation index; NSAIDs, nonsteroidal anti-inflammatory drugs."

"Table 2. Baseline characteristics of Southern China participants included in the study.

	Overall	*CD*	*Control*	*P value*
	n=74	*n=37*	*n=37*	
Age, mean (SD), years	*43.56 (18.31)*	*34.81 (14.88)*	*53.49 (16.94)*	*<0.001*
Sex				*0.093*
Female, n(%)	*26 (36.1)*	*10 (27.0)*	*18 (48.6)*	
Male, n(%)	*46 (63.9)*	*27 (73.0)*	*19 (51.4)*	
BMI, mean (SD), kg/m2	*22.28 (3.37)*	*21.68 (3.61)*	*22.90 (2.93)*	*0.115*
Smoking status				*0.588*
Never smoker, n(%)	*62 (86.1)*	*33 (89.2)*	*31 (83.8)*	
Previous smoker, n(%)	*4 (5.6)*	*1 (2.7)*	*3 (8.1)*	
Current smoker, n(%)	*6 (8.3)*	*3 (8.1)*	*3 (8.1)*	
Alcohol consumption	*8 (11.1)*	*2 (5.4)*	*6 (16.2)*	*0.261*

Note: Data are presented as mean (SD) or n (%). Differences between CD and healthy control groups were assessed using Student's t-test for continuous variables and Pearson's chi-squared test for discrete variables.
Abbreviations: BMI, body mass index; CD, Crohn's disease."

"Supplementary Table 4: Results of ROC analyses.

	UKB testing set (AUC [95%CI])			
	LGBM	*XGBoost*	*RF*	*Extra trees*
9-protein panel	*0.73 (0.72-0.73)*	*0.72 (0.71-0.73)*	*0.71 (0.70-0.73)*	*0.72 (0.71-0.73)*
Demographic	*0.56 (0.55-0.56)*	*0.56 (0.55-0.57)*	*0.58 (0.58-0.58)*	*0.55 (0.55-0.56)*
Serum	*0.63 (0.62-0.64)*	*0.63 (0.61-0.64)*	*0.66 (0.64-0.67)*	*0.62 (0.59-0.65)*
PRS	*0.67 (0.66-0.67)*	*0.67 (0.66-0.68)*	*0.63 (0.61-0.65)*	*0.69 (0.69-0.70)*
Protein+Demographic	*0.72 (0.71-0.73)*	*0.72 (0.72-0.73)*	*0.70 (0.69-0.72)*	*0.73 (0.72-0.74)*
Protein+Serum	*0.72 (0.70-0.74)*	*0.72 (0.70-0.74)*	*0.73 (0.72-0.74)*	*0.76 (0.75-0.76)*
Protein+PRS	*0.75 (0.74-0.76)*	*0.74 (0.73-0.75)*	*0.70 (0.68-0.71)*	*0.75 (0.74-0.77)*
Protein+Demographic+Serum	*0.73 (0.72-0.74)*	*0.73 (0.72-0.74)*	*0.74 (0.72-0.75)*	*0.72 (0.71-0.74)*
Protein+Demographic+PRS	*0.75 (0.75-0.76)*	*0.73 (0.72-0.75)*	*0.72 (0.70-0.73)*	*0.78 (0.76-0.79)*
Protein+Serum+PRS	*0.76 (0.75-0.77)*	*0.76 (0.75-0.77)*	*0.75 (0.74-0.77)*	*0.77 (0.76-0.78)*

Combined	*0.76 (0.75-0.77)*	*0.78 (0.77-0.79)*	*0.75 (0.74-0.76)*	*0.72 (0.68-0.76)*

	Southern China testing set (AUC [95%CI])			
	LGBM	***XGBoost***	***RF***	***Extra trees***
9-protein panel	*0.76 (0.73-0.79)*	*0.79 (0.77-0.81)*	*0.77 (0.73-0.80)*	*0.77 (0.76-0.79)*

	UKB training set (AUC [95%CI])			
	LGBM	***XGBoost***	***RF***	***Extra trees***
9-protein panel	*0.85 (0.82-0.88)*	*0.83 (0.80-0.86)*	*0.84 (0.82-0.87)*	*0.76 (0.74-0.78)*
Demographic	*0.66 (0.66-0.67)*	*0.66 (0.66-0.67)*	*0.67 (0.67-0.68)*	*0.66 (0.65-0.67)*
Serum	*0.77 (0.75-0.78)*	*0.78 (0.75-0.80)*	*0.75 (0.74-0.75)*	*0.72 (0.65-0.78)*
PRS	*0.61 (0.61-0.61)*	*0.64 (0.62-0.65)*	*0.71 (0.69-0.72)*	*0.64 (0.62-0.66)*
Protein+Demographic	*0.82 (0.81-0.83)*	*0.82 (0.80-0.83)*	*0.82 (0.78-0.85)*	*0.77 (0.75-0.79)*
Protein+Serum	*0.90 (0.86-0.93)*	*0.88 (0.85-0.91)*	*0.93 (0.90-0.96)*	*0.79 (0.79-0.80)*
Protein+PRS	*0.88 (0.86-0.90)*	*0.88 (0.84-0.92)*	*0.85 (0.82-0.88)*	*0.79 (0.78-0.79)*
Protein+Demographic+Serum	*0.87 (0.85-0.89)*	*0.89 (0.86-0.92)*	*0.88 (0.83-0.93)*	*0.79 (0.78-0.81)*
Protein+Demographic+PRS	*0.88 (0.85-0.90)*	*0.87 (0.83-0.91)*	*0.80 (0.77-0.82)*	*0.83 (0.81-0.86)*
Protein+Serum+PRS	*0.91 (0.89-0.93)*	*0.93 (0.89-0.98)*	*0.86 (0.81-0.91)*	*0.80 (0.77-0.82)*
Combined	*0.92 (0.90-0.95)*	*0.89 (0.85-0.93)*	*0.87 (0.84-0.91)*	*0.80 (0.77-0.83)*

We randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 3:1 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 6: Results of ROC analyses after splitting the participants into a training and a testing set according to the recruitment centers in a ratio of 8:2.

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.76 (0.75-0.77)*	*0.76 (0.74-0.77)*	*0.77 (0.77-0.78)*	*0.71 (0.70-0.71)*
Demographic	*0.52 (0.51-0.52)*	*0.53 (0.52-0.55)*	*0.52 (0.51-0.53)*	*0.57 (0.56-0.57)*
Serum	*0.60 (0.58-0.62)*	*0.60 (0.58-0.61)*	*0.62 (0.61-0.63)*	*0.54 (0.51-0.57)*
PRS	*0.57 (0.56-0.57)*	*0.58 (0.56-0.59)*	*0.59 (0.58-0.60)*	*0.62 (0.61-0.63)*
Protein+Demographic	*0.76 (0.75-0.77)*	*0.75 (0.74-0.76)*	*0.76 (0.75-0.78)*	*0.73 (0.72-0.75)*
Protein+Serum	*0.73 (0.71-0.75)*	*0.75 (0.73-0.76)*	*0.74 (0.72-0.77)*	*0.72 (0.70-0.73)*
Protein+PRS	*0.78 (0.77-0.80)*	*0.78 (0.78-0.79)*	*0.78 (0.76-0.79)*	*0.76 (0.74-0.77)*
Protein+Demographic+Serum	*0.75 (0.73-0.77)*	*0.75 (0.73-0.76)*	*0.77 (0.76-0.77)*	*0.75 (0.74-0.76)*
Protein+Demographic+PRS	*0.77 (0.75-0.78)*	*0.78 (0.77-0.78)*	*0.78 (0.77-0.79)*	*0.77 (0.76-0.77)*
Protein+Serum+PRS	*0.74 (0.73-0.76)*	*0.77 (0.75-0.80)*	*0.76 (0.75-0.78)*	*0.75 (0.74-0.76)*

Combined	0.75 (0.74-0.77)	0.76 (0.74-0.79)	0.76 (0.75-0.77)	0.76 (0.74-0.77)
Training set				
9-protein panel	0.86 (0.83-0.89)	0.84 (0.79-0.88)	0.81 (0.78-0.83)	0.74 (0.73-0.76)
Demographic	0.68 (0.68-0.68)	0.67 (0.67-0.68)	0.67 (0.67-0.67)	0.64 (0.64-0.65)
Serum	0.81 (0.75-0.88)	0.77 (0.73-0.81)	0.77 (0.73-0.82)	0.70 (0.67-0.73)
PRS	0.68 (0.66-0.70)	0.69 (0.68-0.70)	0.70 (0.69-0.72)	0.64 (0.61-0.66)
Protein+Demographic	0.88 (0.85-0.91)	0.83 (0.79-0.88)	0.83 (0.80-0.87)	0.75 (0.73-0.78)
Protein+Serum	0.91 (0.88-0.94)	0.89 (0.84-0.93)	0.88 (0.84-0.91)	0.78 (0.76-0.81)
Protein+PRS	0.94 (0.92-0.97)	0.89 (0.85-0.93)	0.91 (0.87-0.95)	0.76 (0.76-0.76)
Protein+Demographic+Serum	0.94 (0.91-0.97)	0.87 (0.82-0.92)	0.79 (0.78-0.79)	0.79 (0.77-0.80)
Protein+Demographic+PRS	0.89 (0.86-0.92)	0.86 (0.83-0.89)	0.84 (0.80-0.87)	0.77 (0.76-0.78)
Protein+Serum+PRS	0.97 (0.93-1.00)	0.95 (0.91-0.98)	0.96 (0.94-0.97)	0.80 (0.79-0.82)
Combined	0.96 (0.93-1.00)	0.93 (0.90-0.97)	0.91 (0.87-0.95)	0.79 (0.79-0.80)

We randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 8:2 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 20. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease."

"Supplementary Table 18: Recruitment centers and number of training and validation cohorts in a ratio of 3:1.

Coding	Region	Allocation	Number
11003	Cardiff	Training Cohort	1978
11004	Glasgow	Training Cohort	2036
11006	Stoke	Training Cohort	2030
11007	Reading	Training Cohort	2622
11008	Bury	Training Cohort	3287
11009	Newcastle	Training Cohort	3762
11010	Leeds	Training Cohort	5010
11013	Nottingham	Training Cohort	4129
11014	Sheffield	Training Cohort	3317
11016	Liverpool	Training Cohort	3124
11017	Middlesborough	Training Cohort	2497
11018	Hounslow	Training Cohort	3046
11020	Croydon	Training Cohort	2796
10003	Stockport (pilot)	Validation Cohort	241
11001	Manchester	Validation Cohort	1711
11002	Oxford	Validation Cohort	1328
11005	Edinburgh	Validation Cohort	1671
11011	Bristol	Validation Cohort	3916

11012	*Barts*	*Validation Cohort*	*1676*
11021	*Birmingham*	*Validation Cohort*	*2525*
11022	*Swansea*	*Validation Cohort*	*147*
11023	*Wrexham*	*Validation Cohort*	*47*

"

"Supplementary Table 19: List of the ELISA kits.

Protein	***ELISA kit***	***Catalog Number***	***Manufacturer***
CD274	*Human PDL1(Programmed Cell Death Protein 1 Ligand 1) ELISA Kit*	*ELK3055*	*ELK (Wuhan) Biotechnology CO.,Ltd.*
CHI3L1	*Human CHI3L1(Chitinase-3-Like Protein 1) ELISA Kit*	*ELK10135*	*ELK (Wuhan) Biotechnology CO.,Ltd.*
REG1B	*Human REG1b(Regenerating Islet Derived Protein 1 Beta) ELISA Kit*	*ELK3474*	*ELK (Wuhan) Biotechnology CO.,Ltd.*
ITGAV	*Human ITGAV beta3(ITG alpha V beta3) ELISA Kit*	*ELK8748*	*ELK (Wuhan) Biotechnology CO.,Ltd.*
PRSS8	*Human Prostasin(PRSS8) ELISA kit*	*CSB-EL018825HU*	*Cusabio Technology LLC*
ITGA11	*Human ITGa11(Integrin Alpha 11) ELISA Kit*	*ELK4040*	*ELK (Wuhan) Biotechnology CO.,Ltd.*
GDF15	*Human GDF15(Growth Differentiation Factor 15) ELISA Kit*	*ELK2756*	*ELK (Wuhan) Biotechnology CO.,Ltd.*
DEFA1_DEFA1B	*Human DEFa1(Defensin Alpha 1, Neutrophil) ELISA Kit*	*ELK1775*	*ELK (Wuhan) Biotechnology CO.,Ltd.*
IL6	*Human IL6(Interleukin 6) ELISA Kit*	*ELK1156*	*ELK (Wuhan) Biotechnology CO.,Ltd.*

"

"Supplementary Table 22: Recruitment centers and number of training and validation cohorts in a ratio of 8:2.

Coding	***Region***	***Allocation***	***Number***
11002	*Oxford*	*Training Cohort*	*1328*
11005	*Edinburgh*	*Training Cohort*	*1671*
11006	*Stoke*	*Training Cohort*	*2030*
11007	*Reading*	*Training Cohort*	*2622*
11008	*Bury*	*Training Cohort*	*3287*
11009	*Newcastle*	*Training Cohort*	*3762*
11010	*Leeds*	*Training Cohort*	*5010*
11011	*Bristol*	*Training Cohort*	*3916*
11013	*Nottingham*	*Training Cohort*	*4129*
11014	*Sheffield*	*Training Cohort*	*3317*
11016	*Liverpool*	*Training Cohort*	*3124*
11017	*Middlesborough*	*Training Cohort*	*2497*
11018	*Hounslow*	*Training Cohort*	*3046*

11021	*Birmingham*	*Training Cohort*	*2525*
10003	*Stockport (pilot)*	*Validation Cohort*	*241*
11001	*Manchester*	*Validation Cohort*	*1711*
11003	*Cardiff*	*Validation Cohort*	*1978*
11004	*Glasgow*	*Validation Cohort*	*2036*
11012	*Barts*	*Validation Cohort*	*1676*
11020	*Croydon*	*Validation Cohort*	*2796*
11022	*Swansea*	*Validation Cohort*	*147*
11023	*Wrexham*	*Validation Cohort*	*47*

"

Reference

1. Collins, G. S., Reitsma, J. B., Altman, D. G. & Moons, K. G. M. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. BMJ 350, g7594 (2015).

Reviewer #3 (Remarks to the Author):

The authors did thorough work with suitable statistical modelling.
The noteworthy results are the identification of the serum proteins most associated with the onset of CD with an identification cohort and a replication cohort. This might give significant insight into the biological processes underlying the onset of CD.
As a predictor for CD in high risk populations this is not clinically useful: there are no protective measures against the development of CD. Especially when the signal appears 16 yrs ahead of the actual development of CD, it's not much of a clinical predictor, but more an opportunity to gain insight in pathomechanisms.
The AUC of these 11 proteins plus known disease predictors is bigger than that from similar studies.
For the most extent the work supports the conclusions. The only issue I have is with their conclusion that their predictive model is reliable 16 years before the onset of CD. With only 159 CD patients in the UK Biobank with a large variation in the time to onset I think that this is a bit of a stretch.
With the caveat mentioned above, I think that the methodology is otherwise sound. I again think one should not interpret along the outer borders (outliers) of a small case group, but otherwise the combined phenotype+PRS+protein model seem to have an exceptionally good AUC.
The methodology is sound: machine learning with capped iterations would be the way to go for this work.
The data could be reproduced since UK Biokbank is available on request and authors are transparent in their methods.
My main issue is with the fact that a prediction model such a s proposed here is of extremely limited clinical use, since there are no therapies to prevent CD. This becomes an ever bigger issue if this protein signature arises >10 yrs before disease onset; if it were closer to disease onset one could start monitoring calprotectin on a 6-monthly basis to detect active disease early and start treatment before complications arise.
This protein prediction model is however a gold mine for the biological mechanisms that role into action even before the true onset of CD. I completely understand that the research group is not specialised in mucosal immunology or other wet lab work, but with the UK Biobank data and the other phenotypes

present in this dataset and phenotypes associated with CD risk (depression, obesity etc) it would be worthwhile to check to which extend these proteins also predict depression or whether they are associated to smoking or obesity. To have a protein profile that predicts CD, and that is specifically associated with a "modifiable" phenotype could point into an actual preventative measure for CD with biological underpinnings.

Authors' response:

Thank you very much for your insightful and constructive comments. We truly appreciate the time and effort you've taken to review our work, and your feedback has been invaluable in improving the manuscript.

We are grateful for your recognition of the thorough statistical modeling and the potential biological insights provided by our study. Regarding your concern about the limited direct clinical application of a long-term prediction model due to the current lack of preventive therapies for CD, we fully understand the challenge this presents. In response to your suggestion about exploring associations between our protein signature and modifiable phenotypes, we have conducted additional analyses to further address this important point. Specifically, we examined the relationships between the nine proteins in our model and CD-related phenotypes. After adjusting for age, sex, and ethnicity and applying FDR correction, we found significant associations between these proteins and modifiable risk factors, such as obesity, physical inactivity, smoking, poor diet, and depression The direction of these associations aligns with their previously observed relationship with CD risk, which further supports the role of these proteins in disease development. More importantly, this suggests that modifiable factors may influence these biomarkers, opening up the possibility for early interventions and preventive strategies in high-risk populations. Your insight has helped us better frame the potential clinical value of our findings, and we are grateful for your valuable contribution to our manuscript. We have updated the manuscript accordingly, with the detailed results now presented in: **2.8 Clinical Association Between Protein Levels and CD-Related Phenotypes and Supplementary, 3. Discussion, 4.5 Statistical analysis, Fig. 5, and Supplementary Table 17.**

Moreover, to further strengthen the robustness and generalizability of our findings, we have incorporated additional methodological improvements based on feedback from all reviewers. In particular, we introduced the Southern China external cohort and leveraged geographically distinct UKB cohort for validating, ensuring a more rigorous assessment of model generalizability. Additionally, we refined our machine learning pipeline by performing feature selection on the training set, evaluating multiple algorithms, and implementing 10-fold repeated cross-validation. These enhancements not only reinforce the stability and generalizability of our predictive model but also provide stronger empirical support for its potential translational value.

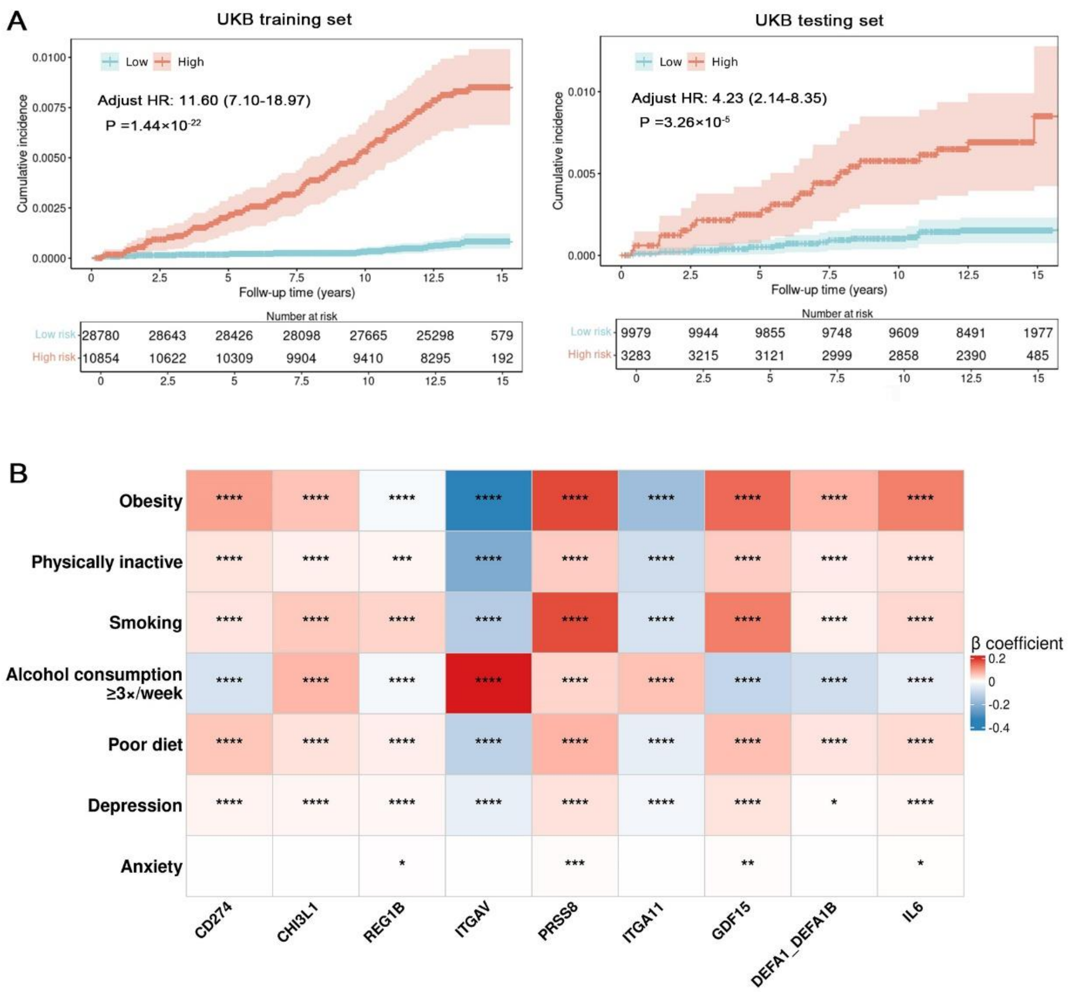
Once again, thank you for your detailed and valuable feedback. It has greatly improved our manuscript, and we sincerely hope that the revisions address your concerns. We truly appreciate the opportunity to refine our work based on your expert suggestions, and we remain open to any further thoughts or recommendations you may have.

*"**Results 2.8 Clinical association between protein levels and CD-related phenotypes** To further explore whether the association between CD and proteins is influenced by modifiable risk factors for CD, we examined the relationships between the nine proteins in the protein model and CD-associated phenotypes. Obesity, physical inactivity, smoking, poor diet, and depression were significantly associated with all nine proteins after false discovery rate (FDR) correction (**Fig. 5B** and **Supplementary Table 17**). Specifically, CD274, CHI3L1, REG1B, PRSS8, GDF15, DEFA1_DEFA1B, and IL6 exhibited positive associations with these modifiable risk factors for CD, whereas ITGAV and ITGA11 showed negative*

associations. Notably, these associations further support the potential of these proteins as early biomarkers for CD, influenced by modifiable risk factors, which may offer opportunities for preventive strategies”

*“**3. Discussion** …To enhance the clinical relevance of our model, we examined the associations between the nine proteins and modifiable CD-related phenotypes, such as obesity, smoking, physical inactivity, poor diet, and depression, and identified significant associations. These findings suggest that the identified protein signature may not only serve as an early biomarker for CD but also as a reflection of modifiable risk factors. This may provide opportunities for early intervention or prevention strategies in high-risk populations through lifestyle modifications….”*

*“**Method 4.5 Statistical analysis…**Lastly, we performed linear regression analyses to assess the cross-sectional associations between the levels of the nine proteins in the protein model and CD-related phenotypes. Protein levels were treated as the outcome variables, while phenotypes such as obesity, physical activity, smoking status, alcohol intake frequency, dietary habits, depression, and anxiety served as explanatory variables. All models were adjusted for age, sex, and ethnicity. The FDR method was applied to correct for multiple testing across all exposures and outcomes….”*



“Fig. 5 Risk stratification and clinical associations of the 9-protein model with Crohn’s disease (CD)

and CD-related phenotypes. A, *Protein model stratifies the risk of CD onset. Unadjusted Kaplan–Meier survival curves illustrate distinct cumulative risk trajectories for incident CD between the stratified subgroups (high-risk subgroup: red line; low-risk subgroup: blue line). The optimal cutoff (0.484) was determined using the Youden index in the training cohort and applied to both the training and testing set. Cox proportional hazards models were used to estimate the association between the protein model and disease risk, adjusting for age, sex, and ethnicity. Hazard ratios (HRs) and P values are presented. The shaded area represents the 95% confidence interval (CI) of the survival curves.* ***B,*** *Clinical association between protein levels and CD-related phenotypes. Heatmap showing the associations between nine proteins in the protein model and modifiable risk factors for CD, including obesity, physical inactivity, smoking, alcohol consumption (⩾3 times per week), poor diet, depression, and anxiety. Linear regression models were used, adjusting for age, sex, and ethnicity. Protein levels were treated as outcome variables, while CD-related phenotypes served as explanatory variables. The β coefficient represents the effect size, with positive associations in red and negative associations in blue. Statistical significance was determined using the false discovery rate (FDR) correction (*P<0.05, **P<0.01, ***P<0.001, ****P<0.0001).”*

“Supplementary Table 17: Clinical Association Between Protein Levels and CD-Related Phenotypes

CD-related phenotypes	Protein	β	Lower CI	Upper CI	P value	P value (FDR-corrected)
Obesity	CD274	0.096	0.086	0.106	3.34E-82	1.32E-81
	CHI3L1	0.062	0.057	0.066	2.12E-178	1.49E-177
	REG1B	-0.013	-0.019	-0.008	5.42E-06	6.97E-06
	ITGAV	-0.334	-0.353	-0.314	3.73E-249	3.92E-248
	PRSS8	0.180	0.172	0.187	<2E-16	<2E-16
	ITGA11	-0.170	-0.180	-0.161	6.03E-277	7.60E-276
	GDF15	0.151	0.144	0.158	<2E-16	<2E-16
	DEFA1_DEFA1B	0.078	0.072	0.085	2.09E-132	1.10E-131
	IL6	0.130	0.126	0.134	<2E-16	<2E-16
Physically inactive	CD274	0.029	0.020	0.038	2.11E-10	2.96E-10
	CHI3L1	0.015	0.011	0.019	1.48E-14	2.25E-14
	REG1B	0.010	0.005	0.015	2.58E-04	3.02E-04
	ITGAV	-0.222	-0.240	-0.205	6.05E-133	3.46E-132
	PRSS8	0.053	0.046	0.060	1.29E-50	3.25E-50
	ITGA11	-0.086	-0.095	-0.078	4.45E-86	1.87E-85
	GDF15	0.053	0.046	0.059	9.31E-56	2.79E-55
	DEFA1_DEFA1B	0.020	0.015	0.026	3.28E-12	4.70E-12
	IL6	0.030	0.026	0.034	1.13E-52	2.96E-52
Smoking	CD274	0.027	0.016	0.038	1.98E-06	2.60E-06
	CHI3L1	0.056	0.051	0.061	1.92E-113	8.62E-113
	REG1B	0.045	0.038	0.051	9.46E-40	2.13E-39
	ITGAV	-0.131	-0.154	-0.109	7.43E-31	1.42E-30
	PRSS8	0.177	0.169	0.186	<2E-16	<2E-16
	ITGA11	-0.072	-0.083	-0.061	4.55E-39	9.89E-39

	GDF15	*0.131*	*0.123*	*0.140*	*5.26E-218*	*4.74E-217*
	DEFA1_DEFA1B	*0.016*	*0.009*	*0.023*	*1.71E-05*	*2.11E-05*
	IL6	*0.039*	*0.034*	*0.044*	*1.48E-55*	*4.23E-55*
	CD274	*-0.072*	*-0.083*	*-0.061*	*2.71E-37*	*5.69E-37*
	CHI3L1	*0.074*	*0.070*	*0.079*	*4.75E-205*	*3.74E-204*
	REG1B	*-0.017*	*-0.024*	*-0.011*	*1.82E-07*	*2.44E-07*
	ITGAV	*0.208*	*0.186*	*0.230*	*1.94E-77*	*7.20E-77*
Alcohol consumption ⩾3×/week	*PRSS8*	*0.043*	*0.034*	*0.052*	*6.47E-23*	*1.13E-22*
	ITGA11	*0.063*	*0.053*	*0.074*	*2.45E-31*	*4.82E-31*
	GDF15	*-0.106*	*-0.114*	*-0.098*	*8.92E-148*	*5.62E-147*
	DEFA1_DEFA1B	*-0.084*	*-0.091*	*-0.077*	*1.81E-121*	*8.77E-121*
	IL6	*-0.044*	*-0.048*	*-0.039*	*2.75E-71*	*9.12E-71*
	CD274	*0.059*	*0.048*	*0.069*	*2.03E-26*	*3.66E-26*
	CHI3L1	*0.030*	*0.025*	*0.035*	*4.91E-36*	*9.97E-36*
	REG1B	*0.018*	*0.011*	*0.024*	*7.28E-08*	*9.97E-08*
	ITGAV	*-0.118*	*-0.140*	*-0.097*	*5.07E-27*	*9.39E-27*
Poor diet	*PRSS8*	*0.078*	*0.069*	*0.086*	*1.54E-73*	*5.39E-73*
	ITGA11	*-0.042*	*-0.053*	*-0.032*	*1.95E-15*	*3.07E-15*
	GDF15	*0.065*	*0.057*	*0.073*	*1.48E-57*	*4.66E-57*
	DEFA1_DEFA1B	*0.027*	*0.020*	*0.034*	*5.20E-14*	*7.63E-14*
	IL6	*0.037*	*0.033*	*0.042*	*1.02E-54*	*2.78E-54*
	CD274	*0.011*	*0.006*	*0.017*	*2.42E-05*	*2.94E-05*
	CHI3L1	*0.010*	*0.008*	*0.012*	*3.95E-18*	*6.72E-18*
	REG1B	*0.007*	*0.004*	*0.010*	*8.49E-06*	*1.07E-05*
	ITGAV	*-0.041*	*-0.052*	*-0.031*	*1.50E-14*	*2.25E-14*
Depression	*PRSS8*	*0.030*	*0.026*	*0.034*	*4.48E-47*	*1.09E-46*
	ITGA11	*-0.021*	*-0.027*	*-0.016*	*2.08E-16*	*3.36E-16*
	GDF15	*0.028*	*0.024*	*0.032*	*1.36E-45*	*3.18E-45*
	DEFA1_DEFA1B	*0.003*	*<0.001*	*0.007*	*4.42E-02*	*4.80E-02*
	IL6	*0.010*	*0.008*	*0.012*	*1.36E-17*	*2.26E-17*
	CD274	*0.001*	*-0.002*	*0.003*	*5.75E-01*	*5.75E-01*
	CHI3L1	*0.001*	*<0.001*	*0.002*	*2.12E-01*	*2.23E-01*
	REG1B	*0.002*	*<0.001*	*0.003*	*3.59E-02*	*3.97E-02*
	ITGAV	*-0.002*	*-0.007*	*0.003*	*4.34E-01*	*4.41E-01*
Anxiety	*PRSS8*	*0.004*	*0.002*	*0.006*	*1.10E-04*	*1.31E-04*
	ITGA11	*-0.002*	*-0.005*	*<0.001*	*9.23E-02*	*9.86E-02*
	GDF15	*0.003*	*0.001*	*0.005*	*9.44E-04*	*1.08E-03*
	DEFA1_DEFA1B	*0.001*	*-0.001*	*0.003*	*2.45E-01*	*2.53E-01*
	IL6	*0.001*	*<0.001*	*0.003*	*1.46E-02*	*1.64E-02*

Protein levels were treated as outcome variables, and phenotypes—including obesity, physical inactivity, smoking, alcohol consumption (≥3 times per week), poor diet, depression, and anxiety—served as explanatory variables. All models were adjusted for age, sex, and ethnicity. The false discovery rate (FDR) method was applied to correct for multiple comparisons across all exposures and outcomes.”

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

The authors thoroughly responded to the comments that were raised. However, some concerns remain.

1. The follow-up duration from baseline to Crohn's disease (CD) diagnosis includes cases with a duration of 0.12 years (approximately 44 days) prior to diagnosis. Given the limitations of relying on the appearance of ICD codes—which may be biased toward a later diagnosis date—it is possible that some of these subjects had already been diagnosed. Therefore, caution is warranted when interpreting the results. The authors performed additional sensitivity analyses excluding individuals with shorter follow-up durations, which showed slightly lower predictive performance. The distribution of follow-up duration should be presented separately for both the training and testing cohorts.

Authors' response:

We thank the reviewer for this important comment. As suggested, we have now provided the distribution of follow-up durations (from baseline to CD diagnosis) separately for the training and testing cohorts. Specifically, the follow-up durations among CD cases ranged from 0.12 to 13.78 years (median: 8.69 years, IQR: 4.75–11.08, mean: 7.87 years) in the training cohort, and from 0.38 to 14.88 years (median: 6.42 years, IQR: 2.70–8.60, mean: 6.36 years) in the testing cohort. As shown in the Kaplan–Meier curves (Figure 5A), the cumulative incidence of CD increased steadily over time, rather than being predominantly concentrated in the initial years of follow-up.

A small number of individuals had short follow-up durations before CD diagnosis. To address potential misclassification due to reliance on ICD-10 codes, we performed sensitivity analyses excluding cases diagnosed within 1 to 5 years of follow-up (with 2-year exclusion results shown in our manuscript). The predictive performance was slightly reduced, but results remained largely consistent with the primary analysis. This limitation has been acknowledged in the revised Discussion section.

These modifications have been made in the corresponding parts of the manuscript, specifically in **Results section (2.1 Study Population)** and **Discussion**.

"Results 2.1 Study population *…When stratified by cohort, follow-up duration among CD cases ranged from 0.12 to 13.78 years in the training set (median: 8.69 years; IQR: 4.75–11.08; mean: 7.87 years), and from 0.38 to 14.88 years in the testing set (median: 6.42 years; IQR: 2.70–8.60; mean: 6.36 years)…"*

"Discussion *…Fourth, CD diagnosis was based on ICD-10 code, which may lag behind the actual clinical diagnosis, leading to potential misclassification—especially for cases with shorter follow-up durations. Nevertheless, sensitivity analyses excluding these cases remained largely consistent with the main findings, supporting the robustness of our model.…"*

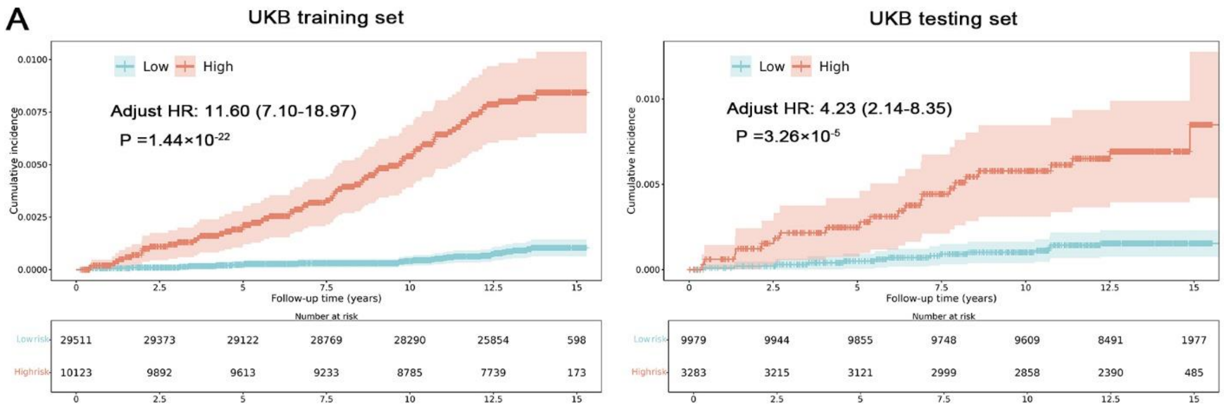


Fig. 5A Protein model stratifies the risk of CD onset.

2. Regarding the external validation cohort, it is unclear whether the Southern China cohort represents a new-onset inception cohort. A description of how this cohort was identified should be included. Additionally, it is important to report the disease duration and medication use for patients with CD. Regardless, it appears that this cohort does not provide external validation that the biomarkers precede CD diagnosis. As such, the validity of this external validation cohort is questionable.

Authors' response:

We sincerely thank the reviewer for raising this important point. To strengthen the external validation of our protein model, we incorporated the EPIC-Norfolk cohort (n = 2,944), which provides long-term follow-up with pre-diagnostic blood samples. The model demonstrated strong predictive performance in this prospective setting (AUC = 0.73 with XGBoost; highest AUC = 0.76 with Extra Trees), supporting its validity and generalizability for long-term risk prediction.

We acknowledge that the Southern China cohort was cross-sectional rather than a new-onset inception cohort, and most CD participants had been diagnosed and were receiving treatment at the time of blood sampling. Nevertheless, this cohort offered an opportunity to assess the model's performance in an ethnically and geographically distinct population, where it achieved an AUC of 0.79, further underscoring its robust case–control discrimination and suggesting potential applicability in diverse clinical populations.

In response to the reviewer's suggestions, we have now included a detailed description of how the Southern China cohort was established, including information on disease duration and medication use among CD patients, and have explicitly acknowledged the limitations of this cohort design. We have also incorporated the details and results of the EPIC-Norfolk cohort into the manuscript. These revisions are reflected throughout the manuscript, including the **Abstract, Results section (2.1 Study population, 2.5 Predictive accuracy of proteomics-based models), Methods section (4.1 Studying population, 4.2 Plasma proteomics, 4.4 Definition of CD, 4.5 Statistical analysis)**, **Discussion**, **Fig 1 and 4, and**

Supplementary Tables 2, 3, 6 and 8.

We hope this clarification and the corresponding manuscript revisions address the reviewer's concerns.

*"**Abstract** …Independent validation was conducted in a geographically distinct UKB cohort (n = 13,262) and in two external cohorts: the EPIC-Norfolk study (n = 2,944) and Southern China cohort (n = 74). … In the geographically distinct UKB cohort, the protein model (area under the curve [AUC] 0.76) outperformed clinical risk models (highest AUC 0.67; P < 0.05) and was externally validated in EPIC-Norfolk (AUC 0.73) and the Southern China cohort (AUC 0.79)…."*

*"**Results 2.1 Study population …**In the European Prospective Investigation into Cancer (EPIC)-Norfolk study (n = 2,944), 16 incident CD cases were identified within 16 years of follow-up. The cohort included 45% males, with a mean age of 61.0 years, and no significant differences in baseline characteristics were observed between cases and controls...*

2.5 Predictive accuracy of proteomics-based models …*In the external validation cohorts, AUCs ranged from 0.70 to 0.76 in EPIC-Norfolk and from 0.76 to 0.79 in the Southern China cohort. Notably, the XGBoost model obtained the highest AUC in the Southern China cohort (0.79, 95% CI 0.77–0.81), good performance in EPIC-Norfolk (0.73, 95% CI 0.71–0.75), and the second-best AUC in the UKB testing set (0.72 for a 75/25 split, 95% CI 0.71–0.73; 0.76 for an 80/20 split, 95% CI 0.74–0.77), indicating its robust generalizability across independent cohorts. Therefore, XGBoost was finally chosen as the optimal model…."*

*"**Methods 4.1 Studying population…**External validation was performed in two independent cohorts. The EPIC-Norfolk study[43] is a population-based prospective cohort that recruited participants aged 40–79 years from Norfolk, UK, between 1993 and 1997. Ethical approval was granted by the Norfolk Research Ethics Committee (ref. 05/Q0101/191), and all participants provided written informed consent. At baseline, participants completed general health questionnaires and underwent a panel of measurements. Following the same inclusion criteria as described above, this study included 2,944 individuals.*

The Southern China cohort comprised 37 patients with CD and 37 non-IBD controls recruited from ×× Hospital between 2024 and 2025. Eligibility criteria for CD cases included individuals aged ≥16 years with a prior or new diagnosis of CD, confirmed by two experienced clinicians based on standard clinical, radiological, endoscopic, and histopathologic criteria[44]. Controls were individuals without IBD and with no clinical or endoscopic evidence of intestinal inflammation. Blood samples were collected at enrollment and centrifuged. Ultimately, 74 participants were enrolled in the study….

***4.2 Plasma proteomics** …For the EPIC-Norfolk study, serum samples were collected at baseline between 1993 and 1997 and immediately stored in liquid nitrogen to preserve protein integrity. Proteomic profiling was performed using the Olink Explore 1536 and Olink Explore Expansion panels, targeting 2,923 unique proteins via 2,941 assays. Participants with failed proteomic quality control or missing data on age, sex, BMI, or smoking status were excluded….*

4.4 Definition of CD…*In the Epic-Norfolk study[43], mortality and hospitalization data were obtained linkage* *to the National Health Service digital database using participants' NHS numbers. These records were coded by trained nosologists according to ICD-10. To ensure comparability with the 16-year observation window in UKB, cases occurring after 16 years of follow-up were treated as censored when evaluating predictive performance…*

4.5 Statistical analysis…*Receiver operating characteristic (ROC) area under the curve (AUC) analyses were calculated to assess the predictive accuracy of the model in predicting CD in the UKB testing set (geographically distinct), the EPIC-Norfolk study and the Southern China cohort."*

"Discussion*…It was further externally validated in EPIC-Norfolk (AUC 0.73) and the Southern China cohort (AUC 0.79)…*

…A simple model based on the nine most important proteins achieved great long-term predictive performance (0.72 for a 75/25 training/testing split; 0.76 for an 80/20 split) in the UKB testing set (geographically distinct), outperforming models based on demographic, serologic, and PRS data (highest AUC 0.67), and was externally validated in EPIC-Norfolk (AUC 0.73) and the Southern China cohort (AUC 0.79)….

…Remarkably, the generalizability of our protein model was validated in the geographically distinct UKB cohort and two independent external cohorts….

…Fifth, successful external validation in two independent cohorts demonstrated the generalizability of our protein model. However, given the cross-sectional design of the Southern China cohort and the limited number of incident CD cases in EPIC-Norfolk, further validation in large-scale, new-onset inception cohorts with pre-diagnostic samples is warranted…"

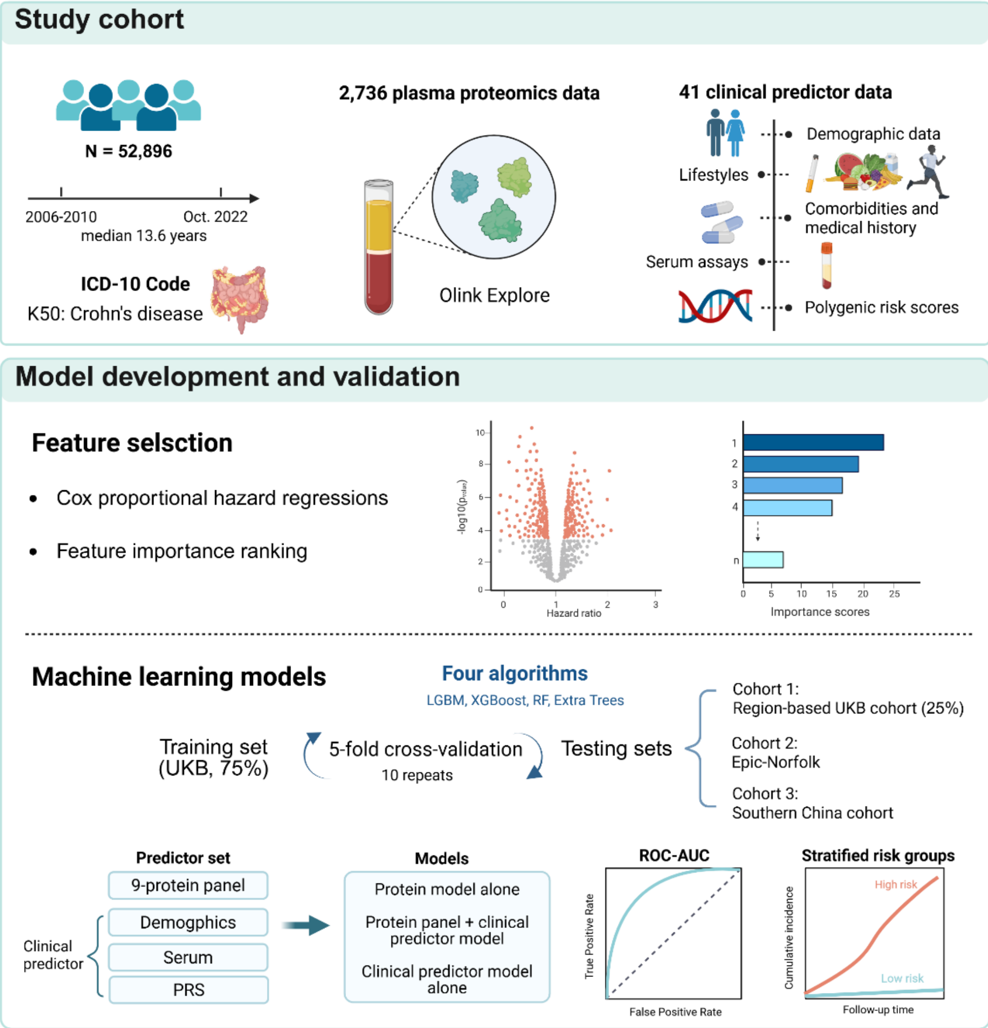


Fig. 1 Study overview. *First, we extracted data from 52,896 UK Biobank participants with a median follow-up time of 13.6 years, including Crohn's disease endpoint defined by ICD10 codes, 2,736 plasma proteomics, and 41 clinical predictors spanning demographic, lifestyles, comorbidity and medication history, serum assays, and polygenic risk scores for Crohn's disease. Next, Cox proportional hazard models and machine learning-based feature importance ranking were performed for further feature selection. Model development proceeded in three phases: 10×5-fold cross-validation on the training set, independent validation in a geographically distinct UK Biobank cohort, and external validation in Epic-Norfolk and the Southern China cohort. Finally, we investigated the predictive performance of the protein model and its risk stratification for Crohn's disease onset.*

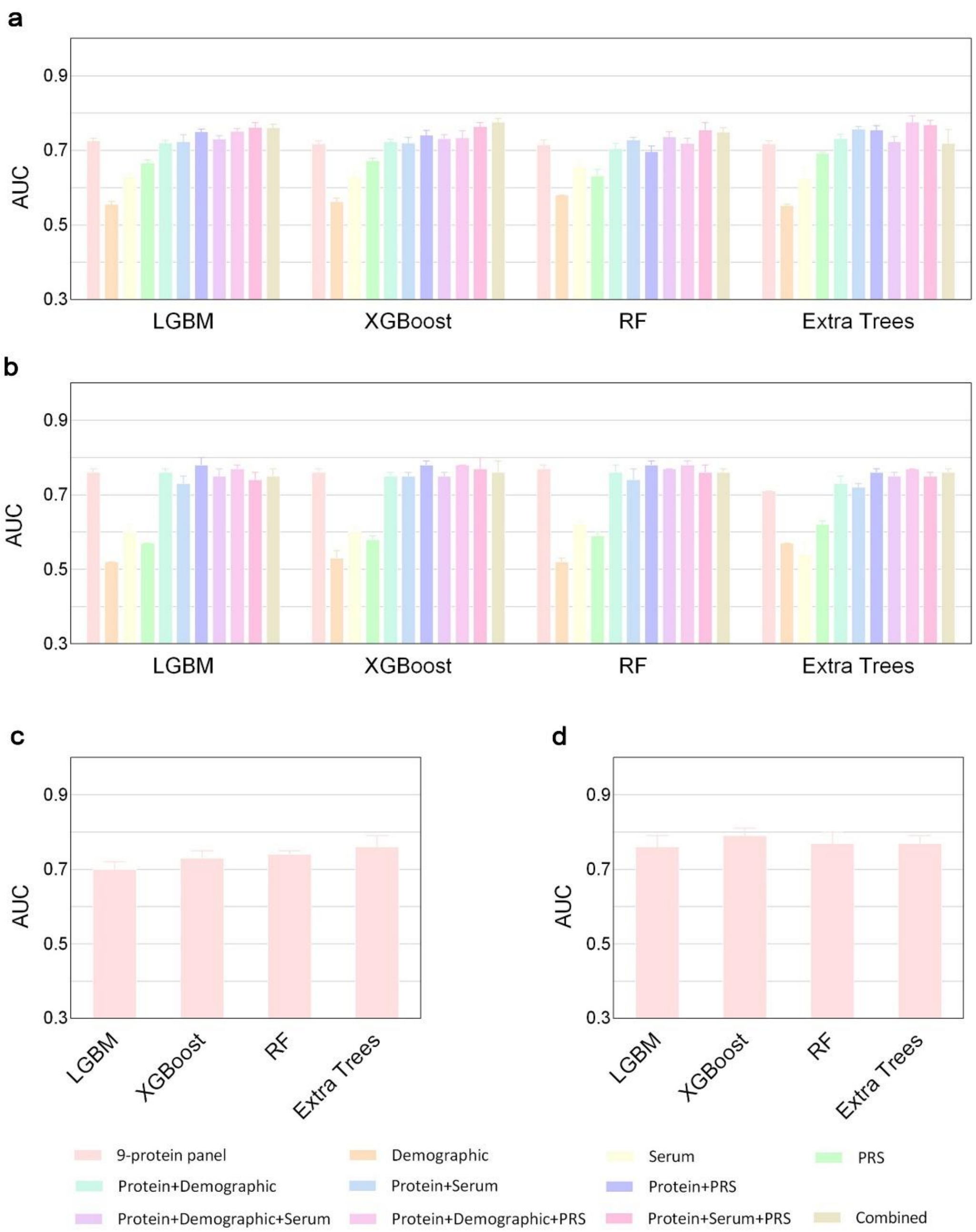


Fig. 4 Predictive accuracy of plasma proteins panel, alone or in combination with clinical variables in region-based UKB and external testing cohorts. *Bar plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease in the geographically distinct UKB testing cohort with 25% (**a**) and 20% (**b**) proportions, and external validation of the protein model in EPIC-Norfolk (**c**) and the Southern China cohort (**d**). Model performance evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 23. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease.*

Supplementary Table 2: Baseline characteristics of the EPIC validation cohort.

	Overall	CD	Control	P value
	n=2,944	n=16	n=2,928	
Age, mean (SD), years	61.01 (9.30)	60.92 (10.90)	61.01 (9.30)	0.97
TDindex, mean (SD)	-2.05 (2.22)	-2.58 (1.04)	-2.04 (2.23)	0.332
BMI, kg/m², mean (SD)	26.50 (3.98)	25.77 (4.37)	26.50 (3.98)	0.469
Sex				0.867
Female, n(%)	1625 (55.2)	8 (50.0)	1617 (55.2)	
Male, n(%)	1319 (44.8)	8 (50.0)	1311 (44.8)	
Physical activity				0.510
No regular physical activity, n(%)	969 (32.9)	7 (43.8)	962 (32.9)	
Regular physical activity, n(%)	1975 (67.1)	9 (56.2)	1966 (67.1)	
Ethnicity				1
Other, n(%)	<10 (<10.0)	<10 (<10.0)	<10 (<10.0)	
White, n(%)	2940 (>90.0)	16 (>90.0)	2924 (>90.0)	
Smoking status				0.294
Never smoker, n(%)	1316 (44.7)	5 (31.2)	1311 (44.8)	
Previous smoker, n(%)	382 (13.0)	4 (25.0)	378 (12.9)	
Current smoker, n(%)	1246 (42.3)	7 (43.8)	1239 (42.3)	
Alcohol consumption				0.137
Previous drinker, n(%)	2753 (93.5)	13 (81.2)	2740 (93.6)	
Never, n(%)	191 (6.5)	3 (18.8)	188 (6.4)	

Note: Data are presented as mean (SD) or n (%). Group comparisons between cases and controls were performed using Student's t-test for continuous variables and Pearson's chi-squared test for categorical variables. Abbreviations: BMI, body mass index; TD index, Townsend deprivation index; CD, Crohn' s disease.

Supplementary Table 3: Baseline characteristics of Southern China participants included in the study.

	Overall	CD	Control	P value
	n=74	n=37	n=37	
Age, mean (SD), years	43.56 (18.31)	34.81 (14.88)	53.49 (16.94)	<0.001
Sex				0.093
Female, n(%)	26 (36.1)	10 (27.0)	18 (48.6)	
Male, n(%)	46 (63.9)	27 (73.0)	19 (51.4)	
Disease duration, mean (SD), years	-	5.16 (4.29)	-	
BMI, mean (SD), kg/m2	22.28 (3.37)	21.68 (3.61)	22.90 (2.93)	0.115
Smoking status				0.588
Never smoker, n(%)	62 (86.1)	33 (89.2)	31 (83.8)	
Previous smoker, n(%)	4 (5.6)	1 (2.7)	3 (8.1)	
Current smoker, n(%)	6 (8.3)	3 (8.1)	3 (8.1)	
Alcohol consumption, n (%)	8 (11.1)	2 (5.4)	6 (16.2)	0.261

Concomitant treatment			
Aminosalicylates, n(%)	-	7 (18.9)	-
Glucocorticoids, n(%)	-	1 (2.7)	-
Immunosuppressants, n(%)	-	5 (13.5)	-
Biologics, n(%)	-	32 (86.5)	-
Upadacitinib, n(%)	-	3 (8.1)	-

Note: Data are presented as mean (SD) or n (%). Differences between CD and healthy control groups were assessed using Student's t-test for continuous variables and Pearson's chi-squared test for discrete variables. Abbreviations: BMI, body mass index; CD, Crohn's disease.

Supplementary Table 6: Results of ROC analyses.

	UKB testing set (AUC [95%CI])			
	LGBM	XGBoost	RF	Extra trees
9-protein panel	0.73 (0.72-0.73)	0.72 (0.71-0.73)	0.71 (0.70-0.73)	0.72 (0.71-0.73)
Demographic	0.56 (0.55-0.56)	0.56 (0.55-0.57)	0.58 (0.58-0.58)	0.55 (0.55-0.56)
Serum	0.63 (0.62-0.64)	0.63 (0.61-0.64)	0.66 (0.64-0.67)	0.62 (0.59-0.65)
PRS	0.67 (0.66-0.67)	0.67 (0.66-0.68)	0.63 (0.61-0.65)	0.69 (0.69-0.70)
Protein+Demographic	0.72 (0.71-0.73)	0.72 (0.72-0.73)	0.70 (0.69-0.72)	0.73 (0.72-0.74)
Protein+Serum	0.72 (0.70-0.74)	0.72 (0.70-0.74)	0.73 (0.72-0.74)	0.76 (0.75-0.76)
Protein+PRS	0.75 (0.74-0.76)	0.74 (0.73-0.75)	0.70 (0.68-0.71)	0.75 (0.74-0.77)
Protein+Demographic+Serum	0.73 (0.72-0.74)	0.73 (0.72-0.74)	0.74 (0.72-0.75)	0.72 (0.71-0.74)
Protein+Demographic+PRS	0.75 (0.75-0.76)	0.73 (0.72-0.75)	0.72 (0.70-0.73)	0.78 (0.76-0.79)
Protein+Serum+PRS	0.76 (0.75-0.77)	0.76 (0.75-0.77)	0.75 (0.74-0.77)	0.77 (0.76-0.78)
Combined	0.76 (0.75-0.77)	0.78 (0.77-0.79)	0.75 (0.74-0.76)	0.72 (0.68-0.76)

	EPIC-Norfolk (AUC [95%CI])			
	LGBM	XGBoost	RF	Extra trees
9-protein panel	0.70 (0.69-0.72)	0.73 (0.71-0.75)	0.74 (0.73-0.75)	0.76 (0.73-0.79)

	Southern China cohort (AUC [95%CI])			
	LGBM	XGBoost	RF	Extra trees
9-protein panel	0.76 (0.73-0.79)	0.79 (0.77-0.81)	0.77 (0.73-0.80)	0.77 (0.76-0.79)

	UKB training set (AUC [95%CI])			
	LGBM	XGBoost	RF	Extra trees
9-protein panel	0.85 (0.82-0.88)	0.83 (0.80-0.86)	0.84 (0.82-0.87)	0.76 (0.74-0.78)
Demographic	0.66 (0.66-0.67)	0.66 (0.66-0.67)	0.67 (0.67-0.68)	0.66 (0.65-0.67)
Serum	0.77 (0.75-0.78)	0.78 (0.75-0.80)	0.75 (0.74-0.75)	0.72 (0.65-0.78)
PRS	0.61 (0.61-0.61)	0.64 (0.62-0.65)	0.71 (0.69-0.72)	0.64 (0.62-0.66)
Protein+Demographic	0.82 (0.81-0.83)	0.82 (0.80-0.83)	0.82 (0.78-0.85)	0.77 (0.75-0.79)
Protein+Serum	0.90 (0.86-0.93)	0.88 (0.85-0.91)	0.93 (0.90-0.96)	0.79 (0.79-0.80)
Protein+PRS	0.88 (0.86-0.90)	0.88 (0.84-0.92)	0.85 (0.82-0.88)	0.79 (0.78-0.79)
Protein+Demographic+Serum	0.87 (0.85-0.89)	0.89 (0.86-0.92)	0.88 (0.83-0.93)	0.79 (0.78-0.81)
Protein+Demographic+PRS	0.88 (0.85-0.90)	0.87 (0.83-0.91)	0.80 (0.77-0.82)	0.83 (0.81-0.86)

Protein+Serum+PRS	*0.91 (0.89-0.93)*	*0.93 (0.89-0.98)*	*0.86 (0.81-0.91)*	*0.80 (0.77-0.82)*
Combined	*0.92 (0.90-0.95)*	*0.89 (0.85-0.93)*	*0.87 (0.84-0.91)*	*0.80 (0.77-0.83)*

We randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 3:1 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 23. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease.

Supplementary Table 8: Results of ROC analyses after splitting the participants into a training and a testing set according to the recruitment centers in a ratio of 8:2.

	AUC (95%CI)			
	LGBM	***XGBoost***	***RF***	***Extra trees***
Testing set				
9-protein panel	*0.76 (0.75-0.77)*	*0.76 (0.74-0.77)*	*0.77 (0.77-0.78)*	*0.71 (0.70-0.71)*
Demographic	*0.52 (0.51-0.52)*	*0.53 (0.52-0.55)*	*0.52 (0.51-0.53)*	*0.57 (0.56-0.57)*
Serum	*0.60 (0.58-0.62)*	*0.60 (0.58-0.61)*	*0.62 (0.61-0.63)*	*0.54 (0.51-0.57)*
PRS	*0.57 (0.56-0.57)*	*0.58 (0.56-0.59)*	*0.59 (0.58-0.60)*	*0.62 (0.61-0.63)*
Protein+Demographic	*0.76 (0.75-0.77)*	*0.75 (0.74-0.76)*	*0.76 (0.75-0.78)*	*0.73 (0.72-0.75)*
Protein+Serum	*0.73 (0.71-0.75)*	*0.75 (0.73-0.76)*	*0.74 (0.72-0.77)*	*0.72 (0.70-0.73)*
Protein+PRS	*0.78 (0.77-0.80)*	*0.78 (0.78-0.79)*	*0.78 (0.76-0.79)*	*0.76 (0.74-0.77)*
Protein+Demographic+Serum	*0.75 (0.73-0.77)*	*0.75 (0.73-0.76)*	*0.77 (0.76-0.77)*	*0.75 (0.74-0.76)*
Protein+Demographic+PRS	*0.77 (0.75-0.78)*	*0.78 (0.77-0.78)*	*0.78 (0.77-0.79)*	*0.77 (0.76-0.77)*
Protein+Serum+PRS	*0.74 (0.73-0.76)*	*0.77 (0.75-0.80)*	*0.76 (0.75-0.78)*	*0.75 (0.74-0.76)*
Combined	*0.75 (0.74-0.77)*	*0.76 (0.74-0.79)*	*0.76 (0.75-0.77)*	*0.76 (0.74-0.77)*
Training set				
9-protein panel	*0.86 (0.83-0.89)*	*0.84 (0.79-0.88)*	*0.81 (0.78-0.83)*	*0.74 (0.73-0.76)*
Demographic	*0.68 (0.68-0.68)*	*0.67 (0.67-0.68)*	*0.67 (0.67-0.67)*	*0.64 (0.64-0.65)*
Serum	*0.81 (0.75-0.88)*	*0.77 (0.73-0.81)*	*0.77 (0.73-0.82)*	*0.70 (0.67-0.73)*
PRS	*0.68 (0.66-0.70)*	*0.69 (0.68-0.70)*	*0.70 (0.69-0.72)*	*0.64 (0.61-0.66)*
Protein+Demographic	*0.88 (0.85-0.91)*	*0.83 (0.79-0.88)*	*0.83 (0.80-0.87)*	*0.75 (0.73-0.78)*
Protein+Serum	*0.91 (0.88-0.94)*	*0.89 (0.84-0.93)*	*0.88 (0.84-0.91)*	*0.78 (0.76-0.81)*
Protein+PRS	*0.94 (0.92-0.97)*	*0.89 (0.85-0.93)*	*0.91 (0.87-0.95)*	*0.76 (0.76-0.76)*
Protein+Demographic+Serum	*0.94 (0.91-0.97)*	*0.87 (0.82-0.92)*	*0.79 (0.78-0.79)*	*0.79 (0.77-0.80)*
Protein+Demographic+PRS	*0.89 (0.86-0.92)*	*0.86 (0.83-0.89)*	*0.84 (0.80-0.87)*	*0.77 (0.76-0.78)*
Protein+Serum+PRS	*0.97 (0.93-1.00)*	*0.95 (0.91-0.98)*	*0.96 (0.94-0.97)*	*0.80 (0.79-0.82)*
Combined	*0.96 (0.93-1.00)*	*0.93 (0.90-0.97)*	*0.91 (0.87-0.95)*	*0.79 (0.79-0.80)*

We randomly split the participants into a training and a testing set according to the UK Biobank recruitment centers in a ratio of 8:2 roughly. To reduce overfitting, internal 5-fold cross-validation (10 repeats) was conducted on a training set and the maximum number of iterations was capped at 100 with early stopping. Balanced class weighting was applied to adjust for class imbalances. We reported the performance metrics using mean AUCs and 95% CIs, which were obtained from 10 repeated runs, each incorporating a bootstrap strategy with 1,000 iterations. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 23. PRS represents the polygenic risk score for Crohn's disease. The combined model included protein, demographics, serum, and PRS of Crohn's disease.

3. As demonstrated by the calibration curves, there appears to be substantial overestimation, likely due to the use of balanced class weighting to address class imbalance. Building the model without adjusting for class imbalance would more accurately reflect predictive performance in the actual target population. This is important contextual information for readers and should be included as supplemental material and acknowledged in the discussion as a limitation, warranting caution when interpreting the results.

Authors' response:

We thank the reviewer for this valuable suggestion. In response, we built additional models without class weighting to more accurately reflect predictive performance in the actual target population.

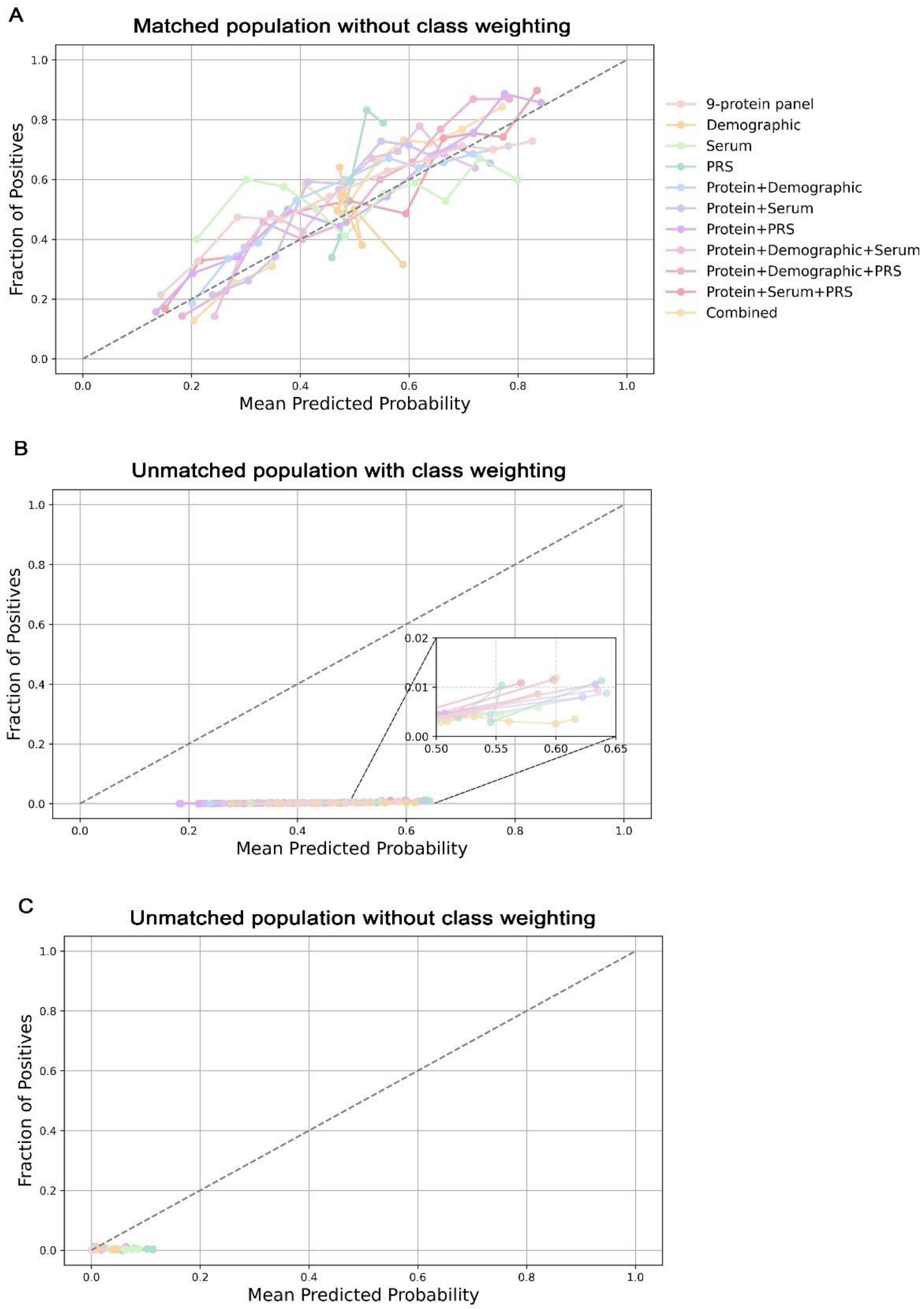
During this re-analysis, we refined our strategy for aggregating calibration results across model repetitions. In the original approach, we interpolated predicted probabilities to a common grid before averaging calibration curves across repeats. In the updated strategy, following Schalkamp et al.[1] (Nature Medicine, 2023), we directly averaged observed outcome frequencies within probability bins across repetitions. This bin-wise aggregation avoids interpolation artifacts and better preserves distributional characteristics. As a result, the updated calibration curves show some visual differences but offer more reliable and statistically grounded estimates.

According to the updated calibration results, models trained on the 1:1 matched population without class weighting exhibited good overall calibration. For the unmatched population, models using class weighting tended to overestimate risk, likely due to the emphasis placed on identifying rare positive cases under class imbalance. Meanwhile, models trained without class weighting showed improved calibration, but the predicted probabilities were concentrated near the origin. While better calibrated mathematically, these models were less sensitive and of limited screening utility.

We have added these calibration analyses to **Supplementary Fig. 10** and **Supplementary Table 26,** and clarified in **Methods section (4.5 Statistical analysis)** and **Discussion** that this trade-off reflects our choice to prioritize sensitivity and early identification of high-risk individuals, consistent with a screening-oriented model design.

"Methods 4.5 Statistical analysis…*We also assessed model calibration using calibration slope and calibration curves[50,51] (Supplementary Table 26 and Supplementary Fig. 10). In the 1:1 matched setting,* *calibration plots showed good agreement between predicted and observed outcomes, with slopes close to 1. In the unmatched population, models trained with class weighting tended to overestimate absolute risk. This is likely because class weighting increases the penalty for false negatives during training, encouraging the model to assign higher predicted probabilities to rare positive cases and potentially leading to calibration bias. Models trained without class weighting showed improved calibration but assigned lower predicted risks to true cases, which may limit sensitivity. Notably, we prioritized class weighting over perfect calibration to enhance sensitivity, consistent with our goal of building a screening-oriented model rather than a diagnostic tool…"*

"Discussion*…A final limitation lies in our modeling strategy. To improve rare case detection, we applied class weighting, which led to some overestimation of risk in the unmatched population. This reflects a trade-off between calibration and sensitivity, aligned with our aim to develop a screening-oriented model rather than a diagnostic tool…"*



Supplementary Fig. 10: Calibration of the prediction models. *Calibration curves based on the XGBoost algorithm are shown for each model under three evaluation settings in the UK Biobank testing set: (A) 1:1 matched population without class weighting, (B) unmatched population with class weighting, and (C) unmatched population without class weighting. Each curve shows the mean predicted probability binned into ten intervals plotted against the observed event rate (fraction of positives). The dashed diagonal line represents perfect calibration.*

Supplementary Table 26: Results of calibration slope.

Model	***Mean calibration slope***	***95% CI***
Matched population without class weighting		
9-protein panel	*0.7458*	*(0.5348, 1.1362)*
Demographic	*-17.9342*	*(-48.4169, -2.941)*
Serum	*0.4193*	*(-0.0251, 1.4775)*
PRS	*7.6830*	*(4.0567, 13.1023)*
Protein+Demographic	*0.9000*	*(0.6635, 1.2879)*
Protein+Serum	*1.3299*	*(0.6121, 3.551)*
Protein+PRS	*1.0176*	*(0.5954, 1.3734)*
Protein+Demographic+Serum	*2.2306*	*(0.3512, 10.2621)*
Protein+Demographic+PRS	*1.5843*	*(0.8243, 4.6175)*
Protein+Serum+PRS	*1.0414*	*(0.3755, 1.6581)*
Combined	*1.3441*	*(0.6705, 1.8575)*
Unmatched population with class weighting		
9-protein panel	*0.1202*	*(0.0145, 0.5627)*
Demographic	*0.0072*	*(0.0033, 0.018)*
Serum	*0.0224*	*(0.0066, 0.0521)*
PRS	*0.1796*	*(0.0271, 0.5096)*
Protein+Demographic	*0.0240*	*(0.0129, 0.0548)*
Protein+Serum	*0.0278*	*(0.0099, 0.0644)*
Protein+PRS	*0.0383*	*(0.0119, 0.1569)*
Protein+Demographic+Serum	*0.0541*	*(0.0117, 0.3004)*
Protein+Demographic+PRS	*0.1063*	*(0.0108, 0.6326)*
Protein+Serum+PRS	*0.0823*	*(0.0121, 0.3111)*
Combined	*0.0418*	*(0.017, 0.1423)*
Unmatched population without class weighting		
9-protein panel	*1.9219*	*(0.9579, 5.4461)*
Demographic	*0.3491*	*(0.083, 0.7567)*
Serum	*1.3804*	*(-0.0391, 5.5888)*
PRS	*5.6606*	*(1.9166, 17.9024)*
Protein+Demographic	*2.7156*	*(0.6296, 6.1093)*
Protein+Serum	*2.3316*	*(0.3075, 6.7387)*
Protein+PRS	*2.6240*	*(0.5934, 7.1394)*
Protein+Demographic+Serum	*1.6616*	*(0.5954, 5.0414)*
Protein+Demographic+PRS	*2.3023*	*(0.8362, 6.3012)*
Protein+Serum+PRS	*1.8026*	*(0.5506, 6.1229)*
Combined	*1.6120*	*(0.6224, 3.0006)*

Calibration slope (mean and 95% confidence interval) was calculated based on the XGBoost algorithm for each model under three evaluation settings in the UK Biobank testing set: (1) 1:1 matched population without class weighting, (2) unmatched population with class weighting, and (3) unmatched population without class weighting.

References

1. Schalkamp, A.-K., Peall, K. J., Harrison, N. A. & Sandor, C. Wearable movement-tracking data identify Parkinson's disease years before clinical diagnosis. *Nat Med* **29**, 2048–2056 (2023).

Reviewer #1 (Remarks on code availability):

Yes, they have provided the code.

Authors' response:

We thank the reviewer for confirming the availability of our code and for their support of our work.

Reviewer #2 (Remarks to the Author):

The authors have made a commendable effort in addressing the reviewers' comments. While the response letter was somewhat dense, it is clear that substantial effort went into revising the manuscript and improving its overall quality. In particular, the addition of external validation significantly strengthens the study by supporting its generalisability. I have no further comments.

Authors' response:

We thank the reviewer for their thoughtful comments and constructive suggestions throughout the review process. Their input was instrumental in improving the quality of our manuscript.

Reviewer #2 (Remarks on code availability):

I reviewed the code to assess the approach used for external validation, specifically to verify whether the models were retrained on the validation data—a common pitfall in such evaluations. I can confirm that the external validation is implemented appropriately, with previously trained models being loaded and tested without retraining, which supports the robustness of the reported results.
That said, the codebase is somewhat disorganized and not easily reusable. Making it more accessible—e.g., through packaging for installation via pip—would improve usability. However, I do not consider this a major limitation.

Authors' response:

We appreciate the reviewer's feedback and have refined the code structure and documentation to improve usability.

Reviewer #4 (Remarks to the Author):

Authors' response to Reviewer 3.

The comments from reviewer 3 have been addressed sufficiently by the authors, especially with respect to the associated phenotypes, e.g., obesity and smoking. However, these results generate one important question as the nine proteins seem to be part of general immunological processes and homeostasis and not specific to Crohn's disease.

1. Patients with CD are more likely to develop other chronic inflammation-driven disease such as rheumatoid arthritis, diabetes, asthma, psoriasis and many more. These diseases can occur years before the CD diagnosis and are typically seen with increasing prevalence in cohorts of elderly patients, like this one. Do the authors have access to this kind of information on potential significant co-morbidities in the UK Biobank? If so, this should be account and adjusted for. If not, this should be mentioned and discussed accordingly, as other chronic inflammatory disease might impose a significant bias in the association between CD and the nine identified proteins.

Author Response:

We thank the reviewer for this important comment. To assess the potential confounding effects of coexisting chronic inflammatory diseases, we conducted an additional Cox regression analysis for the nine proteins included in the predictive model. In this analysis, we further adjusted for seven major chronic inflammatory comorbidities—rheumatoid arthritis, diabetes mellitus, asthma, psoriasis, ankylosing spondylitis, multiple sclerosis, and ischemic heart disease—selected based on previous literature[1,2]. The associations between the nine proteins and incident CD remained significant after this additional adjustment, indicating that the observed associations are robust to potential confounding from these coexisting inflammatory conditions. The updated results are presented in the revised manuscript: **Results section (2.6 Results of replication validation analyses), Methods section (4.5 Statistical analysis)**, **and Supplementary Table 17.**

"Results 2.6 Results of replication validation analyses…*In addition, the associations between the nine proteins and incident CD remained significant after further adjustment for major chronic inflammatory comorbidities (****Supplementary Table 17****)."*

"Methods 4.5 Statistical analysis*…(5) repeating the association analyses between the nine selected proteins and CD using Cox models with additional adjustment for key chronic inflammatory comorbidities[48,49], including rheumatoid arthritis, diabetes mellitus, asthma, psoriasis, ankylosing spondylitis, multiple sclerosis, and ischemic heart disease, to assess the potential confounding by comorbid inflammatory conditions…."*

Supplementary Table 17: Associations of selected proteins with Crohn's disease after adjustment for chronic inflammatory comorbidities

Protein	***Description of protein***	***HR***	***Lower CI***	***Upper CI***	***P value***	***P value (Bonferroni-corrected)***
GDF15	*Growth/differentiation factor 15*	*2.14*	*1.70*	*2.69*	*6.27E-11*	*5.64E-10*
CD274	*Programmed cell death 1 ligand 1*	*2.49*	*1.76*	*3.53*	*2.99E-07*	*2.69E-06*
CHI3L1	*Chitinase-3-like protein 1*	*1.69*	*1.40*	*2.04*	*4.83E-08*	*4.35E-07*
DEFA1_DEFA1B	*Neutrophil defensin 1*	*1.63*	*1.29*	*2.06*	*3.62E-05*	*3.26E-04*
REG1B	*Lithostathine-1-beta*	*1.74*	*1.37*	*2.19*	*3.69E-06*	*3.32E-05*
PRSS8	*Prostasin*	*2.42*	*1.63*	*3.60*	*1.30E-05*	*1.17E-04*
IL6	*Interleukin-6*	*1.50*	*1.32*	*1.71*	*9.13E-10*	*8.22E-09*
ITGA11	*Integrin alpha-11*	*0.36*	*0.22*	*0.60*	*7.74E-05*	*6.97E-04*
ITGAV	*Integrin alpha-V*	*0.13*	*0.05*	*0.35*	*6.48E-05*	*5.84E-04*

Cox proportional hazards regression models were conducted in the training set to evaluate the associations between the nine proteins selected in protein model and incident Crohn's disease, after additional adjustment for chronic inflammatory comorbidities. The models were adjusted for age, sex, ethnicity, BMI, Townsend deprivation index, smoking, alcohol intake frequency, physical activity, dietary habits, anxiety, depression, medication history of antibiotics, medication history of NSAIDs, and seven inflammatory comorbidities (rheumatoid arthritis, diabetes mellitus, asthma, psoriasis, ankylosing spondylitis, multiple sclerosis, and ischemic heart disease). P-values were calculated using two-sided tests. Bonferroni corrections (0.05/9) were applied to correct for multiple comparisons.

References

1. Gordon, H. *et al.* ECCO Guidelines on Extraintestinal Manifestations in Inflammatory Bowel Disease. *Journal of Crohn's and Colitis* **18**, 1–37 (2024).
2. Argollo, M. *et al.* Comorbidities in inflammatory bowel disease: a call for action. *Lancet Gastroenterol Hepatol* **4**, 643–654 (2019).

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

The authors thoroughly responded to the comments that were raised. However, some concerns remain.

The authors have made substantial efforts and responded sufficiently to the comments that were raised. I appreciate the effort they have put into addressing these points. The addition of the EPIC-Norfolk cohort strengthens the paper and enhances the generalizability of predicting the onset of CD. I also thank the authors for showing the calibration curves, presenting the model performance without adjustment for class imbalance, and adding the discussion point regarding the caution needed in interpreting the results.

I have one remaining comment regarding the Southern China cohort. I do not think it is appropriate to state that biomarkers preceding CD were validated in a cross-sectional cohort with no prediagnostic samples. I suggest removing the label of "external validation" for the Southern China cohort, or considering removing this cohort altogether. The EPIC-Norfolk cohort already serves the purpose of external validation.

Authors' response:

We thank the reviewer for this constructive comment. As suggested, we have removed the label of "external validation" for the Southern China cohort in the revised manuscript. These modifications have been made in the corresponding parts of the manuscript, specifically in specifically in **Abstract, Results, Discussion, Methods, Fig. 1** and **Fig. 4**.

*"**Abstract** …It was externally validated in EPIC-Norfolk (n=2,944, AUC 0.73) and confirmed in the Southern China cohort (n=74, AUC 0.79)…"*

*"**Results Study population …**Baseline characteristics of the two cohorts are provided in Supplementary Tables 2 and 3….*

***Predictive accuracy of proteomics-based models**…AUCs ranged from 0.70 to 0.76 in the external EPIC-Norfolk study and from 0.76 to 0.79 in the Southern China cohort.…"*

*"**Discussion** …It was further externally validated in EPIC-Norfolk (AUC 0.73) and confirmed in the Southern China cohort (AUC 0.79).….*

…The model was further externally validated in EPIC-Norfolk (AUC 0.73) and demonstrated high discriminatory capacity in the Southern China cohort (AUC 0.79)….

…Remarkably, the predictive performance of our protein model was confirmed by external validation in an independent cohort…

…Fifth, the predictive performance of our protein model was confirmed in EPIC-Norfolk and the Southern China cohort. However, given the limited number of incident CD cases in EPIC-Norfolk and the cross-sectional design of the Southern China cohort, further validation in large-scale, new-onset inception cohorts with pre-diagnostic samples is warranted…"

"Methods Studying population *…Model performance was evaluated in two additional cohorts…. Following the same inclusion criteria as in the UKB cohort, this study included 2,944 individuals, serving as the external validation cohort…."*

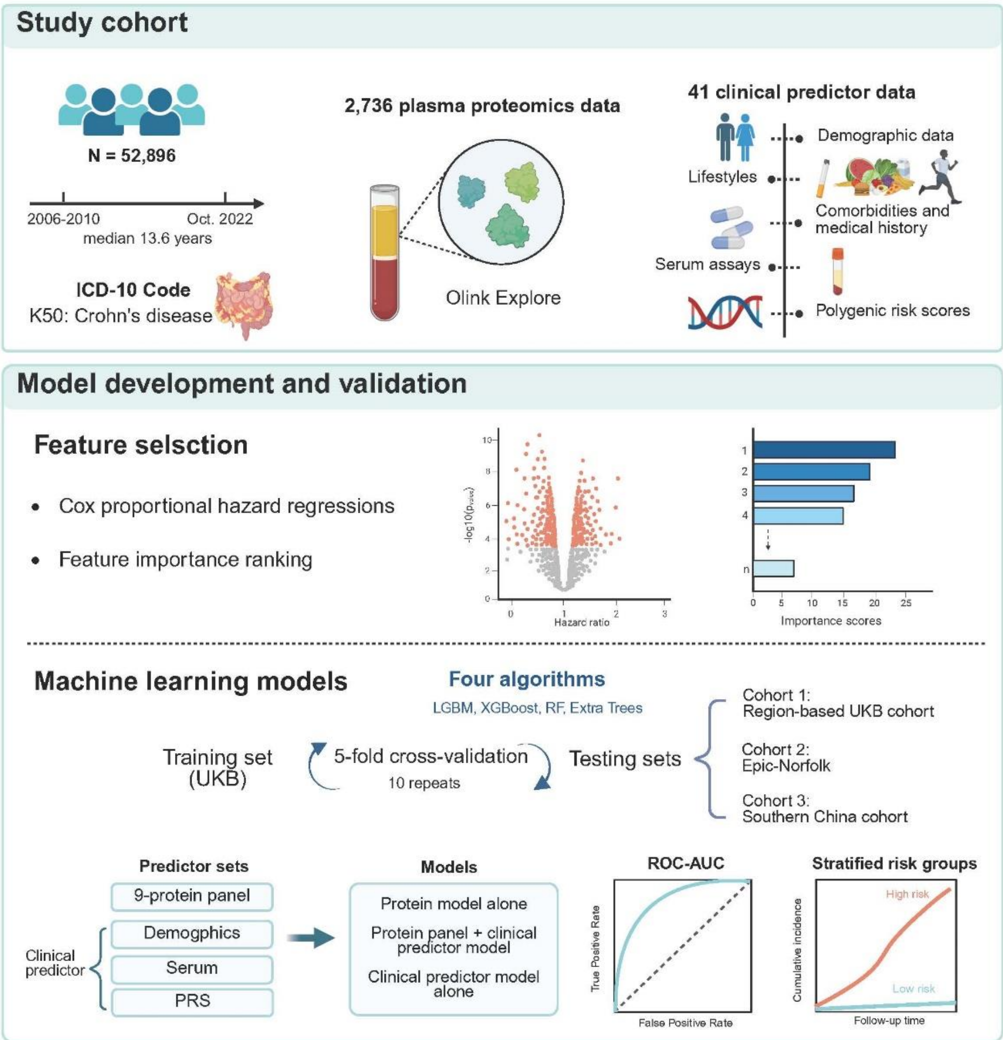


Fig. 1 Study overview. *First, we extracted data from 52,896 UK Biobank (UKB) participants with a median follow-up time of 13.6 years, including Crohn's disease (CD)*

endpoint defined by ICD10 codes, 2,736 plasma proteomics, and 41 clinical predictors spanning demographic, lifestyles, comorbidity and medication history, serum assays, and polygenic risk score (PRS) for CD. Next, Cox proportional hazard models and machine learning-based feature importance ranking were performed for feature selection, followed by model development with 10×5-fold internal cross-validation in the UKB training cohort. The performance of the protein model was then evaluated in a geographically distinct UKB cohort, the external EPIC-Norfolk study, and the Southern China cohort. Finally, we investigated the predictive performance of the protein model and its risk stratification for CD onset. Created in BioRender. Chen, H. (2025) https://BioRender.com/fc0f2j0. Abbreviations: LGBM, Light Gradient Boosting Machine; XGBoost, eXtreme Gradient Boosting; RF, Random Forest; PRS, polygenic risk score; ROC, receiver operating characteristic; AUC, area under the curve.

Fig. 4 Predictive accuracy of plasma proteins panel, alone or in combination with clinical variables. *Bar charts and dot plots show the area under the receiver operating characteristic curve (AUC) of different predictive models for all-time incident Crohn's disease (CD) in the geographically distinct UK Biobank testing cohort with 25% (**a**, n = 13,262) and 20% (**b**, n = 10,632) proportions, and the performance of the protein model in the external EPIC-Norfolk study (**c**, n = 2,944) and the Southern China cohort (**d**, n = 74). Each dot represents one of 10 repeated bootstrap analyses, with model performance estimated as the mean AUC from 1,000 resamplings. Bars show the mean AUC across the 10 runs, and error bars indicate 95% confidence intervals (CIs). Model performance evaluated using four machine learning algorithms: Light Gradient Boosting Machine (LGBM), eXtreme Gradient Boosting (XGBoost), Random Forest (RF), and Extra Trees. Demographic variables included age, while serum markers consisted of 11 indicators detailed in Supplementary Table 23. PRS represents the polygenic risk score for CD. The combined model included protein, demographics, serum, and PRS of CD. Source data are provided as a Source Data file.*

Reviewer #1 (Remarks on code availability):

The authors provided the R and Python scripts that were used for developing and validating the model.

Authors' response:

We thank the reviewer for confirming the availability of our code and for their support of our work.

Reviewer #4 (Remarks to the Author):

Review comments have been sufficiently addressed, thank you!

Authors' response:

We thank the reviewer for their thoughtful comments and constructive suggestions throughout the review process. Their input was instrumental in improving the quality of our manuscript.